Brouillon de manuscrit de thèse

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Introduction en français

Résumé Ce chapitre ne diffère du chapitre 1 que par sa rédaction en langue française. Dans ce chapitre, nous introduisons et motivons les travaux de recherche qui ont été conduits au cours de cette thèse. Les travaux sont motivés principalement par le besoin d'évolution des méthodes d'études sismiques probabilistes de sûreté, mais aussi par le manque de réponse qu'apporte l'état de l'art à la question du choix du prior en inférence bayésienne. Ces problématiques sont détaillées et les travaux sont alors introduits comme s'inscrivant dans celles-ci, le tout donnant lieu à une organisation consistante du manuscrit.

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2.1 Motivation et positionnement de la thèse

2.1.1 Etudes probabilistes de sûreté

Historique

Les études probabilistes de sûreté (EPS) désignent un ensemble de méthodes d'analyses techniques qui permettent de quantifier un risque encouru par une installation lorsqu'elle est sujette à un événement. L'événement peut être d'origine naturelle comme artificielle (technologique, humaine, etc), il peut être de provenance interne comme externe ; il peut s'agir d'un séisme, d'une inondation, d'une combinaison de défaillances internes, entre autres.

Faisant suite au premières recommandations de F. R. Farmer (expert sûreté à la UK atomic authority) dans les années 1960 pour la fiabilité des installations nucléaires, ces méthodes ont pour caractéristique commune l'introduction de la notion d'incertitude dans la qualification de l'événement, ses phénomènes et ses caractéristiques (on parle alors d'aléa). Leur cadre et leur concept ont été rapidement adoptés et développés aux Etats-Unis (cf. le rapport de la Nuclear Regulatory Commission, NRC, 1983). En particulier, de nombreuses études sont venues dès 1968 (Cornell, 1968) y inscrire les analyses de fiabilité parasismique, définissant alors le cadre des études sismiques probabilistes de sûreté (SPRA en anglais).

Le séisme représente en effet un facteur de risque remarquable des études de sûreté. Premièrement, bien que communément caractérisé par sa magnitude et sa distance à la source, son signal est bien plus riche qu'une fonction bivariée et deux séismes de même magnitude et de même source peuvent avoir des caractéristiques (et des conséquences) significativement différentes. Deuxièmement, puisqu'il touche à la fois tous les éléments aussi bien externes qu'internes à l'installation, il est la potentielle source de conséquences lourdes sur les équipements et structures. Le coût potentiel des conséquences d'un aléa sismique peut alors être élevé et critique dans le contexte nucléaire, ce qui en fait un événement d'intérêt majeur et décisif même dans les zones géographiques où il est rare.

Aujourd'hui, la prise en compte de l'aléa sismique sous le cadre des études probabilistes de sûreté est une recommandation internationale dans le contexte de l'industrie nucléaire. Leur cadre d'application dans l'industrie nucléaire française est précisé par l'autorité de sûreté nucléaire et de radioprotection (ASNR)[1] dans la règle fondamentale de sûreté (ASN, 2002).

L'évolution des EPS en France, au CEA, et dans cette thèse

L'évolution des méthodes et des connaissances relatives à la sûreté des installations nucléaires se fait de manière parallèle à l'évolution de la règle et de la norme imposées à celles-ci. Sur le sujet de l'aléa sismique, le rapport à l'évolution de la règle et de la méthode (et donc des EPS) oppose différents points de vue, principalement entre le principal exploitant (EDF) et les experts de l'autorité de sûreté (anciennement l'IRSN, maintenant unifié avec l'autorité de sûreté nucléaire devenue l'ASNR). Pour le premier, la robustesse de l'installation n'est normalement pas remise en question par l'avancée des connaissances et des méthodes puisque l'incertitude sur celles-ci fait parti des marges prises en comptes à la construction de l'équipement. Le second pense au contraire que la robustesse est une question perpétuelle et que la marge n'est pas faite pour être mordue au fil des connaissances qui s'ajoutent (Roger, 2020). L'ASNR se place alors en arbitre dans ce dialogue, entre autre elle définit le "séisme majoré de sécurité"[2] (il s'agit d'une majoration du spectre du "séisme maximal historiquement vraisemblable"[3]) qui produit la marge de référence dans la démonstration de robustesse parasismique d'un équipement.

Cet arbitrage est donc à la fois sensible et critique. L'incident survenu en 2011 à la centrale nucléaire de Fukushima-Daiichi le démontre. L'aléa sismique (qui est la cause du tsunami) de référence a été sous-évalué par le consensus des experts japonais, et c'est cette sous-évaluation qui est au final la cause de l'incident (cf. le rapport de l'agence internationale de l'énergie atomique, IAEA, 2015). Suite à cet événement, l'ASNR a pris la décision de majorer d'un facteur 1.5 le séisme majoré de sécurité en France, imposant une démonstration amplifiée de la robustesse des différentes installations par leur exploitant.

Dans ce cadre, la place du CEA est ambivalente dans l'échiquier nucléaire français. D'une part, il est exploitant d'installations nucléaires de recherche et joue alors son propre rôle quant à la démonstration de robustesse de ses équipements. Aussi, il participe conjointement à des actions de recherche relatives au parc nucléaire civil exploité par EDF. Une partie de la recherche et l'expertise sur les conséquences de l'aléa sismique se fait au CEA au laboratoire EMSI, qui dispose d'une plateforme expérimentale (Tamaris) qui permet de procéder à divers tests mécaniques sur des équipements sous séismes. L'approfondissement des méthodes des études sismiques probabilistes de sûreté est un enjeu du CEA qui s'inscrit dans les missions de ce laboratoire. Le CEA étant responsable devant les autorités de sûreté de ses installations, il cherche à développer des méthodes toujours rigoureuses pour justifier de leur tenue sismique.

Cette thèse s'inscrit dans la démarche d'évolution du SPRA et de démonstration de sûreté. Financée par le CEA dans le cadre d'un contrat de formation par la recherche, elle a pour objectif d'étoffer l'état de l'art en terme d'estimation de fragilité sismique des équipements et des installations. L'objectif est de développer des méthodes qui doivent le plus possible être (i) efficaces devant la complexité des objets étudiés et de l'anticipation de l'aléa, (ii) être robustes dans le temps et devant une potentielle réévaluation des critères de sûreté, et (iii) être transparentes et auditables par les autorités de sûreté.

[1] Anciennement ASN: l'autorité de sûreté nucléaire (ASN), a été unifiée avec l'institut de radioprotection et de sûreté nucléaire (IRSN) le 1er janvier 2025.

[2] Aujourd'hui plutôt appelé "séisme noyau dur".

[3] Aujourd'hui plutôt appelé "séisme de dimensionnement".

2.1.2 La quantification d'incertitudes dans les études probabilistes de sûreté

Principe et étapes de la quantification d'incertitudes

Les études probabilistes de sûreté s'appuient sur la prise en compte et la quantification des incertitudes qui apparaissent dans le procédé d'évaluation d'un risque au sein d'un système physique. La quantification d'incertitudes représente en fait une démarche qui consiste à modéliser et analyser méthodiquement l'incertitude, sa source, et sa propagation au sein de la modélisation du système physique étudié. Elle se construit alors à l'intersection de la physique, de l'ingénierie et des mathématiques appliquées, et s'érige même comme une branche des statistiques à part entière.

Elle concentre son étude sur 3 éléments qui décrivent l'interaction entre le système physique et l'aléa : des paramètres d'entrée $\mathbf{X}$, une réponse physique du système Y, et une modélisation de celle-ci selon : $Y = \mathcal{M}(\mathbf{X})$. Cette fonction $\mathcal{M}$ est liée à la modélisation, souvent complexe, du système et de ses propriétés physiques (il peut s'agir de la résolution d'équations physiques, de simulations numériques, ou même du résultat d'une expérience mécanique). La démarche générale de la quantification d'incertitudes est souvent décrite au travers de différentes étapes clés (voir par ex. Sudret, 2007; Iooss, 2009), à savoir :

1. L'identification des sources d'incertitudes : il est usuel de classifier les incertitudes selon deux grandes catégories. D'une part les incertitudes irréductibles, qui proviennent du hasard "naturel" embarqué dans l'aléa et le système physique. D'autre part, les incertitudes épistémiques, qui existent par manque d'information, et qui sont alors considérées comme réductibles en opposition aux précédentes (Hüllermeier and Waegeman, 2019). Il s'en suit une modélisation probabiliste de l'entrée $\mathbf{X}$.

2. La propagation des incertitudes : cette étape correspondant à l'approximation de la distribution du modèle $\mathcal{M}(\mathbf{X})$, et de l'évaluation d'une quantité d'intérêt qui dépend de celle-ci : comme une variance, une probabilité de défaillance, etc ; plus généralement une quantité de la forme $\mathbb{E}\,\phi(Y)$.

3. L'analyse de sensibilité : son rôle est de comprendre avec plus de recul la manière dont les incertitudes se propagent dans un système, en identifiant l'impact d'un ou de plusieurs des paramètres d'entrées sur la sortie Y (Iooss and Lemaître, 2015).

Des outils mathématiques en quantification d'incertitudes

De nombreux outils mathématiques viennent appuyer les étapes sus-mentionnées. Bien que ce manuscrit n'ait pas pour but d'en décrire un panel exhaustif, ci-dessous sont listés quelques exemples qui reviennent dans les travaux de cette thèse.

- La métamodélisation permet de contrevenir à la complexité d'évaluation du modèle $\mathcal{M}$, en construisant un méta-modèle $\tilde{\mathcal{M}}$, entraîné à partir d'une base de données d'observations $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_k, y_k)$. Le méta-modèle se veut par principe plus simple à évaluer que le modèle réel. Rien ne limite sa définition explicitement qui peut aller d'un simple modèle paramétrique à la sortie d'un réseau de neurones "boîte noire" ; parmi les plus communs on peut évoquer la méta-modélisation par processus gaussien (ou krigeage) et par polynômes du chaos.

- Les indices globaux de sensibilité s'inscrivent pleinement dans l'étape d'analyse de sensibilité du système. Ils constituent depuis les premiers travaux de Sobol' (1993) des outils essentiels pour mesurer statistiquement comment Y est impactée par un ou plusieurs des X_i (où $(X_1, \ldots, X_p) = \mathbf{X}$). Dans ce cadre, l'impact, noté S_i, de l'entrée X_i sur Y s'exprime comme une divergence moyenne entre la distribution $\mathbb{P}_Y$ de Y et sa distribution conditionnellement à $\mathbf{X}$, $\mathbb{P}_{Y|\mathbf{X}}$ (Da Veiga, 2015) :

$$S_i = \mathbb{E}_{X_i}[D(\mathbb{P}_Y || \mathbb{P}_{Y|X_i})], \tag{2.1}$$

  où D est une mesure de dissimilarité entre deux mesures de probabilité. Le choix de D est alors décisif dans l'étude de S_i. On peut noter, par exemple, que choisir D définie par $D(P||Q) = \|\mathbb{E}_{X\sim P}X - \mathbb{E}_{X\sim Q}X\|^2$ revient à étudier un indice de Sobol' du premier ordre. Le choix de D peut être motivé par divers intérêts, par exemple, celui de discriminer l'indépendance entre X_i et Y. Le point suivant détaille un large panel de choix possibles pour D.

- La théorie de l'information est au centre de la comparaison de mesures de probabilité et donc de la quantification d'incertitudes. Par essence, les manières de comparer la façon qu'ont deux mesures de probabilité de distribuer l'information dans un espace sont vastes. Dans la définition de S_i plus haut, le choix de D (et donc le choix de définition de S_i) relève de la théorie de l'information.

  Un choix alors commun, perçu comme une extension de l'entropie de Shannon, est celui des f-divergences de Csiszár (1967). On peut noter qu'elles sont employées dans des domaines variés, au-delà de l'analyse de sensibilité, tel que l'inférence variationnelle (Minka, 2005; Bach, 2023), le design de méta-modèles (Nguyen, Wainwright, and Jordan, 2009), l'apprentissage PAC-bayésien (Picard-Weibel and Guedj, 2022). Lorsque f est convexe et que $f(1) = 0$, elles sont définies selon :

$$D_f(P||Q) = \int_{\mathcal{X}} f\left(\frac{p(x)}{q(x)}\right) q(x) d\omega(x), \tag{2.2}$$

  en notant p, q des densités associées à P et à Q par rapport à une mesure commune ω sur leur ensemble de définition $\mathcal{X}$. Les f-divergences réduisent le choix de D à celui de f, les plus communément employées étant les δ-divergences (H. Zhu and Rohwer, 1995) définies en fixant $f = f_\delta$ où

$$f_\delta(x) = \begin{cases} \frac{x^\delta - \delta x - (1-\delta)}{\delta(\delta-1)} & \text{si } \delta \notin \{0,1\}, \\ x \log x - x + 1 & \text{si } \delta = 1, \\ -\log x + x - 1 & \text{si } \delta = 0. \end{cases} \tag{2.3}$$

  Il est remarquable que la plus connue des mesures de dissimilarité (la divergence de Kullback-Leibler) peut être vue comme un cas particulier des δ-divergences. En effet, rappelons que la divergence de Kullback-Leibler, notée KL, est définie par

$$\mathrm{KL}(Q||P) = \int_{\mathcal{X}} \log\left(\frac{q(x)}{p(x)}\right) q(x) d\omega(x) \tag{2.4}$$

  (en adoptant les mêmes notations qu'en équation (2.2)), si bien que

$$D_{f_\delta}(P||Q) = \mathrm{KL}(P||Q) \text{ si } \delta = 1, \quad D_{f_\delta}(P||Q) = \mathrm{KL}(Q||P) \text{ si } \delta = 0. \tag{2.5}$$

  Bien sûr, il existe aussi de nombreuses mesures de dissimilarité qui ne sont pas des f-divergences, à ce titre, les indices de Sobol' du premier ordre évoqués plus haut font offices d'exemple. Un autre exemple notable présenté ici est celui des normes maximales de discrépances (MMD). Soient P, Q deux mesures de probabilité définies sur un ensemble $\mathcal{X}$, et soit $\mathcal{H}$ un espace de Hilbert à noyau reproduisant (RKHS ; un rappel sur cette notion et ses définitions est proposé en annexe) sur $\mathcal{X}$, dont le noyau reproduisant est noté $k : \mathcal{X} \times \mathcal{X} \longrightarrow \mathbb{C}$. On définit la MMD (Gretton et al., 2012) par

$$\mathrm{MMD}(\mathcal{H}; P||Q) = \sup_{\substack{f \in \mathcal{H} \\ \|f\|_{\mathcal{H}} \leq 1}} |\mathbb{E}_{X\sim P} f(X) - \mathbb{E}_{X\sim Q} f(X)|, \tag{2.6}$$

  ou, dans une forme plus simple :

$$\mathrm{MMD}^2(\mathcal{H}; P||Q) = \mathbb{E}_{X,X'\sim P\otimes P}[k(X,X')] + \mathbb{E}_{Y,Y'\sim Q\otimes Q}[k(Y,Y')] - 2\mathbb{E}_{X,Y\sim P\otimes Q}[k(X,Y)]. \tag{2.7}$$

- L'analyse bayésienne est un outil incontournable de la quantification d'incertitude, puisqu'elle a pour principe fondamental d'introduire une forme d'incertitude dans un ou plusieurs des paramètres du modèle étudié. L'incertitude incorporée par le cadre bayésien peut prendre diverses formes, et se traduit au travers de la définition de l'*a priori*. La propagation d'incertitudes au travers de ce canevas se quantifie en calculant l'*a posteriori*, à partir d'observations statistiques $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_k, y_k)$. La définition de l'*a priori* est critique, elle prend place dans l'étape d'identification des incertitudes, et se révèle impactante sur l'*a posteriori*, et donc, dans un contexte d'étude probabiliste de sûreté, sur le risque.

  Puisque ce dernier outil, qui est en fait un paradigme à lui tout seul, est au centre de cette thèse, la section qui suit lui est pleinement dévouée.

2.1.3 Le choix du prior dans les études Bayésiennes

L'analyse bayésienne est une théorie des statistiques qui se construit selon une interprétation des probabilités qui sont vues comme des degrés de plausibilité des événements, on parle généralement de "crédibilité". Comme toute étude statistique, elle cherche à faire le lien entre des observations statistiques et leur distribution probabiliste. Elle s'appuie sur le théorème de Bayes, en considérant l'inconnue qui permet de définir la distribution des données comme étant aléatoire. Les observations statistiques informent alors la distribution de cette inconnue, ce qui met à jour le degré de crédibilité attribué à une distribution que suivraient les données par rapport à une autre.

Le raisonnement contraire, dit fréquentiste, considère la probabilité comme une fréquence d'occurrence observable au prix de reproduire l'expérience. Sous ce paradigme, l'inconnue existe, et les fréquences données par la répétition de l'expérience sont plus probables selon que l'on considère certaines valeurs de l'inconnue plutôt que d'autres, on parle généralement de "confiance".

Au final, il est important de noter que ces deux raisonnements se complètent, voire parfois se confondent suivant les cas, et ils sont loin d'être antinomiques. Dans un contexte pratique, le cas d'étude et ses spécificités peuvent néanmoins orienter le choix de favoriser l'un par rapport à l'autre.

Dans cette thèse, on se concentre sur l'analyse bayésienne dans le cadre de l'inférence statistique. Dans ce contexte, une variable d'intérêt $Y \in \mathcal{Y}$ est modélisée selon une distribution paramétrique $\mathbb{P}_{Y|\theta}$, $\theta \in \Theta$ étant l'inconnue, alors considérée comme étant une variable aléatoire dans le canevas Bayésien. En toute logique, θ admet une distribution, que l'on appelle le **prior**. Lorsque des réalisations de Y successives et indépendantes conditionnellement à θ sont observées, le **posterior** se définit comme la distribution de θ conditionnellement à ces observations. Une région de crédibilité à un niveau r se définit comme une région de Θ dont la probabilité selon le posterior est au moins égale à r. On notera que dans ce contexte, divers résultats lient les approches bayésienne et fréquentiste d'estimation de θ. Le théorème de Berstein Von-Mises (voir par ex. Vaart, 1992) en est un exemple.

Notons $\mathbf{y} = (y_1, \ldots, y_k)$ les valeurs observées de k réalisations de Y. En supposant que tous ces objets existent et sont bien définis, notons π la densité du prior (selon une mesure ν), $\ell(\cdot|\theta)$ les densités de $\mathbb{P}_{Y|\theta}$ (selon une mesure μ) et $\ell_k(\mathbf{y}|\theta) = \prod_{i=1}^k \ell(y_i|\theta)$, appelée la vraisemblance du modèle. Alors avec ces notations le théorème de Bayes définit la densité du posterior (par rapport à ν) selon :

$$p(\theta|\mathbf{y}) = \frac{\ell_k(\mathbf{y}|\theta)\pi(\theta)}{\int_\Theta \ell_k(\mathbf{y}|\theta)\pi(\theta)d\nu(\theta)}. \tag{2.8}$$

Cette formule fait le lien entre le prior, le modèle (représenté par la vraisemblance), les données, et le posterior qui traduit la prédiction crédible. Elle permet de schématiser le canevas bayésien comme une chaîne de transmission d'information vers l'*a posteriori*, depuis deux sources : le "réel" (les observations) et l'*a priori*. Ce schéma est celui présenté ci-dessous.

$$\begin{array}{rcl} \text{Base de données } \mathbf{y} = (y_i)_{i=1}^k \in \mathcal{Y}^k & \longrightarrow & \text{vraisemblance } \ell_k(\mathbf{y}|\theta) \\ \text{Paramètre } \theta \in \Theta & \longrightarrow & \text{prior } \pi(\theta) \end{array} \quad \longrightarrow \quad \text{posterior } p(\theta|\mathbf{y}) \propto \ell_k(\mathbf{y}|\theta)\pi(\theta)$$

Si l'information issue des données ne pose normalement pas question, c'est l'information *a priori* qui concentre l'attention. Par essence, elle est la source de l'incertitude dans le résultat, sa construction doit se faire de manière attentive lors de l'étape de modélisation et ne peut être arbitraire. Cependant, la formulation bayésienne lui attribue un impact significatif sur le posterior, particulièrement dans des cas où la taille de la base de données est limitée. Une illustration de ce phénomène est proposée en figure 2.1, où différents priors, en apparence proches, donnent lieu à des posteriors très différents, à partir des mêmes observations. Bien sûr, le diable est dans les détails, et ici, ce sont les queues de distribution des priors, peu discernables sur l'illustration, qui produisent cette différence. Malgré tout cet exemple questionne, à juste titre, sur le choix du prior.

La question de la construction du prior reste une question ouverte dans la littérature. Pour certains, elle est l'opportunité d'ajouter de l'information provenant d'une source extérieure, il peut s'agir d'un jugement

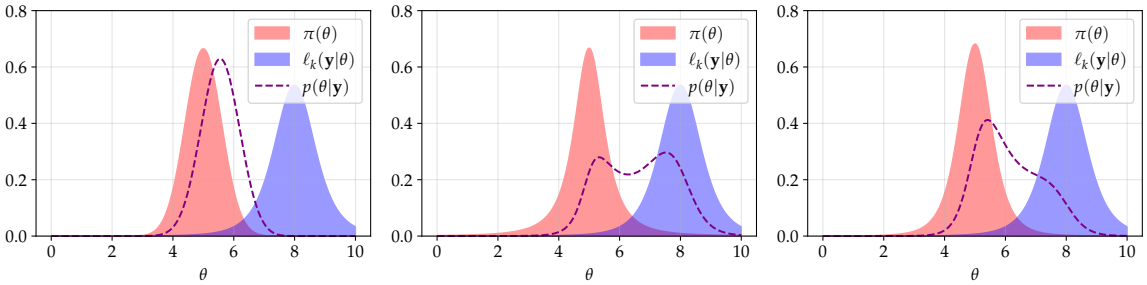

Figure 2.1: Différents calculs de posteriors (en pointillé) à partir d'une même vraisemblance (bleu) mais de différents priors (rouge).

d'expert, de données historiques, ou de connaissances sur certaines propriétés attendues du résultat. Leurs priors sont plutôt qualifiés d'informatifs. Pour d'autres, elle est au contraire le moyen d'introduire une incertitude dans le modèle, l'*a priori* est alors dans un tel cas une absence de connaissance, qui vient laisser l'information provenir essentiellement des données. Ces derniers ont confiance en la modélisation de la vraisemblance, et leur prior sera qualifié de non-informatif.

On comprend au travers de ces paragraphes introductifs que cette question est insoluble : il ne peut pas exister de méthode unanime de sélection de prior. Elle illustre même l'un des "trous" du workflow bayésien selon Gelman and Yao (2020) : les priors trop informés ne permettent pas d'avoir confiance en le moindre résultat tandis que les priors trop peu informés donnent lieu a des zones de crédibilité parfois trop larges et inutilisables. Ce n'est pas pour autant que l'approche bayésienne perde de son intérêt et cette section n'est pas vouée à discréditer la démarche, la suite du manuscrit le prouvera. Mais elle doit laisser celui ou celle qui l'emploie conscient ou consciente des impacts et des limites de ses choix. Le choix d'un prior est aussi critique que le choix du modèle lui-même.

2.1.4 Ce qui motive la recherche sur l'élicitation de priors pour les études sismiques probabilistes de sûreté

L'analyse bayésienne a gagné en popularité dans de nombreux domaines, y compris dans des études de fiabilité ou de sûreté. En effet, elle est plébiscitée pour sa capacité (i) à introduire et propager une incertitude dans un problème d'inférence, et (ii) à être plus régulière que beaucoup de méthodes fréquentistes, en particulier lorsque le nombre de données observées est limité.

Le cadre de l'inférence bayésienne s'introduit en effet bien dans la démarche de quantification d'incertitudes. L'identification des incertitudes sur l'entrée $\mathbf{X}$ ou sur le modèle même $\mathcal{M}$ (ou bien s'il y a lieu le méta-modèle) peut se faire au travers d'une paramétrisation de celles-ci, en donnant lieu à une distribution paramétrique de Y : $\mathbb{P}_{Y|\mathbf{X},\theta}$. La variable θ, aléatoire sous le paradigme bayésien, peut alors représenter diverses sources d'incertitudes, de différentes natures selon les cas (Bousquet, 2024a). Cette approche a été employée à pléthore dans divers types de modèles, on peut citer par exemple les cas des processus gaussiens (par ex. Gu and Berger, 2016) ou des réseaux de neurones (Arbel et al., 2023).

Parfois, certaines études adoptent la démarche pour la possibilité qu'elle offre d'introduire un jugement ou une connaissance *a priori*. C'est pourtant ce même point qui fait aussi débat puisqu'il est la potentielle source d'une subjectivité difficile à justifier. Dans le cadre des études sismiques probabilistes de sûreté en industrie nucléaire, la démonstration de robustesse est centrale. Les cas d'études y étant souvent très complexes, on trouvera alors parfois autant de priors qu'il y a d'experts en suivant cette démarche. De tels cas caricaturaux mettent en péril l'auditabilité de la moindre région de crédibilité *a posteriori*.

On comprend en fait que dans une étude de sûreté qui se veut auditable, une "bonne" région de crédibilité sur un paramètre estimé n'est pas une région nécessairement étroite, même si elle suggère que la sûreté est satisfaite. A l'inverse, une région trop large n'est pas pour autant "bonne" car elle ne permet pas de démontrer la robustesse attendue de l'équipement. L'équilibre est complexe, et le "bon" prior est celui qui, bien que démontrant au mieux la capacité ou l'incapacité de l'équipement à résister à l'aléa, produit un

résultat inspirant confiance.

Toutes ces réflexions mettent en lumière l'intérêt d'une construction méthodique du prior dans le cadre des études sismiques probabilistes de sûreté, au travers d'un cadre rigoureux qui vise à éviter, dans la mesure du possible, l'introduction de la moindre subjectivité dans la démarche.

2.2 Esquisse du manuscrit et contributions

2.2.1 Problématiques et plan

Plusieurs questions principales se détachent de cette introduction et de ces premières réflexions. Elles sont listées ci-dessous, et décrivent les problématiques majeures qui se sont manifestées au cours de cette thèse et auxquelles les travaux présentés dans ce manuscrit cherchent à répondre.

Question i Comment peut-on définir et appuyer l'objectivité d'un prior ?

Question ii Quelles sont les limites de l'emploi d'un tel prior, peu informatif par essence, et comment concilier son usage avec les besoins pratiques ?

Question iii Comment construire et implémenter de tels priors dans un cas pratique ?

Question iv Dans le contexte des études sismiques probabilistes de sûreté, comment s'implémentent ces priors objectifs dans un modèle concret d'estimation de fiabilité sismique ?

Question v Quelles conséquences peut avoir le manque d'information *a priori* ou provenant des données dans ce modèle et comment les solutionner ?

Question vi Comment peut-on finalement bénéficier au mieux des différentes sources d'information dans le workflow bayésien du modèle étudié dans son ensemble ?

Dans cette thèse, nous tentons de répondre à ces six problématiques au travers d'une étude selon deux axes. Le premier axe a une dimension qualifiée de théorique, en adressant les problématiques liées à la construction de prior dits "objectifs" dans son ensemble. Il se concentre sur le développement d'une théorie appelée la théorie des priors de référence. Le second axe est alors plutôt qualifié de pratique, en étudiant la mise en œuvre des développements théoriques sur des cas d'études réels, en se concentrant sur l'estimation de courbes de fragilité sismique, qui représentent un outil important du cadre des études sismiques probabilistes de sûreté. Il reste important de rappeler que, bien qu'indépendants, les deux grands axes de travail sus-mentionnés s'alimentent l'un l'autre. Ce sont autant les problématiques pratiques qui ont motivé les recherches théoriques, que les découvertes théoriques qui ont été mises au service des études pratiques.

La **partie I** de ce manuscrit est alors dévouée au développement de la théorie des priors de référence.

Dans le chapitre 3, on introduit la théorie et on développe son état de l'art. Il permet d'introduire la notion de prior de référence telle qu'elle est communément définie dans la littérature, et sa place parmi les priors objectifs. Ce chapitre permet aussi de formaliser le cadre mathématique de travail pour la suite de la partie.

Dans le chapitre 4, on observe que la définition des priors de référence s'appuie sur une mesure de dissimilarité. On cherche alors dans ce chapitre à généraliser la définition de cette dernière afin d'appuyer le caractère objectif des priors de référence. Ce chapitre apporte une réponse à la **question i**.

Dans le chapitre 5, on cherche une ouverture au cadre des priors de référence, lorsqu'il donne lieu à des priors difficilement utilisables en pratique. On propose alors une définition de ce qu'on appelle priors de référence contraints. Les solutions de cette définition apportent une réponse à la **question ii**.

Dans le chapitre 6, on développe une méthode numérique d'approximation des priors de référence. Cette méthode s'appuie sur l'inférence variationnelle en définissant le prior comme la sortie d'un réseau de neurones, afin d'éviter un calcul explicite de celui-ci. Ce chapitre apporte une réponse à la **question iii**.

La **partie II** de ce manuscrit aborde le sujet de l'estimation bayésienne dite objective des courbes de fragilité sismique.

Dans le chapitre 7, on introduit les courbes de fragilité sismique, leur définition, leur historique et leur intégration dans le cadre des études sismiques probabilistes de sûreté. Un état de l'art des méthodes d'estimation de ces courbes suivant les différents types de données à disposition y est proposé.

Dans le chapitre 8, on propose un calcul et une étude en profondeur du prior de référence pour un modèle classique employé pour l'estimation de ces courbes de fragilité. L'inférence bayésienne via l'emploi de ce prior y est comparée à d'autres méthodes. Ce chapitre apporte une réponse à la **question iv**.

Dans le chapitre 9, on s'intéresse en profondeur aux limites du modèle considéré pour l'estimation des courbes de fragilité. On propose une solution qui est appuyée par les développements théoriques de la première partie. Ce chapitre apporte une réponse à la **question v**.

Dans le chapitre 10, on termine notre étude en proposant une méthode qui vient appuyer l'estimation bayésienne en optimisant l'information issue des données non plus seulement lors du choix du prior mais également lors de la sélection de celles-ci, au travers d'une méthode de planification d'expériences. Ce chapitre apporte une réponse à la **question vi**.

Une conclusion générale viendra clore le manuscrit dans une **partie III** finale.

2.2.2 Liste des contributions

La recherche conduite au cours de cette thèse a donné lieu à plusieurs contributions dans la littérature scientifique. Ci-après sont listés les travaux publiés pendant celle-ci.

Articles de revues

- **A. Van Biesbroeck**, C. Gauchy, C. Feau & J. Garnier (2025). "Robust a posteriori estimation of probit-lognormal seismic fragility curves via sequential design of experiments and constrained reference prior". *arXiv* 2503.07343. DOI: 10.48550/arXiv.2503.07343.
- N. Baillie, **A. Van Biesbroeck** & C. Gauchy (2025). "Variational inference for approximate reference priors using neural networks". *arXiv* 2502.02364. DOI: 10.48550/arXiv.2502.02364
- **A. Van Biesbroeck** (2024). "Properly constrained reference priors decay rates for efficient and robust posterior inference", *arXiv* 2409.13041. DOI: 10.48550/arXiv.2409.13041
- **A. Van Biesbroeck**, C. Gauchy, C. Feau & J. Garnier (2025). "Design of experiments based on a low fidelity model for seismic fragility curves estimation". *ESAIM: ProcS* (to appear). HAL: hal-04719458v1
- **A. Van Biesbroeck** (2023). "Generalized mutual information and their reference priors under Csizar f-divergence". *arXiv* 2310.10530. DOI: 10.48550/arXiv.2310.10530
- **A. Van Biesbroeck**, C. Gauchy, C. Feau & J. Garnier (2024). "Reference prior for Bayesian estimation of seismic fragility curves". *Probabilistic Engineering Mechanics*, 76, pp 103622. DOI: 10.1016/j.probengmech.2024.103622

Articles de conférences

- **A. Van Biesbroeck**, C. Feau & J. Garnier (2025). "Design of experiments for efficient and conform Bayesian learning of seismic fragility curves". *Proceedings of the 28th conference on Structural Mechanics in Reactor Techology (SMiRT).*
- N. Baillie, **A. Van Biesbroeck**, C. Feau & C. Gauchy (2025). "Bayesian estimation of seismic fragility curves based on variational reference priors using neural networks". *Proceedings of the 6th Thematic Conference on Uncertainty Quantification in Computational Sciences and Engineering (UNCECOMP).*

- **A. Van Biesbroeck**, C. Gauchy, J. Garnier & C. Feau (2023). "Connections between reference prior theory and global sensitivity analysis, an illustration with f-divergences". *Proceedings des 54èmes Journées de Statistiques (JdS)*. HAL: hal-04171446
- **A. Van Biesbroeck**, C. Gauchy, C. Feau & J. Garnier (2023). "Influence of the choice of the seismic intensity measure on fragility curves estimation in a Bayesian framework based on reference prior". *Proceedings of the 5th Thematic Conference on Uncertainty Quantification in Computational Sciences and Engineering (UNCECOMP)*, pp. 94-111. DOI: 10.7712/120223.10327.19899

I

Contribution to the reference prior theory

3

Review of the reference prior theory

Abstract The reference prior theory is a prominent tool when one researches for a prior embedded with an objective construction. It is designed to ensure the information in the posterior comes promptly from the observations rather than from the prior itself. In this chapter, the theory is reviewed, and its formalism is rigorously introduced. We recall the main results about the reference priors that are proven in the literature. Additionally, because we find the usual Bayesian framework limiting for the study of reference priors, we propose a novel Bayesian formalism that provides a rigorous incorporation of improper priors. The latter fixes the notations we use in rest of the manuscript.

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3.1 Introduction

Bayesian analysis offers a coherent framework for integrating prior information and updating beliefs with data. The incorporation of prior knowledge can be motivated by various needs, such as enhancing interpretability, introducing uncertainty, or embedding existing knowledge into the analysis. Actually, the selection of this a priori centralizes a lot of attention in Bayesian inference and can be a critical aspect of the method. As a matter of fact, it can significantly influence the posterior distribution and, consequently, the conclusions drawn from the analysis. Thus, an inconsistent or insufficiently justified choice of prior can jeopardize the validity of the entire study, a concern now well identified in the modern Bayesian workflow described in Gelman, Vehtari, et al., 2020, and echoed across applied domains.

A plethora of approaches exist in the literature to address the prior elicitation problem. They offer different strategies for different purposes. Comprehensive reviews of such methods are proposed in Mikkola et al., 2023 and in Consonni et al., 2018, for instance. Among the different challenges that are commonly addressed by these methods, we can emphasize (i) the incorporation of information, (ii) the ease of posterior inference, or (iii) the research for interpretability.
The former refers to the so-called "informative priors", because their distributions tend to concentrate the information. They aim to incorporate genuine prior knowledge about parameters, often arising from past data, domain expertise, physical constraints, or theoretical understanding. The difficulty lies in doing so transparently, reproducibly, and justifiably, especially when seeking to avoid subjective bias.

Regarding the second, we can recall conjugate priors (see e.g. Robert, 2007) that are designed to provide a simple formulation of the posterior. Penalized Complexity (PC) priors (Simpson et al., 2017) and sparse priors (Castillo, Schmidt-Hieber, and Vaart, 2015) are made to be suitable for high-dimensional problems; and g-priors (Liang et al., 2008) are used for variable selection in normal regression models.

Thirdly, in order to provide a better interpretability of the information transmission process, several studies favor a hierarchical design, in which the prior is the result of a latent prior modeling. This approach allows the use of historical data, as in power priors (Chen and Ibrahim, 2006) and Information Matrix priors (M. Gupta and Ibrahim, 2009); or even imaginary data (Pérez and Berger, 2002; Spitzner, 2011). Posterior priors, developed for sensitivity analysis studies (Bousquet, 2024b), are also part of this approach. These hierarchical constructions are appreciated for their ability to tackle the improper aspect (they integrate to infinity) of most low-informative priors (priors whose distribution is more diffuse through the parameter's space, in opposition with informative priors).

Many of these methods involve subjective choices that remain open to criticism. Often, they are based on the selection of a class and hyperparameters that remain to be tuned.

The research for so-called "objective priors" arose in contexts where the incorporation of uncritical beliefs is impossible (either by lack of available beliefs, or by lack of confidence within the existing ones). In those contexts, there is consensus that an appropriate prior belongs within the non-informative ones. In this respect, Lindley (1956) already suggested the maximization of the Shannon's entropy, as a way to choose a prior that would provide the least possible information. The same philosophy has been the source of several rules, which lead to different definitions of objective priors. A review of the existing rules and their application on simple examples is proposed by Kass and Wasserman, 1996 and by Berger, Bernardo, and Sun (2015). We also mention the work of Datta and M. Ghosh (1996), in which the question of invariance by re-parametrization of several of these rules is thoroughly focused.

In this chapter and in this thesis, we focus on the reference prior theory, first introduced by Bernardo (1979a). Based on an "expected utility maximization", the idea was to shift from minimizing the information in the prior to maximizing the knowledge brought by the observations over prior distribution itself. This notion of expected information, is defined with the support of the mutual information which allows for a formal construction of priors that are designed to let the data dominate the inference.

The reference prior theory has been extensively studied since its deepen formalization by Berger, Bernardo, and Sun (2009). The derivation of reference priors and the study of their properties in different contexts has been extensively focused. We can mention the works in Berger and Sun, 2008 for bivariate normal models, Berger and Bernardo, 1992b for multinomial models, and Gu and Berger, 2016 for Gaussian process models, among others. More recent studies on the theory are scarce, yet we can we cite the work of Bodnar and Elster (2014), who suggest the derivation of reference priors using sequential maximization of Shannon's entropy, and the work of Gao, Ramesh, and Chaudhari (2022) who consider reference priors to the training of deep neural networks. We mention that Muré (2018) provides a comprehensive review of the theory, and introduces original results about the uniqueness of the reference priors.

This chapter proposes a novel brief review of the reference prior theory. We fix the framework and define important main tools for the theory in section 3.2. The reference priors are formally defined in section 3.3, and we review their different properties that are proven in the literature. Eventually, this chapter is the occasion to discuss the limitations of the standard framework of the reference priors, and to suggest a solution based on an original formalization of the Bayesian inference tools. The latter is done in section 3.4 and fixes the

mathematical elements and notations that we will manipulate throughout the whole manuscript. A short conclusion terminates the chapter in section 3.5.

3.2 The standard Bayesian framework

3.2.1 A minimal framework

The Bayesian framework is commonly defined from the given of a statistical model. Let $(\Omega, \mathscr{P}, \mathbb{P})$ be a probability space, and define Y, a random variable defined on Ω and taking values on a measurable space $(\mathcal{Y}, \mathscr{Y})$. A statistical model is characterized by a collection of parameterized probability measures $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$ on $(\mathcal{Y}, \mathscr{Y})$. The Bayesian viewpoint considers a random variable $T : \Omega \longrightarrow \Theta$ taking values in a measurable space $(\Theta, \mathscr{T})$, following a **prior distribution** denoted Π, and which is such that the conditional distribution of Y to $T = \theta$ is $\mathbb{P}_{Y|\theta}$. In other terms:

$$\forall A \in \mathscr{Y},\, B \in \mathscr{T},\, \mathbb{P}(Y \in A,\, T \in B) = \int_B \mathbb{P}_{Y|\theta}(A) d\Pi(\theta). \tag{3.1}$$

In practice, k realizations $\mathbf{y} = (y_1, \dots, y_k)$ of the r.v. Y are observed, they are supposed to be identically distributed and independent conditionally to T, meaning they are the realization of a random vector $\mathbf{Y}_k = (Y_1, \dots, Y_k)$, whose distribution $\mathbb{P}_{\mathbf{Y}_k}$ is defined on $(\mathcal{Y}^k, \mathscr{Y}^{\otimes k})$ by:

$$\forall A \in \mathscr{Y}^{\otimes k},\, \mathbb{P}_{\mathbf{Y}_k}(A) = \int_\Theta \mathbb{P}_{\mathbf{Y}_k|\theta}(A) d\Pi(\theta), \tag{3.2}$$

where $\mathbb{P}_{\mathbf{Y}_k|\theta} := \mathbb{P}_{Y|\theta}^{\otimes k}$. The distribution $\mathbb{P}_{\mathbf{Y}_k}$ is called the **marginal distribution**.

Given the observations $\mathbf{y}$, the **posterior distribution**, denoted $\mathbb{P}_{T|\mathbf{y}}$, is then defined as the $\mathbb{P}_{\mathbf{Y}_k}$-a.s. unique one verifying

$$\forall A \in \mathscr{Y}^{\otimes k},\, \forall B \in \mathscr{T},\, \int_A \mathbb{P}_{T|\mathbf{y}}(B) d\mathbb{P}_{\mathbf{Y}_k}(\mathbf{y}) = \int_B \mathbb{P}_{\mathbf{Y}_k|\theta}(A) d\Pi(\theta). \tag{3.3}$$

We can notice that the posterior distribution is always absolutely continuous w.r.t. the prior distribution. Its existence and $\mathbb{P}_{\mathbf{Y}_k}$-a.s. uniqueness are ensured by the disintegration theorem (see e.g. Chang and Pollard, 1997), while $(\mathcal{Y}, \mathscr{Y})$ and $(\Theta, \mathscr{T})$ are Borel spaces.

3.2.2 Likelihood and densities

It is common to assume that the statistical model admits a likelihood: there exists a collection of density functions $(\ell(\cdot|\theta))_{\theta\in\Theta}$ w.r.t. a common σ-finite measure μ on $\mathcal{Y}$ such that for Π-a.e. $\theta \in \Theta$:

$$\forall A \in \mathscr{Y},\, \mathbb{P}_{Y|\theta}(A) = \int_A \ell(y|\theta) d\mu(y). \tag{3.4}$$

Example 3.1 (Multinomial model)**.** The multinomial model is an example of statistical model where $\theta \in (0,1)^d$, $\mathcal{Y} = \{y \in \{0, \dots, n\}^d, \sum_{i=1}^d y^i = n\}$ for a fixed $n \in \mathbb{N}^*$, and with $\ell(y|\theta) = \frac{n!}{y^1!\dots y^d!} \prod_{i=1}^d \theta_i^{y^i}$.

This way, $\mathbb{P}_{\mathbf{Y}_k}$ admits a **marginal density** $p_{\mathbf{Y}_k}$ w.r.t. $\mu^{\otimes k}$ defined as

$$\forall \mathbf{y} \in \mathcal{Y}^k,\, p_{\mathbf{Y}_k}(\mathbf{y}) = \int_\Theta \prod_{i=1}^k \ell(y_i|\theta) d\Pi(\theta) = \int_\Theta \ell_k(\mathbf{y}|\theta) d\Pi(\theta), \tag{3.5}$$

where $\ell_k(\mathbf{y}|\theta)$ denotes $\prod_{i=1}^k \ell(y_i|\theta)$ and is called the likelihood.

Also, we can suppose that Π admits a **prior density** π w.r.t. a σ-finite measure ν on Θ (usually, it is simply assumed that $\Theta \subset \mathbb{R}^d$ and ν is the Lebesgue measure). The posterior distribution admits a **posterior density** $p(\cdot|\mathbf{y})$ w.r.t. ν defined by

$$\forall \theta \in \Theta,\, p(\theta|\mathbf{y}) = \frac{\ell_k(\mathbf{y}|\theta)\pi(\theta)}{p_{\mathbf{Y}_k}(\mathbf{y})} \text{ if } p_{\mathbf{Y}_k}(\mathbf{y}) \neq 0,\; p(\theta|\mathbf{y}) = 1 \text{ otherwise.} \tag{3.6}$$

To conclude this section, we acknowledge the usual confusion between the notation for the random variable T and notation for the values it takes θ. Thus, we might refer as "the distribution of θ" or the "distribution conditionally to θ" to respectively mention the prior distribution or the conditional distributions to $T = \theta$.

3.2.3 Improper priors

It is common when dealing with non-informative priors in Bayesian analysis to define improper ones. Improper priors are defined as distributions admitting a density π that integrates to infinity: $\int_\Theta \pi(\theta)d\nu(\theta) = \infty$.

Of course, such a prior is not a probability distribution anymore, so that the framework elucidated in sections 3.2.1 and 3.2.2 does not stand. A way-around to be able to use improper priors is to focus on restrictions of the model, as formalized in the definition below.

Definition 3.1. As ν is a σ-finite measure, there exists an increasing sequence (B_n) of elements in Θ such that $\Theta = \bigcup_{n\in\mathbb{N}} B_n$ and $\nu(B_n) < \infty$. An improper prior is said to be admissible if for any n, $\int_{B_n} \pi(\theta)d\nu(\theta) < \infty$, and if there exists n_0 such that $\int_{B_{n_0}} \pi(\theta)d\nu(\theta) > 0$.

In this case, on any $\tilde{\Theta} \in \mathscr{T}$ such that there exists $N \in \mathbb{N}$ verifying $\tilde{\Theta} \subset \bigcup_{n\geq n_0}^N B_n$ and such that $B_{n_0} \subset \tilde{\Theta}$, one can define the probability space $(\tilde{\Theta}, \tilde{\mathscr{T}}, \tilde{\Pi})$ with:

$$\tilde{\mathscr{T}} = \{B \cap \tilde{\Theta},\, B \in \mathscr{T}\}, \quad \forall \tilde{B} \in \tilde{\mathscr{T}},\, \tilde{\Pi}(\tilde{B}) = \frac{\int_{\tilde{B}} \pi(\theta)d\nu(\theta)}{\int_{\tilde{\Theta}} \pi(\theta)d\nu(\theta)}. \tag{3.7}$$

That modeling is the restricted modeling driven by $\tilde{\Theta}$. The probability $\tilde{\Pi}$ (respectively its density) is referred as the renormalized restriction of the prior Π (resp. of the prior density π) to $\tilde{\Theta}$.

On any restricted model, the posterior and the marginal distributions exist. Given two restrictions driven by $\tilde{\Theta}_1$ and $\tilde{\Theta}_2$, the renormalized restricted prior densities $\tilde{\pi}_1$ and $\tilde{\pi}_2$ are equal on $\tilde{\Theta}_1 \cap \tilde{\Theta}_2$ up to a constant:

$$\tilde{\pi}_1(\theta) = K\tilde{\pi}_2(\theta), \quad \text{so that} \quad \forall \mathbf{y} \in \mathcal{Y}^k,\, \tilde{p}_{1,\mathbf{Y}_k}(\mathbf{y}) = K\tilde{p}_{2,\mathbf{Y}_k}(\mathbf{y}), \tag{3.8}$$

where $\tilde{p}_{1,\mathbf{Y}_k}$ (reps. $\tilde{p}_{2,\mathbf{Y}_k}$) denotes the marginal density in the restricted model driven by $\tilde{\Theta}_1$ (resp. $\tilde{\Theta}_2$). Thus, calling $\tilde{p}_1(\cdot|\mathbf{y})$ (resp. $\tilde{p}_2(\cdot|\mathbf{y})$) the posterior density given the observations $\mathbf{y}$ and under the restricted model driven by $\tilde{\Theta}_1$ (resp. $\tilde{\Theta}_2$), we have

$$\forall \theta \in \tilde{\Theta}_1 \cap \tilde{\Theta}_2,\, \tilde{p}_1(\theta|\mathbf{y}) = \tilde{p}_2(\theta|\mathbf{y}). \tag{3.9}$$

We conclude that the posterior density is always well-defined on Θ when the improper prior is admissible. If the posterior density integrates to ∞, the posterior is said to be improper.

This construction is called a way-around because the framework remains incomplete: T is not embedded with a probability distribution anymore, and the marginal distribution $\mathbb{P}_{\mathbf{Y}_k}$ is not appropriately defined. Later in this chapter, we will introduce a novel construction that authorizes the definition of improper priors in a more suitable way. Before that, the current construction is enough to define the reference priors among the admissible priors: priors that are either proper, either admissible improper priors.

3.3 Mutual information and reference priors

3.3.1 Mutual information

The principle of the theory is to minimize the role of the prior within the posterior, in order to maximize the information gain from the data. Thus, it amounts to maximize how "away" is the prior to the posterior.

When it resorts to compare information between probability distributions, we generally consider dissimilarity measures. The most common one is the Kullback-Leibler divergence, it is the one that is considered in the historical settings of the reference prior theory. We recall below the expression of the Kullback-Leibler divergence between two distributions P and Q on $\mathcal{X}$, which admit densities p, q w.r.t. a common measure ω:

$$\mathrm{KL}(Q||P) = \int_{\mathcal{X}} \log\left(\frac{q(x)}{p(x)}\right) q(x)d\omega(x). \tag{3.10}$$

We extend the definition of the Kullback-Leibler divergence when P and Q are not probability distributions, but are such that $P(\mathcal{X}), Q(\mathcal{X}) < \infty$ as $\mathrm{KL}(Q||P) := \mathrm{KL}(\tilde{Q}||\tilde{P})$ where $\tilde{P}$, $\tilde{Q}$ respectively equal $P/P(\mathcal{X})$ and $Q/Q(\mathcal{X})$.

Thus, the idea is to maximize the divergence $\mathrm{KL}(\mathbb{P}_{T|\mathbf{y}}||\Pi)$. As the observed sample $\mathbf{y}$ is unknown, the average value of the divergence is considered:

Definition 3.2 (Mutual information)**.** Given a Bayesian framework defined as in section 3.2 with a proper prior Π, and given a number of observations k, the mutual information is defined as the quantity:

$$\mathscr{I}^k(\Pi) := \mathbb{E}_{\mathbf{Y}_k \sim \mathbb{P}_{\mathbf{Y}_k}}[\mathrm{KL}(\mathbb{P}_{T|\mathbf{Y}_k}||\Pi)] = \int_{\mathcal{Y}^k} \mathrm{KL}(\mathbb{P}_{T|\mathbf{y}}||\Pi) p_{\mathbf{Y}_k}(\mathbf{y}) d\mu^{\otimes k}(\mathbf{y}). \tag{3.11}$$

We recall that the definition of the mutual information is not limited to the scope of Bayesian analysis. In information theory it is common to refer to the mutual information $\mathscr{I}(X, Z)$ between a random variable X and a random variable Z (see e.g. MacKay, 2003). The quantity in definition 3.2 aligns with their definition of the mutual information $\mathscr{I}(T, \mathbf{Y}_k)$ between the random parameter T and the random data $\mathbf{Y}_k$.

One can apply Fubini-Lebesgue's theorem to write the mutual information as

$$\mathscr{I}^k(\Pi) = \mathbb{E}_{\mathbf{Y}_k \sim \mathbb{P}_{\mathbf{Y}_k}}[\mathrm{KL}(\mathbb{P}_{T|\mathbf{Y}_k}||\Pi)] = \mathbb{E}_{T \sim \Pi}[\mathrm{KL}(\mathbb{P}_{\mathbf{Y}_k|T}||\mathbb{P}_{\mathbf{Y}_k})] \tag{3.12}$$

This second expression allows an interpretation of the mutual information from another viewpoint: it measures how the parameter T impacts the distribution of $\mathbf{Y}_k$.

3.3.2 Reference priors

Reference priors are defined as the priors that are expected to maximize the role of the observed data within the posterior distribution. Consequently, they are historically sought as maximizers of the mutual information.

However, formally maximizing the mutual information $\mathscr{I}^k(\Pi)$ w.r.t. Π presents three main issues:

1. First, its expression depends on the value of the number of observations k, and k alters the form of its maximal argument. This problem is tackled arguing that $\mathscr{I}^k$ only measures a limited quantity of information brought by T onto the distribution of the data $\mathbf{Y}$. The Bayesian framework does considerate the data $(Y_1, \ldots, Y_k)$ to be non-independent, so that the distribution of $\mathbf{Y}$ is enriched as k rises. According to Bernardo and Smith (1994), the limit $\mathscr{I}^\infty$ (if it exists) measures the knowledge missing from the prior. Thus, the formal definition tends to maximize $\mathscr{I}^k$ as k diverges to ∞.

2. The second issue comes with the fact that the quantity $\mathscr{I}^k(\Pi)$ generally does not admit a finite limit as k diverges to ∞.

3. The third issue is that the mutual information is only correctly defined when the prior is proper, while many non-informative priors used in Bayesian analysis are improper.

The commonly admitted definition for the reference priors, which takes into account the three issues aforementioned is the following, adapted from Bernardo, 2005.

Definition 3.3 (Reference priors)**.** Let $\mathcal{P}$ be a set of priors on Θ, a prior $\Pi \in \mathcal{P}$ is a reference prior over $\mathcal{P}$ if there exists an appropriate increasing sequence $(\Theta_i)_{i \in \mathbb{N}}$ such that $\bigcup_{i \in \mathbb{N}} \Theta_i = \Theta$ with for any i: $0 < \Pi(\Theta_i) < \infty$ and

$$\lim_{k \to \infty} \mathscr{I}_i^k(\Pi_i) - \mathscr{I}_i^k(P_i) \geq 0 \text{ for all } P \in \mathcal{P}_i \tag{3.13}$$

with Π_i (resp. P_i) being the renormalized restriction of Π (resp. P) on Θ_i, $\mathscr{I}_i^k$ denoting the mutual information on the restricted modeling driven by Θ_i, and $\mathcal{P}_i = \{P \in \mathcal{P}, 0 < P(\Theta_i) < \infty\}$.

This first definition considers a set $\mathcal{P}$ called a set of priors. We provide a thorough definition of such a class in section 3.4. Until then, it refers to a set of priors that are admissible as described in the section 3.2.3.

This definition also requires the sequence (Θ_i) to be "appropriate". Actually, the original definition does not restrict the choices of that sequence, so that this qualifying adjective can be ignored in this section. However, we will see in the sequel that some studies on reference priors require to impose some assumptions on $(\Theta_i)_i$.

Example 3.2. If Θ is finite with $\mathscr{T} = \mathscr{P}(\Theta)$, and if $\mathbb{P}_{Y|\theta} \neq \mathbb{P}_{Y|\theta'}$ for any $\theta' \neq \theta$ then the reference prior Π over the set of all priors is the uniform prior: $\Pi(B) = \Pi(B')$ for all $B, B' \subset \Theta$. A proof of this result is proposed in Muré, 2018.

We must note that other definitions of reference priors exist. In the genesis of the reference prior theory, Bernardo (1979b) suggested maximizing the mutual information $\mathscr{I}^k$, and derive the limit of the maximizers when $k \longrightarrow \infty$ to define reference priors. In Berger, Bernardo, and Sun, 2009, the authors prove that in the real and one parameter case (i.e. $\Theta \subset \mathbb{R}$), this strategy is consistent with definition 3.3. Their result is the following, sometimes taken as a definition.

Theorem 3.1 (Explicit form of the reference prior)**.** *Assume $\Theta \subset \mathbb{R}$ with ν being the Lebesgue measure. Call $\mathcal{P}_s$ the set of admissible priors that are positive and continuous, and that issue proper posterior distributions. Let $\Pi^* \in \mathcal{P}_s$ we call $p^*(\cdot|\cdot)$ its posterior and define for any interior point $\theta_0 \in \Theta$,*

$$f_k(\theta) = \exp\left(\int_{\mathcal{Y}^k} \ell_k(\mathbf{y}|\theta)\log(p^*(\theta|\mathbf{y}))d\mu^{\otimes k}(\mathbf{y})\right) \quad \textit{and} \quad f(\theta) = \lim_{k\to\infty}\frac{f_k(\theta)}{f_k(\theta_0)}. \tag{3.14}$$

If: (i) $\forall\theta, \varepsilon > 0$, $\int_{|\tilde{\theta}-\theta|<\varepsilon} p^*(\tilde{\theta}|\mathbf{y})d\tilde{\theta} \xrightarrow[k\to\infty]{\mathbb{P}_{\mathbf{Y}|\theta}} 1$,
(ii) each f_k is continuous with $\forall k, \theta$, $\frac{f_k(\theta)}{f_k(\theta_0)} < h(\theta)$ such that h is integrable on any compact set, and
(iii) f is the density of $F \in \mathcal{P}_s$ and is such that for any increasing sequence of compact sets $(\Theta_i)_{i\in\mathbb{N}}$ that covers Θ, calling g_i the restricted posterior density of F on Θ_i, we have $\forall\mathbf{y}$, $\int_{\Theta_i} g_i(\theta|\mathbf{y})\log\frac{g_i(\theta|\mathbf{y})}{g(\theta|\mathbf{y})}d\theta \xrightarrow[i\to\infty]{} 0$ where g is F's posterior,
then F is a reference prior over $\mathcal{P}_s$.

The form of the functions f_k in the above theorem comes from maximizing the mutual information (expressed in definition 3.2) when k is fixed. Indeed, it is possible to research for a maximal argument of the mutual information form a constrained optimization viewpoint. A Lagrange multipliers theorem should give an implicit expression of the density whose prior maximizes the mutual information for a fixed k: it is proportional to $\exp\left(\int_{\mathcal{Y}_k} \ell_k(\mathbf{y}|\theta)\log(p(\theta|\mathbf{y}))d\mu^{\otimes k}(\mathbf{y})\right)$. This result is rigorously proven in the appendix for a particular case. It was intuited by Bernardo and Smith (1994), with the idea that, asymptotically, the posterior becomes independent of the chosen prior. The theorem formalizes the intuition, as the posterior in f_k's expression comes from any initial prior $\Pi^* \in \mathcal{P}_s$.

Hierarchical construction in high dimensional settings

Definition 3.3 and theorem 3.1 both define reference priors as maximizers of the mutual information. While there is consensus that such an approach leads to appropriate objective priors in low-dimensional cases, it is not always the case in high-dimensional settings. Indeed, the prior maximizing the mutual information has often non-desirable properties in such scenarios, according to Berger, Bernardo, and Sun (2015). In Berger and Bernardo, 1992a, the authors detail their suggestion to define the reference priors hierarchically, by considering an ordering of the parameters:

$$\theta = (\theta_1, \ldots, \theta_r) \in \Theta = \Theta_1 \times \cdots \times \Theta_r. \tag{3.15}$$

Typically, it is recommended to assume $\Theta_j \subset \mathbb{R}^{d_j}$ with small dimensions d_j (e.g., lower or equal than 2) for any $j \in \{1, \ldots, r\}$, and to sequentially build a reference prior on the Θ_j, $j \in \{1, \ldots, r\}$:

1. initially fix $\ell_k^1 = \ell_k$;
2. for any values of $\theta_{j+1}, \ldots, \theta_r \in \Theta_{j+1} \times \cdots \times \Theta_r$, compute a reference prior (in the sense of definition 3.3) under the model with likelihood $\theta_j \mapsto \ell_k^j(\mathbf{y}|\theta_j, \ldots, \theta_r)$, denote $\pi_j(\cdot|\theta_{j+1}, \ldots, \theta_r)$ its normalized density;

3. derive ℓ_k^{j+1} such as

$$\ell_k^{j+1}(\mathbf{y}|\theta_{j+1},\dots,\theta_r) = \int_{\Theta_j} \ell_k^j(\mathbf{y}|\theta_j,\dots,\theta_r) d\pi_j(\theta_j|\theta_{j+1},\dots,\theta_r). \tag{3.16}$$

This construction is often considered in the literature when it resorts to define reference priors in multidimensional context. However, it relies on the hierarchization of the parameter's coordinates fixed in eq. (3.15), and the recommended one depends on the model of interest. There are a few models, such as the multinomial (defined in example 3.1) one, for which the different reference priors resulting from this construction have been extensively studied. In the case of the multinomial model, it is claimed that this construction with an ordering where $r = d$ when $\Theta \subset \mathbb{R}^d$ must be favored (Berger and Bernardo, 1992b; Berger, Bernardo, and Sun, 2015). In this case, the authors show that the density π^* of this reference prior given by this construction is given by

$$\pi^*(\theta) = K \left[\prod_{j=1}^{r}\prod_{i=1}^{d_j} \theta_j^{(i)-1/2}\right] \left[\prod_{j=1}^{r-1}\left(1 - \sum_{j'=1}^{j}\sum_{i=1}^{d_j}\theta_{j'}^{(i)}\right)^{-d_{j+1}/2}\right] \left[1 - \sum_{j=1}^{r}\sum_{i=1}^{d_j}\theta_j^{(i)}\right]^{-1/2}, \tag{3.17}$$

where K is a normalization constant.

3.3.3 Properties and Jeffreys prior

The elucidation of the reference priors remains an open question for some restricted sets of priors $\mathcal{P}$ or when we lack of regularity assumptions on the model. In most studies, $\mathcal{P}$ remains as large as possible and is restricted only by some mild regularity assumptions, such as that the priors admitting continuous densities.

In the general case, Muré (2018) studied the uniqueness of reference priors. His study requires to define the sequence $(\Theta_i)_i$ in definition 3.3 "appropriate" if and only if the Θ_i are open subsets of Θ. We summarize his main result as follows:

Theorem 3.2. *Let $\mathcal{P}$ be the set of admissible priors. Call an "appropriate" increasing sequence of subsets $(\Theta_i)_i$, an increasing sequence such that each for any i, Θ_i is open. Under the assumption that there exists a weakly estimator of θ, the reference prior over $\mathcal{P}$, if it exists, is unique up to a multiplicative constant.*

This result allows referring to "the reference prior" instead to "reference priors" in some cases. Also, there does not exist to our knowledge a result about the existence of the reference prior in the general case.

Actually, the reference priors are mostly studied in what are called regular cases. The following assumption is a regularity assumption on the statistical model, it ensures useful tools to be well-defined.

Assumption 3.1. Θ is a subset of $\mathbb{R}^d$ with non-empty interior, and the likelihood is such that

1. for μ-a.e. y, $\theta \mapsto \ell(y|\theta)$ is twice differentiable, with $\nabla_\theta^2 \ell(y|\theta)$ being negative definite,
2. for any $\theta \in \Theta$, there exists $\tau > 0$ such that

$$\mathbb{E}_{Y\sim\mathbb{P}_{Y|\theta}}\left[\sup_{\theta',\,\|\theta-\theta'\|<\tau} \|\nabla_\theta^2 \log \ell(Y|\theta')\|\right] \quad \text{and} \quad \mathbb{E}_{Y\sim\mathbb{P}_{Y|\theta}}\left[\sup_{\theta',\,\|\theta-\theta'\|<\tau} \|\nabla_\theta \log \ell(Y|\theta')\|^2\right], \tag{3.18}$$

are continuous functions of θ on the interior of Θ.

Under assumption 3.1, the Fisher information matrix and the Jeffreys prior are well-defined (see Lehmann, 1999). The Fisher information matrix $\mathcal{I}$ is expressed as

$$\mathcal{I}(\theta) = (\mathcal{I}(\theta)_{i,j})_{i,j=1}^d \quad \text{with} \quad \mathcal{I}(\theta)_{i,j} = \int_{\mathcal{Y}} [\partial_{\theta_i} \log \ell(y|\theta)][\partial_{\theta_j} \log \ell(y|\theta)]\, \ell(y|\theta) d\mu(y), \tag{3.19}$$

under assumption 3.1 its coefficients can be expressed as follows

$$\mathcal{I}(\theta)_{i,j} = -\int_{\mathcal{Y}} [\partial^2_{\theta_i\theta_j} \log \ell(y|\theta)]\, \ell(y|\theta) d\mu(y). \tag{3.20}$$

The Jeffreys prior refers to the prior whose density is J, defined by

$$J(\theta) = \sqrt{\det \mathcal{I}(\theta)}. \tag{3.21}$$

Under assumption 3.1, J is a continuous and positive function on Θ. Here are a few examples of Jeffreys priors in different modelings.

Example 3.3 (A proper Jeffreys prior in the multinomial case)**.** The multinomial model is defined as $\theta \in (0,1)^d$, $\mathcal{Y} = \{y \in \{0,\dots,n\}^d, \sum_{i=1}^d y^i = n\}$ for a fixed $n \in \mathbb{N}^*$. And with

$$\ell(y|\theta) = \frac{n!}{y^1!\dots y^d!}\prod_{i=1}^d \theta_i^{y^i}. \tag{3.22}$$

In this model, the Jeffreys prior corresponds to a Dirichlet$(1/2,\dots,1/2)$ distribution:

$$J(\theta) \propto \prod_{i=1}^d \theta_i^{-1/2}(1-\theta_i)^{-1/2}. \tag{3.23}$$

It is a proper prior and corresponds to the prior given in eq. (3.17) when $r = d$.

Example 3.4 (Jeffreys prior in the Gaussian case)**.** The Gaussian model is defined by $\mathcal{Y} = \mathbb{R}$, $\ell(y|\theta) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}$, with $\theta = \mu \in \mathbb{R}$ in the known variance case, or $\theta = (\mu, \sigma^2) \in \mathbb{R} \times (0,\infty)$ in the unknown variance case. In both cases, the Jeffreys prior has a density:

$$J(\theta) \propto \frac{1}{\sigma}. \tag{3.24}$$

In the known variance case the prior for μ corresponds to a uniform distribution on $\mathbb{R}$. In both cases, it is improper. However, its posteriors are always proper.

Clarke and Barron (1994) have shown that the Jeffreys prior is the reference prior over the priors with continuous and positive densities on Θ. Their result can be expressed as follows.

> **Theorem 3.3.** *Under assumption 3.1. Call an "appropriate" increasing sequence of set $(\Theta)_i$, an increasing sequence such that for any i Θ_i is compact. Assume ν is the Lebesgue measure and let $\mathcal{P}$ be the set of priors admitting a continuous and positive density. Then the Jeffreys prior is the unique reference prior over $\mathcal{P}$.*

Muré (2018) has shown that the above theorem still stands when "appropriate" refers to open subsets of Θ. This result can be anticipated by reminding that the form of the density maximizing the mutual information for a fixed k is $\exp\left(\int_{\mathcal{Y}_k} \ell_k(\mathbf{y}|\theta)\log(p(\theta|\mathbf{y}))d\mu^{\otimes k}(\mathbf{y})\right)$. Indeed, according to the Bernstein von Mises theorem (see e.g. Vaart, 1992), under appropriate assumptions, the posterior density $p(\theta|\mathbf{y})$ is asymptotically close to a centered Gaussian density whose variance is $\mathcal{I}(\theta)^{-1/2}/\sqrt{k}$ when $\mathbf{y}$ are distributed w.r.t. $\mathbb{P}_{\mathbf{Y}_k|\theta}$. This convergence does not depend on the prior. We thus asymptotically have

$$\log p(\theta|\mathbf{y}) \simeq \log\frac{k}{2\pi} + \frac{1}{2}\log\det\mathcal{I}(\theta), \tag{3.25}$$

and we approximate

$$\int_{\mathcal{Y}^k} \ell_k(\mathbf{y}|\theta)\log p(\theta|\mathbf{y})d\mu^{\otimes k}(\mathbf{y}) \simeq \log\frac{k}{2\pi} + \frac{1}{2}\log\det\mathcal{I}(\theta). \tag{3.26}$$

Calling $f_k(\theta) = \exp\left(\int_{\mathcal{Y}^k} \ell_k(\mathbf{y}|\theta)\log(p(\theta|\mathbf{y}))d\mu^{\otimes k}(\mathbf{y})\right)$, the above heuristic gives that f_k is asymptotically proportional to $\exp\left(\frac{1}{2}\log\det\mathcal{I}(\theta)\right)$, which is the density of the Jeffreys prior.

We conclude this section by noticing that the Jeffreys prior arises as a cornerstone in the reference prior theory. We recall that this prior has been introduced by Jeffreys (1946). It is a prior that is already praised in Bayesian statistics for its main property to be invariant by re-parametrization of the model: define $\phi = g(\theta)$

with g being a diffeomorphism, and define the model induced considering ϕ instead of θ. Then the Jeffreys prior for this re-parameterized model admits the density J_g:

$$J_g(\phi) = J(g^{-1}(\phi))|\det d_\phi g^{-1}(\phi)|. \tag{3.27}$$

This ensures that inference is not affected by the choice of parameterization, preserving coherence. We also recall that the Jeffreys prior is known to be improper in most cases.

3.4 Limits of the classical framework of reference priors and solution

During this brief review, we mentioned some limits in the framework of the reference priors definition. They are related to the incorporation of improper priors in the admissible priors. As a matter of fact, when dealing with non-informative distributions, it is common to find one of the main assumptions on probability distributions (that is, their total mass is 1) limiting. The fact is that in some cases, improper distribution seems really "close" to proper ones. As an example, we could mention that the improper uniform distribution on $\mathbb{R}$ can be seen as a Gaussian distribution with a variance that diverges to infinity. A logarithmic re-parametrization of the latter gives the improper distribution of density proportional to $x \mapsto 1/x$.

Additionally, when dealing with densities, nothing strongly forbid the definition of the posterior density when the prior is improper (see section 3.2.3). It became therefore common to normalize the use of improper prior, and to carry on *a posteriori* inference from them when the issued posterior is proper.

Even though the way-around works, we lose the existence of a latent probability space with random variables expressing formally the existence of a path that links the "chaos" of a universe to the phenomena observed and modeled. This section is dedicated to the elucidation of such construction, in a framework that allows priors to be improper.

3.4.1 Novel framework for Bayesian inference with improper priors

Since the early work of Kolmogorov (1933), it is appreciated to be reminded that there exists an abstract, underlying structure from which result the probabilistic description and tools that are used to perform inference. This structure, which takes the form of a probability space in the axiomatic of Kolmogorov, represents the indescribable phenomena that induce randomness in the observations.

However, the usual Kolmogorov's axiomatic does not take into account improper distributions to construct the underlying space that provides the usual statistical modeling. In this section, we construct from scratch a framework for Bayesian statistics, in a way that allows priors to be improper. We re-define the objects defined in section 3.2.

Our suggestion is based on the work of Taraldsen and Lindqvist (2016), who adapted the propositions of Renyi (1970). Their idea is to extend the notion of a probability space when the probability is only a σ-finite measure. Up to a multiplicative constant, their framework is shown to coincide with standard probability spaces when the measure has a finite total mass, or when conditioning on a set with finite mass.

We let $(\Theta, \mathscr{T})$ be a measurable space. If Π_1 and Π_2 are two non-null, non-negative and σ-finite measures we can define the relation $\simeq$ by

$$\Pi_1 \simeq \Pi_2 \iff \exists t > 0,\ \Pi_1 = t\Pi_2. \tag{3.28}$$

Define by $\mathscr{M}$ the space of non-null, non-negative and σ-finite measures on Θ. The following properties are verified:

- The relation $\simeq$ is an equivalence relation on $\mathscr{M}$. We note $[\Pi]$ the set of a $\Pi \in \mathscr{M}$.
- Denote by $\mathscr{M}^\nu$ the space of absolutely continuous measures w.r.t. $\nu \in \mathscr{M}$. It is stable by $\simeq$.
- Define by $\mathscr{R}$ the space of non-negative non-null measurable functions from Θ to $\mathbb{R}$. The relation

$$f \propto g \iff \exists t > 0,\ f = tg \tag{3.29}$$

is an equivalence relation on $\mathscr{R}$. Let $\nu \in \mathscr{M}$, the Radon-Nikodym Theorem defines a natural surjection from the quotient space $\mathscr{R}/\propto$ to $\mathscr{M}^\nu/\simeq$. A class in $\mathscr{R}/\propto$ is called a density class of the unique class in $\mathscr{M}^\nu/\simeq$ that it induces by this mapping. If two elements $[f]$, $[g]$ of $\mathscr{R}/\propto$ are density classes of a same element in $\mathscr{M}^\nu/\simeq$, they are equal ν-a.e. in the sense $\exists t > 0$, $f = tg$ ν-a.e.

Bioche and Druilhet (2016) consider such a definition of σ-finite measures equal up-to-a constant to formalize the notion of prior and posterior approximations, whether they are proper or improper. Notably, they define a topology induced by a convergence in $\mathscr{M}/\simeq$: the Q-vague convergence.

A prior is then generally defined as a class $[\Pi]$ in $\mathscr{M}/\simeq$. The tuple $(\Theta, \mathscr{T}, [\Pi])$ corresponds to what Taraldsen and Lindqvist (2016) call a conditional probability space for any $[\Pi] \in \mathscr{M}/\simeq$. For any $U \in \mathscr{T}$ such that $\Pi(U) \in (0, +\infty)$, that space issues a unique probability space on U, with the probability $\Pi(\cdot|U) := \Pi(\cdot \cap U)/\Pi(U)$, it is independent of the choice of the representative of $[\Pi]$. This framework gives what is necessary to construct a general conditional probability space that models our problem, as we express below.

Proposition 3.1. *We assume that the observations take values in a Polish space $(\mathcal{Y}, \mathscr{Y})$ and are statistically modeled by the collection of conditional probabilities $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$ with $(\Theta, \mathscr{T})$ being a Polish space and such that $\forall A \in \mathscr{Y}$, $\theta \mapsto \mathbb{P}_{Y|\theta}(A)$ is measurable. We consider $\Pi \in \mathscr{M}$.*
Then there exist a conditional probability space $(\mathbf{\Omega}, \mathbf{\Xi}, [\mathbf{\Pi}])$, a measurable process $\overline{Y} = (Y_i)_{i\in\mathbb{N}}$, $Y_i : \mathbf{\Omega} \longrightarrow \mathcal{Y}$, and a measurable function $T : \mathbf{\Omega} \longrightarrow \Theta$ such that

1. *for any $U \in \mathscr{T}$, $\mathbf{\Pi}(T \in U) = \Pi(U)$;*
2. *if $\Pi(U) \in (0, +\infty)$ then the unique probability space conditioned on $\{T \in U\}$ is such that the conditional probability of any $(Y_{i_l})_{l=1}^k$ to $T = \theta$ is $\mathbb{P}_{Y|\theta}^{\otimes k}$. More explicitly, if $A_{j_1}, \dots, A_{j_k} \in \mathscr{Y}$ then*

$$\mathbf{\Pi}\Big(\bigcap_{l=1}^k Y_{j_l} \in A_{j_l} | T \in U\Big) = \int_\Theta \mathbb{P}_{Y|\theta}^{\otimes k}(A_{j_1} \times \cdots \times A_{j_k}) d\Pi(\theta|U). \tag{3.30}$$

Proposition 3.1 is proven in section 3.4.3. It ensures the existence of an underlying abstract space $\mathbf{\Omega}$ from only the given of the statistical model (the collection of probability measures $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$) and the prior (which is a measure $\Pi \in \mathscr{M}$). When Π is improper, the tuple $(\mathbf{\Omega}, \mathbf{\Xi}, [\mathbf{\Pi}])$ is not a probability space anymore, but they still describe the randomness that goes beyond the observations and the parameter. In the following section, we develop the compatibility of this framework with the usual one.

3.4.2 Notations and compatibility with the usual framework

Given the construction proposed in section 3.4.1, the Bayesian framework is defined from the given of probability measure $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$ and a prior Π which is a representative of an element $[\Pi] \in \mathscr{M}/\simeq$, under the assumptions in the proposition 3.1. The consideration of another representative $\tilde{\Pi}$ of $[\Pi]$ does not alter the construction.

That framework allows the construction of a marginal distribution $\mathbb{P}_{\mathbf{Y}_k}$, defined as the pushforward measure of $\mathbf{\Pi}$ by $\mathbf{Y}_k$. In the same way as Π, it is unique up to a constant. The posterior distribution $\mathbb{P}_{T|\mathbf{y}}$ is well-defined for $\mathbb{P}_{\mathbf{Y}_k}$-a.e. $\mathbf{y}$ by

$$\forall A \in \mathscr{Y}^{\otimes k}, \forall B \in \mathscr{T}, \int_A \mathbb{P}_{T|\mathbf{y}}(B) d\mathbb{P}_{\mathbf{Y}_k}(\mathbf{y}) = \int_B \mathbb{P}_{\mathbf{Y}_k|\theta}(A) d\Pi(\theta). \tag{3.31}$$

It belongs to $\mathscr{M}$, and its class $[\mathbb{P}_{T|\mathbf{y}}]$ in $\mathscr{M}/\simeq$ is uniquely defined.

Densities

From now on, we assume that the model admits a likelihood: there exists a collection of density functions $(\ell(\cdot|\theta))_{\theta\in\Theta}$ w.r.t. a common σ-finite measure μ on $\mathcal{Y}$ such that for Π-a.e. θ:

$$\forall A \in \mathscr{Y}, \mathbb{P}_{Y|\theta}(A) = \int_A \ell(y|\theta) d\mu(y). \tag{3.32}$$

We also assume that Π belongs to $\mathscr{M}^\nu$ (it is absolutely continuous w.r.t. ν), with $\nu \in \mathscr{M}$. We denote by $\pi \in \mathscr{R}$ a density of Π w.r.t. ν. For $\mathbf{y} \in \mathcal{Y}^k$ there exists a posterior density w.r.t. ν defined in the class $[p(\cdot|\mathbf{y})] \in \mathscr{R}/\propto$ where

$$p(\theta|\mathbf{y}) \propto \ell_k(\mathbf{y}|\theta)\pi(\theta) = \prod_{i=1}^{k} \ell(y_i|\theta)\pi(\theta) \tag{3.33}$$

If the posterior is proper for $\mathbb{P}_{\mathbf{Y}_k}$-a.e. $\mathbf{y}$, a marginal density can be defined $\mathbb{P}_{\mathbf{Y}_k}$-a.e. by

$$p_{\mathbf{Y}_k}(\mathbf{y}) = \int_\Theta \ell_k(\mathbf{y}|\theta)\pi(\theta)d\theta. \tag{3.34}$$

Impact on the framework of reference priors

The elucidating of the densities in this novel Bayesian construction is consistent with the usual construction. When restricting the framework to a $\tilde{\Theta} \in \mathscr{T}$ such that $0 < \Pi(\tilde{\Theta}) < \infty$, the restricted model driven by $\tilde{\Theta}$ (in the sense given in section 3.2.3) corresponds to the model on $\tilde{\Theta}$ considering the prior $\Pi(\cdot|\tilde{\Theta})$.

Therefore, the notations and definitions written in the previous sections still stand. The modeling remains the same for any prior that equals Π up to a positive multiplicative constant, so that if $\Pi \in \mathscr{M}$ is a reference prior, any prior in $[\Pi]$ also is. For this reason it becomes appropriate to talk about a reference prior class. We will authorize however the slight abuse of language referring to "the reference prior" instead of "reference prior in the reference prior class", when the latter is unique. We formalize this terminology in the following definitions.

Definition 3.4. A set of priors refers to a subset of $\mathscr{M}^\nu$.
When ν is the Lebesgue measure, a set of continuous prior refers to a subset $\mathcal{P} \subset \mathscr{M}^\nu$ where for any $\Pi \in \mathcal{P}$, Π admits a ν-a.e. continuous and locally bounded density π. We call $\mathscr{M}^\nu_{\mathcal{C}}$ the set of continuous priors.
The subset of $\mathscr{R}$ composed by ν-a.e. continuous and locally bounded functions is denoted $\mathscr{R}_{\mathcal{C}^b}$.

Proposition 3.2. *$\mathscr{M}^\nu_{\mathcal{C}}$ is stable by $\simeq$ and $\mathscr{R}_{\mathcal{C}^b}$ is stable by $\propto$.*
If $\Theta \subset \mathbb{R}^d$, and if f is ν-a.e. continuous, f is ν-a.e. locally bounded.
Any f in $\mathscr{R}_{\mathcal{C}^b}$ is bounded on every compact.

Proof. 1. The stability of $\mathscr{R}_{\mathcal{C}^b}$ to $\propto$ comes since multiplying a function by a constant does not affect its continuity nor its locally bounded characteristic.
We consider M the mapping assigning a density in $\mathscr{R}$ to its associated measure in $\mathscr{M}^\nu$ as

$$M : f \in \mathscr{R} \longmapsto \left(B \in \mathscr{T} \mapsto \int_B f(\theta)d\nu(\theta) \right). \tag{3.35}$$

We have $f \propto g \implies M(f) \simeq M(g)$, so that $M(\mathscr{R}_{\mathcal{C}^b})$ is stable by $\simeq$. The second statement of the proposition comes since $M(\mathscr{R}_{\mathcal{C}^b}) = \mathscr{M}^\nu_{\mathcal{C}}$ by definition.
2. For the second statement, we consider f ν-a.e. continuous. We note $E = \{\theta, \forall\tau > 0, f \text{ is not bounded on } B(\theta,\tau)\}$. Let $\theta \in E$, for any $n > 0$, f is not bounded on $B(\theta, 1/n)$, so that there exists $\theta_n \in B(\theta, 1/n)$ such that $f(\theta_n) > n$. That leading to $f(\theta_n) \underset{n\to\infty}{\longrightarrow} \infty$ while $\theta_n \underset{n\to\infty}{\longrightarrow} \theta$, f is discontinuous on θ. We deduce that $\nu(E) = 0$.
3. If $\tilde{\Theta}$ is a compact subset of Θ, we can cover $\tilde{\Theta}$ by open balls on which f is bounded (consequently to statement 2.). Since $\tilde{\Theta}$ is compact, the open cover admits a finite subcover. f being bounded on this finite subcover, we deduce that it is bounded on $\tilde{\Theta}$. □

Definition 3.5 (Advanced definition of reference priors)**.** Let $\mathcal{P}$ be a set of priors. A prior $\Pi \in \mathcal{P}$ is a reference prior over $\mathcal{P}$ if there exists an openly increasing sequence of compact sets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$ with for any i: $0 < \Pi(\Theta_i) < \infty$ and

$$\lim_{k\to\infty} \mathscr{I}^k(\Pi(\cdot|\Theta_i)) - \mathscr{I}^k(P(\cdot|\Theta_i)) \geq 0 \text{ for all } P \in \mathcal{P} \text{ verifying } 0 < P(\Theta_i) < \infty. \tag{3.36}$$

Let $\mathcal{Q}$ be a subset of $\mathscr{M}^\nu / \simeq$. And denote by q the surjection $q : \Pi \in \mathscr{M}^\nu \mapsto [\Pi]$. A class $[\Pi] \in \mathcal{Q}$ is called a reference prior class over $\mathcal{Q}$ if Π is a reference prior over $q^{-1}(\mathcal{Q})$.

The above definition is congruous with the definition 3.3, apart from the necessary condition on the sequence $(\Theta_i)_i$. The definition of an openly increasing sequence of compacts is stated below. It is, according to ourselves, an appropriate condition on this sequence, as it allows the consistency between the compact case and the general case.

Definition 3.6. A sequence $(\Theta_i)_{i\in\mathbb{N}}$ of subsets of Θ is said to be openly increasing if there exist $i_0 \geq 0$ and a sequence $(V_i)_{i\geq i_0}$ of open subsets of Θ such that for any $i \geq i_0$:

$$\Theta_i \subset V_i \subset \Theta_{i+1}. \tag{3.37}$$

Such sequence does not necessarily exist if Θ has an "unusual" form. In most cases, one would work on sets Θ that are "almost" open or "almost" closed, i.e. $\nu(\partial\Theta) = 0$. In this case, one can restrict the study to $\overline{\Theta}$ or to $\mathring{\Theta}$. If Θ is moreover a subset of $\mathbb{R}^d$ then there always exist openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$, $(\Theta'_i)_{i\in\mathbb{N}}$ with $\bigcup_{i\in\mathbb{N}} \Theta_i = \mathring{\Theta}$ and $\bigcup_{i\in\mathbb{N}} \Theta'_i = \overline{\Theta}$. We express that statement in the proposition below.

Proposition 3.3. *1. If Θ, is an open or a closed subset of $\mathbb{R}^d$ there exist an openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$.*
2. If Θ is a countable intersection of open subsets of $\mathbb{R}^d$, there exist an openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$.

Proof. If Θ is open, then we can write Θ as an increasing countable union of open balls:

$$\Theta = \bigcup_{n\in\mathbb{N}^*} \bigcup_{\theta\in\Theta\cap\mathbb{R}^d} B(\theta, \min(R(\theta) - \frac{1}{n}, n)) \quad \text{with} \quad R(\theta) = \sup\{\tau > 0,\, B(\theta,\tau) \subset \Theta\}. \tag{3.38}$$

We write $\Theta \cap \mathbb{Q}^d = \{\theta_i,\, i \in \mathbb{N}^*\}$, and write $\Theta = \bigcup_{n\in\mathbb{N}^*} V_n$ with $V_n = \bigcup_{i=1}^n B(\theta_i, \hat{R}_n^i)$, where $\hat{R}_n^i = \min(R(\theta_i) - \frac{1}{n}, n)$. This way, each V_n is a finite union of closed balls, so that $\overline{V_n}$ is compact. Also, V_{n+1} contains balls with the same center as the one that constitute V_n, yet with higher radius. Thus, $\overline{V_n} \subset V_{n+1}$. Therefore, setting $\Theta_i = \overline{V_i}$ for all i makes $(\Theta_i)_{i\in\mathbb{N}^*}$ being an openly incresing sequence of compact subsets that cover Θ.

If Θ is closed, we recall that $\mathcal{F} = \{\Theta \cap B(\theta,\tau),\, \theta \in \mathbb{Q}^d,\, \tau \in \mathbb{Q}_+^*\}$ is a subbase of the induced topology of $\mathbb{R}^d$ on Θ. We can thus cover Θ by elements in $\mathcal{F}$: $\Theta = \bigcup_{i\in I} U_i$. It is possible to index this cover by $\mathbb{N}$. We now construct the sequences $(\Theta_i)_i \in \mathbb{N}$ and $(V_i)_{i\in\mathbb{N}}$ by induction. Starting from $V_0 = U_0$ and $\Theta_0 = \overline{V_0}$, we then take for all i, $V_{i+1} = \bigcup_{j=1}^{m(i)} U_j$, where $m(i) = \min\{j',\, \bigcup_{j=0}^{j'} U_j \supset \Theta_i\}$. By induction, Θ_i is compact, so that $m(i) < \infty$ and $\Theta_{i+1} = \overline{V_{i+1}}$ also is. That makes $(\Theta_i)_{i\in\mathbb{N}}$ being an openly increasing sequence of subsets that covers Θ.

To prove the second statement, we write $\Theta = \bigcup_{n\in\mathbb{N}} U_n$ where the $(U_n)_{n\in\mathbb{N}}$ are open subsets of $\mathbb{R}^d$. Using the first statement, there exists an openly increasing cover for each of the $(U_n)_{n\in\mathbb{N}}$: $U_n = \bigcup_{j\in\mathbb{N}} \Theta_j^n$, with $\Theta_j^n \subset V_j^n \subset \Theta_{j+1}^n$ for some open subsets $(V_j^n)_{j\in\mathbb{N}}$ of Θ. Let us define $\Theta_i = \bigcup_{j,n\leq i} \Theta_j^n$. We verify that Θ_i is compact and that $\Theta_i \subset \bigcup_{j,n\leq i} V_j^n$. Since for all n, j V_j^n is an open subset of U_n, it is an open subset of Θ. This concludes the proof since $\Theta = \bigcup_{i\in\mathbb{N}} \Theta_i$. □

Corollary 3.1. *If Θ is an open subset of $\mathbb{R}^d$ and $f : \Theta \longrightarrow \mathbb{R}$ is a ν-a.e. continuous function of Θ, there exists an openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta \setminus E$. Where E is the negligible set of points on which f is discontinuous.*

Proof. The corollary follows from the previous proposition recalling that E is a countable intersection of closed subsets of Θ. □

Proposition 3.4. *Assume Θ is compact. Then if Π is a reference prior over a set of priors $\mathcal{P}$, it verifies*

$$\lim_{k\to\infty} \mathscr{I}^k(\Pi) - \mathscr{I}^k(P) \geq 0 \text{ for all } P \in \mathcal{P}. \tag{3.39}$$

Proof. We consider an openly increasing sequence of compact sets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$. By definition there exist $i_0 \geq 0$ and a sequence $(V_i)_{i\geq i_0}$ of open subsets of Θ such that for any $i \geq i_0$

$$\Theta_i \subset V_i \subset \Theta_{i+1}. \tag{3.40}$$

This way, $\bigcup_i V_i = \Theta$ and the compacity of Θ imposes it to be a finite union, so that $\Theta_i = \Theta$ for any $i \geq i_1$ for some $i_1 \geq 0$. Since Π is a reference prior over $\mathcal{P}$ its renormalized restriction to Θ_{i_1} verifies the eq. (3.39). Hence the result. □

3.4.3 Proof of proposition 3.1

Let $1 \leq \mathbf{i} \leq \infty$, $I = \{i \in \mathbb{N}, i < \mathbf{i}\}$, and $(U_i)_{i\in I}$ be a sequence of disjoint non-empty sets in $\mathscr{T}$ that covers Θ such that $\Pi(U_i) < \infty\ \forall i$. When $\Pi(U_i) = 0$, we denote by $\Pi(\cdot|U_i)$ an arbitrary probability distribution on U_i (for instance a Dirac distribution); otherwise, it is defined by $\Pi(\cdot|U_i) = \Pi(\cdot \cap U_i)/\Pi(U_i)$. Fix firstly $i \in I$ and for any $k \geq 1$ let us construct $\mathbb{P}^i_{\mathbf{Y}_k,T}$ on $\mathscr{Y}^{\otimes k} \otimes \mathscr{T}$ such that

$$\mathbb{P}^i_{\mathbf{Y}_k,T}(A_1 \times \cdots \times A_k \times B) = \int_B \mathbb{P}^{\otimes k}_{Y|\theta}(A_1 \times \cdots \times A_k) d\Pi(\theta|U_i), \tag{3.41}$$

so that $\mathbb{P}^i_{\mathbf{Y}_k,T}$ is a probability distribution on $\mathcal{Y}^k \times U_i$. Therefore, using Kolmogorov extension theorem, there exist a probability space $(\Omega^i, \mathscr{X}^i, \mathbb{P}^i)$, a random process $\overline{\mathbf{Y}^i} = (Y^i_j)_{j\geq 1}$, $Y^i_j : \Omega^i \longrightarrow \mathcal{Y}$, and a random variable $T^i : \Omega^i \longrightarrow \Theta$ such as their joint distribution is uniquely defined by the distribution $\mathbb{P}^i_{\mathbf{Y}_k,T}$ of $(T^i, \mathbf{Y}^i_k)$ for any k, with $\mathbf{Y}^i_k = (Y^i_j)^k_{j=1}$.

Denote $\mathbf{\Omega} = I \times \prod_{i\in I} \Omega^i$. A natural σ-algebra on $\mathbf{\Omega}$ is the one generated by the cylinder sets:

$$\mathbf{\Xi} := \sigma\big(N \times E_0 \times \cdots \times E_{k-1} \times \prod_{k\leq i<\mathbf{i}} \Omega^i,\ N \in \mathscr{P}(I),\ E_0, \ldots, E_k \in \mathscr{X}^i,\ k < \mathbf{i}\big). \tag{3.42}$$

We can define $\overline{\Pi}$ by

$$\overline{\Pi}\Big(N \times \prod_{i\in I} E_i\Big) = \sum_{n\in N} \mathbb{P}^n(E_n)\Pi(U_n), \tag{3.43}$$

for any $N \in \mathscr{P}(I)$, $(E_i)_{i\in I} \in \prod_{i\in I} \mathscr{X}^i$ with $E_i = \Omega^i$ for any $i \geq k$ for some $k < \mathbf{i}$. The existence of a measure Π on $\mathbf{\Xi}$, that coincides with $\overline{\Pi}$ on the cylinder sets is guaranteed by Carathéodory's extension Theorem.

Now, we call $p^n : (m, (\omega^i)_i) \in \mathbf{\Omega} \mapsto \omega^n$ for any $n \in I$, $c : (m, (\omega^i)_i) \in \mathbf{\Omega} \mapsto m$, and $p : \mathbf{w} \in \mathbf{\Omega} \mapsto p^{c(\mathbf{w})}(\mathbf{w})$. This way, we can define $T : \mathbf{\Omega} \longrightarrow \Theta$ by $T(\mathbf{w}) = T^{c(\mathbf{w})}(p(\mathbf{w}))$ and $\overline{\mathbf{Y}} = (Y_j)_{j\geq 1}$ with for any $j \geq 1$, $Y_j : \mathbf{\Omega} \longrightarrow \mathscr{Y}$ by $Y_j(\mathbf{w}) = Y^{c(\mathbf{w})}_j(p(\mathbf{w}))$.
We can verify that T is a measurable: if $B \in \mathscr{T}$,

$$T^{-1}(B) = \bigcup_{i\in I} \{\mathbf{w} \in \mathbf{\Omega},\ c(\mathbf{w}) = i,\ T^i(p(\mathbf{w})) \in B\} = \bigcup_{i\in I} c^{-1}(\{i\}) \cap (T^i(p^i))^{-1}(B) \tag{3.44}$$

which is measurable. The same arguments stand for the measurability of Y_j for any j.

We have, for any $V \in \mathscr{T}$:

$$\begin{aligned}\mathbf{\Pi}(T \in V) &= \overline{\Pi}\Big(\bigcup_{i\in I}\{i\} \times \Omega^1 \times \cdots \times \{T^i \in V\} \times \Omega^{i+1} \times \dots\Big) \\ &= \sum_{i\in I} \mathbb{P}^i(T^i \in V)\Pi(U_i) = \sum_{i\in I}\Pi(V|U_i)\Pi(U_i) = \Pi(V);\end{aligned} \tag{3.45}$$

and if $\Pi(V) < \infty$ then for any $(A_{jl})_{l=1}^k \in \mathscr{Y}^k$:

$$\begin{aligned}&\mathbf{\Pi}\left((Y_{jl})_{l=1}^k \in (A_{jl})_{l=1}^k | T \in V\right)\mathbf{\Pi}(T \in V) \\ &\qquad = \mathbf{\Pi}\Big(\bigcup_{i\in I}\{i\} \times \Omega^1 \times \dots \{(Y_{jl}^i)_{l=1}^k \in (A_{jl})_{l=1}^k\} \cap \{T^i \in V\} \times \Omega^{i+1} \dots\Big),\end{aligned} \tag{3.46}$$

so that

$$\begin{aligned}&\mathbf{\Pi}\left((Y_{jl})_{l=1}^k \in (A_{jl})_{l=1}^k | T \in V\right) = \sum_{i\in I}\mathbb{P}^i\left(\{(Y_{jl}^i)_{l=1}^k \in (A_{jl})_{l=1}^k\} \cap \{T^i \in V\}\right)\frac{\Pi(U_i)}{\Pi(V)} \\ &= \sum_{i\in I}\int_{V\cap U_i} \mathbb{P}_{Y|\theta}^{\otimes k}\left((A_{jl})_{l=1}^k\right)\frac{d\Pi(\theta)}{\Pi(V)} = \int_\Theta \mathbb{P}_{Y|\theta}^{\otimes k}\left((A_{jl})_{l=1}^k\right)d\Pi(\theta|V).\end{aligned} \tag{3.47}$$

We precise that the above expressions rely on the fact that $\mathbf{\Pi}(I \times \prod_{i\in I} E_i) = \sum_{i\in I}\mathbb{P}^i(E_i)\Pi(U_i)$, for any $E_i \in \mathscr{X}^i$, for every $i \in I$. To state that equality, we write $I \times \prod_{i\in I} E_i = \bigcup_{n\geq 1}\bigcap_{k<\mathbf{i}}(I^n \times \prod_{i\in I} E'_{i,k})$ with $I^n = \{i \in I, i < n\}$ and $E'_{i,k} = E_i$ if $i \leq k$, $E'_{i,k} = \Omega^i$ otherwise. This way for any $n \geq 1$: $I^n \times \prod_{i\in I} E'_{i,k} \subset I^{n+1} \times \prod_{i\in I} E'_{i,k}$ for any $k < \mathbf{i}$, and $\bigcap_{k<\mathbf{i}} I^n \times \prod_{i\in I} E'_{i,k} \subset \bigcap_{k<\mathbf{i}} I^{n+1} \times \prod_{i\in I} E'_{i,k}$. Thus,

$$\mathbf{\Pi}\Big(I \times \prod_{i\in I} E_i\Big) = \lim_{n\to\infty}\mathbf{\Pi}\Big(\bigcap_{k<\mathbf{i}}(I^n \times \prod_{i\in I} E'_{i,k})\Big). \tag{3.48}$$

For any $n \geq 1$, $(I^n \times \prod_{i\in I} E'_{i,k})$ is a decreasing sequence, because $E'_{i,k+1} \subset E'_{i,k}$ for any $k < \mathbf{i} - 1$. Also, $\mathbf{\Pi}(I^n \times \prod_{i\in I} E'_{i,0}) = \sum_{i=0}^{n-1}\Pi(U_i) < \infty$. Therefore, we can write:

$$\begin{aligned}\mathbf{\Pi}\Big(\bigcap_{k<\mathbf{i}}(I^n \times \prod_{i\in I} E'_{i,k})\Big) &= \lim_{k\to\infty}\mathbf{\Pi}\Big(I^n \times \prod_{i<k,\mathbf{i}} E_i \times \prod_{k\leq i<\mathbf{i}}\Omega^i\Big) \\ &= \lim_{k\to\infty}\Big[\sum_{i<k,n,\mathbf{i}}\mathbb{P}^i(E_i)\Pi(U_i) + \sum_{k\leq i<n,\mathbf{i}}\Pi(U_i)\Big] = \sum_{i<n,\mathbf{i}}\mathbb{P}^i(E_i)\Pi(U_i),\end{aligned} \tag{3.49}$$

because the sums are all finite. Eventually, the limit in eq. (3.48) is equal to $\sum_{i<\mathbf{i}}\mathbb{P}^i(E_i)\Pi(U_i)$ as expected, hence the result.

3.5 Conclusive remarks

Reference priors are designed as a solution to the search for objective priors. To do so, they are constructed on informatic-theoretic foundations, under a framework though to ensure they maximize the role of the data in the definition of the posterior distribution.

The theory is provided with a formal definition and several works propose rigorous studies of reference priors, in different contexts. That let them be elected for their lack of subjectivity in various practical studies (e.g. Chen, Ibrahim, and S. Kim, 2008; Gu and Berger, 2016; D'Andrea et al., 2021) and various statistical models, such as Gaussian process-based models (Paulo, 2005; Gu and Berger, 2016), generalized linear models (Natarajan and Kass, 2000) or general exponential family models (Clarke and Ghosal, 2010).

Moreover, the framework that we have introduced allows a rigorous incorporation of improper priors in Bayesian modelings. This provides a better interpretability of the reference priors when it leads to an improper prior distribution.

However, some limits remain. First, the theory is still built on some choices, such as the choice of the dissimilarity measure to quantify how different are the prior and the posterior. Second, the incorporation of improper priors is appealing yet can represent an issue in cases where they issue improper posteriors. Actually, reference priors are known to lead to improper posteriors in some cases. Last, the theoretical developments limit their studies to sets of priors that are the more general possible, and effects of its restriction are rarely focused.

4

Generalized mutual information and their reference priors

This chapter draws mainly on the submitted work: Antoine Van Biesbroeck (2024a). *Generalized mutual information and their reference priors under Csizar f-divergence*. DOI: 10.48550/arXiv.2310.10530

Abstract This chapter complements the reference prior theory. We propose an interpretation of the theory from a sensitivity analysis viewpoint. This leads us to propose a new way of defining mutual information, using different dissimilarity measures between probability distributions. Our construction creates a new framework for reference priors, which are rigorously studied. Our main result gives a limit of the generalized mutual information when the dissimilarity measure considered resembles to a δ-divergence. It makes it easier to derive reference priors under constraints or within specific sets. In the absence of constraints, we prove that the Jeffreys prior maximizes the generalized mutual information, reinforcing its objective characteristic.

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4.1 Introduction and motivations

The reference prior theory aims at defining priors that are the most objective possible. A review of the theory can be found in the chapter 3. It is built on the maximization of the mutual information, which is designed to measure the information brought by the data in the posterior distribution.

Although there are infinite ways to compare probability distributions, the mutual information is classically defined as an expected Kullback-Leibler divergence between the prior and the posterior. The settings under which the reference priors are usually defined result from the aforementioned choice. Since the aim is to define priors that are not impacted by potentially subjective choice, we ask how the results about reference priors change when the definition of the mutual information is modified.

Extensions of the mutual information in the reference prior definition have already been explored in the literature (e.g. Chen, Dey, et al., 2010; Liu et al., 2014; Le, 2014), and some do not necessarily lead to the Jeffreys prior (Hashimoto, 2021; Clarke and Sun, 1997; M. Ghosh, Mergel, and Liu, 2011). They are all based on the expression of mutual information as an average divergence between the posterior and the prior distributions.

In this chapter, we contribute to the reference prior theory with an original derivation of the mutual information from a Global Sensitivity Analysis (GSA) based viewpoint. GSA, whose principle is to measure how the uncertainty of an output is impacted by that of some of its input (Iooss and Lemaître, 2015), allows us indeed to provide an interpretation of the reference prior as a maximizer of such sensitivity influence that the observations get from the parameter of interest. This suggestion leads to defining what we call generalized mutual information, by analogy with global sensitivity indices (which are introduced for instance in Da Veiga et al., 2021). It relies on the wide range of existing dissimilarity measures between two probability distributions. An example is the f-divergence subclass (Csiszár, 1967), commonly employed as an extension of Shannon's entropy for various purposes in statistics, such as, variational inference (Minka, 2005; Bach, 2023), surrogate model design (Nguyen, Wainwright, and Jordan, 2009), PAC Bayesian learning (Picard-Weibel and Guedj, 2022) and Differential Privacy (Mironov, 2017). A study of those divergences within our generalized mutual information is also a main contribution of this chapter, with the goal of deriving what one is invited to call generalized reference priors. We provide an accomplished formalization for the generalized reference prior settings, and results based on classical f-divergences for sensitivity analysis such as δ-divergences. Our main result takes the form of a limit w.r.t. the number of data of the mutual information. Its analytical expression as a function of the prior permits a simple expression of our reference priors: they are its maximal arguments. It opens the path of easier theoretical derivation of reference priors among the ones that satisfy different kinds of constraints, or that belong to some particular sets of priors.

In the next section, we elucidate our motivation from a Global Sensitivity Analysis viewpoint for an enrichment of the mutual information. The generalized mutual information that we define is constructed from any possible dissimilarity measure. That definition is used in section 4.3 to define the generalized reference priors. We show in that section that the usual properties satisfied by reference priors still stand under our framework. A deeper study is afterward conducted in section 4.4 in the case where the dissimilarity measure is an f-divergence resembling to a δ-divergence. These results, alongside open thoughts about their robustness when the dissimilarity measure is not limited to such f-divergences, are discussed in section 4.5. After proving our statements in section 4.6, we conclude this chapter in section 4.7.

4.2 Generalized mutual information

4.2.1 Mutual information and definitions

We consider a statistical modeling characterized by a collection of distributions $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$. The Bayesian framework is constructed as described in the chapter 3 (section 3.4). We suppose that $\Theta \subset \mathbb{R}^d$ and denote by ν the Lebesgue measure on $\mathbb{R}^d$. The priors considered are absolutely continuous distributions w.r.t. ν.

We consider the same notations as in chapter 3: $\mathbf{Y}_k$ denote a random vector of k data items whose distribution conditionally to $T = \theta$ is $\mathbb{P}_{\mathbf{Y}_k|\theta} := \mathbb{P}_{Y|\theta}^{\otimes k}$, where T is an r.v. whose distribution is the prior Π

(which is not necessarily a probability distribution, see section 3.4.1). The density of Π is denoted by π.

We suppose that the modeling admits a likelihood, denoted by ℓ with $\forall \theta \in \Theta, \mathbf{y} \in \mathcal{Y}^k$, $\ell_k(\mathbf{y}|\theta) = \prod_{i=1}^k \ell(y_i|\theta)$. For all θ, $\ell(\cdot|\theta)$ is the density of $\mathbb{P}_{T|\theta}$ w.r.t. a common measure μ on $\mathcal{Y}$. We denote by $p_{\mathbf{Y}^k}$ the marginal density of $\mathbf{Y}_k$, and by $p(\cdot|\mathbf{y})$ the posterior density given the observations $\mathbf{y} \in \mathcal{Y}^k$. Furthermore, we suppose the modeling to be regular (i.e. assumption 3.1 in chapter 3 is verified), so that the Fisher information matrix (that we denote $\mathcal{I}$) and the Jeffreys prior exist. A density of the Jeffreys prior is denoted by J.

Under this framework, when the prior Π is proper, we recall that the mutual information is classically defined as the following quantity:

$$\mathscr{I}^k(\Pi) = \mathbb{E}_{\mathbf{Y}_k \sim \mathbb{P}_{\mathbf{Y}_k}}[\mathrm{KL}(\mathbb{P}_{T|\mathbf{Y}_k}||\Pi)]. \tag{4.1}$$

We recall that in the above expression, the posterior and marginal distributions depend on the prior considered. Applying Fubini-Lebesgue's theorem allows to also write the mutual information as follows

$$\mathscr{I}^k(\Pi) = \mathbb{E}_{T \sim \Pi}[\mathrm{KL}(\mathbb{P}_{\mathbf{Y}_k|T}||\mathbb{P}_{\mathbf{Y}_k})]. \tag{4.2}$$

That last quantity expresses the mutual information as a measure of the influence that the stochastic parameter T has on the data $\mathbf{Y}_k$. That interpretation aligns with a sensitivity analysis viewpoint. Actually, the expression in eq. (4.2) can be seen as a sensitivity index, where the impact of the input T on the output $\mathbf{Y}_k$ is studied. We develop this interpretation in the following section.

> *Proof that eq. (4.1) and eq. (4.2) are equal.* One can remark that for any $x \in (1, \infty)$, $|\log x| \leq x - \log x$. That statement allows applying proposition 4.3, which is stated later on, with $f = -\log$. It provides the desired equality whether the involved quantities are finite or infinite. □

4.2.2 Mutual information as a sensitivity index

The integration of the Bayesian formalism in the settings of uncertainty quantification resorts in modeling $\mathbf{Y}_k$ as the output of a stochastic system whose inputs include T. In other terms, we write $\mathbf{Y}_k = \mathcal{M}(T, \epsilon)$. Here, T embeds uncertainty on the input parameters and ϵ represents an unknown uncertain variable, sometimes referred to as a nuisance parameter.

A well-designed Bayesian modeling parameterized by T would be such that the input between ϵ and T whose impact on the output is the highest is T. Such a modeling should reduce the impact of the irreducible uncertainty embedded in ϵ.

This impact is measured by a sensitivity index as follows, considering a dissimilarity measure D,

$$S = \mathbb{E}_{T \sim \Pi}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})]. \tag{4.3}$$

It corresponds to the mutual information $\mathscr{I}^k(\Pi)$ when D is defined as $D(P||Q) = \mathrm{KL}(Q||P)$.

Actually, different choices for dissimilarity measures D can be done to define sensitivity indices (Da Veiga, 2015). For instance, setting $D(P||Q) = |\mathbb{E}_{X \sim P} X - \mathbb{E}_{X \sim Q} X|^2$ gives the un-normalized Sobol' index (Sobol', 1993). Among the most common ones, we recall the f-divergences, defined from a real and measurable function f that is convex and maps 1 to 0. The f-divergence is denoted D_f and defined as

$$D_f(P||Q) = \int_{\mathcal{X}} f\left(\frac{p(x)}{q(x)}\right) q(x) d\omega(x) \tag{4.4}$$

where p, q respectively are densities of P and Q w.r.t. a common measure ω on $\mathcal{X}$.

4.2.3 Generalized mutual information definition

The same way as sensitivity indices are supported by various dissimilarity measures, our suggestion is to define the mutual information using other dissimilarity measures than the Kullback-Leibler divergence. Our novel definition is the following.

Definition 4.1 (D-mutual inforamation)**.** Let D be a dissimilarity measure, for a fixed k, the D-mutual information of a proper prior Π with total mass equal to 1 under k observations is defined as

$$\mathscr{I}_D^k(\Pi) := \mathbb{E}_{T\sim\Pi}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})]. \tag{4.5}$$

In the case where $D = D_f$ is an f-divergence, the mutual information equals

$$\mathscr{I}_{D_f}^k(\Pi) = \mathbb{E}_{T\sim\pi}[D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})] = \int_\Theta \int_{\mathcal{Y}^k} f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right) \ell_k(\mathbf{y}|\theta) d\mu^{\otimes k}(\mathbf{y})\pi(\theta)d\theta. \tag{4.6}$$

It corresponds to the classical definition in the case where $f = -\log$. Actually, the classical definition is a subcase of the study of a particular class of the f-divergences: the δ-divergences. They are defined setting $f = f_\delta$ where

$$f_\delta(x) = \begin{cases} \frac{x^\delta - \delta x - (1-\delta)}{\delta(\delta-1)} & \text{if } \delta \notin \{0,1\}, \\ x\log x - x + 1 & \text{if } \delta = 1, \\ -\log x + x - 1 & \text{if } \delta = 0. \end{cases} \tag{4.7}$$

When $\delta = 0$, we retrieve the original form (KL form) of the mutual information.

4.3 Generalized reference priors

Reference priors are defined as asymptotic maximizers of the mutual information. Using the same formalism as in their original definition (that is expressed in the chapter 3), we suggest the following definition of generalized reference priors using our generalized mutual information.

Definition 4.2 (Generalized reference prior)**.** Let D be a dissimilarity measure and $\mathcal{P}$ a set of priors on Θ. A prior $\Pi \in \mathcal{P}$ is called a D-reference prior over $\mathcal{P}$ with rate $\varphi(k)$ if there exists an openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$ and for any i: $0 < \Pi(\Theta_i) < \infty$ and

$$\lim_{k\to\infty} \varphi(k)[\mathscr{I}_D^k(\Pi(\cdot|\Theta_i)) - \mathscr{I}_D^k(P(\cdot|\Theta_i))] \geq 0 \text{ for all } P \in \mathcal{P} \text{ verifying } 0 < P(\Theta_i) < \infty; \tag{4.8}$$

where $\varphi(k)$ is a positive and monotonous function of k.

This definition matches with the original one when the dissimilarity measure is the Kullback-Leibler divergence. Nevertheless, it introduces the definition of an associated rate proposed by ourselves. In fact, our work provides elements showing that such rate exists and may vary as a function of the dissimilarity measure considered. The idea behind φ is to look for an asymptotic expansion of $\mathscr{I}_D^k(P_1) - \mathscr{I}_D^k(P_2)$ that stands for any tuple of probability distributions (P_1, P_2):

$$\mathscr{I}_D^k(P_1) - \mathscr{I}_D^k(P_2) \underset{k\to\infty}{=} \varphi(k)(L(P_1) - L(P_2)) + o(\varphi(k)), \tag{4.9}$$

where L is a mapping that does not depend on P_1 or P_2. When it resorts to the original mutual information, the rate φ is constant.

The results provided below ensure that the generalized reference priors verify similar properties as the original reference priors.

Proposition 4.1 (Invariance by reparametrization)**.** *Consider a reparametrization $\phi = g(\theta)$ with g being a diffeomorphism and $\mathcal{P}$ a set of priors on Θ. Then if Π is a reference prior over $\mathcal{P}$ for the model parameterized by θ, $\tilde{\Pi}$ is a reference prior over $\tilde{\mathcal{P}}$ for the model parameterized by ϕ, where:*

$$\tilde{\mathcal{P}} = \{M(\pi|\det g_\phi^{-1}|),\ M(\pi) \in \mathcal{P})\}, \tag{4.10}$$

with $M : f \in \mathscr{R} \longmapsto (B \mapsto \int_B f(\theta)d\theta) \in \mathscr{M}^\nu$.

Proof. The proposition comes from the fact that $\mathscr{I}_D^k$ is clearly invariant by reparametrization: for any prior Π having a density π on Θ, and any $\tilde{\Theta} \subset \Theta$ with $\Pi(\tilde{\Theta}) \in (0, \infty)$, calling $\tilde{\Xi} = g(\tilde{\Theta})$,

$$\mathscr{I}_D^k(\Pi(\cdot|\tilde{\Theta})) = \mathbb{E}_{T\sim\Pi(\cdot|\tilde{\Theta})}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})] = \mathbb{E}_{\Phi\sim\tilde{\Pi}(\cdot|\tilde{\Xi})}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|g^{-1}(\Phi)})] =: \tilde{\mathscr{I}}_D^k(\tilde{\Pi}(\cdot|\tilde{\Xi})) \tag{4.11}$$

where $\tilde{\Pi}$ is defined by its density $\tilde{\pi} = \pi|\det d_\phi g^{-1}|$, and $\tilde{\mathscr{I}}_D^k$ denoted the D-mutual information on the reparameterized model. □

Proposition 4.2 (Consistency with compact cases)**.** *Assume Θ is compact. Let $\mathcal{P}$ be a set of continuous priors. If Π is a reference prior over $\mathcal{P}$ with rate φ, then*

$$\lim_{k\to\infty} \varphi(k)[\mathscr{I}_D^k(\Pi) - \mathscr{I}_D^k(P)] \geq 0 \textit{ for all } P \in \mathcal{P}. \tag{4.12}$$

Proof. As the priors are continuous, their restrictions are proper on any compact. Thus, the quantities involved in the proposition are well-defined. As Π is a reference prior over $\mathcal{P}$ we can consider an associated openly increasing sequence of compacts $(\Theta_i)_{i\in\mathbb{N}}$ that covers Θ. Since Θ is compact there exists i such that $\Theta_i = \Theta$. Hence the result since the renormalized restriction $\Pi(\cdot|\Theta_i)$ verifies the statement of the proposition by definition. □

Proposition 4.3 (Invertibility)**.** *Consider that Π is a continuous prior and let f be a convex function.*

1. *If for a compact set $\tilde{\Theta}$,*
$$\sup_{\theta\in\tilde{\Theta}} \sup_{\theta'\in\tilde{\Theta}} \int_{\mathcal{Y}^k} f\left(\frac{\ell_k(\mathbf{y}|\theta')}{\ell_k(\mathbf{y}|\theta)}\right) \ell_k(\mathbf{y}|\theta) d\mu^{\otimes k}(\mathbf{y}) < \infty \tag{4.13}$$
Then the D_f-mutual information value of $\Pi(\cdot|\tilde{\Theta})$ is finite.

2. *If there exist $K \geq 0$ such that for any $x > 0$, $|f(x)\mathbb{1}_{f(x)<0}| \leq Kx$ then the D_f-mutual information restricted to compact expression can be inverted back:*
$$\mathbb{E}_{T\sim\Pi(\cdot|\tilde{\Theta})}[D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})] = \mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k}}[D(\Pi(\cdot|\tilde{\Theta})||\mathbb{P}_{T|\mathbf{Y}_k})]. \tag{4.14}$$

Proof. We denote $\tilde{\Pi} = \Pi(\cdot|\tilde{\Theta})$ and $\tilde{\pi}$ the latter's density. We have

$$\begin{aligned}\mathscr{I}_{D_f}^k(\tilde{\Pi}) &= \int_{\tilde{\Theta}} \int_{\mathcal{Y}^k} f\left(\frac{p_{\mathbf{Y}^k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right) \ell_k(\mathbf{y}|\theta) d\mu^{\otimes k}(\mathbf{y}) \tilde{\pi}(\theta) d\theta \\ &\leq \sup_{\theta\in\tilde{\Theta}} \int_{\mathcal{Y}^k} \int_{\tilde{\Theta}} f\left(\frac{\ell_k(\mathbf{y}|\theta')}{\ell_k(\mathbf{y}|\theta)}\right) \ell_k(\mathbf{y}|\theta) \tilde{\pi}(\theta') d\theta' d\mu^{\otimes k}(\mathbf{y}),\end{aligned} \tag{4.15}$$

using the convexity of f. Since $\tilde{\Theta}$ is compact, $\tilde{\pi}$ is bounded. Thus, we obtain $\mathscr{I}_{D_f}^k(\tilde{\Pi}) < \infty$.

For inverting the integrals, we notice that the assumption on f allows writing

$$\int_{\tilde{\Theta}} \int_{\mathcal{Y}^k} \left| f\left(\frac{p_{\mathbf{Y}^k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right)\right| \ell_k(\mathbf{y}|\theta) d\mu^{\otimes k}(\mathbf{y}) \tilde{\pi}(\theta) d\theta \leq 2K + \int_{\tilde{\Theta}} \int_{\mathcal{Y}^k} f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right) \ell_k(\mathbf{y}|\theta) d\mu^{\otimes k}(\mathbf{y}) \tilde{\pi}(\theta) d\theta \tag{4.16}$$

using $|f(x)| \leq 2Kx\mathbb{1}_{f(x)<0} + f(x)$. We deduce that, if $\mathbb{E}_{T\sim\tilde{\Pi}}[D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})]$ is finite, then Fubini-Lebesgue's theorem can be applied, leading to the equality (4.14). The same way, we have

$$\int_{\mathcal{Y}^k} \int_{\tilde{\Theta}} \left| f\left(\frac{\tilde{\pi}(\theta)}{p(\theta|\mathbf{y})}\right)\right| p(\theta|\mathbf{y}) d\theta p_{\mathbf{Y}_k}(\mathbf{y}) d\mu^{\otimes k}(\mathbf{y}) \leq 2K + \int_{\mathcal{Y}^k} \int_{\tilde{\Theta}} f\left(\frac{\tilde{\pi}(\theta)}{p(\theta|\mathbf{y})}\right) p(\theta|\mathbf{y}) d\theta p_{\mathbf{Y}_k}(\mathbf{y}) d\mu^{\otimes k}(\mathbf{y}), \tag{4.17}$$

so that if $\mathbb{E}_{\mathbf{Y}_k \sim \mathbb{P}_{\mathbf{Y}_k}}[D_f(\tilde{\Pi}||\mathbb{P}_{T|\mathbf{Y}_k})]$ is finite, the integrals can also be inverted and equality (4.14) still stands. We conclude by deducing that, if one of the quantity in eq. (4.14) is infinite, the other cannot be finite, letting the equality to stand as well. □

Proposition 4.4 (Compatibility with sufficient statistics)**.** *Suppose D is an f-divergence such that there exists $K \geq 0$ with for all $x > 0$, $|f(x)\mathbb{1}_{f(x)<0}| \leq Kx$. Consider $Z := z(\mathbf{Y}_k) \in \mathcal{Z}$ a sufficient statistic of the model built on $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$. The D_f-mutual information remains stable by considering Z instead of $\mathbf{Y}_k$:*

$$\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) = \mathbb{E}_{T\sim\Pi(\cdot|\tilde{\Theta})}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})] = \mathbb{E}_{T\sim\Pi(\cdot|\tilde{\Theta})}[D(\mathbb{P}_Z||\mathbb{P}_{Z|T})] =: \tilde{\mathscr{I}}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) \tag{4.18}$$

for all $\Pi \in \mathscr{M}^\nu$; where $\tilde{\mathscr{I}}^k$ denotes the mutual information considering the second model.

Proof. If Z is a sufficient statistic, the posterior verifies $p(\theta|z(\mathbf{y})) = p(\theta|\mathbf{y})$. Thus, the terms$D_f(\Pi(\cdot|\tilde{\Theta})||\mathbb{P}_{T|\mathbf{Y}_k})$ and $D_f(\Pi(\cdot|\tilde{\Theta})||\mathbb{P}_{T|Z})$ are equal. Hence the result since

$$\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) = \mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k}}[D_f(\Pi(\cdot|\tilde{\Theta})||\mathbb{P}_{T|\mathbf{Y}_k})] \quad \text{and} \quad \tilde{\mathscr{I}}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) = \mathbb{E}_{Z\sim\mathbb{P}_Z}[D_f(\Pi(\cdot|\tilde{\Theta})||\mathbb{P}_{T|Z})]. \tag{4.19}$$

□

To conclude this section, we mention the work of Le (2014). The author proposes a formulation of D_{f_δ}-reference priors as limit of priors when $\Theta \subset \mathbb{R}$, in the same fashion as in Berger, Bernardo, and Sun, 2009 (see chapter 3, theorem 3.1).

Theorem 4.1 (Le, 2014)**.** *Consider a δ-divergence D_{f_δ} with $\delta \in [0,1]$. Assume $\Theta \subset \mathbb{R}$. Let $\mathcal{P}_s$ be the set of continuous positive priors admitting proper posteriors. Let $\Pi^* \in \mathcal{P}_s$ we call $p^*(\cdot|\cdot)$ its posterior density and define for any interior point $\theta_0 \in \Theta$,*

$$f_k(\theta) = \exp\left(\int_{\mathcal{Y}^k} \ell_k(\mathbf{y}|\theta)\log(p^*(\theta|\mathbf{y}))d\mu^{\otimes k}(\mathbf{y})\right) \quad \textit{and} \quad f(\theta) = \lim_{k\to\infty}\frac{f_k(\theta)}{f_k(\theta_0)}. \tag{4.20}$$

Under mild assumptions that are similar to the ones of Berger, Bernardo, and Sun, 2009, the prior admitting f (if it exists) as density is a reference prior over $\mathcal{P}_s$.

4.4 Towards δ-divergences-reference priors

The objective of this section is to study the D_f-reference priors for certain classes of functions f that include δ-divergence. We precise that from now on, the priors will always be considered continuous (they admit ν-a.e. continuous and locally bounded density functions). We recall that the set of such priors is denoted by $\mathscr{M}^\nu_C$, and they all admit a density in the set $\mathscr{R}_{C^b}$.

We recall that the random vectors are given by a process $\overline{Y}$ defined on a conditional probability space $(\mathbf{\Omega}, \Xi, [\mathbf{\Pi}])$ (see proposition 3.1). If $\tilde{\Theta}$ is a compact subset of Θ, the restriction $(\tilde{\mathbf{\Omega}}, \tilde{\Xi}, \mathbf{\Pi}(\cdot|T \in \tilde{\Theta}))$ of the conditional probability space to $\{T \in \tilde{\Theta}\}$ is a probability space. For some $\theta \in \tilde{\Theta}$ we will adopt the notation $\mathbb{P}_\theta$ for the conditional probability on $\tilde{\mathbf{\Omega}}$ to $\{T = \theta\}$. That means, for any k, any $A \in \mathscr{Y}^{\otimes k}$,

$$\mathbb{P}_\theta(\mathbf{Y}_k \in A) = \mathbb{P}_{\mathbf{Y}_k|\theta}(A). \tag{4.21}$$

We denote by $\mathbb{E}_\theta$ the associated expectation. The proofs of the results that are stated in this section are given in section 4.6.

4.4.1 A first result

A first useful result comes when we consider an asymptotic expansion of f in the neighborhood of 0:

$$f(x) \underset{x\to 0}{=} g_1(x) + o(g_2(x)) \tag{4.22}$$

for some $g_1 : (0,\infty) \longrightarrow \mathbb{R}$, $g_2 : (0,\infty) \longrightarrow (0,\infty)$.

Proposition 4.5. *Let $\tilde{\Theta}$ be a compact subset of Θ. Consider a prior $\Pi \in \mathscr{M}_{\mathcal{C}}^{v}$ and denote by $\pi \in \mathscr{R}_{\mathcal{C}^b}$ a density of $\Pi(\cdot|\tilde{\Theta})$. For almost every θ in the interior of* Supp π *there exists $c = c_\theta$, $C = C_\theta$ positive constants such that the following convergence in probability holds:*

$$\tilde{g}_2^{C,c}(k^{-d/2})^{-1}\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - g_1\left(k^{-d/2}\pi(\theta)(2\pi)^{d/2}|\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right)\right| \xrightarrow[k\to\infty]{\mathbb{P}_\theta} 0, \tag{4.23}$$

where $\tilde{g}_2^{C,c}(k^{-d/2}) = \sup_{x\in[c,C]} g_2(xk^{-d/2})$, and S_k denotes $\frac{1}{\sqrt{k}}\sum_{i=1}^k \nabla_\theta \log \ell(Y_i|\theta)$.

There are three things we can say about this first asymptotic result. First, it emphasizes the asymptotic link between two ratios: (i) the marginal over the likelihood on the one hand, (ii) the chosen prior density over Jeffreys one on the other hand. The intuition comes from noticing that $S_k^T\mathcal{I}(\theta)^{-1}S_k$ converges in distribution to a standard Gaussian, which does not depend on θ. Second, this result also highlights that it is the behavior of f in the neighborhood of 0 that is decisive. Third, this result is two steps away from giving an asymptotic expansion of the D_f-mutual information: (i) stating that this limit stands in $L^1(\mathbb{P}_\theta)$, (ii) verifying the limit can be switched with the expectation w.r.t. $T \sim \Pi(\cdot|\tilde{\Theta})$.

One can notice that this proposition can be applied with $f = -\log$, $g_1 = f$ and $g_2 = 1$. In that case we obtain the convergence in $\mathbb{P}_\theta$-probability already stated in Clarke and Barron, 1990:

$$\log\frac{\ell_k(\mathbf{Y}_k|\theta)}{p_{\mathbf{Y}_k}(\mathbf{Y}_k)} - \frac{d}{2}\log k + \log\pi(\theta) - \frac{1}{2}\log|\mathcal{I}(\theta)| + \frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k \xrightarrow[k\to\infty]{\mathbb{P}_\theta} 0. \tag{4.24}$$

4.4.2 Results when $\delta < 0$

In what follows, we precise the asymptotic form of f in the neighborhood of 0:

$$f(x) \underset{x\to 0}{=} ax^\delta + o(x^\delta). \tag{4.25}$$

We also require f to be locally bounded and to be controlled as $x \longrightarrow \infty$:

$$f \underset{x\to\infty}{=} O(x). \tag{4.26}$$

We invite the reader to remark that such function f aligns with translated δ-divergences, among others.

In this section we suppose $\delta \in (-1, 0)$. An asymptotic limit of the D_f-mutual information can be given once the following assumption is satisfied.

Assumption 4.1. For every compact subset $\tilde{\Theta}$ of Θ, there exist K_0 such that the quantity

$$\mathbb{E}_\theta\left[\sup_{\theta\in\tilde{\Theta}}\|\nabla_\theta^2 \log \ell(Y_1|\tilde{\theta})\|\right] \leq K_0. \tag{4.27}$$

Theorem 4.2. *Under assumption 4.1, let $\tilde{\Theta} \subset \Theta$ be a compact set and $\Pi \in \mathscr{M}_{\mathcal{C}}$ such that $\Pi(\cdot|\tilde{\Theta})$ admits a positive density π. The quantity $k^{d\delta/2}\mathscr{I}_{D_f}^k(\Pi(\cdot|\tilde{\Theta}))$ has a positive limit when $k \longrightarrow \infty$:*

$$\lim_{k\to\infty} k^{d\delta/2}\mathscr{I}_{D_f}^k(\Pi(\cdot|\tilde{\Theta})) = aC_\delta\int_{\tilde{\Theta}}\pi(\theta)^{1+\delta}|\mathcal{I}(\theta)|^{-\delta/2}d\theta, \tag{4.28}$$

where $C_\delta = (2\pi)^{d\delta/2}(1-\delta)^{-d/2}$.

Proposition 4.6. *With the assumptions of the above theorem, if* $a(\delta+1) > 0$, *then*

$$\lim_{k\to\infty} k^{d\delta/2}(\mathscr{I}^k_{D_f}(\mathbf{J}(\cdot|\tilde{\Theta})) - \mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta}))) \geq 0, \tag{4.29}$$

where $\mathbf{J}(\cdot|\tilde{\Theta})$ *is the restricted Jeffreys prior on* $\tilde{\Theta}$, *its density is* $J(\theta) = |\mathcal{I}(\theta)|^{1/2} / \int_{\tilde{\Theta}} |\mathcal{I}(\tilde{\theta})|^{1/2} d\tilde{\theta}$. *The equality in eq. (4.29) stands if and only if* $\pi = J$ *a.e.*

The above proposition makes the Jeffreys prior class being the unique D_f-reference prior class over $\mathscr{M}^v_{\mathcal{C}^*}$, where $\mathscr{M}^v_{\mathcal{C}^*}$ represents the subset of $\mathscr{M}^v_{\mathcal{C}}$ of priors admitting positive densities on Θ. We formalize this statement in the following theorem.

Theorem 4.3. *Let* $b \in \mathbb{R}$, *define* $\overline{f} = f + b$ *and assume* $a(\delta+1) > 0$. *Under assumption 4.1, the Jeffreys prior class is the unique* $D_{\overline{f}}$*-reference prior class over* $\mathscr{M}^v_{\mathcal{C}^*} / \simeq$. *That means the Jeffreys prior* $\mathbf{J}$ *is a* $D_{\overline{f}}$*-reference prior over* $\mathscr{M}^v_{\mathcal{C}^*}$ *and if* Π *is another reference prior over* $\mathscr{M}^v_{\mathcal{C}^*}$, *then* $\Pi \simeq \mathbf{J}$.

4.4.3 Results when $\delta > 0$

As in the previous section, we suppose that f takes the following form in the neighborhood of 0:

$$f(x) \underset{x\to 0}{=} ax^\delta + o(x^\delta). \tag{4.30}$$

And we require as well f to be locally bounded and to be controlled as $x \longrightarrow \infty$:

$$f \underset{x\to\infty}{=} O(x). \tag{4.31}$$

In comparison with the preceding section, we suppose here that $\delta \in (0,1)$. The elucidating of a similar result than the one in the previous section requires the satisfaction of stronger assumptions.

Assumption 4.2. For any compact subset $\tilde{\Theta} \subset \Theta$, there exists $m > 0$ such that for all $\theta \in \tilde{\Theta}$,

$$\mathbb{P}_\theta\left(\forall x \in \mathbb{R}^d,\ \inf_{\tilde{\theta}\in\tilde{\Theta}} -x^T\nabla^2_{\tilde{\theta}} \log \ell(Y_1|\tilde{\theta})x > m\|x\|^2\right) = 1. \tag{4.32}$$

Assumption 4.3. For any compact subset $\tilde{\Theta} \subset \Theta$, the random variables $S_k = \frac{1}{\sqrt{k}}\sum_{i=1}^k \nabla_\theta \log \ell(y_i|\theta)$ are sub-Gaussians: there exist $\xi > \delta/2$ and $K_1 > 0$ such that for all $\theta \in \tilde{\Theta}$, and all k, $\mathbb{E}_\theta e^{\frac{\xi}{m}\|S_k\|^2} < K_1$.

Theorem 4.4. *Suppose assumptions 4.2 and 4.3. Let* $\tilde{\Theta} \subset \Theta$ *be a compact set and* $\Pi \in \mathscr{M}^v_{\mathcal{C}}$. *We denote* $\pi \in \mathscr{R}_{\mathcal{C}^b}$ *a density of* $\Pi(\cdot|\tilde{\Theta})$. *Then* $k^{d\delta/2}\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta}))$ *admits a finite limit when* $k \longrightarrow \infty$:

$$\lim_{k\to\infty} k^{d\delta/2}\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) = l(\pi) = aC_\delta \int_{\tilde{\Theta}} \pi(\theta)^{1+\delta}|\mathcal{I}(\theta)|^{-\delta/2} d\theta, \tag{4.33}$$

where $C_\delta = (2\pi)^{d\delta/2}(1-\delta)^{-d/2}$.

Proposition 4.7. *With the assumptions of the above theorem, if* $a(\delta+1) > 0$, *then*

$$\lim_{k\to\infty} k^{d\delta/2}(\mathscr{I}^k_{D_f}(\mathbf{J}(\cdot|\tilde{\Theta})) - \mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta}))) \geq 0, \tag{4.34}$$

where $\mathbf{J}(\cdot|\tilde{\Theta})$ *is the restricted Jeffreys prior on* $\tilde{\Theta}$, *its density is* $J(\theta) = |\mathcal{I}(\theta)|^{1/2} / \int_{\tilde{\Theta}} |\mathcal{I}(\tilde{\theta})|^{1/2} d\tilde{\theta}$. *The equality in eq. (4.34) stands if and only if* $\pi = J$ *a.e.*

In the same way as in the previous subsection, this result implies that the Jeffreys prior is the up-to-a-constant-unique D_f-reference prior over $\mathscr{M}_{\mathcal{C}}^{\nu}$.

Theorem 4.5. *Let $b \in \mathbb{R}$, define $\overline{f} = f + b$ and assume $a(\delta+1) > 0$. Under assumptions 4.2 and 4.3, the Jeffreys prior class is the unique $D_{\overline{f}}$-reference prior class over $\mathscr{M}_{\mathcal{C}}^{\nu} / \simeq$.*

4.5 Discussions

4.5.1 About the results

Regarding the optimal characteristic of the Jeffreys prior, we argue that it actually supports the relevance of our construction, in continuity with the original reference prior framework which we rely on. Indeed, we remind that Jeffreys prior is optimal for KL-mutual information (as shown in Clarke and Barron, 1994). Although its multivariate derivation is criticized, its expression remains robust and appreciated in low-dimensional cases.

For a practical implementation when the dimension is high, some authors would recommend referring to hierarchical strategies such as the one proposed by Berger, Bernardo, and Sun (2015) and evoked in the chapter 3 (section 3.3.2). As these strategies are based on the sequential derivation of reference priors on sequentially conditioned models, they take the form of hierarchically derived low dimensional Jeffreys priors. Consequently, since our result supports the consideration of Jeffreys prior as a universal maximizer of the mutual information, they left that methodology unchanged.

Additionally, we draw attention to the open-ended nature of the results we express. Indeed, theorems 4.2 and 4.4 provide an analytical limit of the mutual information that takes the form of a functional l of the prior. One can notice that a reference prior is a prior that maximizes l. Thus, a maximization of our function l over a well-chosen set of priors constitutes an easy path to express reference priors that could differ from Jeffreys, as expressed in the theorem below.

Theorem 4.6. *Let $f \underset{x\to 0}{=} ax^{\delta} + o(x^{\delta})$, $f \underset{x\to\infty}{=} O(x)$. Assume $\delta \in (-1,1)\setminus\{0\}$, $a(\delta+1) > 0$. Assume assumption 4.1 if $\delta < 0$ or assumptions 4.2 and 4.3 if $\delta > 0$. For some $\pi \in \mathscr{R}_{\mathcal{C}^b}$, define*

$$l(\pi) = aC_{\delta} \int_{\Theta} \pi(\theta)^{1+\delta} |\mathcal{I}(\theta)|^{-\delta/2} d\theta \tag{4.35}$$

with $C_{\delta} = (2\pi)^{d\delta/2}(1-\delta)^{-d/2}$.
Let $\mathcal{P}$ be a set of continuous priors. Denote by $\mathcal{R} \subset \mathscr{R}_{\mathcal{C}^b}$ the set of their associated densities. Let $\Pi \in \mathcal{P}$ whose density is denoted by $\pi \in \mathcal{R}$. If there exists an openly increasing sequence $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$ with for any i: $\Pi \in (0,\infty)$ and $\pi \cdot \mathbb{1}_{\Theta_i} \propto \pi_i$ where:

$$\pi_i \in \underset{p_i \in \mathcal{R}_i}{\arg\max}\, l(p_i), \text{ with } \mathcal{R}_i = \left\{ p : \theta \mapsto \frac{\tilde{p}(\theta)}{\int_{\Theta_i} \tilde{p}(\tilde{\theta}) d\tilde{\theta}}, \tilde{p} \in \mathcal{R}, \int_{\Theta_i} \tilde{p} \in (0,\infty) \right\}; \tag{4.36}$$

then Π is a D_f-reference prior over $\mathcal{P}$.

One is invited to notice that when Θ is compact, maximizing l over the normalized densities on Θ is sufficient. The following subsection suggests an example of a constrained problem.

4.5.2 About the introduction of simple constraints

Here we illustrate with a simple example an application of theorem 4.6 to the derivation of constrained reference priors. Consider a Bernoulli modeling: $Y|(T=\theta) \sim \mathcal{B}(\theta)$, $\mathcal{B}(\theta)$ being the Bernoulli distribution

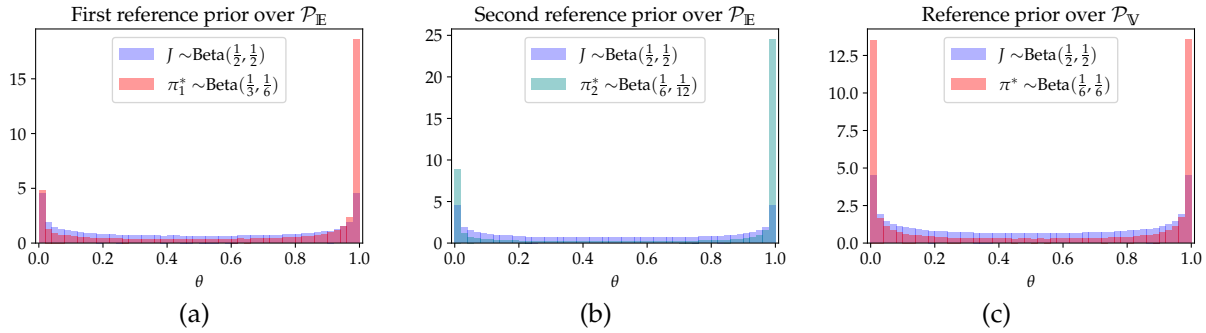

Figure 4.1: Comparison of histograms of samples from Jeffreys prior and from the D_f-reference priors over a set $\mathcal{P}_{\mathbb{E}} = \{\pi = \text{Beta}, \mathbb{E}_{T\sim\pi} T = \frac{2}{3}\}$ (fig. (a) and (b)), and over a set $\mathcal{P}_{\mathbb{V}} = \{\pi = \text{Beta}, \mathbb{V}_{T\sim\pi} T = \frac{3}{16}\}$ (fig. (c)). In this example, their exist 2 solutions π_1^* (in fig. (a)) and π_2^* (in fig. (b)) of reference priors over $\mathcal{P}_{\mathbb{E}}$. δ has been set to $1/2$ in this example.

with parameter θ. We take $\Theta = [0, 1]$ and write $\ell(y|\theta) = \theta^y (1-\theta)^{1-y}$. Within this modeling, the Jeffreys prior takes the form of a Beta distribution: Beta$(1/2, 1/2)$ (see e.g. Robert, 2007).

A possible type of constraints is the restriction of the moments of the prior. For instance, we denote $\mathcal{P}_{\mathbb{E}}$ the class of priors $\mathcal{P}_{\mathbb{E}} = \{\pi = \text{Beta}(\lambda_1, \lambda_2), \mathbb{E}_{T\sim\pi}[T] = 1/c\}$. A prior in $\mathcal{P}_{\mathbb{E}}$ can be parameterized as $\pi_\lambda = \text{Beta}(\lambda, \lambda(c-1))$, $\lambda \in (0, 1)$. Then, solving $\lambda^* = \arg\max_\lambda l(\pi_\lambda)$ can be carried out numerically or analytically to issue the reference prior π_{λ^*} over $\mathcal{P}_{\mathbb{E}}$. The same way, some class $\mathcal{P}_{\mathbb{V}}$ could be considered this time to fix the variance of the prior.

In fig. 4.1 are plotted the D_f-reference priors over different sets of Beta priors with constrained expectations first, and with constrained variances second. We notice that, when it concerns the class $\mathcal{P}_{\mathbb{E}}$, two priors are reference priors. Actually, the considered class is not convex in this case, which makes the uniqueness not ensured. An additional constraint on the researched reference prior should be set to make it unique. For instance, with the idea of constructing non-informative priors, the one with maximal Shanon's entropy could be chosen. In our example, π_1^* (fig. 4.1.(a)) has a higher entropy than π_2^* (fig. 4.1.(b)).

4.5.3 About the assumptions and their limitations

The sub-Gaussian assumption over the likelihood, which is required for the statement of the result in the case $\delta < 0$ remains somehow restrictive. For instance, the univariate Gaussian model with unknown variance does not satisfy this assumption. However, it is still verified by other regular statistical models, such as the univariate normal model with known variance, or continuous compact models ($\mathcal{Y}$ compact and $(y, \theta) \mapsto \ell(y|\theta)$ continuously differentiable). Note that due to the asymptotic normality of the r.v. $\mathcal{I}(\theta)^{-1/2} S_k$ induced by the central limit theorem, the parameter ξ of its assumed Gaussian tail cannot exceed $1/2$. This restricts δ to be smaller than 1.

Furthermore, in opposition with the classical settings for f-divergences which tend to suppose the convexity of f, the Jeffreys prior optimally within our results is actually implied by the concavity of $x \mapsto f(1/x)$ in the neighborhood of 0. This statement is showcased by the heuristic developed in the next subsection.

That gives some insight about the potential use of other functions f than the ones we have considered. Note that Clarke and Sun (1997) proved under some assumptions that another prior than Jeffreys prior is the D-reference prior when D is a chi-square distance. This dissimilarity measure corresponds to the case where $f(x) = x^{-1}$, making $x \mapsto f(1/x)$ both convex and concave. A conclusion about the optimum of such generalized mutual information thus requires stronger regularity assumptions on the likelihood to push the asymptotic analysis further.

4.5.4 About the robustness of Jeffreys prior with different divergences

When D_f is a $\delta = 0$-divergence

The results elucidated in the previous section explicitly address f-divergences which are δ-divergences. It is important to recall that when $\delta = 0$, the δ-divergence is usually defined as a KL divergence, more precisely it is an f-divergence with $f = -\log$.

This way, the $\delta = 0$-divergence leads to the original definition of the mutual information. In this case, results similar to those of theorems 4.3 and 4.5 exist (see chapter 3, section 3.3.2).

A heuristic for other f- divergences

We provide a heuristic for the derivation of the limit of a D_f-mutual information when $k \longrightarrow \infty$. It is similar to discussions conducted by Muré (2018) and Xie and Barron (1997).

The result of proposition 4.5 takes the form of asymptotic connection between $p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta)$ and $e^{\frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k}$. This connection can be anticipated from the second order Taylor expansion written as follows: for every $\mathbf{y}, \theta, \tilde{\theta}$, there exists a θ' in the segment between θ and $\tilde{\theta}$ such that

$$\log\left(\frac{\prod_{i=1}^k \ell(y_i|\tilde{\theta})}{\prod_{i=1}^k \ell(y_i|\theta)}\right) = (\tilde{\theta}-\theta)^T \sum_{i=1}^k \nabla_\theta \log \ell(y_i|\theta) + \frac{1}{2}(\tilde{\theta}-\theta)^T \sum_{i=1}^k \nabla_\theta^2 \log \ell(y_i|\theta')(\tilde{\theta}-\theta). \tag{4.37}$$

Indeed, $p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta) = \int_\Theta \frac{\prod_{i=1}^k \ell(Y_i|\tilde{\theta})}{\prod_{i=1}^k \ell(Y_i|\theta)}\pi(\tilde{\theta})d\tilde{\theta}$. The terms $\nabla_\theta^2 \log \ell_k(y_i|\theta)$ are definite negative, so that when we exponentialize the Taylor expansion, they vanish quickly when $\tilde{\theta}$ is not close to θ.

We deduce that the mass is concentrated around θ in the integrals that define $p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta)$, so that

$$\begin{aligned} p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta) &\approx \pi(\theta)\int_\Theta \exp\left(\sqrt{k}(\tilde{\theta}-\theta)^T S_k - \frac{k}{2}(\tilde{\theta}-\theta)^T\mathcal{I}(\theta)(\tilde{\theta}-\theta)\right)d\tilde{\theta} \\ &= \pi(\theta)(2\pi)^{d/2}|k\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right). \end{aligned} \tag{4.38}$$

To go toward an asymptotic expression of the mutual information, we must evaluate the last term with f and compute its expectation w.r.t. S_k. Conditionally to θ, the terms $\nabla_\theta \log \ell(Y_i|\theta)$ are i.i.d. and a central limit theorem gives that $(S_k)_k$ converges in distribution to a Gaussian with mean $\mathbb{E}_\theta \nabla_\theta \log \ell(Y|\theta) = 0$ and covariance matrix $\mathcal{I}(\theta)$.

That leads to approximating the D_f-mutual information as

$$\mathscr{I}_{D_f}^k(\Pi) \approx \int_\Theta \int_{\mathbb{R}^k} f\left(\pi(\theta)(2\pi)^{d/2}|k\mathcal{I}(\theta)|^{-1/2}e^{\frac{\|x\|^2}{2}}\right)e^{-\frac{\|x\|^2}{2}}(2\pi)^{-d/2}dx\pi(\theta)d\theta. \tag{4.39}$$

We notice that if $x \mapsto f(1/x)$ is concave, the right-hand term in the above equation is smaller than

$$\int_{\mathbb{R}^k} f\left(\left(\int_\Theta (2\pi)^{-d/2}|k\mathcal{I}(\theta)|^{1/2}e^{-\frac{\|x\|^2}{2}}d\theta\right)^{-1}\right)e^{-\frac{\|x\|^2}{2}}(2\pi)^{-d/2}dx, \tag{4.40}$$

which equals the right-hand term in eq. (4.39) evaluated with $\pi(\theta) = |k\mathcal{I}(\theta)|^{1/2}/\int_\Theta |k\mathcal{I}(\tilde{\theta})|^{1/2}d\tilde{\theta}$.

This simple heuristic emphasizes that the decisive assumption on f is that $x \mapsto f(1/x)$ is concave, rather than f being convex. Regarding our developments in section 4.4, the final results on the optimally of Jeffreys stood while $a(\delta+1) > 0$. In our cases, this condition implies the convexity of f in the neighborhood of 0, that is equivalent in these cases to $x \mapsto f(1/x)$ being concave in the neighborhood of 0.

A simple development with a Maximum Mean Discrepancy divergence

We also investigate briefly the D-mutual information in a case where D is not an f-divergence. The dissimilarity measure we have chosen to study is a squared maximum mean discrepancy (MMD):

$$\mathrm{MMD}^2(\mathcal{H}; P||Q) = \mathbb{E}_{X,X'\sim P\otimes P}[K(X,X')] + \mathbb{E}_{Y,Y'\sim Q\otimes Q}[K(Y,Y')] - 2\mathbb{E}_{X,Y\sim P\otimes Q}[K(X,Y)]; \tag{4.41}$$

where $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS), and K is a reproducing kernel.

We study the simple Gaussian model with unknown mean: $\mathcal{Y} = \mathbb{R}$, $\Theta \subset \mathbb{R}$, and $\ell(y|\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\theta)^2}{2\sigma^2}}$. We consider the kernels $K_k : \mathbf{y}, \mathbf{y}' \in \mathcal{Y}^k$, $K_k(\mathbf{y}, \mathbf{y}') = \frac{1}{(\sqrt{2\pi}h)^k} e^{-\frac{\|\mathbf{y}-\mathbf{y}'\|^2}{2h^2}}$, and $\mathcal{H}_k$ their associated RKHS. For any k it is thus possible to define the MMD between $\mathbb{P}_{\mathbf{Y}_k}$ and $\mathbb{P}_{\mathbf{Y}_k|\theta}$ by considering the RKHS $\mathcal{H}_k$. Let $\tilde{\Theta} \subset \Theta$ be compact. For some prior $\Pi \in \mathscr{M}_{\mathcal{C}}^{v}$, denoting $\pi \in \mathscr{R}_{\mathcal{C}^b}$ a density of $\Pi(\cdot|\tilde{\Theta})$ we then have

$$\begin{aligned}\mathscr{I}_{\mathrm{MMD}^2}(\Pi(\cdot|\tilde{\Theta})) = &\frac{1}{(2\pi)^{3k/2}h^k\sigma^{2k}} \int_{\tilde{\Theta}} \int_{\mathbb{R}^k\times\mathbb{R}^k} \prod_{i=1}^{k} e^{-\frac{(y_i-y_i')^2}{2h^2}} e^{-\frac{(y_i-\theta)^2}{2\sigma^2}} e^{-\frac{(y_i'-\theta)^2}{2\sigma^2}} d\mathbf{y}d\mathbf{y}'\pi(\theta)d\theta \\ &- \frac{1}{(2\pi)^{3k/2}h^k\sigma^{2k}} \int_{\tilde{\Theta}\times\tilde{\Theta}} \int_{\mathbb{R}^k\times\mathbb{R}^k} \prod_{i=1}^{k} e^{-\frac{(y_i-y_i')^2}{2h^2}} e^{-\frac{(y_i-\theta)^2}{2\sigma^2}} e^{-\frac{(y_i'-\theta')^2}{2\sigma^2}} d\mathbf{y}d\mathbf{y}'\pi(\theta)\pi(\theta')d\theta d\theta'. \quad (4.42)\end{aligned}$$

In this very specific case, it is possible to derive and study the MMD^2-reference priors. The proposition below summarizes our results.

Proposition 4.8. *Let $\tilde{\Theta} \subset \Theta$ be a compact set.*

1. *Let Π, $\tilde{\Pi} \in \mathscr{M}_{\mathcal{C}}^{v}$, we denote respectively π and $\tilde{\pi}$ the densities of $\Pi(\cdot|\tilde{\Theta})$ and $\tilde{\Pi}(\cdot|\tilde{\Theta})$. We have:*

$$\lim_{k\to\infty} C_1^{-k+1}\sqrt{k}(\mathscr{I}_{MMD^2}(\Pi(\cdot|\tilde{\Theta})) - \mathscr{I}_{MMD^2}(\tilde{\Pi}(\cdot|\tilde{\Theta}))) = -\int_{\tilde{\Theta}} \pi(\theta)^2 d\theta + \int_{\tilde{\Theta}} \tilde{\pi}(\theta)^2 d\theta, \quad (4.43)$$

   with $C_1 = \frac{(2\sigma^2+h^2)^{-1/2}}{\sqrt{2\pi}}$. And the Jeffreys prior class is the unique MMD^2-reference prior class over $\mathscr{M}_{\mathcal{C}}^{v}$.

2. *For a density π we denote by $M(\pi)$ its associated normalized prior: $M(\pi) = B \mapsto \int_B \pi / \int_{\tilde{\Theta}} \pi$. Let π_k^* be continuous densities such that for any k, $\mathscr{I}_{MMD^2}(M(\pi_k^*)) \geq \mathscr{I}_{MMD^2}^{k}(M(\pi))$ for any other $\pi \in \mathscr{M}_{\mathcal{C}}^{v}$. If $(\pi_k^*)_k$ converges uniformly to a function π^* on $\tilde{\Theta}$, then π^* is the Jeffreys prior density: $\pi^* \propto 1$.*

Proof. First we derive for any $\theta, \theta' \in \tilde{\Theta}$

$$\begin{aligned}&\int_{\mathbb{R}\times\mathbb{R}} e^{-\frac{(y-y')^2}{2h^2}} e^{-\frac{(y-\theta)^2}{2\sigma^2}} e^{-\frac{(y'-\theta')^2}{2\sigma^2}} dydy' \\ &= \sqrt{2\pi\tilde{\sigma}^2} \int_{\mathbb{R}} e^{-\frac{y'^2}{2h^2}} e^{-\frac{\theta^2}{2\sigma^2}} e^{\frac{1}{2\tilde{\sigma}^2}\left(\frac{\tilde{\sigma}^2}{h^2}y' + \frac{\tilde{\sigma}^2}{\sigma^2}\theta\right)^2} e^{-\frac{(y'-\theta')^2}{2\sigma^2}} dy' && \text{with } \tilde{\sigma}^{-2} = \frac{1}{h^2} + \frac{1}{\sigma^2} \\ &= 2\pi\tilde{\sigma}\overline{\sigma} e^{-\frac{\theta^2}{2\sigma^2}} e^{-\frac{\theta'^2}{2\sigma^2}} e^{\frac{1}{2}\frac{\tilde{\sigma}^2}{\sigma^4}\theta^2} e^{\frac{1}{2\tilde{\sigma}^2}\left(\frac{\tilde{\sigma}^2}{\sigma^2}\theta' - \frac{\tilde{\sigma}^2\tilde{\sigma}^2}{h^2\sigma^2}\theta\right)^2} && \text{with } \overline{\sigma}^{-2} = \frac{1}{h^2} - \frac{\tilde{\sigma}^2}{h^4} + \frac{1}{\sigma^2} = \frac{1}{\sigma^2+h^2} + \frac{1}{\sigma^2} \quad (4.44) \\ &= 2\pi\tilde{\sigma}\overline{\sigma} e^{-\frac{1}{2}\frac{1}{2\sigma^2+h^2}(\theta'-\theta)^2}.\end{aligned}$$

That allows us to write

$$\mathscr{I}_{\mathrm{MMD}^2}(\Pi(\cdot|\tilde{\Theta})) = -C_1^k \int_{\tilde{\Theta}\times\tilde{\Theta}} e^{-\frac{k}{2}\frac{1}{2\sigma^2+h^2}(\theta'-\theta)^2} \pi(\theta)\pi(\theta')d\theta d\theta' + C_1^k, \quad (4.45)$$

with $C_1 = \frac{\tilde{\sigma}\overline{\sigma}}{\sqrt{2\pi}h\sigma^2} = \frac{(2\sigma^2+h^2)^{-1/2}}{\sqrt{2\pi}}$.

1. To prove the first statement of the theorem, we start by computing the limit as $k \longrightarrow \infty$ of the integrals $C_1\sqrt{k}\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{1}{2\sigma^2+h^2}(\theta'-\theta)^2}\pi(\theta)d\theta$ for a fixed θ' in which π is continuous. The scheme of this proof is standard as this integral is the convolution of π with an approximate Dirac distribution. Let $\varepsilon > 0$, by continuity of π there exists a neighborhood V of θ' such that for any $\theta \in V$, $|\pi(\theta) - \pi(\theta')| <$

$\varepsilon/2$. Additionally, it is possible to choose a k_0 such that for all $k \geq k_0$, $C_1\sqrt{k}\int_{\mathbb{R}\setminus V} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}d\theta < \varepsilon/(4\sup_{\tilde{\Theta}}\pi)$. This way, for all $k \geq k_0$,

$$\left|C_1\sqrt{k}\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{1}{2\sigma^2+h^2}(\theta'-\theta)^2}\pi(\theta)d\theta - \pi(\theta')\right| \leq C_1\sqrt{k}\int_{\mathbb{R}} e^{-\frac{k}{2}\frac{1}{2\sigma^2+h^2}(\theta'-\theta)^2}|\pi(\theta)\mathbb{1}_{\theta\in\tilde{\Theta}} - \pi(\theta')|d\theta < \varepsilon. \tag{4.46}$$

That being stated, we can apply a dominated convergence theorem since for ν-a.e. θ' and for all k, the terms $|C_1\sqrt{k}\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta-\theta')^2}{2\sigma^2+h^2}}\pi(\theta)\pi(\theta')d\theta| \leq (\sup_{\tilde{\Theta}}\pi)^2$, which is summable on $\tilde{\Theta}$. We obtain

$$\lim_{k\to\infty} C_1^{-k+1}\sqrt{k}(\mathscr{I}_{\mathrm{MMD}^2}(\Pi(\cdot|\tilde{\Theta})) - \mathscr{I}_{\mathrm{MMD}^2}(\tilde{\Pi}(\cdot|\tilde{\Theta}))) = -\int_{\tilde{\Theta}}\pi(\theta)^2d\theta + \int_{\tilde{\Theta}}\tilde{\pi}(\theta)^2d\theta. \tag{4.47}$$

We recall that the Jeffreys prior $\mathbf{J}$ for this model is the uniform prior: $J \propto 1$. Since Chauchy-Schwarz inequality implies $\int_{\tilde{\Theta}}\tilde{\pi}(\theta)d\theta \leq \sqrt{\nu(\tilde{\Theta})}\left(\int_{\tilde{\Theta}}\pi(\theta)^2d\theta\right)^{1/2}$, with equality if and only if $\tilde{\pi} \propto 1$, we deduce

$$\lim_{k\to\infty} C_1^{-k+1}\sqrt{k}(\mathscr{I}_{\mathrm{MMD}^2}(\mathbf{J}(\cdot|\tilde{\Theta})) - \mathscr{I}_{\mathrm{MMD}^2}(\tilde{\Pi}(\cdot|\tilde{\Theta}))) \geq 0 \text{ with equality iff } \mathbf{J}(\cdot|\tilde{\Theta}) = \tilde{\Pi}(\cdot|\tilde{\Theta}). \tag{4.48}$$

2. We consider the vectorial space $\mathcal{C}(\tilde{\Theta})$ of continuous real functions defined on $\tilde{\Theta}$. It is a Banach space when equipped with the norm $\|p\| = \sup_{\tilde{\Theta}}|p|$. The expression of $\mathscr{I}_{\mathrm{MMD}^2}(\Pi(\cdot|\tilde{\Theta}))$ elucidated in eq. (4.45) can be seen as a functional of π of the form $\pi \mapsto B(\pi,\pi)$, B being a continuous bilinear form on E^2. This function is differentiable w.r.t. π. We thus apply a Lagrange multipliers theorem to write that if $\pi_k^* \in E$ maximizes $\pi \mapsto \mathscr{I}^k_{\mathrm{MMD}^2}(\Pi(\cdot|\tilde{\Theta}))$ on the constrained space $E \cap \{p \in E, \int_{\tilde{\Theta}} p = 1\}$, then there exists $\lambda \in \mathbb{R}$ such that for all $p \in E$:

$$-2C_1^k\int_{\Theta\times\Theta} e^{-\frac{k}{2}\frac{1}{2\sigma^2+h^2}(\theta-\theta')^2}p(\theta)\pi_k^*(\theta')d\theta d\theta' - \lambda\int_{\Theta}p(\theta)d\theta = 0. \tag{4.49}$$

This being true for any $p \in E$, we deduce for all $\theta' \in \tilde{\Theta}$,

$$-2C_1^k\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}\pi_k^*(\theta)d\theta = \lambda \quad \text{and} \quad -2C_1^k\int_{\tilde{\Theta}\times\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}\pi_k^*(\theta)\pi_k^*(\theta')d\theta d\theta' = \lambda. \tag{4.50}$$

We now apply a similar reasoning as in 1. to derive the limits of the above integrals. The π_k^* are assumed to form a uniformly convergent sequence of functions. That ensures its limit π^* is continuous on $\tilde{\Theta}$. We fix $\theta' \in \tilde{\Theta}$. Let $\varepsilon > 0$, we fix a neighborhood V of θ' such that for all $\theta \in V$, $|\pi^*(\theta) - \pi^*(\theta')| < \varepsilon/4$. By uniform convergence of (π_k^*), we can fix k_0 such that for all $k \geq k_0$, $\|\pi_k^* - \pi^*\| < \varepsilon/4$. It is also possible to fix k_1 such that for all $k \geq k_1$, $C_1\sqrt{k}\int_{\mathbb{R}\setminus V} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}d\theta < \varepsilon/(8\sup_{k'}\|\pi_{k'}^*\|)$. We thus have for all $k \geq \max(k_1,k_0)$

$$\begin{aligned}\left|C_1\sqrt{k}\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}\pi_k^*(\theta)d\theta - \pi^*(\theta')\right| &\leq C_1\sqrt{k}\int_{\mathbb{R}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}|\pi_k^*(\theta)\mathbb{1}_{\theta\in\Theta} - \pi^*(\theta')|d\theta \\ \leq C_1\sqrt{k}\int_{\mathbb{R}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}|\pi_k^*(\theta)\mathbb{1}_{\theta\in\Theta} - \pi^*(\theta)\mathbb{1}_{\theta\in\Theta}|d\theta &+ C_1\sqrt{k}\int_{\mathbb{R}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}|\pi^*(\theta)\mathbb{1}_{\theta\in\Theta} - \pi^*(\theta')|d\theta < \varepsilon.\end{aligned} \tag{4.51}$$

As the integrated terms are all bounded, a dominated converge theorem gives:

$$\begin{aligned}&\lim_{k\to\infty} C_1\sqrt{k}\int_{\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}\pi_k^*(\theta)d\theta = \pi^*(\theta') \\ \text{and} \quad &\lim_{k\to\infty} C_1\sqrt{k}\int_{\tilde{\Theta}\times\tilde{\Theta}} e^{-\frac{k}{2}\frac{(\theta'-\theta)^2}{2\sigma^2+h^2}}\pi_k^*(\theta)\pi_k^*(\theta')d\theta d\theta' = \int_{\tilde{\Theta}}\pi^*(\theta)^2d\theta.\end{aligned} \tag{4.52}$$

Since the above terms are equal, we conclude that π^* is constant.

□

4.6 Proofs of the main results

4.6.1 Proof of the convergence in probability

Proof of proposition 4.5. We fix θ in the interior of Supp π such that π is continuous in θ. Let $\varepsilon, \tilde{\varepsilon} > 0$. We want to show that there exist k_0, such that for all $k \geq k_0$

$$\mathbb{P}_\theta\left(\tilde{g}_2^{C,c}(k^{-d/2})^{-1}\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - g_1\left(k^{-d/2}\pi(\theta)(2\pi)^{d/2}|\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right)\right| < \tilde{\varepsilon}\right) > 1-\varepsilon, \tag{4.53}$$

for some $C, c = C_\theta, c_\theta$. We start by defining some events that will be asymptotically negligible:

1. Remind the definition of the r.v. $S_k = \frac{1}{\sqrt{k}}\sum_{i=1}^k \nabla_\theta \log \ell(Y_i|\theta)$. The central limit theorem ensures that when $\mathbf{Y}_k \sim \mathbb{P}_{\mathbf{Y}_k|\theta}$, S_k converges in distribution to $\mathcal{N}(0, \mathcal{I}(\theta))$ as $k \longrightarrow \infty$. Therefore, one can choose M_θ such that the events $\underline{\mathbf{A}}_k(\theta)$ defined as $\underline{\mathbf{A}}_k(\theta) = \{\|S_k\| > M_\theta\}$ verify
$$\mathbb{P}_\theta(\underline{\mathbf{A}}_k(\theta)) \underset{k\to\infty}{\longrightarrow} \varepsilon/5 \tag{4.54}$$

2. Denote for any $i = 1, \ldots, k$, and for any $\mathbf{y} \in \mathcal{Y}^k$,
$$\kappa_i(\mathbf{y}) = \{-x^T\nabla_\theta^2 \log \ell(y_i|\theta')x,\ \theta' \in \tilde{\Theta},\ \|x\| = 1\}. \tag{4.55}$$
They are compact subsets of $(0, +\infty)$. The law of large numbers gives
$$\frac{1}{k}\sum_{i=1}^k \inf \kappa_i(\mathbf{Y}_k) \underset{k\to\infty}{\longrightarrow} \mathbb{E}_\theta\left[\inf_{\theta'\in\tilde{\Theta}, \|x\|=1} -x^T\nabla_\theta^2 \log \ell(Y_1|\theta')x\right], \tag{4.56}$$
which is positive as the infima are positive $\mathbb{P}_\theta$-a.s. We choose an $\hat{m}_\theta > 0$ smaller than this limit, and we denote
$$\underline{\mathbf{B}}_k(\theta) = \left\{\frac{1}{k}\sum_{i=1}^k \inf \kappa_i(\mathbf{Y}_k) < \hat{m}\right\}. \tag{4.57}$$
Then $\mathbb{P}_\theta(\underline{\mathbf{B}}_k(\theta)) \underset{k\to\infty}{\longrightarrow} 0$.

3. Beforehand, denote
$$C_\theta = \exp\left(M_\theta^2/(2\hat{m}_\theta)\right) \quad \text{and} \quad c_\theta = \pi(\theta)(2\pi)^{d/2}|\mathcal{I}(\theta)|^{-1/2}, \tag{4.58}$$
which are finite and positive. Using the assumed local behavior of f around 0, one can choose $w, \eta > 0$ such that for any $c_\theta < |x| < C_\theta$, $|l| < w$, and $|t| < \eta$,
$$|f(xt + lt) - g_1(xt)| \leq \tilde{\varepsilon} g_2(xt) \leq \tilde{\varepsilon}\tilde{g}_2^{C_\theta, c_\theta}(t). \tag{4.59}$$
Using the $\mathbb{P}_\theta$-a.s. convergence of $-\frac{1}{k}\sum_{i=1}^k \nabla_\theta^2 \log \ell(Y_i|\theta)$ to $\mathcal{I}(\theta)$, we have $\mathbb{P}_\theta(\underline{\mathbf{C}}_k(\theta)) \underset{k\to\infty}{\longrightarrow} 0$ where
$$\begin{aligned}\underline{\mathbf{C}}_k(\theta) = \Bigg\{\forall \|x\| \leq M,\ \Bigg|&\pi(\theta)(2\pi)^{d/2}|k\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}x^T\mathcal{I}(\theta)^{-1}x\right)\\ &-\pi(\theta)(2\pi)^{d/2}|k\hat{\mathcal{I}}_k(\theta)|^{-1/2}\exp\left(\frac{1}{2}x^T\hat{\mathcal{I}}_k(\theta)^{-1}x\right)\Bigg| \geq wk^{-d/2}\Bigg\},\end{aligned} \tag{4.60}$$
with $\hat{\mathcal{I}}_k(\theta')$ denoting $-\frac{1}{k}\sum_{i=1}^k \nabla_\theta^2 \log \ell(Y_i|\theta')$.

4. Let us choose $R_\theta > 0$ large enough to have

$$(\sup_{\tilde{\Theta}} \pi) \int_{\|x\|>R_\theta} e^{M_\theta x} e^{-\frac{\hat{m}_\theta}{2}x^2} dx < w, \tag{4.61}$$

and $u > 0$ such that $\forall |x| < uR_\theta^2$, $|e^x - 1| < w/(2C_\theta R_\theta^d \sup_{\tilde{\Theta}} \pi)$.

From assumption 3.1, the function $\theta' \mapsto \|\nabla_\theta^2 \log \ell(y|\theta') - \nabla_\theta^2 \log \ell(y|\theta)\|$ is dominated on the Euclidean ball $B(\theta, \tau_1)$ by a variable that has a finite expectation $\mathbb{E}_\theta$. Therefore, a dominated convergence theorem gives that the function

$$\theta' \mapsto \mathbb{E}_\theta \left[\|\nabla_\theta^2 \log \ell(Y_1|\theta') - \nabla_\theta^2 \log \ell(Y_1|\theta)\| \right] \tag{4.62}$$

converges to 0 when $\theta' \longrightarrow \theta$ as a consequence of the continuity of the second order partial derivatives of the log-likelihood.

Thus, if we choose $\tilde{\tau}_\theta$ such that the above quantity is smaller than $u\varepsilon$ and if we define $\underline{\mathbf{D}}_k(\theta)$ as the event

$$\underline{\mathbf{D}}_k(\theta) = \left\{ \sup_{\theta' \in \tilde{\Theta},\, \|\theta'-\theta\|<\tilde{\tau}_\theta} \|\hat{\mathcal{I}}_k(\theta') - \hat{\mathcal{I}}_k(\theta)\| > u \right\}, \tag{4.63}$$

one gets by applying Markov's inequality that $\mathbb{P}_\theta(\underline{\mathbf{D}}_k(\theta)) < \varepsilon/4$.

Now, notice that for any $\tilde{\theta} \in \tilde{\Theta}$, $\mathbf{y} \in \mathcal{Y}^k$, there exists $\hat{\theta}$ on the segment between θ and $\tilde{\theta}$ such that

$$\log \left(\frac{\prod_{i=1}^k \ell(y_i|\tilde{\theta})}{\prod_{i=1}^k \ell(y_i|\theta)} \right) = (\tilde{\theta} - \theta)^T \sum_{i=1}^k \nabla_\theta \log \ell(y_i|\theta) + \frac{1}{2}(\tilde{\theta} - \theta)^T \sum_{i=1}^k \nabla_\theta^2 \log \ell(y_i|\hat{\theta})(\tilde{\theta} - \theta). \tag{4.64}$$

Let us write $p_{\mathbf{Y}_k}(\mathbf{y})/\ell_k(\mathbf{y}|\theta) = \int_\Theta h_{\mathbf{y}}(\tilde{\theta}, \theta)\pi(\tilde{\theta})d\tilde{\theta}$ where

$$h_{\mathbf{y}}(\tilde{\theta}, \theta) = \exp\left((\tilde{\theta} - \theta)^T \nabla_\theta \log \ell_k(\mathbf{y}|\theta) \right) \exp\left(\frac{1}{2}(\tilde{\theta} - \theta)^T \nabla_\theta^2 \log \ell_k(\mathbf{y}|\hat{\theta})(\tilde{\theta} - \theta) \right). \tag{4.65}$$

We fix k and work under the event $\underline{\mathbf{A}}_k(\theta)^c \cap \underline{\mathbf{B}}_k(\theta)^c \cap \underline{\mathbf{C}}_k(\theta)^c \cap \underline{\mathbf{D}}_k(\theta)^c$. i.e. we consider $\varpi \in \underline{\mathbf{A}}_k(\theta)^c \cap \underline{\mathbf{B}}_k(\theta)^c \cap \underline{\mathbf{C}}_k(\theta)^c \cap \underline{\mathbf{D}}_k(\theta)^c$ and denote $\mathbf{y} = \mathbf{Y}_k(\varpi)$, $S_k^{\mathbf{y}} = \frac{1}{\sqrt{k}} \nabla_\theta \log \ell_k(\mathbf{y}|\theta)$, $\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta) = -\frac{1}{k} \nabla_\theta^2 \log \ell_k(\mathbf{y}|\theta)$. We can thus bound $h_{\mathbf{y}}(\tilde{\theta}, \theta)$ as

$$h_{\mathbf{y}}(\tilde{\theta}, \theta) \leq \exp\left(M_\theta \sqrt{k} \|\tilde{\theta} - \theta\| \right) \exp\left(-\frac{k}{2} \hat{m}_\theta \|\tilde{\theta} - \theta\|^2 \right) \leq C_\theta. \tag{4.66}$$

The rest of the proof only consists in bounding four quantities by $wk^{-d/2}$.

- Define $\tilde{R}_k = R_\theta k^{-1/2}$. Inequality in eq. (4.61) implies

$$\mathbf{I}_1 := \int_{\|\tilde{\theta}-\theta\|>\tilde{R}_k} \exp\left((\tilde{\theta} - \theta)^T \sqrt{k} S_k^{\mathbf{y}} \right) \exp\left(-\frac{k}{2} (\tilde{\theta} - \theta)^T \hat{\mathcal{I}}_k^{\mathbf{y}}(\theta)(\tilde{\theta} - \theta) \right) \pi(\theta) d\tilde{\theta} < wk^{-d/2}. \tag{4.67}$$

- Recalling the upper bound in eq. (4.66), we have the same way

$$\mathbf{I}_2 := \int_{\|\tilde{\theta}-\theta\|>\tilde{R}_k} h_{\mathbf{y}}(\tilde{\theta}, \theta)\pi(\tilde{\theta})d\tilde{\theta} < wk^{-d/2}. \tag{4.68}$$

- As π is continuous in θ, there exist k_1 such that if $k \geq k_1$ then for all $\|\tilde{\theta} - \theta\| < \tilde{R}_k$, $|\pi(\tilde{\theta}) - \pi(\theta)| \leq w/(2C_\theta R_\theta^d)$. We write:

$$\mathbf{I}_3 := \int_{\|\tilde{\theta}-\theta\|<\tilde{R}_k} h_{\mathbf{y}}(\tilde{\theta}, \theta)\pi(\tilde{\theta})d\tilde{\theta} - \int_{\|\tilde{\theta}-\theta\|<\tilde{R}_k} \exp\left((\tilde{\theta} - \theta)^T \sqrt{k} S_k^{\mathbf{y}} \right) \exp\left(-\frac{k}{2} (\tilde{\theta} - \theta)^T \hat{\mathcal{I}}_k^{\mathbf{y}}(\theta)(\tilde{\theta} - \theta) \right) \pi(\theta) d\tilde{\theta}$$

$$|\mathbf{I}_3| \leq \int_{\|\tilde{\theta}-\theta\|<\tilde{R}_k} |\hat{h}_{\mathbf{y}}(\tilde{\theta}, \theta, \hat{\theta}) - \hat{h}_{\mathbf{y}}(\tilde{\theta}, \theta, \theta)|\pi(\tilde{\theta})d\tilde{\theta} + \int_{\|\tilde{\theta}-\theta\|<\tilde{R}_k} \hat{h}_{\mathbf{y}}(\tilde{\theta}, \theta, \theta)|\pi(\tilde{\theta}) - \pi(\theta)|d\tilde{\theta} \tag{4.69}$$

where $\hat{h}_{\mathbf{y}}(\tilde{\theta},\theta,\hat{\theta}) = \exp((\tilde{\theta}-\theta)^T\sqrt{k}S_k^{\mathbf{y}})\exp\left(-\frac{k}{2}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k^{\mathbf{y}}(\hat{\theta})(\tilde{\theta}-\theta)\right)$. Notice that for any $\theta',\tilde{\theta}$ in the Euclidean ball $B(\theta,\tilde{R}_k)$,

$$|\hat{h}_{\mathbf{y}}(\tilde{\theta},\theta,\theta') - \hat{h}_{\mathbf{y}}(\tilde{\theta},\theta,\theta)| \leq \hat{h}_{\mathbf{y}}(\tilde{\theta},\theta,\theta')\left|\exp\left(-\frac{k}{2}(\tilde{\theta}-\theta)^T(\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta')-\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta))(\tilde{\theta}-\theta)\right)-1\right|. \tag{4.70}$$

Thus, if $k \geq k_2$ where k_2 is such that $\tilde{R}_{k_2} < \tilde{\tau}$, we obtain $(\tilde{\theta}-\theta)^T(\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta')-\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta))(\tilde{\theta}-\theta) < u$ consequently to $\mathbf{y} = \mathbf{Y}_k(\varpi)$ with $\varpi \in \underline{\mathbf{D}}_k(\theta)^c$. It then comes $|\mathbf{I}_3| < wk^{-d/2}$.

- By recognizing a Gaussian density in the integral below, we get the equality

$$\begin{aligned}&\int_{\mathbb{R}^d}\exp((\tilde{\theta}-\theta)^T\sqrt{k}S_k^{\mathbf{y}})\exp\left(\frac{1}{2}(\tilde{\theta}-\theta)^T\nabla_\theta^2\ell_k(\mathbf{y}|\theta)(\tilde{\theta}-\theta)\right)\pi(\theta)d\tilde{\theta}\\ &= \pi(\theta)(2\pi)^{d/2}|k\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^{\mathbf{y}T}\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta)^{-1}S_k^{\mathbf{y}}\right).\end{aligned} \tag{4.71}$$

Defining $\mathbf{I}_4 = (4.71) - \pi(\theta)(2\pi)^{d/2}|k\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^{\mathbf{y}T}\mathcal{I}(\theta)^{-1}S_k^{\mathbf{y}}\right)$, we obtain, using that $\varpi \in \underline{\mathbf{C}}_k(\theta)^c$,

$$|\mathbf{I}_4| \leq wk^{-d/2}. \tag{4.72}$$

If we define k_3 such that $k^{-d/2} < \eta/4$, and if $k \geq \max(k_1,k_2,k_3)$, then the four inequalities elucidated above allows evaluating eq. (4.59) with $t = k^{-d/2}$,

$$\begin{aligned}x(\mathbf{y}) &= \int_{\tilde{\Theta}}\exp\left((\tilde{\theta}-\theta)^T\sqrt{k}S_k^{\mathbf{y}}\right)\exp\left(-\frac{k}{2}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k^{\mathbf{y}}(\theta)(\tilde{\theta}-\theta)\right)\pi(\theta)d\tilde{\theta},\\ \text{and}\quad l(\mathbf{y}) &= x(\mathbf{y}) - \frac{p_{\mathbf{y}}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}k^{d/2} = \mathbf{I}_1+\mathbf{I}_2+\mathbf{I}_3+\mathbf{I}_4.\end{aligned} \tag{4.73}$$

All in all, we have proven that for $k \geq \max(k_1,k_2,k_3)$ the inclusion

$$\underline{\mathbf{A}}_k(\theta)^c \cap \underline{\mathbf{B}}_k(\theta)^c \cap \underline{\mathbf{C}}_k(\theta)^c \cap \underline{\mathbf{D}}_k(\theta)^c \subset \left\{\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - g_1(k^{-d/2}x(\mathbf{Y}_k))\right| < \tilde{g}_2^{C_\theta,c_\theta}(k^{-d/2})\tilde{\varepsilon}\right\} \tag{4.74}$$

stands. To conclude we denote k_4 such that for any $k \geq k_4$,

$$\mathbb{P}_\theta(\underline{\mathbf{A}}_k(\theta)) < \varepsilon/4,\quad \mathbb{P}_\theta(\underline{\mathbf{B}}_k(\theta)) < \varepsilon/4,\quad \mathbb{P}_\theta(\underline{\mathbf{C}}_k(\theta))\varepsilon/4,\quad \mathbb{P}_\theta(\underline{\mathbf{D}}_k(\theta)) < \varepsilon/4. \tag{4.75}$$

This way, defining $k_0 = \max(k_1,k_2,k_3,k_4)$, the following inequality is true for any $k \geq k_0$:

$$\mathbb{P}_\theta\left(\tilde{g}_2^{C_\theta,c_\theta}(k^{-d/2})^{-1}\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - g_1\left(k^{-d/2}\pi(\theta)(2\pi)^{d/2}|\mathcal{I}(\theta)|^{-1/2}\exp\left(\frac{1}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right)\right| < \tilde{\varepsilon}\right) > 1-\varepsilon. \tag{4.76}$$

□

4.6.2 Proof of the results when $\delta < 0$

Proof of theorems 4.2 and 4.3 and proposition 4.6. We suppose that assumption 4.1 is verified. We recall that in this proof $\delta < 0$. The following lemma is required for the proof. It is proven later on.

Lemma 4.1. *Let $\zeta < 0$. There exists a constant K independent of θ such that*

$$k^{d\zeta/2}\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] \leq K. \tag{4.77}$$

We recall that proposition 4.5 states the following convergence in probability for a.e. θ:

$$k^{d\delta/2}\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)-ak^{-d\delta/2}\pi(\theta)^{\delta}(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\exp\left(\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right|\xrightarrow[k\to\infty]{\mathbb{P}_\theta}0. \tag{4.78}$$

The first goal in this prove will be to show that this convergence is a convergence in $L^1(\mathbb{P}_\theta)$. This is shown by proving that the sequence of r.v. is uniformly integrable. Let $\rho>0$ and choose an even $\underline{\mathbf{A}}$ such that $\mathbb{P}_\theta(\underline{\mathbf{A}})=\rho$. Firstly, we have

$$\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{A}}}\pi(\theta)^{\delta}(2\pi)^{d\delta/2}e^{\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k}\right]\leq\pi(\theta)^{\delta}(2\pi)^{d\delta/2}\rho, \tag{4.79}$$

and secondly, considering t, $\hat{C}$ and $\hat{C}'$ such that $|f(x)|\leq\hat{C}x$ if $x>t$ and $|f(x)|\leq\hat{C}'x^{\delta}$ otherwise; we obtain using $\underline{\mathbf{G}}_k=\underline{\mathbf{G}}_k(\theta)\{p_{\mathbf{Y}}(\mathbf{y})>\gamma\ell_k(\mathbf{y}|\theta)\}$ and Hölder's inequality:

$$\begin{aligned}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right]&\leq\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{G}}_k}\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}\hat{C}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right]+\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{G}}_k^c}\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}\hat{C}'\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^{\delta}\right]\\&\leq\hat{C}\rho^{1/2}+\hat{C}'\rho^{1/p}K^{1/q},\end{aligned} \tag{4.80}$$

with $1/p+1/q=1$. The above ensures the required uniform-integrability. That allows us to write:

$$\mathbb{E}_\theta k^{d\delta/2}\left|f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)-ak^{-d\delta/2}\pi(\theta)^{\delta}(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\exp\left(\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right|\xrightarrow[k\to\infty]{}0. \tag{4.81}$$

This limit stands for every θ in the interior of Supp π within which π is continuous. For such a θ it comes as $k\to\infty$

$$k^{d\delta/2}D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|\theta})=a\pi(\theta)^{\delta}(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|\mathcal{I}(\theta)^{-1/2}S_k\|^2}\right]+o(1). \tag{4.82}$$

The sequence of r.v. $\mathcal{I}(\theta)^{-1/2}S_k$ converges in distribution to a standard normal r.v. denoted X. Thus, since $\delta<0$, $\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|\mathcal{I}(\theta)^{-1/2}S_k\|^2}\right]\xrightarrow[k\to\infty]{}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\right]$. Therefore,

$$k^{d\delta/2}D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|\theta})\underset{k\to\infty}{=}a\pi(\theta)^{\delta}(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\right]+o(1). \tag{4.83}$$

The second step of the proof it to integrate this limit w.r.t. π, using a dominated convergence theorem. It is possible since the following domination stands:

$$\begin{aligned}k^{d\delta/2}D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|\theta})&=k^{d\delta/2}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{G}}_k}f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right]+k^{d\delta/2}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{G}}_k^c}f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right]\\&\leq\hat{C}+\hat{C}'K.\end{aligned} \tag{4.84}$$

The domination above makes the following licit:

$$\lim_{k\to\infty}k^{d\delta/2}\mathscr{I}_{D_f}^k(\Pi(\cdot|\tilde{\Theta}))=a(2\pi)^{d\delta/2}(1-\delta)^{-d/2}\int_{\tilde{\Theta}}\pi(\delta)^{\delta+1}|\mathcal{I}(\theta)|^{-\delta/2}d\theta. \tag{4.85}$$

This concludes the proof of theorem 4.2.

The results of proposition 4.6 and theorem 4.3 comes by concavity of $x\mapsto ax^{-\delta}$, which results from $a(\delta+1)>0$ and gives

$$\begin{aligned}&a\int_{\tilde{\Theta}}\pi(\theta)^{\delta}|\mathcal{I}(\theta)|^{-\delta/2}\pi(\theta)d\theta\\&\leq a\left(\int_{\tilde{\Theta}}\pi(\theta)^{-1}|\mathcal{I}(\theta)|^{1/2}\pi(\theta)d\theta\right)^{-\delta}=a\left(\int_{\tilde{\Theta}}|\mathcal{I}(\theta)|^{1/2}d\theta\right)^{-\delta}.\end{aligned} \tag{4.86}$$

The concavity of $x\mapsto ax^{-\delta}$ being strict, the above inequality is an equality if and only if $\pi(\theta)^{\delta}|\mathcal{I}(\theta)|^{-\delta/2}$ is constant with respect to θ, i.e. $\pi\propto J$.

□

Proof of lemma 4.1. Set θ that belongs in the interior of Supp π. We write,

$$\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] \leq \mathbb{E}_\theta\left[\left(\int_{\|\tilde{\theta}-\theta\|\leq \tilde{R}_k} \exp((\tilde{\theta}-\theta)^T\sqrt{k}S_k)\exp\left(-\frac{k}{2}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)\right)\pi(\tilde{\theta})d\tilde{\theta}\right)^\zeta\right], \tag{4.87}$$

where $\hat{\theta}$ belongs to the segment between θ and $\tilde{\theta}$, and with $\hat{\mathcal{I}}_k(\theta')$ denoting $-\frac{1}{k}\sum_{i=1}^k \nabla^2_\theta \log \ell(y_i|\theta')$. Denote $H_k = \sup_{\tilde{\theta}\in\tilde{\Theta},\|x\|=1} x^T\hat{\mathcal{I}}_k(\tilde{\theta})x$, and $m_\pi = \inf_{\tilde{\Theta}} \pi > 0$. We have

$$\begin{aligned}\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] &\leq \mathbb{E}_\theta\left[m_\pi^\zeta\left(k^{-d/2}\int_{\mathbb{R}^d} e^{x^TS_k}e^{-\frac{H_k}{2}\|x\|^2}dx\right)^\zeta\right] \\ &= \mathbb{E}_\theta\left[m_\pi^\zeta k^{-d\zeta/2}(2\pi)^{d\zeta/2}H_k^{-\zeta/2}e^{\frac{\zeta}{2}H_k\|S_k\|^2}\right].\end{aligned} \tag{4.88}$$

We now bound H_k as below:

$$H_k = \sup_{\tilde{\Theta}\in\tilde{\Theta},\,\|x\|=1} -\frac{1}{k}\sum_{i=1}^k x^T\nabla^2_\theta \log \ell(y_i|\theta)x \leq \frac{1}{k}\sum_{i=1}^k \sup_{\tilde{\Theta}} \|\nabla^2_\theta \ell(y_i|\theta)\|. \tag{4.89}$$

That allows to conclude using Jensen's inequality:

$$k^{d\zeta/2}\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] \leq m_\pi^\zeta(2\pi)^{d\zeta/2}K_0^{-\zeta/2}. \tag{4.90}$$

□

4.6.3 Proof of the results when $\delta > 0$

Proof of theorems 4.4 and 4.5 and proposition 4.7. We suppose that assumptions 4.2 and 4.3 are verified, we recall that in this proof $\delta > 0$. The following lemmas are required for the proof. They are proven later on.

Lemma 4.2. *For any $\zeta < 2\xi$, there exists a constant $K_2 > 0$ independent of $\theta \in \tilde{\Theta}$ such that for any $k > 0$,*

$$k^{d\zeta/2}\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] \leq K_2. \tag{4.91}$$

Lemma 4.3. *There exists a constant $K_3 > 0$ independent of $\theta \in \tilde{\Theta}$ such that for any $k > 0$ and any $\rho > 0$,*

$$k^{d\delta/2}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] \leq \rho^{1/2}K_3 \tag{4.92}$$

with $\underline{\mathbf{E}}_k = \underline{\mathbf{E}}_k(\theta) = \{\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)} > \rho^{-1/\delta}\}$.

We start by proving the following claims about the consequences of the sub-Gaussianity of S_k (assumption 4.3).

- **Claim 1.** Let $\zeta < \xi$, then if $X_k = \mathcal{I}(\theta)^{-1/2}S_k$, then $\mathbb{E}_\theta e^{\zeta\|X_k\|^2} < K_1$.
  Proof. Assumption 4.2 makes $\|\mathcal{I}(\theta)^{-1/2}\| \leq m^{-1/2}$, i.e. $\|X_k\|^2 \leq \|S_k\|^2/m$. Hence the result since $\mathbb{E}_\theta e^{\frac{\xi}{m}\|S_k\|^2} < K_1$.

- **Claim 2.** Let $t \in \mathbb{R}$ and $v \in \mathbb{R}^d$ with $\|v\| = 1$. We have $\mathbb{E}_\theta e^{tv^TS_k} \leq K_1 e^{\frac{t^2\chi^2}{2}}$ with $\chi^2 = \xi/(2m)$.

Proof. For any $s > 0$, we have $|tv^T S_k| \leq \frac{s}{2}t^2 + \frac{1}{2s}|v^T S_k|^2 \leq \frac{s}{2}t^2 + \frac{1}{2s}\|S_k\|^2$. Choosing $s = m/(2\xi)$, we get

$$\mathbb{E}_\theta e^{tv^T S_k} \leq e^{\frac{mt^2}{4\xi}} \mathbb{E}_\theta e^{\frac{\xi}{m}\|S_k\|^2} \leq K_1 e^{\frac{t^2\chi^2}{2}}. \tag{4.93}$$

We recall that proposition 4.5 states the following convergence in probability for a.e. θ:

$$k^{d\delta/2}\left| f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - ak^{-d\delta/2}\pi(\theta)^\delta(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\exp\left(\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right| \xrightarrow[k\to\infty]{\mathbb{P}_\theta} 0. \tag{4.94}$$

The first goal in this prove will be to show that this convergence is a convergence in $L^1(\mathbb{P}_\theta)$. This is shown by proving that the sequence of r.v. is uniformly integrable. Let $\rho > 0$ and choose an even $\underline{\mathbf{A}}$ such that $\mathbb{P}_\theta(\underline{\mathbf{A}}) = \rho$. Denoting $X_k = \mathcal{I}(\theta)^{-1/2}S_k$, using Hölder's inequality it comes

$$\begin{aligned}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{A}}}\pi(\theta)^\delta(2\pi)^{d\delta/2}e^{\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k}\right] &\leq \pi(\theta)^\delta(2\pi)^{d\delta/2}\rho^{1/p}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}q\|X_k\|^2}\right]^{1/q}\\ &\leq \rho^{1/p}(\sup_\Theta \pi)^\delta(2\pi)^{d\delta/2}K_1^{1/q}\end{aligned} \tag{4.95}$$

for any p, q such that $1/p + 1/q = 1$ and $q\delta/2 < \xi$. Second, consider $\hat{C}$ and $\hat{C}'$ such that $|f(x)| \leq \hat{C}x$ if $x > \rho^{-1/\delta}$ and $|f(x)| \leq \hat{C}'x^\delta$ otherwise. Using $\underline{\mathbf{E}}_k = \underline{\mathbf{E}}_k(\theta) = \{p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta) > \rho^{-1/\delta}\}$ and Hölder's inequality, we compute:

$$\begin{aligned}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right] &\leq \mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}\hat{C}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] + \mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k^c}\mathbb{1}_{\underline{\mathbf{A}}}k^{d\delta/2}\hat{C}'\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\delta\right]\\ &\leq \hat{C}\rho^{1/2}K_3 + \hat{C}'\rho^{1/p}K_2^{1/q},\end{aligned} \tag{4.96}$$

where $1/p = 1 - 1/q$ with $q\delta/2 < \xi$. Equations (4.95) and (4.96) ensure the required uniform-integrability. That allows us to write:

$$\mathbb{E}_\theta k^{d\delta/2}\left| f\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right) - ak^{-d\delta/2}\pi(\theta)^\delta(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\exp\left(\frac{\delta}{2}S_k^T\mathcal{I}(\theta)^{-1}S_k\right)\right| \xrightarrow[k\to\infty]{} 0. \tag{4.97}$$

This limit stands for every θ in the interior of Supp π within which π is continuous. For such a θ it comes as $k \to \infty$

$$k^{d\delta/2}D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|\theta}) = a\pi(\theta)^\delta(2\pi)^{d\delta/2}|\mathcal{I}(\theta)|^{-\delta/2}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X_k\|^2}\right] + o(1). \tag{4.98}$$

The second step of the proof is to precise the above asymptotic expansion by studying the limit of $\mathbb{E}_\theta e^{\frac{\delta}{2}\|X_k\|^2}$. The sequence of r.v. X_k converges in distribution to a standard normal r.v. denoted X. This implies that for any $A > 0$, the expectation $\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X_k\|^2}\mathbb{1}_{\|X_k\|\leq A}\right]$ converges to $\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\mathbb{1}_{\|X\|\leq A}\right]$. Moreover, under Assumption 4.3 there exists a constant K' independent of k such that for any k

$$\begin{aligned}\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X_k\|^2}\left|1 - \mathbb{1}_{\|X_k\|\leq A}\right|\right] &\leq \mathbb{E}_\theta\left[e^{\frac{\delta q}{2}\|X_k\|^2}\right]^{1/q}\mathbb{P}_\theta\left(\|X_k\| \geq A\right)^{1/p}\\ &\leq K_1^{1/q}\mathbb{P}_\theta(\exp(t^2\|X_k\|^2) \geq \exp(t^2A))\\ &\leq K'e^{-t^2A^2/p}\end{aligned} \tag{4.99}$$

using Hölder inequality with $1/p + 1/q = 1$ such as $\delta q/2 \leq \xi$ first, and Markov inequality with $t^2 < \xi$ second. Thus, for any $\varepsilon > 0$, one can choose A such that (4.99) $\leq \varepsilon/3$ and $\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\left|1 - \mathbb{1}_{\|X\|\leq A}\right|\right] < \varepsilon/3$ as well to get

$$\left|\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X_k\|^2}\right] - \mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\right]\right| < 2\varepsilon/3 + \left|\mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X_k\|^2}\mathbb{1}_{\|X_k\|\leq A}\right] - \mathbb{E}_\theta\left[e^{\frac{\delta}{2}\|X\|^2}\mathbb{1}_{\|X\|\leq A}\right]\right|. \tag{4.100}$$

Eventually, k can be chosen large enough such that the right-hand side of above equation is smaller than ε, which states the convergence $\mathbb{E}_\theta e^{\frac{\delta}{2}\|X_k\|^2} \underset{k\to\infty}{\longrightarrow} \mathbb{E}_\theta e^{\frac{\delta}{2}\|X\|^2}$. This lets us write

$$k^{d\delta/2} D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|\theta}) \underset{k\to\infty}{=} a\pi(\theta)^\delta (2\pi)^{d\delta/2} |\mathcal{I}(\theta)|^{-\delta/2}(1-\delta)^{-d/2} + o(1). \tag{4.101}$$

The third and last step of the proof it to carry out a dominated convergence theorem to elucidate the limit of $\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta}))$. We denote $\underline{\mathbf{E}}_k = \underline{\mathbf{E}}_k(\theta) = \{p_{\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta) > 1\}$, we write

$$\begin{aligned} k^{d\delta/2}|D_f(\mathbb{P}_{\mathbf{Y}}||\mathbb{P}_{\mathbf{Y}|\theta})|(a(2\pi)^{d/2}\sup_{\tilde{\Theta}} \pi|\mathcal{I}|^{-1/2})^{-\delta} &\leq k^{d\delta/2}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\hat{C}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] + \mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k^c} k^{d\delta/2}\hat{C}'\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\delta\right] \\ &\leq \hat{C}K_3 + \hat{C}'K_2. \end{aligned} \tag{4.102}$$

The domination above makes the following licit:

$$\lim_{k\to\infty} k^{d\delta/2}\mathscr{I}^k_{D_f}(\Pi(\cdot|\tilde{\Theta})) = a(2\pi)^{d\delta/2}(1-\delta)^{-d/2}\int_{\tilde{\Theta}} \pi(\delta)^{\delta+1}|\mathcal{I}(\theta)|^{-\delta/2}d\theta. \tag{4.103}$$

This concludes the proof of theorem 4.4. The result of proposition 4.7 and theorem 4.5 comes by concavity of $x \mapsto ax^{-\delta}$, which results from $a(\delta+1) > 0$ and gives

$$\begin{aligned} &a\int_{\tilde{\Theta}} \pi(\theta)^\delta |\mathcal{I}(\theta)|^{-\delta/2}\pi(\theta)d\theta \\ &\leq a\left(\int_{\tilde{\Theta}} \pi(\theta)^{-1}|\mathcal{I}(\theta)|^{1/2}\pi(\theta)d\theta\right)^{-\delta} = a\left(\int_{\tilde{\Theta}} |\mathcal{I}(\theta)|^{1/2}d\theta\right)^{-\delta}. \end{aligned} \tag{4.104}$$

The concavity of $x \mapsto ax^{-\delta}$ being strict, the above inequality is an equality if and only if $\pi(\theta)^\delta|\mathcal{I}(\theta)|^{-\delta/2}$ is constant with respect to θ, i.e. $\pi \propto J$.

□

Proof of lemma 4.2. We work under the assumptions of theorem 4.4. We write

$$\begin{aligned} \mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] &= \mathbb{E}_\theta\left(\int_{\tilde{\Theta}} e^{(\tilde{\theta}-\theta)^T\sqrt{k}S_k} e^{-\frac{k}{2}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)}\pi(\tilde{\theta})d\tilde{\theta}\right)^\zeta \\ &\leq \mathbb{E}_\theta k^{-d\zeta/2}\left(\int_{\mathbb{R}^d} e^{u^TS_k}e^{-\frac{1}{2}u^T\hat{\mathcal{I}}_k(\hat{\theta})u}du \sup_{\tilde{\Theta}}|\pi|\right)^\zeta \\ &\leq \mathbb{E}_\theta k^{-d\zeta/2}\left(\int_{\mathbb{R}^d} e^{u^TS_k}e^{-\frac{\|u\|^2}{2}\frac{1}{k}\sum_{i=1}^k \inf_{\tilde{\Theta}} L^{Y_i}}du \sup_{\tilde{\Theta}}|\pi|\right)^\zeta \end{aligned} \tag{4.105}$$

with $\hat{\theta}$ being a point in $\mathbb{R}^d$ within the segment between θ and $\tilde{\theta} = u/\sqrt{k} + \theta$; with $L^y(\theta)$ being the smallest eigenvalue of $(\nabla^2_\theta \log \ell(y|\theta))^{1/2}$. Therefore, $\inf_{\tilde{\Theta}} L^{Y_i} > m$ $\mathbb{P}_\theta$ a.s. and

$$\mathbb{E}_\theta\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)^\zeta\right] \leq k^{-d\zeta/2}K''\mathbb{E}_\theta\left[m^{-d\zeta/2}e^{\frac{\zeta}{2m}\|S_k\|^2}\right] \tag{4.106}$$

for some constant $K'' > 0$. By considering the sub-Gaussianity of S_k, the proof of the Lemma is complete.

□

Proof of lemma 4.3. We work under the assumptions of theorem 4.4. Notice

$$\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] = \int_{\mathcal{Y}^k} \mathbb{1}_{\underline{\mathbf{E}}_k} d\mathbb{P}_{\mathbf{Y}_k}(\mathbf{y}). \tag{4.107}$$

According to Markov's inequality,

$$\begin{aligned}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right] &\leq \rho^{\tilde{\delta}/\delta}\int_{\mathcal{Y}^k}\left(\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right)^{\tilde{\delta}} p_{\mathbf{Y}_k}(\mathbf{y})d\mu^{\otimes k}(\mathbf{y})\\ &\leq \rho^{\tilde{\delta}/\delta}\int_{\mathcal{Y}^k}\left(\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right)^{\tilde{\delta}}\int_{\tilde{\Theta}}\ell_k(\mathbf{y}|\tilde{\theta})\pi(\tilde{\theta})d\tilde{\theta}d\mu^{\otimes k}(\mathbf{y})\\ &\leq \rho^{\tilde{\delta}/\delta}\int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\tilde{\theta})}\right)^{\tilde{\delta}}\left(\frac{\ell_k(\mathbf{y}|\tilde{\theta})}{\ell_k(\mathbf{y}|\theta)}\right)^{\tilde{\delta}}\right]\pi(\tilde{\theta})d\tilde{\theta}, \end{aligned} \tag{4.108}$$

for some $\tilde{\delta} > 0$ that we determine later. Using the expansion of the likelihood quotient expressed in eq. (4.64) we can write, for some $\hat{\theta}$ in the segment between θ and $\tilde{\theta}$:

$$\begin{aligned}\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] &\leq \rho^{\tilde{\delta}/\delta}\int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\tilde{\theta})}\right)^{\tilde{\delta}} e^{\tilde{\delta}(\tilde{\theta}-\theta)^T\sqrt{k}S_k}e^{-\frac{k}{2}\tilde{\delta}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)}\right]\pi(\tilde{\theta})d\tilde{\theta}\\ &\leq \rho^{\tilde{\delta}/\delta}\left(\int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\tilde{\theta})}\right)^{p\tilde{\delta}}\right]\pi(\tilde{\theta})d\tilde{\theta}\right)^{1/p}\\ &\quad\times\left(\int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[e^{q\tilde{\delta}(\tilde{\theta}-\theta)^T\sqrt{k}S_k}e^{-q\frac{k}{2}\tilde{\delta}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)}\right]\pi(\tilde{\theta})d\tilde{\theta}\right)^{1/q}, \end{aligned} \tag{4.109}$$

the second inequality being a consequence of Hölder's inequality (necessarily $1/p + 1/q = 1$). The first integral in the right-hand side of the equation verifies

$$\int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[\left(\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\tilde{\theta})}\right)^{p\tilde{\delta}}\right]\pi(\tilde{\theta})d\tilde{\theta} \leq K_2 k^{-p\tilde{\delta}d/2} \tag{4.110}$$

when $\tilde{\delta}p/2 < \xi$ according to Lemma 4.2.

Now, we look for an upper-bound of the second integral:

$$I = \int_{\tilde{\Theta}}\mathbb{E}_{\tilde{\theta}}\left[e^{q\tilde{\delta}(\tilde{\theta}-\theta)^T\sqrt{k}S_k}e^{-q\frac{k}{2}\tilde{\delta}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)}\right]\pi(\tilde{\theta})d\tilde{\theta}. \tag{4.111}$$

Using the claim 2. on the consequence of the sub-Gaussianity of S_k, we get for any $\tilde{\theta} \in \tilde{\Theta}$:

$$\mathbb{E}_{\tilde{\theta}}\left[e^{q\tilde{\delta}(\tilde{\theta}-\theta)^T\sqrt{k}S_k}e^{-q\frac{k}{2}\tilde{\delta}(\tilde{\theta}-\theta)^T\hat{\mathcal{I}}_k(\hat{\theta})(\tilde{\theta}-\theta)}\right] \leq K_1 e^{\frac{c^2}{2}q^2\tilde{\delta}^2 k\|\tilde{\theta}-\theta\|^2}e^{-q\frac{k}{2}\tilde{\delta}m\|\tilde{\theta}-\theta\|^2}. \tag{4.112}$$

Therefore,

$$\begin{aligned} I &\leq K_1 k^{-d/2}\int_{\mathbb{R}^d} e^{\frac{c^2}{2}q^2\tilde{\delta}^2\|x\|^2}e^{-\frac{q}{2}\tilde{\delta}m\|x\|^2}\sup_{\tilde{\Theta}}\pi dx\\ &\leq K_1 k^{-d/2}\int_{\mathbb{R}^d} e^{\frac{m}{4\xi}q^2\tilde{\delta}^2\|x\|^2}e^{-\frac{q}{2}\tilde{\delta}m\|x\|^2}\sup_{\tilde{\Theta}}\pi dx. \end{aligned} \tag{4.113}$$

Let us suppose the integral of the right hand term above to be a finite constant that we denote L; for this to be true, the inequality $q\tilde{\delta} < 2\xi$ must be satisfied. In that case, one gets

$$\mathbb{E}_\theta\left[\mathbb{1}_{\underline{\mathbf{E}}_k}\frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right] \leq \rho^{\tilde{\delta}/\delta}K_2^{1/p}L^{1/q}K_1^{1/q}k^{-d(\tilde{\delta}+1/q)/2}. \tag{4.114}$$

Choose now $\tilde{\delta} = \delta/2$, and $q = p = 2$. Thus $p\tilde{\delta}/2 < \xi$ and $q\tilde{\delta}/2 < \xi$ as required. Moreover, $\tilde{\delta} + 1/q > \delta$ given $1 > \delta$. This way,

$$\mathbb{E}_\theta \left[\mathbb{1}_{\underline{\mathbf{E}}_k} \frac{p_{\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)} \right] \leq \rho^{1/2} K_3 k^{-d\delta/2}, \tag{4.115}$$

with $K_3 = K_2^{1/2} L^{1/2} K_1^{1/2}$.

□

4.7 Conclusion and prospects

This chapter contributes to the reference prior theory on multiple fronts.

First, we interpreted reference priors as maximizers of sensitivity indices. This interpretation reinforces the trustworthiness of their selection in practice: they are priors that minimize the impact of irreducible uncertainty on the studied systems.

Second, building on the aforementioned interpretation we provided a formalized framework for a generalization of the mutual information. We rigorously proved that usual properties of reference priors are still verified under that novel formalism. These generalized reference priors have been elucidated, and our main results show that they are Jeffreys priors in all the different cases that we study, namely when the mutual information is defined using a δ-divergence with $\delta \in (-1, 1)$.

These results push further the robustness of reference priors that are constructed from Jeffreys —or hierarchical versions of Jeffreys— and support the objectivity of their choice in practice.

We conclude by emphasizing that our results provide an analytical criterion that aids in the definition of some generalized reference priors: they maximize a concave functional. That criterion constitutes an open path for the derivation of generalized reference priors over less general sets of priors. While this thought is slightly explored in this chapter, we note that it is furthered studied in the following chapter.

5

Properly constrained generalized reference priors

This chapter draws mainly on the submitted work: Antoine Van Biesbroeck (2024b). *Properly constrained reference priors decay rates for efficient and robust posterior inference*. DOI: 10.48550/arXiv.2409.13041

Abstract Reference priors are widely recognized for their objective nature. Yet, they often lead to intractable and improper priors, which complicates their application. Besides, informed prior elicitation methods are penalized by the subjectivity of the choices they require. In this chapter, we aim at proposing a reconciliation of the aforementioned aspects. Leveraging the objective aspect of the reference prior theory, we introduce two strategies of constraint incorporation to build tractable reference priors. One provides a simple and easy-to-compute solution when the improper aspect is not questioned, and the other introduces constraints to ensure the reference prior is proper, or it provides proper posterior. Our methodology emphasizes the central role of Jeffreys prior decay rates in this process, and the practical applicability of our results is demonstrated using an example taken from the literature.

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5.1 Introduction

The reference prior theory represents a widely elected theory for constructing priors that are qualified as "objective". The theory has been thoroughly introduced in the chapter 3. The theory provides a formal mechanism to incorporate prior information in a way that maximizes the information gained from the data within the issued *a posteriori* quantities. In opposition with a plethora of existing methods for building a prior (see e.g. Mikkola et al., 2023), this process is tuned to prevent the incorporation of subjective beliefs in the workflow.

However, despite their objective nature, their implementation is often cumbersome and not always recommended in high dimensions (Berger, Bernardo, and Sun, 2015). Moreover, the low-informative nature of these priors is associated with their common improper aspect, necessitating careful handling to ensure valid statistical inference. Thus, the construction of priors is expected to strike a balance between several criteria. While many works restrict their sets of priors to ones that are tractable, proper, or suitable for high dimensions, others seek to minimize any source of subjectivity.

This chapter aims to reconcile all these viewpoints to improve the prior elicitation. Our contribution takes the form of an enrichment of the reference prior theory to leverage the objective aspect that it provides to reference priors. Building on the developments presented in chapter 4, we restrict priors to the ones that belong to well-chosen —and not too restrictive— sets, and we introduce two strategies to define convenient reference priors. Our first strategy provides a simple, tractable solution for constraining reference priors when the improper aspect is not questioned. The second, by contrast, introduces constraints that lead to reference priors that are proper, or lead to proper posteriors. For both strategies, we try to examine the potential loss of objectivity induced by the constraints, and we discuss their limits. Our results emphasize the central role of Jeffreys prior decay rates when they are improper. Additionally, we draw attention to the fact that our methodology opens a way to define various reference priors on the basis of constraints that could result from any other motivation.

This chapter is organized as follows. In section 5.2, after reviewing briefly the notations for the generalized reference priors framework, we develop our motivation and the objective of this work. This section is also the occasion to introduce a novel definition that is useful for the rest of the study: the quasi D-reference priors. Our main results on constrained reference priors are presented in section 5.3, and discussed in section 5.4. Then, the practical aspect of our work is studied by the application of our method to an example taken from the literature in section 5.5. Detailed mathematical proofs are compiled in section 5.6. Section 5.7 terminates the chapter with a conclusion.

5.2 Definitions, notations and motivation

5.2.1 Notations

In this work we consider a statistic model characterized by a collection of probability distributions $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$ on a measurable set $(\mathcal{Y}, \mathscr{Y})$. We consider the same construction of the Bayesian framework as in the chapter 3 (section 3.4): considering any prior Π (that is a σ-finite measure on Θ) we denote $\mathbf{Y}_k$ a random vector of k observations whose distribution conditionally to $T = \theta$ is $\mathbb{P}_{\mathbf{Y}^k|\theta} = \mathbb{P}_{Y|\theta}^{\otimes k}$, where T is a r.v. whose distribution is the prior Π (see section 3.4.1).

The modeling is supposed to be regular: we assume $\Theta \subset \mathbb{R}^d$ with ν being the Lebesgue measure on $\mathbb{R}^d$ and every prior Π is supposed to admit a density π w.r.t. ν (i.e. $\Pi \in \mathscr{M}^\nu$). We also assume that the model admits a likelihood, denoted by ℓ with for any $\theta \in \Theta$ and $\mathbf{y} \in \mathcal{Y}^k$, $\ell_k(\mathbf{y}|\theta) := \prod_{i=1}^k \ell(\mathbf{y}|\theta)$. We suppose that it verifies assumption 3.1 in chapter 3, making the Fisher information matrix (denoted $\mathcal{I}$) and the Jeffreys prior (whose density is denoted J) being well-defined. The marginal distribution (resp. density) is denoted by $\mathbb{P}_{\mathbf{Y}_k}$ (resp. $p_{\mathbf{Y}_k}$) and the posterior distribution (resp. density) given the observations $\mathbf{y} \in \mathcal{Y}^k$ is denoted by $\mathbb{P}_{T|\mathbf{y}}$ (resp. $p(\cdot|\mathbf{y})$).

Given these notations, we recall the expression of the generalized mutual information, defined in the

chapter 4 when Π is proper:

$$\mathscr{I}_D^k(\Pi) := \mathbb{E}_{T\sim\Pi}[D(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})], \tag{5.1}$$

with D being a dissimilarity measure. In this chapter, we mostly focus on such generalized mutual information when D is a δ divergence with $\delta \in (0,1)$. For some $\delta \in (0,1)$, the notation D_δ will refer to the δ-divergence whose expression is reminded below:

$$D_\delta(P||Q) = \int_{\mathcal{X}} f_\delta\left(\frac{p(x)}{q(x)}\right) q(x)d\omega(x) \quad \text{with} \quad f_\delta(x) = \frac{x^\delta - \delta x - (1-\delta)}{\delta(1-\delta)}, \tag{5.2}$$

where p, q respectively are densities of P and Q w.r.t. a common measure ω on $\mathcal{X}$. When evoking reference priors in a general way, we will refer to generalized reference priors, as proposed in the chapter 4. We remind below their definition:

Definition 5.1 (Generalized reference prior)**.** Let D be a dissimilarity measure and $\mathcal{P}$ a set of priors on Θ. A prior $\Pi \in \mathcal{P}$ is called a D-reference prior over $\mathcal{P}$ with rate $\varphi(k)$ if there exists an openly increasing sequence of compact subsets $(\Theta_i)_{i\in\mathbb{N}}$ such that $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$ and for any i: $0 < \Pi(\Theta_i) < \infty$ and

$$\lim_{k\to\infty} \varphi(k)[\mathscr{I}_D^k(\Pi(\cdot|\Theta_i)) - \mathscr{I}_D^k(P(\cdot|\Theta_i))] \geq 0 \text{ for all } P \in \mathcal{P} \text{ verifying } 0 < P(\Theta_i) < \infty; \tag{5.3}$$

where $\varphi(k)$ is a positive and monotonous function of k. It is said to be unique if for any other D-refernce prior Π', $\Pi \simeq \Pi'$.

We also remind the following result on the D_δ-mutual information and their reference priors (see chapter 4).

Theorem 5.1. *Suppose Θ to be compact and $\Pi \in \mathscr{M}_{\mathcal{C}}^v$ be a prior with $\Pi(\Theta) = 1$. The D_δ-mutual information admits a limit:*

$$\lim_{k\to\infty} k^{d\delta/2}\mathscr{I}_{D_\delta}(\Pi) = l(\pi) - (\delta(1-\delta))^{-1}, \quad l(\pi) = C_\delta \int_\Theta \pi(\theta)^{1+\delta}|\mathcal{I}(\theta)|^{-\delta/2}d\theta, \tag{5.4}$$

where π is the density of Π, and with $C_\delta = (2\pi)^{d\delta/2}(1-\delta)^{-d/2}/(\delta(\delta-1))$.
Call $\mathcal{R} \subset \mathscr{R}_{\mathcal{C}^b}$ a set of densities such that $M(\mathcal{R}) = \mathcal{P}$, with M mapping a density to its associated prior. Then Π is a D_δ-reference prior over $\mathcal{P}$ iff π maximizes l over $\mathcal{R}$.

5.2.2 Objective and motivation

We already know that the definition of the reference prior and the D_δ-reference prior is satisfied by the Jeffreys prior over the large set of priors $\mathscr{M}_{\mathcal{C}}^v$ in most cases (see chapters 3 and 4).

This result is, however, limiting and disappointing in some cases. The reasons are the following ones: (i) the Jeffreys prior is not recommended in high-dimensional problems as it is known to be "either too diffuse or too concentrated" (Berger, Bernardo, and Sun, 2015); moreover (ii) when the expression of the likelihood is itself complex, the computation of the Jeffreys prior can become intractable; also (iii) the Jeffreys prior is known to often lead to an improper prior, which does not necessarily issue a proper posterior distribution, essential for practical *a posteriori* inference and sampling.

To tackle these limitations, we propose in this work to restrict the set of priors over which we derive the reference priors. Indeed, the reference prior definition is usually considered with very large sets of priors, which are constrained only by some regularity assumptions imposed to the priors (such as continuity, positivity). These regularity assumptions do not generally discriminate the Jeffreys prior from the studied set of priors. In this chapter, different restricted sets of priors will be suggested, they are sets that are though to counter the limitations (ii) and (iii) aforementioned. In most cases, they will not include the Jeffreys prior.

The tackling of limitation (i) mentioned above is not a purpose of this work. We recall that it is actually frequently tackled by a sequential construction of the reference prior as suggested by Bernardo (1979b). On the condition that an ordering of the parameters is set:

$$\theta = (\theta_1, \ldots, \theta_r) \in \Theta = \Theta_1 \times \cdots \times \Theta_r, \tag{5.5}$$

this construction considers a hierarchical construction of the reference prior. It is already described in chapter 3 (section 3.3.2). We remind below the steps of the sequential construction:

1. initially fix $\ell_k^1 = \ell_k$;
2. for any values of $\theta_{j+1}, \dots, \theta_r \in \Theta_{j+1} \times \dots \times \Theta_r$, compute a reference prior (in the sense of definition 3.3) under the model with likelihood $\theta_j \mapsto \ell_k^j(\mathbf{y}|\theta_j, \dots, \theta_r)$, denote $\pi_j(\cdot|\theta_{j+1}, \dots, \theta_r)$ its normalized density;
3. derive ℓ_k^{j+1} such as

$$\ell_k^{j+1}(\mathbf{y}|\theta_{j+1}, \dots, \theta_r) = \int_{\Theta_j} \ell_k^j(\mathbf{y}|\theta_j, \dots, \theta_r) d\pi_j(\theta_j|\theta_{j+1}, \dots, \theta_r). \tag{5.6}$$

In this work, the reference priors will be derived only given their formal definition (definition 5.1). Yet, our results can be incorporated in this sequential construction. Indeed, step 2 of the method depicted above consists of the derivation of a reference prior w.r.t. the variable θ_j. Additionally, we invite to note that this construction does not solve the limitations (ii) and (iii) previously evoked. Actually, it makes them essential. Indeed, step 2 requires, firstly, a derivation of a reference prior, so that it would lead to a low-dimensional Jeffreys prior if the set of priors is not constrained. Also step 3 necessitates, secondly, that the latter leads to a proper posterior so that the integral involved does not diverge.

We note that this last issue is taken into account by Berger and Bernardo (1992a) with the suggestion of such construction on an increasing sequence of compact subsets of Θ: $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$. The hierarchical reference prior can then be chosen as a limit of the ones obtained under Θ_i when $i \longrightarrow \infty$. However, this limit can be cumbersome to derive in practice. Another solution suggested by Muré (2018) is to restrict the σ-algebra $\mathscr{Y}$ until the reference prior derived in step 2 leads to a proper posterior. It is still imperfect, as there is no guarantee that such a restricted σ-algebra exists outside the trivial one.

5.2.3 A useful definition: quasi-reference priors

While we aim at restricting the set of priors $\mathcal{P}$ in definition 5.1, we must notice that such a restriction leaves really unsure the existence of a reference prior. Indeed, the definition is itself restrictive, as to admit a reference prior, the set $\mathcal{P}$ must contain a prior whose restrictions are optimal on any compact subsets of Θ. In this section, we suggest an extension of the definition of reference priors in the case where in the set $\mathcal{P}$, the optimal priors on compact subsets of Θ are not renormalization of each other, but converge to a prior in $\mathcal{P}$. Such convergence is considered in the sense of the Q-vague convergence (Bioche and Druilhet, 2016) on $\mathscr{M}_C^v$. The Q-vague convergence of a sequence $(\Pi_n)_n$ to a limit Π is equivalent to the convergence of $([\Pi_n])_n$ to $[\Pi]$ in $\mathscr{M}_C^v / \simeq$ for the quotient topology of the vague convergence on $\mathscr{M}_C^v$.

Definition 5.2 (Quasi reference prior)**.** Let $\mathcal{P}$ be a set of priors. We call $\Pi \in \mathcal{P}$ a quasi D-reference prior if it exists an openly increasing sequence $(\Theta_i)_{i\in\mathbb{N}}$ of compact sets with $\bigcup_{i\in\mathbb{N}} \Theta_i = \Theta$ such that

(i) for any $i \in \mathbb{N}$, there exists a D-reference prior Π_i over $\mathcal{P}_i = \{P(\cdot|\Theta_i),\ P \in \mathcal{P},\ P(\Theta_i) \in (0, \infty)\}$,

(ii) Π is the Q-vague limit of the sequence $(\Pi_i)_{i\in\mathbb{N}}$.

It is said to be unique if for any other quasi D-reference prior Π', $\Pi \simeq \Pi'$.

Proposition below ensures that this definition properly extends definition 5.1 in the case of δ-divergences.

Proposition 5.1.
- *If Π is a D_δ-reference prior over a set $\mathcal{P}$, then it is a quasi D_δ-reference prior.*
- *If $\mathcal{P}$ is a set of priors convex and stable by multiplication by indicator functions over measurable sets, then the quasi D_δ-reference prior over $\mathcal{P}$ is the unique D_δ-reference prior over $\mathcal{P}$.*

- *If $\mathcal{P}$ is a convex set of priors and if the sequence of subsets $(\Theta_i)_i$ in definition 5.1 is fixed, then the quasi-reference prior over $\mathcal{P}$ is unique.*

Proof. The first statement of the proposition is clear given the definition of a D_δ-reference prior.
For the second, let us adopt the notations of theorem 5.1 and notice that if Θ is compact and if π^* is the maximal argument of l over $\mathcal{R}$, then its renormalized restriction π_1^* on a compact subset U maximizes l over the set of all renormalized densities $\mathcal{R}_U$. Indeed, if we suppose that $\pi_1 \in \mathcal{R}_U$ maximizes l then, denoting π_0^* the renormalized restriction of π^* to $\Theta \setminus U$, $t = \int_U \pi^*$, and $\pi = t\pi_1 + (1-t)\pi_0^*$, $\pi \in \mathcal{R}$ and

$$l(\pi) = t^{\delta+1} l(\pi_1) + (1-t)^{\delta+1} l(\pi_0^*) > t^{\delta+1} l(\pi_1^*) + (1-t)^{\delta+1} l(\pi_0^*) = l(\pi^*). \tag{5.7}$$

Hence π^* does not maximize l over $\mathcal{R}$, which is absurd.
Therefore, in our problem, considering two sequences $(\pi_i^{(1)})_i$ and $(\pi_i^{(2)})_i$ respectively defined on $(\Theta_i^{(1)})_i$ and $(\Theta_i^{(2)})_i$, we will get that for any i, $\pi_i^{(1)}(\theta) = \pi_i^{(2)}(\theta)$ for all $\theta \in \Theta_i^{(1)} \cap \Theta_i^{(2)}$. Eventually, they are identical on every compact subsets of Θ, and equal to their Q-vague limits which are the same.
Finally, the third statement of the proposition results from the strict concavity of l. Indeed, for any i, the set $\mathcal{R}_i$ of renormalized restricted densities on Θ_i is convex so that the maximal argument of l over $\mathcal{R}_i$ is unique. Hence the uniqueness of the quasi-reference prior over $\mathcal{P}$. □

5.3 Constrained D_δ-reference priors

In this section, we present results of two different constraint incorporations on the reference priors. In section 5.3.1, the constraint limits the set of priors to exponentiation of coordinates of θ. The result suggests a simpler reference priors in a case where the improper Jeffreys prior's decay rates are not an issue. In section 5.3.2, we introduce a well-chosen linear constraint on the set of priors that ensures the constraint reference prior is proper, or leads to proper posteriors.

5.3.1 Constrained D_δ-reference priors based on Jeffreys' asymptotic decay rates

In this section, we tackle the computational cost of the reference prior. As mentioned in section 5.2.2, the Jeffreys prior expression is often complex to derive even in low dimensional models. This is even more a problem in practical studies where the prior must be evaluated a numerous number of times, when it resorts to MCMC simulations to provide posterior samples of θ for instance.

In some works of the literature, the Jeffreys prior is replaced by its decay rates at the boundary of the domain. For instance, the reference prior for Gaussian processes suggested by Gu (2019) is built on the basis of the decay rates of a Jeffreys prior sequentially computed on the different variables that compose θ (following the construction presented in section 5.2.2). Their idea is that, in particular when it is improper, the prior provides the most information from its asymptotic rates, and variations of them are noticed to have a strong influence on the posterior distribution. The result that follows provides a formalization of this intuition, focusing on the case where the Jeffreys prior asymptotically behaves like exponentiation of coordinates of θ.

Theorem 5.2. *Suppose $\Theta \subset \mathbb{R}$ is an interval of the form $[c, b)$ (or $(b, c]$). Call $M : \pi \in \mathscr{R}_{\mathcal{C}^b} \mapsto (B \mapsto \int_B \pi d\nu) \in \mathscr{M}_{\mathcal{C}}^{\nu}$.*

- *If $b \in \mathbb{R}$ and $J(\theta) \underset{\theta \to b}{\sim} C|\theta - b|^a$ for constants $C \in \mathbb{R}$ and $a \leq -1$, then $M(\pi^*)$ where $\pi^*(\theta) \propto |\theta - b|^a$ is the unique quasi D_δ-reference prior over $M(\hat{\mathcal{R}})$ where $\hat{\mathcal{R}} = \{\pi(\theta) \propto |\theta - b|^u,\ u \in \mathbb{R}\}$.*
- *If $|b| = \infty$ and $J(\theta) \underset{\theta \to b}{\sim} C\theta^a$ for constants $C \in \mathbb{R}$ and $a \geq -1$, then $M(\pi^*)$ where $\pi^*(\theta) \propto \theta^a$ is the unique quasi D_δ-reference prior over $M(\hat{\mathcal{R}})$ where $\hat{\mathcal{R}} = \{\pi(\theta) \propto \theta^u,\ u \in \mathbb{R}\}$.*

Proof. The proof is technical and detailed in section 5.6. The idea is that $l(\pi)$ can be seen as a negative divergence between π and J. However, when J is improper at the boundary of the domain, the maximization of $l(\pi)$ gets closer to the minimization of a divergence between π and the improper decay rate of J. □

Remark 5.1. Theorem 5.2 still stands when $\Theta = (b, c)$ (or (c, b)) if $c \neq \infty$ and if $J(\theta)$ admits a non-null and finite limit when $\theta \longrightarrow b$.

This theorem serves the statement of two conclusions: (i) it emphasizes that when Jeffreys prior is improper, its improper decay rates contain the most relevant information, and (ii) it proposes to choose this asymptotic expansion of Jeffreys as a quasi D_δ-reference prior when we look for an easy prior to compute.

5.3.2 Properly constrained D_δ-reference priors

In section 5.3.1, we have provided some elements to construct a tractable reference prior on the coordinates over which Jeffreys prior is improper. The reference prior proposed by our theorem keeps the improper characteristic of Jeffreys prior on the same coordinates. This improper aspect can, however, remain an issue in some cases, especially when the resulting posterior is improper as well.

For this reason, it might happen that some asymptotic rates in some directions still have to be tackled. The work in this section is concluded by results that allow defining a D_δ-reference prior (or quasi D_δ-reference prior), which benefits from adjusted decay rates from Jeffreys prior. The proposition below constitutes a preliminary result that gives the form of a D_δ-reference prior over a set of priors with linear constraints.

Assumption 5.1. A family of functions from Θ to $\mathbb{R}$ $(g_j)_{j=1}^p$ is said to satisfy assumption 5.1 if $g_0, \ldots, g_p$ are linearly independent in the space of a.e. continuous functions from Θ to $\mathbb{R}$, where $g_0 := \theta \mapsto 1$.

Proposition 5.2. *Suppose Θ to be a compact subset of $\mathbb{R}^d$. Let $g_1, \ldots, g_p$ be functions in $\mathscr{R}_{\mathcal{C}^b}$ that satisfy assumption 5.1. Define $\tilde{\mathcal{P}}$ the set of priors Π on Θ such that $\forall 1, \ldots, p$, $\int_\Theta g_j d\Pi = c_j$, for some $c_j \in \mathbb{R}$. If there exists a D_δ-reference prior over $\tilde{\mathcal{P}}$, it is unique. If it is positive, its density π verifies*

$$\pi(\theta) = J(\theta) \left(\lambda_0 + \sum_{j=1}^p \lambda_j g_j(\theta) \right)^{1/\delta}, \tag{5.8}$$

for some $\lambda_j \in \mathbb{R}$. Reciprocally, if there exists a prior whose density verifies the above equation for some $\lambda_j \in \mathbb{R}$, it is the D_δ-reference prior over $\tilde{\mathcal{P}}$.

Proof. This proposition results from a Lagrange multipliers theorem. A detailed proof is proposed in section 5.6. □

Remark 5.2. While it is not the subject of this work, we let the reader notice that this proposition opens the way to the introduction of constraints based on expert judgments in prior elicitation. They can take the form of moment constraints or predictive constraints.

Remark 5.3. The expression of the reference prior given by proposition 5.2 depends on the chosen δ-divergence. While this work considers only the framework of reference priors under δ-divergences as a dissimilarity measure, a version of this theorem could be written in the original framework of the reference prior theory that uses the Kullback-Leibler divergence. The expression of the resulting reference prior would be impacted. In the appendix we prove that the expression using the Kullback-Leibler divergence would take the form:

$$\pi^* \propto J \cdot \exp\left(\sum_{j=1}^p \lambda_j g_j \right), \tag{5.9}$$

for some λ_j that remain to be determined. This expression was already intuited by Bernardo and Smith (1994).

Below, given a function g that is selected to adjust the asymptotics of Jeffreys prior, is stated the expression of a proper D_δ-reference prior.

Theorem 5.3. *Let $g : \Theta \longrightarrow (0, \infty)$ be a function in $\mathscr{R}_{\mathcal{C}^b}$ such that*

$$\int_\Theta J(\theta) g^{1/\delta}(\theta) d\theta < \infty \quad \text{and} \quad \int_\Theta J(\theta) g^{1/\delta+1}(\theta) d\theta < \infty, \tag{5.10}$$

and suppose that g is bounded in the neighborhood of b for an element $b \in \partial\Theta$. We denote by $\overline{\mathcal{P}}$ the set of positive priors Π on Θ such that $\int_\Theta g d\Pi < \infty$, and we define $\Pi \in \overline{\mathcal{P}}$ as the prior whose density π verifies

$$\pi(\theta) \propto J(\theta) g(\theta)^{1/\delta}. \tag{5.11}$$

If Jg is non-integrable in the neighborhood of b, then Π is a D_δ-reference prior over $\overline{\mathcal{P}}$. Otherwise, and if J is improper in the neighborhood of b, Π is a D_δ-reference prior over the set of proper priors in $\overline{\mathcal{P}}$.

Proof. The statement of this theorem results from the sequential use of proposition 5.2 on an increasing sequence of compact subsets of Θ. A detailed proof is written in section 5.6. □

To improve the above theorem, one would like to relax the first assumption in eq. (5.10), i.e., to let $\int_\Theta J g^{1/\delta}$ be infinite. Indeed, in this way, the result would provide a reference prior Π —non-necessarily proper— but such that $\pi g \in L^1$, where π is a density of Π. With a good choice of g, π^* could be built as a prior that provides a proper posterior. It is the purpose of the next theorem. The cost of this relaxation is the provision of a quasi-reference prior instead of a reference prior.

Theorem 5.4. *Let $g : \Theta \longrightarrow (0, \infty)$ be in $\mathscr{R}_{\mathcal{C}^b}$ such that*

$$\int_\Theta J(\theta) g(\theta) d\theta = \infty \quad \text{and} \quad \int_\Theta J(\theta) g^{1/\delta+1}(\theta) d\theta < \infty, \tag{5.12}$$

and suppose that $g(\theta) \underset{\theta \to b}{\longrightarrow} 0$ for an element $b \in \partial\Theta$ such that J is non-integrable in the neighborhood of b.
Let $(\Theta_i)_{i \in \mathbb{N}}$ be an openly increasing sequence of compact sets that covers Θ and $(c_i)_i$ be a bounded sequence in $(0, \infty)$. Define the set of priors $\overline{\mathcal{P}}' = \{\Pi, \forall i, \int_{\Theta_i} g d\Pi = c_i \int_{\Theta_i} d\Pi\}$.
Denote for any i $\overline{\mathcal{P}}'_i$ the set of renormalized restrictions to Θ_i of priors in $\overline{\mathcal{P}}'$. If for any i there exists a positive maximum of l over $\overline{\mathcal{P}}'_i$, then Π whose density is denoted π is a quasi D_δ-reference prior over $\overline{\mathcal{P}}'$ with

$$\pi(\theta) \propto J(\theta) g(\theta)^{1/\delta}. \tag{5.13}$$

This prior is such that $\int_\Theta \pi g d\nu < \infty$.

5.4 Discussion

The knowledge of Jeffreys prior's decay rates is central in the results presented in this work. These results indicate that in common scenarios where Jeffreys prior is improper, these rates must be explicitly considered in order to construct a reference prior.

We let the reader note that theorems 5.3 and 5.4 introduce results that also depend on the chosen dissimilarity measure. Therefore, a balance must be found between the subjective influence of the constraint and the quest for an informed prior to facilitate possible sampling from the posterior. This is illustrated in the example we address in the following section. However, it is important to observe that using the KL-divergence instead of a δ-divergence would result in a stronger influence of the constraint on the final prior. As noted in remark 5.3, the exponentialization of the function g could lead to a prior with distribution tails that are significantly negligible beyond those of Jeffreys, thereby jeopardizing its objective nature.

Generally, the results we propose in sections 5.3.1 and 5.3.2 address different problems and are thus fundamentally different in nature. In one case, the improper aspect of Jeffreys prior is not necessarily challenged, and an efficient construction of the latter is proposed. In the other case, the goal is to significantly attenuate its improper aspect while maintaining as much objectivity as possible. In this latter case, however, the expressions of the proposed reference priors still depend on the expression of Jeffreys prior. Nevertheless, when Jeffreys prior is proper, there is no guarantee that a straightforward construction inspired by its convergence rates at the domain boundaries will be relevant. Indeed, although improper tails concentrate an infinite mass that constitutes all the information at the boundaries, when they are proper, the information of interest may need to be sought elsewhere. In this case, a calculation or approximation of the 'proper' Jeffreys prior remains to be considered.

Finally, regarding theorem 5.2, although the result is limited to parameter power distribution tails, it is observed that, in practice, these include a wide range of improper Jeffreys priors. For example, this includes Jeffreys priors derived from various Gaussian models, such as those introduced by Neyman and Scott (1948); Jeffreys priors related to specific parameters within Gaussian process models (Gu and Berger, 2016); and those arising in more specialized contexts, like the one that we develop in the part II of this manuscript. Moreover, the invariance of Jeffreys priors under re-parameterization can sometimes allow us to return to this case. Specifically, if J can be asymptotically written as a power of a function f, where f is differentiable, monotone and with bounded derivative (from above and from below), then the re-parameterization $\vartheta = f(\theta)$ should allow us to recover the reference prior among those expressible as powers of f.

In the following section, we illustrate an application of our work with an example taken from the literature.

5.5 An example

In their work, Rubio and Steel (2014) prove that the two piece location-scale model they proposed has an improper Jeffreys prior, which issues an improper posterior. The model is parameterized by $\theta = (\mu, \sigma_1, \sigma_2) \in \mathbb{R} \times (0, \infty)^2$, inferred over observations in $\mathcal{Y} = \mathbb{R}$. It has the following likelihood:

$$\ell(y|\theta) = \frac{2}{\sigma_1 + \sigma_2} \left[f\left(\frac{y-\mu}{\sigma_1}\right) \mathbb{1}_{(-\infty,\mu)}(y) + f\left(\frac{y-\mu}{\sigma_2}\right) \mathbb{1}_{(\mu,\infty)}(y) \right], \tag{5.14}$$

where f is a density function with support on $\mathbb{R}$, assumed to be symmetric with a single mode at zero, and with a few integrability assumptions that are detailed in Rubio and Steel, 2014. The choice of f is open and we can take, for instance, the standard Gaussian density function. Under this construction, the Fisher information matrix of this model takes the form:

$$\mathcal{I}(\theta) = \begin{pmatrix} \frac{\alpha_1}{\sigma_1\sigma_2} & -\frac{2\alpha_3}{\sigma_1(\sigma_1+\sigma_2)} & \frac{2\alpha_3}{\sigma_2(\sigma_1+\sigma_2)} \\ * & \frac{\alpha_2}{\sigma_1(\sigma_1+\sigma_2)} + \frac{\sigma_2}{\sigma_1(\sigma_1+\sigma_2)^2} & -\frac{1}{(\sigma_1+\sigma_2)^2} \\ * & * & \frac{\alpha_2}{\sigma_2(\sigma_1+\sigma_2)} + \frac{\sigma_1}{\sigma_2(\sigma_1+\sigma_2)^2} \end{pmatrix} \tag{5.15}$$

for some positive constants α_1, α_2 and α_3.

The full Jeffreys prior can be computed as

$$J(\theta) \propto \frac{1}{\sigma_1\sigma_2(\sigma_1+\sigma_2)}, \tag{5.16}$$

it is improper and leads to an improper posterior. In the following, we construct different priors based on the suggestions developed in this chapter.

Proper priors based on a moment constraint Considering results in section 5.3.2 and the decay rates of the Jeffreys prior above, a simple correction can be done to issue a proper reference prior w.r.t. σ_1, σ_2, which results in a proper posterior. We consider $\delta \in (0, 1)$; given $\varepsilon \in (0, \frac{1}{1+1/\delta})$ we have

$$\int J(\theta)(\sigma_1\sigma_2)^{\varepsilon/\delta+1} d\sigma_1 d\sigma_2 < \infty, \tag{5.17}$$

so that the associate proper D_δ-reference prior density π, which is such that $\pi(\mu,\sigma_1,\sigma_2)\sigma_1^\varepsilon\sigma_2^\varepsilon$ is integrable w.r.t. σ_1,σ_2, is

$$\pi(\mu,\sigma_1,\sigma_2) \propto \frac{(\sigma_1\sigma_2)^{\varepsilon/\delta-1}}{\sigma_1+\sigma_2}. \tag{5.18}$$

Overview and sensitivity on the parameters Our prior densities are compared with the Jeffreys prior one in fig. 5.1.(a), w.r.t. the parameter σ_2 the others being fixed to 1. For this comparison, a multiplicative constant had to be chosen on J, we have chosen the one such that $J(1,1,1) = 2$. Our priors differ as a function of $\gamma = \varepsilon/\delta \in (0, \frac{1}{1+\delta}) \subset (0,1)$. On the one hand, when γ becomes close to 0, the prior —which we denote by π_γ from now on— becomes close to the Jeffreys prior, i.e., the most objective prior w.r.t. the mutual information criterion. However, in this case, π_γ becomes close to an improper prior, and its posterior becomes close to an improper posterior. On the other hand, setting γ away from 0 rearranges the quantity of information in the prior. Its referential nature decreases in favor of an increase in its entropy. Therefore, a trade-off has to be made between suitability for inference and objectivity. Finally, note that in fig. 5.1.(a) is also drawn the *independent Jeffreys prior* (IJ) proposed by the authors in Rubio and Steel, 2014 for this model as an alternative to Jeffreys:

$$\mathrm{IJ}(\mu,\sigma_1,\sigma_2) \propto \frac{\sqrt{\sigma_1+\alpha_2(\sigma_1+\sigma_1)}\sqrt{\sigma_2+\alpha_2(\sigma_1+\sigma_2)}}{\sqrt{\sigma_1\sigma_2}(\sigma_1+\sigma_2)^2}. \tag{5.19}$$

Its decay rates are actually the same as the ones of π_γ when $\gamma = 1/2$, and the two priors are hard to distinguish. Note that this latter prior $\pi_{1/2}$ equals the hierarchical reference prior (as described in section 5.2.2) constructed from the ordering $\pi(\theta) = \pi_1(\sigma_1,\sigma_2|\mu)\pi_2(\mu)$.

To evaluate a bit further our method, we propose a visualization of the posterior sensitivity to the priors, i.e., to γ. Such influence quantification constitutes a critical step of the *Bayesian workflow*, as expressed by Gelman, Vehtari, et al. (2020). Several methods exist in the literature for this purpose (e.g., Berger, 1990; Nott et al., 2020). An approach is to compare the variations of an *a posteriori* quantity as a function of the parameter (Kallioinen, Paananen, and Bürkner, 2023). In this example, this methodology is yet limited by the improper aspect of Jeffreys posterior. It cannot be considered for any comparison. In fig. 5.1.(b), (c) and (d) are plotted, for numerous γ and several data set sizes k, the posterior densities that result from a sample of data and from the prior densities π_γ. As expected, the influence of the prior appears for small values of k. Indeed, we can notice that for high values of γ, the posterior is slightly more shifted to the right and seems to be a little flatter. It is remarkable that this observation becomes limited when k increases. When $k = 50$, the difference between the posterior densities is hard to distinguish. This indicates that the little losses of objectivity should induce small variations in the resulting inference in this example.

For practical details, the chosen f is the standard Gaussian density, and the different data sets have been generated according to the likelihood in eq. (5.14) conditionally to $\theta^* = (2,2,2)$.

5.6 Detailed proofs

5.6.1 Proof of theorem 5.2

To prove this theorem, we consider to simplify the derivations that $b = 0$ with $\Theta = (0,1]$. We will show later how to extend the result to the other cases. As the Jeffreys prior can be defined up to a positive multiplicative constant with no incidence on the definition of a D_δ-reference prior, we will simplify its decay rate assuming $J(\theta) \underset{\theta\to 0}{\sim} \theta^a$.

Let us define the increasing sequence of compact subsets Θ: $\Theta_i = [\theta_i, 1]$, $i \geq 0$, with $\theta_i \underset{i\to\infty}{\longrightarrow} 0$. We denote by ψ_i and $\tilde{\psi}_i$ the functions defined as follow:

$$\psi_i(u) = -\frac{\int_{\Theta_i} J(\theta)^{-\delta}\theta^{u(1+\delta)}d\theta}{\left(\int_{\Theta_i}\theta^u d\theta\right)^{1+\delta}}, \qquad \tilde{\psi}_i(u) = -\frac{\int_{\Theta_i}\theta^{-a\delta}\theta^{u(1+\delta)}d\theta}{\left(\int_{\Theta_i}\theta^u d\theta\right)^{1+\delta}}. \tag{5.20}$$

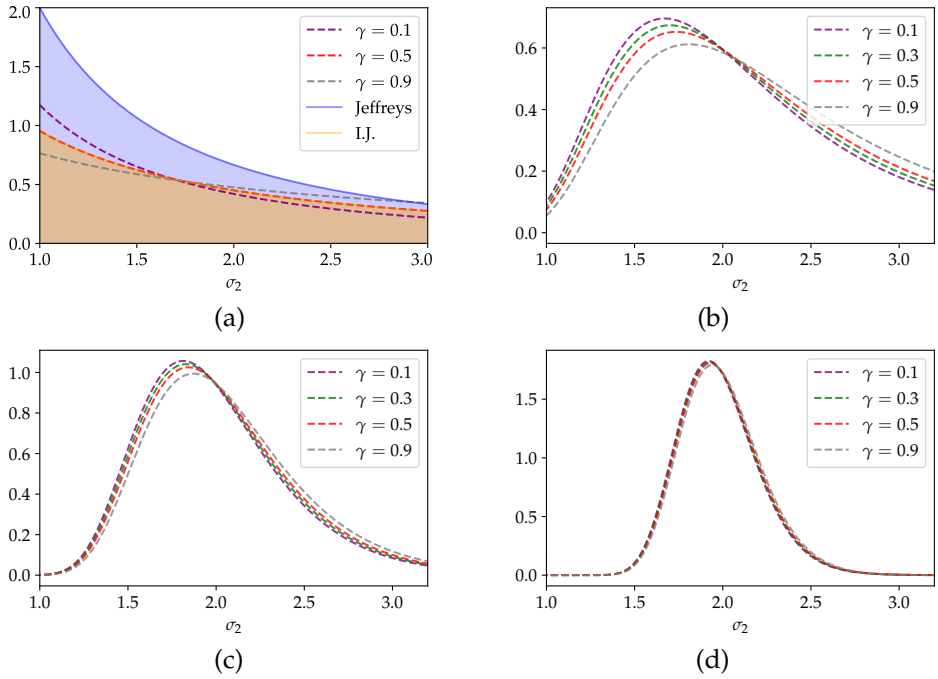

Figure 5.1: In (a), different prior densities π_γ (in dashed line) w.r.t. σ_2 along with a Jeffreys prior density (that delimits the blue area) and the *Independent Jeffreys* of Rubio and Steel, 2014 (that delimits the orange area). In (b), (c) and (d), posterior densities for different values of γ are plotted. One observed sample was used to calculate all the posteriors in a figure. The observed sample has a size of $k = 5$ in (b), $k = 15$ in (c), and $k = 50$ in (d).

The quantity $\psi_i(u)$ corresponds —up to a positive constant— to the parametrization w.r.t. $u \in \mathbb{R}$ of $l(\pi_i)$; where $\pi(\theta) \propto \theta^u$, π_i is re-normalized restriction of π to Θ_i and where l is the function defined in theorem 5.1 that we seek to maximize. Therefore, if we call $u_i \in \arg\max_{u\in\mathbb{R}} \psi_i(u)$ for any i, and if the associated sequence of priors $(M(\pi_i))_i$ converge Q-vaguely to a prior Π^* over Θ, that would prove that Π^* is a quasi D_δ-reference prior.

The case $a < -1$ Firstly, we assume that $a < -1$. Denote $U = (-\infty, u_m := \frac{\delta a - 1}{1+\delta})$, we want to derive an asymptotic equivalent when $i \longrightarrow \infty$ of $\psi_i(u)$, uniformly w.r.t. $u \in U$. More precisely, we are about to show that for any $\varepsilon > 0$ there exists a i_1 such that for any $u \in U$ and $i > i_i$, $|\psi_i(u) - \tilde{\psi}_i(u)| < \varepsilon|\tilde{\psi}_i(u)|$.

Let $\varepsilon > 0$, using that $J(\theta)^{-\delta} \underset{\theta\to 0}{\sim} \theta^{-\delta a}$, there exists i_0 such that for any $\theta < \theta_{i_0}$, $|J(\theta)^{-\delta} - \theta^{-\delta a}| < \varepsilon\theta^{-\delta a}$. Now, we consider an i_1 such that:

$$\frac{\int_{\Theta_{i_0}} J(\theta)^{-\delta} d\theta}{\int_{\theta_{i_1}/\theta_{i_0}}^1 J(\theta_{i_0}\theta)^{-\delta}\theta^{u_m(1+\delta)} d\theta} < \theta_{i_0}\varepsilon/2 \quad \text{and} \quad \frac{\int_{\Theta_{i_0}} \theta^{-\delta a} d\theta}{\int_{\theta_{i_1}/\theta_{i_0}}^1 (\theta_{i_0}\theta)^{-\delta a}\theta^{u_m(1+\delta)} d\theta} < \theta_{i_0}\varepsilon/2. \tag{5.21}$$

Thus, for any $i > i_1$ and any $u \in U$:

$$\frac{\int_{\Theta_{i_0}} J(\theta)^{-\delta}\theta^{u(1+\delta)} d\theta}{\int_{\Theta_i\setminus\Theta_{i_0}} J(\theta)^{-\delta}\theta^{u(1+\delta)} d\theta} = \frac{\theta_{i_0}^{-1}\int_{\theta_{i_0}}^1 J(\theta)^{-\delta}(\frac{\theta}{\theta_{i_0}})^{u(1+\delta)} d\theta}{\int_{\theta_i/\theta_{i_0}}^1 J(\theta_{i_0}\theta)^{-\delta}\theta^{u(1+\delta)} d\theta} < \frac{\theta_{i_0}^{-1}\int_{\Theta_{i_0}} J(\theta)^{-\delta} d\theta}{\int_{\theta_{i_1}/\theta_{i_0}}^1 J(\theta_{i_0}\theta)^{-\delta}\theta^{u_m(1+\delta)} d\theta} < \varepsilon/2 \tag{5.22}$$

and

$$\frac{\int_{\Theta_{i_0}} \theta^{-\delta}\theta^{u(1+\delta)} d\theta}{\int_{\Theta_i\setminus\Theta_{i_0}} \theta^{-\delta}\theta^{u(1+\delta)} d\theta} = \frac{\theta_{i_0}^{-1}\int_{\theta_{i_0}}^1 \theta^{-\delta}(\frac{\theta}{\theta_{i_0}})^{u(1+\delta)} d\theta}{\int_{\theta_i/\theta_{i_0}}^1 (\theta_{i_0}\theta)^{-\delta}\theta^{u(1+\delta)} d\theta} < \frac{\theta_{i_0}^{-1}\int_{\Theta_{i_0}} \theta^{-\delta} d\theta}{\int_{\theta_{i_1}/\theta_{i_0}}^1 (\theta_{i_0}\theta)^{-\delta}\theta^{u_m(1+\delta)} d\theta} < \varepsilon/2, \tag{5.23}$$

so that $|\psi_i(u) - \tilde{\psi}_i(u)| < \tilde{\varepsilon}|\tilde{\psi}_i(u)|$ as expected.

Now we want to use this asymptotic equivalence to bound the difference $|a - u_i|$, where u_i is defined as a maximal argument of ψ_i. The next step is thus to show that such $(u_i)_i$ exists.

There exist $\tilde{K}$, $\tilde{K}'$ such that $\tilde{K}\theta^{-\delta} \leq J(\theta)^{-\delta} \leq \tilde{K}'\theta^{-\delta}$. Let $i \geq 0$, we can write

$$\tilde{K}|\tilde{\psi}_i(u)| \leq |\psi_i(u)| \leq \tilde{K}'|\tilde{\psi}_i(u)| \tag{5.24}$$

with

$$|\tilde{\psi}_i(u)| = \theta_i^{-\delta(a+1)} \frac{|u+1|^{1+\delta}}{|\delta(u-a)+u+1|} \frac{1-\theta_i^{-\delta(u-a)-u-1}}{(1-\theta_i^{-u-1})^{1+\delta}} \underset{u\to-\infty}{\longrightarrow} +\infty. \tag{5.25}$$

That makes $|\psi_i|$ being a coercive and continuous function on U, so that it admits minimal arguments in U. We denote by u_i one of them:

$$u_i \in \arg\max_{u\in U} \psi_i(u). \tag{5.26}$$

We recall that, by concavity of $x \mapsto -x^{-\delta}$, we find that a is the only maximal argument of $\tilde{\psi}_i$ for any i. This way, for $i > i_1$, we write

$$\begin{aligned} |\tilde{\psi}_i(u_i) - \tilde{\psi}_i(a)| &\leq |\psi_i(u_i) - \tilde{\psi}_i(u_i)| + |\psi_i(a) - \tilde{\psi}_i(a)| + |\psi_i(u_i) - \psi_i(a)| \\ \tilde{\psi}_i(a) - \tilde{\psi}_i(u_i) &\leq \varepsilon(|\tilde{\psi}_i(u_i)| + |\tilde{\psi}_i(a)|) + \psi_i(u_i) - \psi_i(a), \end{aligned} \tag{5.27}$$

which leads to

$$\begin{aligned} 2(\tilde{\psi}_i(a) - \tilde{\psi}_i(u_i)) &\leq \varepsilon(|\tilde{\psi}_i(u_i)| + |\tilde{\psi}_i(a)|) + \psi_i(u_i) - \tilde{\psi}_i(u_i) + \tilde{\psi}_i(a) - \psi_i(a) \\ \tilde{\psi}_i(a) - \tilde{\psi}_i(u_i) &\leq \varepsilon(|\tilde{\psi}_i(u_i)| + |\tilde{\psi}_i(a)|). \end{aligned} \tag{5.28}$$

Consequently to the convergence of $(\theta_i^{\delta(a+1)}\tilde{\psi}_i(a))_i$ toward a positive limit when $i \longrightarrow \infty$, we deduce that $\theta_i^{\delta(a+1)}(\tilde{\psi}_i(a) - \tilde{\psi}_i(u_i))/|\theta_i^{\delta(a+1)}\tilde{\psi}_i(u_i)|$ is asymptotically null. This prevents the sequence $(\theta_i^{\delta(a+1)}\tilde{\psi}_i(u_i))_i$ to admit a non finite subsequential limit, meaning it has to be bounded and to converges to the same limit as $(\theta_i^{\delta(a+1)}\tilde{\psi}_i(a))_i$, i.e. $-|a+1|^\delta$.

On another hand, we notice that for any $M > 0$, there exist a M' such that for any $u < M'$,

$$\frac{|u+1|^{1+\delta}}{|\delta(u-a)+u+1|} > M \tag{5.29}$$

and $|\theta^{\delta(a+1)}\tilde{\psi}_i(u)| > M$ for any $i \geq 0$. Thus, as $(\theta^{\delta(a+1)}\tilde{\psi}_i(u_i))_i$ has been proven to be bounded, so must be $(u_i)_i$.

To conclude on that sequence, we denote by ρ a finite subsequential limit of $(u_i)_i$, if $\rho \neq u_m$ then deriving the limit of $\theta^{\delta(a+1)}\tilde{\psi}_i(u_i)$ leads to

$$\begin{aligned} -\frac{|\rho+1|^{1+\delta}}{\delta(\rho-a)+\rho+1} &= |a+1|^\delta \\ \text{i.e.} \quad -|\rho+1|(|\rho+1|^\delta - |a+1|^\delta) &= |a+1|^\delta\delta(\rho-a); \end{aligned} \tag{5.30}$$

necessarily, $\rho = a$. It remains to prove that $\rho = u_m$ is absurd. Indeed, in this case the integrals $\int_{\Theta_i} \theta^{-\delta a + u_i(1+\delta)} d\theta$ converge either to 0, either to $+\infty$. Therefore, that would make $(\theta_i^{\delta(a+1)}\tilde{\psi}_i(u_i))_i$ converging either to $-\infty$, either to 0, which in both case is different to $-|a+1|^\delta$.

Let us now work beyond the subset U of $\mathbb{R}$. First, if $u \in (-1, +\infty)$, the integrals that compose $\psi_i(u)$ both admit finite and positive limits when $i \longrightarrow \infty$. The limit of $(|\psi_i(u)|)_i$ is moreover bounded from below as a consequence of eq. (5.24):

$$|\psi_i(u)| \geq \tilde{K}\frac{(u+1)^{1+\delta}}{\delta(u-a)+u+1} \frac{1-\theta_i^{\delta(u-a)+u+1}}{(1-\theta_i^{u+1})^{1+\delta}} \geq \tilde{K}'|\log\theta_i|^{-1-\delta}. \tag{5.31}$$

Thus, there exists $i_2 \geq 0$ such that for any $i > i_2$ $|\psi_i(a)| < \tilde{K}'|\log\theta_i|^{-1-\delta}$, consequently to $\psi_i(a) \underset{i\to\infty}{\sim} |a+1|^\delta \theta_i^{-\delta(a+1)} \underset{i\to\infty}{=} o(|\log\theta_i|^{-\delta-1})$. As a result, for any $i > i_2$:

$$\sup_{u\in(-1,+\infty)} \psi_i(u) < \psi_i(a) \leq \psi_i(u_i). \tag{5.32}$$

Finally, if $u \in (u_m, -1)$, analogously than in eq. (5.31), we can write

$$|\psi_i(u)| \geq \tilde{K}(u+1)^{1+\delta} \frac{|\log\theta_i|}{(1-\theta_i^{u+1})^{1+\delta}} \geq \tilde{K}''\theta_i^{-(u_m+1)(1+\delta)}|\log\theta_i|^{-\delta}. \tag{5.33}$$

Once again, we have $\psi_i(a) \underset{i\to\infty}{=} o(\theta_i^{-(u_m+1)(1+\delta)}|\log\theta_i|^{-\delta})$ and we can consider $i_3 \geq 0$ such that for any $i > i_3$:

$$\sup_{u\in(u_m,-1)} \psi_i(u) < \psi_i(a) \leq \psi_i(u_i). \tag{5.34}$$

All the work that precedes proves that any sequence $(v_i)_i$ defined by $v_i \in \arg\max_{\mathbb{R}} \psi_i$ converges to a.

The case a $= -1$ In this case, we easily get that $\psi_i(a) \underset{i\to\infty}{\sim} -|\log\theta_i|^{-\delta}$. When $u < \eta < a$,

$$|\psi_i(u)| \geq \tilde{K}\frac{|u+1|^\delta}{\delta+1}\frac{1-\theta_i^{-(\delta+1)(u+1)}}{(1-\theta_i^{-u-1})^{1+\delta}} \geq \hat{K}|\eta+1|^\delta(1-\theta_i^{-(\eta-1)(1+\delta)}) \tag{5.35}$$

and when $u > \tilde{\eta} > a$,

$$|\psi_i(u)| \geq \tilde{K}\frac{|u+1|^\delta}{\delta+1}\frac{1-\theta_i^{(\delta+1)(u+1)}}{(1-\theta_i^{u+1})^{1+\delta}} \geq \hat{K}'|\tilde{\eta}+1|^\delta(1-\theta_i^{(\tilde{\eta}-1)(1+\delta)}). \tag{5.36}$$

Thus, for any $\varepsilon > 0$, the equations above with $\eta = a-\varepsilon$ and $\tilde{\eta} = a+\varepsilon$ let state that there exists an $i_1 \geq 0$ such that for any $i > i_1$:

$$\sup_{u\in(-\infty,a-\varepsilon)} \psi_i(u) < \psi_i(a) \quad \text{and} \quad \sup_{u\in(a+\varepsilon,\infty)} \psi_i(u) < \psi_i(a) \tag{5.37}$$

so that $\arg\max_{\mathbb{R}} \psi_i \subset (a-\varepsilon, a+\varepsilon)$, which let the definition of a sequence $(u_i)_i$ of maximal arguments of ψ_i which converges to a.

Q-vague convergence The conclusion concerning the Q-vague convergence of the $M(\pi_i^*)$ defined from the u_i constructed in the work above: $\pi_i^*(\theta) \propto \theta^{u_i}$ is a direct result of Bioche and Druilhet, 2016, Proposition 2.16. Indeed, the convergence of sequence $(u_i)_i$ toward a, implies that the sequence of our priors converges Q-vaguely to Π^* such that $\Pi^* = M(\pi^*)$ with $\pi^*(\theta) \propto \theta^a$.

Extension to other set Θ We shall now demonstrate that the proven result extends itself to the general case: $\Theta = [c, b)$ or $(b, c]$, $b \in \mathbb{R} \cup \{-\infty, \infty\}$.

We first consider $\Theta = (b, c]$ with $b \in \mathbb{R}$. We denote by $\mathcal{Q}$ the set of priors densities after the substitution $\vartheta = (\theta - b)/(c-b) \in T = (0,1)$: $\mathcal{Q} = \{\tilde{\pi}(\vartheta) = (b-c)\pi(\vartheta(b-c)+b), \pi \in \hat{\mathcal{R}}\}$. Thus, $\mathcal{Q} = \{\pi(\theta) \propto \vartheta^u, u \in \mathbb{R}\}$. We define the increasing sequence of compact sets $(\Theta_i)_i$ by $\Theta_i = [b+t_i, c]$ with $t_i \underset{i\to}{\longrightarrow} 0$. Therefore, for $\pi \in \hat{\mathcal{P}}$, calling π_i the renormalized restriction of π to Θ_i gives

$$l(\pi_i) = \tilde{l}(\tilde{\pi}_i) = C_\delta \int_{T_i} \tilde{\pi}_i(\vartheta)^{1+\delta}\tilde{J}(\vartheta)^{-\delta}d\vartheta \tag{5.38}$$

with $T_i = [t_i, 1]$, $\tilde{\pi}_i(\vartheta) = (b-c)\pi_i(\vartheta(b-c)+b) \in \mathcal{Q}$ and $\tilde{J}(\vartheta) = (b-c)J(\vartheta(b-c)+b) \underset{\vartheta\to 0}{\sim} \vartheta^a$. Thus, the work done above states that the family of maximal arguments $\tilde{\pi}_i^*$ of $\tilde{l}$ over $\mathcal{Q}_i$ provides a sequence of

priors $(M(\tilde{\pi}_i^*))_i$ that converges Q-vaguely toward $M(\tilde{\pi}^*)$ where $\tilde{\pi}^*(\vartheta) \propto \vartheta^a$. Thus, the associate densities π_i^* maximize l over $\mathcal{P}_i$ and issues a sequence of priors $(M(\pi_i^*))_i$ that converges Q-vaguely toward $M(\pi^*)$ where $\pi^*(\theta) \propto |\theta - b|^a$.

To treat the other cases, other substitutions with analogous work permit to conclude: (i) the substitution $\vartheta = (b-\theta)/(b-c)$ when $\Theta = (c,b)$, $b \in \mathbb{R}$; (ii) the substitution $\vartheta = 1/(|\theta - c| + 1)$ when $\Theta = [c,\infty)$ or $(-\infty, c]$.

5.6.2 Proof of proposition 5.2

Let us start by the uniqueness. A prior $\Pi^* \in \mathcal{P}$ is a D_δ-reference prior if and only if its density π^* maximizes l over $\mathcal{R}$, where $M(\mathcal{R}) = \mathcal{P}$. As the set $\mathcal{P}$ is convex, $\mathcal{R}$ is convex as well. Also, we notice that the function $\pi \mapsto l(\pi)$ is strictly concave, so that its maximizer is unique in the sense that two maximizer are equal ν-a.e. That ensures the D_δ-reference prior is unique.

Regarding the expression of π^*, let us call E the space of bounded a.e. continuous functions from Θ to $\mathbb{R}$ and we equip E with the supremum norm over E: $\|f\| = \sup_\Theta |f|$. The set Θ being supposed compact, the pair $(E, \|\cdot\|)$ constitutes a Banach vector space whose restriction U composed by the positive functions of E is an open and convex subset. It is possible to see l as being a concave function defined on U.

Let us compute the differentiate of l over U. One can write $l = \phi_2 \circ \phi_1$ with

$$\phi_1 : \pi \in E \longmapsto \pi^{1+\delta} \in E; \qquad \phi_2 : \pi \in E \longmapsto C_\delta \int_\Theta \pi(\theta)|\mathcal{I}(\theta)|^{-\delta/2} d\theta. \tag{5.39}$$

As ϕ_2 is a continuous linear mapping from E to $\mathbb{R}$, l is differentiate while ϕ_1 is, with for any $h \in E$:

$$dl(\pi) = \phi_2 \circ d\phi_1(\pi). \tag{5.40}$$

Fix $\pi \in U$. For an $\varepsilon > 0$ there exists $\tilde{\varepsilon} > 0$ such that while $|x| < \|\pi\|$ and $|u| < \tilde{\varepsilon}$ then $|(x+u)^{1+\delta} - x^\delta - (1+\delta)x^\delta u| < \varepsilon|u|$. Thus for any $h \in E$ such that $\|h\| < \tilde{\varepsilon}$, we have

$$\|\phi_1(\pi + h) - \phi_1(\pi) - (1+\delta)\pi^\delta h\| < \varepsilon\|h\|. \tag{5.41}$$

We conclude that ϕ_1 differentiable on U with $d\phi_1(\pi)h = (1+\delta)\pi^\delta h$, for any $\pi \in U$, $h \in E$. This differentiate is additionally continuous, which makes l continuously differentiable as well.

Thus, considering the additional constraint $\int_\Theta \pi(\theta)d\theta = 1$, this problem can be treated applying the Lagrange multipliers theorem (see e.g. (Zeidler, 2012)) to state that there exist $\lambda_0, \ldots, \lambda_p \in \mathbb{R}^{p+1}$ such that

$$dl(\pi^*)h - \lambda \int_\Theta h(\theta)d\theta - \sum_{i=1}^p \lambda_i \int_\Theta h(\theta)g_i(\theta)d\theta = 0 \tag{5.42}$$

for any $h \in E$. Eventually, as $dl(\pi)h = C_\delta(1+\delta)\int_\Theta \pi(\theta)^\delta|\mathcal{I}(\theta)|^{-\delta/2}h(\theta)d\theta$, we get

$$\pi^*(\theta) = J(\theta)\left(\lambda_0 + \sum_{i=1}^p \lambda_i g_i(\theta)\right)^{1/\delta}. \tag{5.43}$$

Moreover, as l is strictly concave and the constraints are linear, the second order condition for the Lagrangian states the reciprocal aspect of this result: calling C the space of functions satisfying the constraints, if $\pi^* \in U \cap C$ satisfies the last equation above, it maximizes l.

To conclude, the elements in $\tilde{\mathcal{P}}$ are associated to densities which belong to E because Θ is compact. They are non necessarily everywhere positive, but only supposed to be non-negative. This set of bounded a.e. continuous and non-negative functions satisfying the constraints is actually the closure of $U \cap C$ over which l is continuous. Therefore, calling $\tilde{\mathcal{P}}^*$ the set of positive priors in $\tilde{\mathcal{P}}$, we have $\arg\max_{M(\pi)\in\tilde{\mathcal{P}}} l(\pi) = \arg\max_{M(\pi)\in\tilde{\mathcal{P}}^*} l(\pi)$ which is π^*.

5.6.3 Proof of theorem 5.3

Assuming that it exists, denote by π^* the density of the D_δ-reference prior Π^* over $\overline{\mathcal{P}}$, denote as well

$$c = \int_\Theta \pi^*(\theta) g(\theta) d\theta, \quad Z_i = \int_{\Theta_i} \pi^*(\theta) d\theta \tag{5.44}$$

for any $i \in \mathbb{N}$, considering an openly increasing sequence $(\Theta_i)_{i\in\mathbb{N}}$ of compact sets that cover Θ over which $1 > \Pi^*(\Theta_i) > 0$ for any i such that $\Theta_i \neq \Theta$ (we show later that they exist). In particular, Π^* must be the D_δ-reference prior over the set $\mathcal{P}^c = \{\Pi \in \mathscr{M}_{\mathcal{C}}^{\gamma}, \int_\Theta g d\Pi = c\}$.

Let $i \in \mathbb{N}$, if π_i is a density on Θ_i such that

$$\int_{\Theta_i} \pi_i g + \frac{1}{Z_i} \int_{\Theta\setminus\Theta_i} \pi^* g = \frac{c}{Z_i}, \tag{5.45}$$

then the prior whose density π on Θ defined by $\pi = Z_i \pi_i + \pi^* \mathbb{1}_{\Theta\setminus\Theta_i}$ belongs to $\mathcal{P}^c$. Therefore, denoting π_i^*, the renormalized restriction of π^* to Θ_i, $l(\pi_i^*)$ must be larger than $l(\pi_i)$ on Θ_i by definition of the reference prior.

Thus, π_i^* is a maximal argument of l under the constraints

$$\int_{\Theta_i} \pi_i = 1, \qquad \int_{\Theta_i} \pi_i g = \frac{c}{Z_i} - \frac{1}{Z_i} \int_{\Theta\setminus\Theta_i} \pi^* g. \tag{5.46}$$

Given the result of proposition 5.2, π_i^* takes the form of:

$$\pi_i^* = J \cdot (\lambda_i^{(1)} + \lambda_i^{(2)} g)^{1/\delta} \tag{5.47}$$

for some $\lambda_i^{(1)}$ and $\lambda_i^{(2)}$.

Then, denoting as well $\pi_{i+1}^* = J \cdot (\lambda_{i+1}^{(1)} + \lambda_{i+1}^{(2)} g)^{1/\delta}$, and reminding that $\pi_{i+1}^* \propto \pi_i^*$ on Θ_i, we deduce that $\lambda_{i+1}^{(1)} = \lambda_i^{(1)}$ as well as $\lambda_{i+1}^{(2)} = \lambda_i^{(2)}$. Eventually, $\pi^* \propto J \cdot (\lambda_1 + \lambda_2 g)^{1/\delta}$. Thus, in the neighborhood of b, $\pi^*(\theta) \underset{\theta\to b}{\sim} K\lambda_1^{1/\delta}$ for some $K \neq 0$ if λ_1 is non-null, which is discordant with the satisfaction of the constraint $\int_\Theta \pi^* g < \infty$ in the case where $\int_\Theta Jg = \infty$ or with the constraint $\int_\Theta \pi^* = 1$ otherwise.

Finally, π^* is proportional to $Jg^{1/\delta}$ and the value of c must be fixed by

$$c = \left(\int_\Theta J \cdot g^{1+1/\delta}\right) \cdot \left(\int_\Theta J \cdot g^{1/\delta}\right)^{-1}. \tag{5.48}$$

To finish the proof, we still have to show that Π^* is not null on any of the Θ_i, or on any of the $\Theta \setminus \Theta_i$. First, there must exist i_0 such that $\Pi^*(\Theta_{i_0}) > 0$, so that $\Pi^*(\Theta_i) > 0$ for any $i \geq i_0$ and the sequence $(\Theta_i)_{i\geq i_0}$ can be considered instead of the initial one. Second, for any i, if $\Theta \setminus \Theta_i$ is non-empty then it has a non-empty interior, as a consequence of the definition of openly increasing sequences of compact sets. Thus, as Π^* is assumed to be positive, $\Pi^*(\Theta \setminus \Theta_i)$ is non-null. Hence the result.

5.6.4 Proof of theorem 5.4

Consider an openly increasing sequence of compact sets $(\Theta_i)_{i\in\mathbb{N}}$ that covers Θ. Let $i \in \mathbb{N}$, for Π to be a D_δ-reference prior over $\mathcal{P}_i$, its normalized density π_i must maximize the function l under the constraints $\int_{\Theta_i} \pi_i = 1$ and $\int_{\Theta_i} \pi_i g = c_i$. Thus, according to proposition 5.2, π_i can be written as:

$$\pi_i(\theta) \propto J(\theta)(\lambda_i^{(1)} + \lambda_i^{(2)} g(\theta))^{1/\delta}, \tag{5.49}$$

for some $\lambda_i^{(1)}$ and $\lambda_i^{(2)}$. Considering, if required, a subsequence of $(\pi_i)_{i\in\mathbb{N}}$, we can assume that $\lambda_i^{(1)}$ and $\lambda_i^{(2)}$ have constant signs. Also, by convexity we write:

$$\begin{aligned} c_i^\delta &\geq \int_{\Theta_i} J(\theta)(\lambda_i^{(1)} + \lambda_i^{(2)} g(\theta))g(\theta)d\theta \left(\int_{\Theta_i} Jg\right)^{\delta-1} \\ &\geq \lambda_i^{(1)} \left(\int_{\Theta_i} Jg\right)^{\delta} + \lambda_i^{(2)} \frac{\int_{\Theta_i} Jg^2}{\left(\int_{\Theta_i} Jg\right)^{1-\delta}}, \end{aligned} \tag{5.50}$$

and

$$\begin{aligned} 1 &\geq \int_{\Theta_i} J(\theta)(\lambda_i^{(1)} + \lambda_i^{(2)} g(\theta))d\theta \left(\int_{\Theta_i} J\right)^{\delta-1} \\ &\geq \lambda_i^{(1)} \left(\int_{\Theta_i} J\right)^{\delta} + \lambda_i^{(2)} \frac{\int_{\Theta_i} Jg}{\left(\int_{\Theta_i} J\right)^{1-\delta}}. \end{aligned} \tag{5.51}$$

We want to identify the possible subsequential limits of $\lambda_i^{(1)}$. We assume that one exists in $(0,\infty]$, we can consider if required a subsequence to assume that it is the actual the limit of $\lambda_i^{(1)}$. This way, eq. (5.51) does not allow $\lambda_i^{(2)}$ to be non-negative. Thus, $\lambda_i^{(2)}$ is negative and by convexity,

$$c_i^\delta \geq \lambda_i^{(1)} \left(\int_{\Theta_i} Jg\right)^{\delta} + \lambda_i^{(2)} \left(\int_{\Theta_i} Jg^{1+1/\delta}\right)^{\delta}. \tag{5.52}$$

Given that $(c_i)_i$ is bounded, and that $\int_{\Theta_i} Jg \underset{i\to\infty}{\longrightarrow} \infty$ while $\int_{\Theta_i} Jg^{1+1/\delta}$ has a finite limit, we deduce that $\lambda_i^{(2)}$ diverges to $-\infty$. However, to ensure the expression of π_i to be well defined, we must have $\lambda_i^{(1)} + \lambda_i^{(2)} g(\theta) \geq 0$ for any i and any θ. As g is non-null, it necessary comes that $(\lambda_i^{(2)}/\lambda_i^{(1)})_i$ is a bounded sequence. Therefore, $\lambda_i^{(2)} \underset{i\to\infty}{=} o\left(\lambda_i^{(1)} \left(\int_{\Theta_i} Jg\right)^{\delta}\right)$ and the right hand side in eq. (5.52) diverges to ∞, which is a contradiction with $(c_i)_i$ being bounded.

As a consequence, any subsequential limit of $(\lambda_i^{(1)})_i$ must belong to $[-\infty, 0]$. Without changing the notations, we can now assume that $(\lambda_i^{(1)})_i$ has a limit, which we suppose to be strictly negative firstly, also, $(\lambda_i^{(1)})_i$ can then be assumed to be always negative. Thus, still to ensure that $\lambda_i^{(1)} + \lambda_i^{(2)} g(\theta) \geq 0$ for any i, θ, and reminding that $g(\theta) \underset{\theta\to b}{\longrightarrow} 0$ we deduce $\frac{\lambda_i^{(2)}}{|\lambda_i^{(1)}|} \underset{i\to\infty}{\longrightarrow} \infty$. Then, the sequence of functions $\left(\pi_i/\lambda_i^{(2)}\right)_i$ converges pointwisely to π^*, defined by $\pi^*(\theta) \propto J(\theta)g(\theta)^{1/\delta}$ with $|\frac{1}{\lambda_i^{(2)}}\pi_i(\theta)| \leq C + g(\theta)$ for some C, with g being bounded on all compact sets. According to Bioche and Druilhet, 2016, Proposition 2.15, what precedes implies that the sequence of priors $(M(\pi_i))_i$ converges Q-vaguely to $M(\pi^*)$.

Eventually, if $(\lambda_i^{(1)})_i$ admits 0 as a subsequential limit, we can assume that it is its actual limit. Consequently, the sequence of functions $(\pi_i/\lambda_i^{(2)})_i$ converges pointwisely to the density π^* while being bounded by g. Therefore, the sequence of priors $(M(\pi_i))_i$ converges Q-vaguely to $M(\pi^*)$. In any cases, $M(\pi^*)$ is a quasi-reference prior over $\overline{\mathcal{P}}'$.

5.7 Conclusion

Prior elicitation remains an open topic in Bayesian analysis, with no solution satisfying all criteria (objectivity, computational feasibility, property, ...) simultaneously. In this work, we have focused on the objectivity and exploitability of priors. Specifically, we have provided some solutions for practitioners seeking non-subjective priors, addressing the practical challenges of traditional reference priors, particularly in terms of complexity

and proper aspect. Through the example we presented, we demonstrated that simple criteria can lead to the construction of reference priors that are either straightforward to formulate or proper, depending on the case.

Moreover, our results show that one cannot completely dispense with a consideration of the asymptotic properties of Jeffreys prior. This emphasizes the importance of studying this prior for a complete construction of objective priors. Indeed, although one of our results offers a way to define in a simpler way the reference prior when it has improper decay rates, Jeffreys' rate still have to be elucidated. Our other result requires to derive the whole Jeffreys prior.

The approximation of the latter thus becomes a problem of a prior importance. Of course, one can thoroughly focus its explicit expression, yet it can be cumbersome to derive. In the chapter 6 we propose a numerical solution for the approximation of the Jeffreys prior.

6

Variational approximation of generalized reference priors using neural networks

This chapter develops the work done by Nils Baillie, during his end-of-study internship that I supervised at CEA Saclay. It draws mainly from the submitted paper: Nils Baillie, Antoine Van Biesbroeck, and Clément Gauchy (2025). *Variational inference for approximate objective priors using neural networks*. DOI: 10.48550/arXiv.2502.02364

Abstract The computation of priors yielded by the framework of the reference prior theory remains a difficult task in various statistical models. In this chapter, we develop a flexible algorithm based on variational inference to compute approximations of reference priors from a set of distributions parameterized using neural networks. The algorithm is implemented to approximate the maximizer of generalized mutual information. It allows the introduction of constraints in the optimization problem that ensure the resulting prior is proper or issues proper posteriors. Also, our method incorporates the possibility to approximate the posterior distribution using Markov Chain Monte Carlo methods. Numerical experiments on two statistical models are presented, and we evaluate the performance of the algorithm by comparing both prior and posterior distributions. Our results show the usefulness of this approach to recover the target distribution.

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6.1 Introduction

The reference prior theory has been used in various statistical models, and the reference priors are recognized for their objective nature in several practical studies. However, while the theoretical expression of the

reference priors is known in most cases (see the chapter 3 for a comprehensive review of the reference prior theory), they suffer from their low computational feasibility. Indeed, they generally take the form of a Jeffreys prior or hierarchical Jeffreys priors, whose expression necessitate a heavy numerical cost to be derived, and that becomes even more cumbersome as the dimensionality of the problem increases. The developments done in chapters 4 and 5 make the tackling of this problem essential since the expression of the generalized reference prior suggested lead (with or without constrained) to a prior that is derived from Jeffreys'. It is an issue that jeopardizes their applicability since, in many applications, *a posteriori* estimates are obtained using Markov Chain Monte Carlo (MCMC) methods, which require many prior evaluations, further compounding the computational burden.

In general, when we look for sampling or approximating a probability distribution, several approaches arise and may be used within a Bayesian framework. In this work, we focus on variational inference methods. Variational inference seeks to approximate a complex target distribution p, —e.g. a posterior— by optimizing over a family of simpler parameterized distributions q_λ. The goal then is to find the distribution q_{λ^*} that is the best approximation of p by minimizing an objective function, such as a dissimilarity measure. Variational inference methods have been widely adopted in various contexts, including popular models such as Variational Autoencoders (VAEs; Kingma and Welling, 2019), which are a class of generative models where one wants to learn the underlying distribution of data samples. We can also mention normalizing flows (see e.g. Papamakarios et al., 2021; Kobyzev, Prince, and Brubaker, 2021), which consider diffeomorphism transformations to recover the density of the approximated distribution from the simpler one taken as input.

When it resorts to approximate reference priors, it is possible to leverage that they maximize the mutual information, instead of directly maximizing a divergence between a target and an output. Indeed, the mutual information does not depend on the target distribution that we want to reach, so that iterative derivations of the theoretical solution are not necessary. In Nalisnick and Smyth, 2017, the authors propose a variational inference procedure to approximate reference priors using a lower bound of the mutual information as an optimization criterion. By building on these foundations, this chapter proposes a novel variational inference algorithm to compute reference priors.

In this work, the reference priors are approximated from a parametric family of probability distributions implicitly defined by the push-forward probability distribution through a nonlinear function that takes the form of a neural network. The algorithm is thought to handle the maximization of mutual information that is defined using f-divergence (as suggested in the chapter 4), instead of the traditional Kullback-Leibler divergence. Additionally, building on the developments conducted in the chapter 5, we extend the framework to allow the integration of constraints on the prior. That last feature permits handling situations where the Jeffreys prior is improper. Improper priors represent a particular challenge in this work because they cannot be sampled from, yet the prior constructed from our methodology is not known explicitly and can only be used for sampling. In comparison with the previous works, we benchmark extensively our algorithm on statistical models of different complexity and nature to ensure its robustness.

In the following, we start by explaining the algorithmic construction of the variational approximation of the reference prior in section 6.2. We define in that section the optimization problem, its adaptation for incorporating constraints, and how the output prior is built and used for posterior sampling. The methodology is afterward implemented in section 6.3 on two different models: the multinomial model and the normal model. A conclusion terminates the chapter in section 6.4.

6.2 Variational approximation of the reference prior (VA-RP)

6.2.1 Reference priors, notations

In this study, we consider classical and regular statistic models defined by a collection of probability distributions $(\mathbb{P}_{Y|\theta})_{\theta\in\Theta}$ on a measure set $\mathcal{Y}$, which admit likelihoods $(\ell(\cdot|\theta))_{\theta\in\Theta}$ w.r.t. a common measure μ on $\mathcal{Y}$. Given a prior Π on Θ, one can construct the Bayesian framework as in the chapter 3 (section 3.4) to define the D_f-mutual information when k data are observed:

$$\mathscr{I}^k_{D_f}(\Pi) = \mathbb{E}_{T\sim\Pi}[D_f(\mathbb{P}_{\mathbf{Y}_k}||\mathbb{P}_{\mathbf{Y}_k|T})]; \tag{6.1}$$

where $\mathbb{P}_{\mathbf{Y}_k|\theta} := \mathbb{P}_{Y|\theta}^{\otimes k}$, $\mathbb{P}_{\mathbf{Y}_k}$ being the marginal distribution and D_f denoting an f-divergence. This definition of the mutual information using f-divergence is an extension of the original definition that uses a Kullback-Leibler divergence instead, we refer to the chapter 4 where this extension is proposed and developed. In this chapter, the regularity of the model will not be questioned and always be assumed sufficient for our mathematical derivations to stand. In particular the Fisher information matrix (denoted by $\mathcal{I}$) exists as well as the Jeffreys prior (whose density is denoted J).

Note that in the expression of the D_f-mutual information above, the involved distributions do not always exist when Π is improper. In this case, the quantity is not always well-defined. That is why in the context of reference priors this quantity is considered taking restrictions of Π on compact subsets of Θ. Reference priors are then defined as asymptotic maximizer of the restricted mutual information as $k \longrightarrow \infty$.

In this work, we aim at "globally" maximizing the D_f-mutual information, i.e. considering its expression written as in eq. (6.1) for priors defined on the whole space Θ. Moreover, our method limits itself to the maximization of the mutual information as expressed in eq. (6.1), i.e. for a fixed value of k. All in all, we focus the solving of an optimization problem that takes the form:

$$\text{find} \quad \Pi^* \in \underset{\Pi \in \mathcal{P}_N}{\arg\max}\, \mathscr{I}_{D_f}^k(\Pi), \tag{6.2}$$

where $\mathcal{P}_N$ is a set of normalized proper continuous priors: $\mathcal{P}_N = \{\Pi \in \mathscr{M}_{\mathcal{C}}^{\nu}, \Pi(\Theta) = 1\}$. In this work, we will always treat cases where $\Theta \subset \mathbb{R}^d$ and ν is the Lebesgue measure. Actually, restricting the research of reference priors to the optimization problem written above is not very limiting. Indeed, the set $\mathcal{P}_N$ contains priors that are "very close" to any improper priors: considering the topology on $\mathscr{M} / \simeq$ induced by the Q-vague convergence (see Bioche and Druilhet, 2016), $\mathcal{P}_N$ is dense in $\mathscr{M}_{\mathcal{C}}^{\nu}$. Also, we expect that if k is large enough, the solution of the optimization problem should get closer to the reference prior. That intuition has already been developed by Berger, Bernardo, and Sun (2009) or Le, 2014 who proved it in the one dimensional case.

Furthermore, as mentioned in the introduction, this work aims also at approximating "properly constrained reference priors", i.e. priors that are maximizer of the mutual information under some constraints. The idea is to leverage the development conducted in the chapter 5 to make reference priors being proper. We process by addressing the resolution of the following optimization problem:

$$\text{find} \quad \tilde{\Pi}^* \in \underset{\substack{\Pi \in \mathcal{P}_N \\ \text{s.t. } C(\Pi) < \infty}}{\arg\max}\, \mathscr{I}_{D_f}^k(\Pi), \tag{6.3}$$

where $C(\Pi)$ defines a constraint of the form $\int_\Theta a(\theta) d\Pi(\theta)$, a being a positive function. We remind that, when the mutual information in the above optimization problem is defined from a δ-divergence, and when a verifies

$$\int_\Theta J(\theta) a(\theta)^{1/\delta} d\theta < \infty \quad \text{and} \quad \int_\Theta J(\theta) a(\theta)^{1+1/\delta} d\theta < \infty, \tag{6.4}$$

the developments done in the chapter 5 state that the constrained solution $\tilde{\Pi}^*$ admits a density $\tilde{\pi}^*$ that asymptotically takes the form:

$$\tilde{\pi}^*(\theta) \propto J(\theta) a(\theta)^{1/\delta}, \tag{6.5}$$

which is proper.

6.2.2 Implicit expression of the prior using neural networks

In this work, the goal is to solve the optimization problems expressed in eqs. (6.2) and (6.3) by variational inference. The idea is to restrict the study to a parametric set of priors $\{\Pi_\lambda, \lambda \in \Lambda\}$, $\Lambda \subset \mathbb{R}^L$, reducing the optimization problem to a finite dimensional one: find $\arg\max_{\lambda \in \Lambda} \mathscr{I}_{D_f}^k(\Pi_\lambda)$. The construction of this parametric set comes by defining our priors implicitly, from a simpler random variable ε and a non-linear function g:

$$\theta \sim \Pi_\lambda \iff \theta = g(\lambda, \varepsilon) \quad \text{and} \quad \varepsilon \sim \mathbb{P}_\varepsilon. \tag{6.6}$$

The function g is a measurable function parameterized by some $\lambda \in \Lambda \subset \mathbb{R}^L$. Typically, g is a neural network with λ corresponding to its weights and biases, and g is assumed to be differentiable w.r.t. λ. The variable ε can be seen as a latent variable. It has an easy-to-sample distribution $\mathbb{P}_\varepsilon$ with a simple density function. In practice, we use the centered multivariate Gaussian $\mathcal{N}(0, I_{p\times p})$. If one chooses a dense and heavy parameterized function g, then the family $\mathcal{P}_\Lambda = \{\Pi_\lambda, \lambda \in \Lambda\}$ will be expected to be vast, so that many priors could be approached using this method. However, this implicit construction comes with one inconvenient: except in very simple cases, the density of π_λ is not known and cannot be evaluated. Only samples of $\theta \sim \Pi_\lambda$ can be obtained.

We note that in Nalisnick and Smyth, 2017, this implicit sampling method is compared to several other algorithms used to learn reference priors in one-dimensional cases. Among these methods, we can mention an algorithm proposed in Berger, Bernardo, and Sun, 2009 which does not sample from the reference prior but only evaluates it for specific points, or an MCMC-based approach in Lafferty and Wasserman, 2001, which is inspired from the previous one but can sample from the reference prior. According to this comparison, implicit sampling is, in the worst case, competitive with the other methods, but achieves state-of-the-art results in the best case. Hence, computing the variational approximation of the reference prior, which we will refer to as the VA-RP, seems to be a promising technique.

6.2.3 Objective functions and their gradient

The VA-RP is formally formulated as the solution of an optimization problem of this such:

$$\text{find} \quad \Pi_{\lambda^*} \quad \text{where} \quad \lambda^* = \arg\max_{\lambda\in\Lambda} \mathcal{O}^k_{D_f}(\Pi_\lambda), \tag{6.7}$$

where Π_λ refers to the implicit parametrization described in section 6.2.2. The function $\mathcal{O}^k_{D_f}$ is called the objective function, it is maximized using stochastic gradient optimization. It is intuitive to fix $\mathcal{O}^k_{D_f}$ to equal $\mathscr{I}^k_{D_f}$, in order to maximize the mutual information of interest. Yet, in this section, we suggest alternative objective functions that can be considered to compute the VA-RP. Our method is adaptable to any objective function $\mathcal{O}^k_{D_f}$ admitting a gradient w.r.t. λ that takes the form of the following definition.

Definition 6.1. We recall that $\mathcal{P}_\Lambda$ denotes the set of priors defined by the implicit parametrization described in section 6.2.2. A mapping $\Pi_\lambda \in \mathcal{P}_\Lambda \mapsto \mathcal{O}^k_{D_f}(\Pi_\lambda)$ is said to be an acceptable objective function if it admits a gradient w.r.t. λ that takes the form

$$\nabla_\lambda \mathcal{O}^k_{D_f}(\Pi_\lambda) = \mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}\left[\sum_{j=1}^{d} \partial_{\theta_j} \tilde{\mathcal{O}}^k_{D_f}(g(\lambda,\varepsilon))\nabla_\lambda g_j(\lambda,\varepsilon)\right] \tag{6.8}$$

where $\tilde{\mathcal{O}}^k_{D_f} : \Theta \longrightarrow \mathbb{R}$ is a function that is independent of λ.

Such objective functions allow for flexible implementation, as it permits the separation of sampling and differentiation operations:

- The gradient of $\tilde{\mathcal{O}}^k_{D_f}$ mostly relies on random sampling and depends only on the likelihood ℓ_k and the function f.
- The gradient of g is computed independently. In practice, it is possible to leverage usual automatic differentiation techniques for the neural network. An example is PyTorch's automatic differentiation feature "autograd".

This separation is advantageous as automatic differentiation tools —such as autograd— are well-suited to differentiating complex networks but struggle with functions incorporating randomness. This way, the optimization problem can be addressed using stochastic gradient optimization, approximating at each step the gradient in eq. (6.8) via Monte Carlo estimates. In our experiments, the implementation of the algorithm is done using Adam optimizer.

Several acceptable objective functions exist, such as the D_f-mutual information $\mathscr{I}^k_{D_f}$, as stated in the proposition below.

Proposition 6.1. *The D_f-mutual information $\mathscr{I}^k_{D_f}$ is an acceptable objective function. Its gradient equals*

$$\begin{aligned}\nabla_\lambda \mathscr{I}^k_{D_f}(\Pi_\lambda) =& \mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}\left[\sum_{j=1}^d \partial_{\theta_j}\tilde{I}(g(\lambda,\varepsilon))\nabla_\lambda g_j(\lambda,\varepsilon)\right] \\ &+ \mathbb{E}_{\theta\sim\Pi_\lambda}\left[\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[\frac{1}{\ell_k(\mathbf{Y}_k|\theta)}\nabla_\lambda p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right]\right],\end{aligned} \tag{6.9}$$

where:

$$\partial_{\theta_j}\tilde{I}(\theta) = \mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[\partial_{\theta_j}\log\ell_k(\mathbf{Y}_k|\theta)F\left(\frac{p_\lambda(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right], \tag{6.10}$$

with $F(x) = f(x) - xf'(x)$ and $p_{\lambda,\mathbf{Y}_k}$ denoting the marginal density when the prior is Π_λ.

Proof. We remind that $\mathscr{I}^k_{D_f}(\Pi_\lambda) = \mathbb{E}_{\theta\sim\Pi_\lambda}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}} f\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)$. We compute its gradient w.r.t. λ as follows:

$$\nabla_\lambda \mathscr{I}^k_{D_f}(\Pi_\lambda) = \nabla_\lambda\left[\mathbb{E}_{\theta\sim\Pi_\lambda}\tilde{I}(\theta)\right] + \mathbb{E}_{\theta\sim\Pi_\lambda}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[\frac{1}{\ell_k(\mathbf{Y}_k|\theta)}\nabla_\lambda p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)f'\left(\frac{p_{\mathbf{Y}_k,\lambda}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right], \tag{6.11}$$

with $\tilde{I}(\theta) = \mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[f\left(p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)/\ell_k(\mathbf{Y}_k|\theta)\right)\right]$ so that for any j its derivative w.r.t. θ_j is

$$\partial_{\theta_j}\tilde{I}(\theta) = \int_{\mathcal{Y}^k}\frac{-p_{\lambda,\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right)\partial_{\theta_j}\ell_k(\mathbf{y}|\theta)d\mu^{\otimes k}(\mathbf{y}) + \int_{\mathcal{Y}^k} f\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})}{\ell_k(\mathbf{y}|\theta)}\right)\partial_{\theta_j}\ell_k(\mathbf{y}|\theta)d\mu^{\otimes k}(\mathbf{y}); \tag{6.12}$$

that being equal to the expression in eq. (6.10). Using the trick $\nabla_\lambda\mathbb{E}_{\theta\sim\Pi_\lambda}\tilde{I}(\theta) = \mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}\nabla_\lambda\tilde{I}(g(\lambda,\varepsilon))$, we recovered the expression of the gradient stated in the proposition.
Now we have to show that the second term of the gradient can be written in the same way as the first one to ensure the gradient is acceptable. We call $\mathcal{G}$ this second term:

$$\mathcal{G} = \mathbb{E}_\varepsilon\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|g(\lambda,\varepsilon)}}\frac{1}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon))}\nabla_\lambda p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon))}\right), \tag{6.13}$$

with $\mathbb{E}_\varepsilon$ being a shortcut for $\mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}$. Remarking that $\nabla_\lambda p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k) = \mathbb{E}_{\varepsilon_2}\sum_{j=1}^d\partial_{\theta_j}\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_2))\nabla_\lambda g_j(\lambda,\varepsilon_2)$, we develop $\mathcal{G}$ as:

$$\begin{aligned}\mathcal{G} =& \mathbb{E}_{\varepsilon_1}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\cdot|g(\lambda,\varepsilon_1)}}\mathbb{E}_{\varepsilon_2}\sum_{j=1}^d\frac{1}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y_k})}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}\right)\partial_{\theta_j}\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_2))\nabla_\lambda g_j(\lambda,\varepsilon_2) \\ =& \mathbb{E}_{\varepsilon_2}\sum_{j=1}^d\nabla_\lambda g_j(\lambda,\varepsilon_2)\mathbb{E}_{\varepsilon_1}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\cdot|g(\lambda,\varepsilon_1)}}\frac{1}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}\right)\partial_{\theta_j}\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_2)).\end{aligned} \tag{6.14}$$

The above can be written as $\mathcal{G} = \mathbb{E}_{\varepsilon_2}\sum_{j=1}^d\partial_{\theta_j}\tilde{\mathcal{G}}(g(\lambda,\varepsilon_2))\nabla_\lambda g_j(\lambda,\varepsilon_2)$, with $\tilde{\mathcal{G}}$ being defined as

$$\tilde{\mathcal{G}} : \theta \mapsto \tilde{\mathcal{G}}(\theta) = \mathbb{E}_{\varepsilon_1}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|g(\lambda,\varepsilon_1)}}\left[\frac{1}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}f'\left(\frac{p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k)}{\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_1))}\right)\ell_k(\mathbf{Y}_k|\theta)\right]. \tag{6.15}$$

Eventually, we obtain that $\mathscr{I}^k_{D_f}$ is acceptable with $\tilde{\mathscr{I}}^k_{D_f} = \tilde{I} + \tilde{\mathcal{G}}$. □

In their work, Nalisnick and Smyth (2017) propose an alternative objective function to optimize, that we denote $\mathcal{B}^k_{D_f}$ and call the lower-bound objective function. This function corresponds to a lower bound

of the mutual information. It is derived from an upper bound on the marginal distribution and relies on maximizing the likelihood. Their approach is only presented for $f = -\log$, we generalize the lower bound for any decreasing function f:

$$\mathcal{B}^k_{D_f}(\Pi_\lambda) = \mathbb{E}_{\theta\sim\Pi_\lambda}\mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[f\left(\frac{\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}})}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right] \tag{6.16}$$

where $\hat{\theta}^{\mathrm{MLE}}$ refers to the maximum likelihood estimator (MLE), i.e. $\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}}) = \sup_{\theta\in\Theta}\ell_k(\mathbf{Y}_k|\theta)$. It only depends on the likelihood and not on λ, making the gradient computation simpler:

$$\nabla_\lambda\mathcal{B}^k_{D_f}(\Pi_\lambda) = \mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}\left[\sum_{j=1}^d \partial_{\theta_j}\tilde{\mathcal{B}}^k_{D_f}(g(\lambda,\varepsilon))\nabla_\lambda g_j(\lambda,\varepsilon)\right], \tag{6.17}$$

where:

$$\partial_{\theta_j}\tilde{\mathcal{B}}^k_{D_f}(\theta) = \mathbb{E}_{\mathbf{Y}_k\sim\mathbb{P}_{\mathbf{Y}_k|\theta}}\left[\partial_{\theta_j}\log\ell_k(\mathbf{Y}_k|\theta)F\left(\frac{\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}})}{\ell_k(\mathbf{Y}_k|\theta)}\right)\right]. \tag{6.18}$$

This gradient is such that the following results stands.

Proposition 6.2. *The lower-bound objective function $\mathcal{B}^k_{D_f}$ is an acceptable objective function. Its gradient is expressed in eq. (6.17).*

We emphasize that, given $p^\lambda_{\mathbf{Y}_k}(\mathbf{Y}_k) \leq \ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}})$ for all λ, we have $\mathcal{B}^k_{D_f}(\Pi_\lambda) \leq \mathscr{I}^k_{D_f}(\Pi_\lambda)$ while f is decreasing. We recall that, in this work, we aim at using δ-divergences (with $\delta \in (0,1)$), i.e. $f = f_\delta$ with $f_\delta(x) = \frac{x^\delta - \delta x - (1-\delta)}{\delta(\delta-1)}$. Unfortunately, f_δ is not a decreasing function, yet we can circumvent this by replacing it with $\hat{f}_\delta$ defined hereafter, because $D_{f_\delta} = D_{\hat{f}_\delta}$:

$$\hat{f}_\delta(x) = \frac{x^\delta - 1}{\delta(\delta-1)} = f_\delta(x) + \frac{1}{\delta-1}(x-1). \tag{6.19}$$

The use of this function results in a more stable computation overall. Moreover, one argument for the use of δ-divergences rather that the KL-divergence, is that we have a universal and explicit upper bound on the mutual information:

$$\mathscr{I}^k_{D_{f_\delta}}(\Pi) = \mathscr{I}^k_{D_{\hat{f}_\delta}}(\Pi) \leq \hat{f}_\delta(0) = \frac{1}{\delta(1-\delta)}. \tag{6.20}$$

This bound can be an indicator on how well the mutual information is optimized, although there is no guarantee that it can be attained in general.

Approximation of the gradients

The gradients of the objective functions proposed in this section can be approximated via successive Monte-Carlo, sampling from $\mathbb{P}_\varepsilon$ and from the conditional distributions $\mathbb{P}_{Y|\theta}$. Concerning the marginal density $p_{\lambda,\mathbf{Y}_k}$ and the maximum likelihood $\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}})$, they can be approximated using:

$$\begin{aligned}
&p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k) = \mathbb{E}_{\varepsilon\sim\mathbb{P}_\varepsilon}\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon)) \approx \frac{1}{M}\sum_{m=1}^M \ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_m)) \quad \text{where} \quad \varepsilon_1,\dots,\varepsilon_M \sim \mathbb{P}_\varepsilon;\\
&\nabla_\lambda p_{\lambda,\mathbf{Y}_k}(\mathbf{Y}_k) \approx \sum_{m=1}^M\sum_{j=1}^d \partial_{\theta_j}\ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_m))\nabla_\lambda g_j(\lambda,\varepsilon_m) \quad \text{where} \quad \varepsilon_1,\dots,\varepsilon_M \sim \mathbb{P}_\varepsilon;\\
&\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}}) \approx \max_{m\in\{1,\dots,M\}} \ell_k(\mathbf{Y}_k|g(\lambda,\varepsilon_m)) \quad \text{where} \quad \varepsilon_1,\dots,\varepsilon_M \sim \mathbb{P}_\varepsilon.
\end{aligned} \tag{6.21}$$

6.2.4 Adaptation for the constrained VA-RP

In this section, we suggest an adaptation of the variational optimization problem to incorporate some linear constraints on the prior. As mentioned in section 6.2.1 and developed in chapter 5, there exist specific constraints that would make the theoretical solution proper. This is also a way to incorporate expert knowledge to some extent. We consider R constraints of the form:

$$\forall r \in \{1, \dots, R\},\ \mathcal{C}_r(\Pi_\lambda) = \mathbb{E}_{\theta \sim \Pi_\lambda}[a_r(\theta)] - b_r, \tag{6.22}$$

with $a_r : \Theta \mapsto \mathbb{R}^+$ integrable and linearly independent functions, and $b_r \in \mathbb{R}$. We then adapt the optimization problem in eq. (6.7) to propose the following constrained optimization problem:

$$\begin{aligned} \text{find} \quad \Pi^C_{\lambda^*} \quad \text{where} \quad & \lambda^* \in \arg\max_{\lambda \in \Lambda} \mathcal{O}^k_{D_f}(\Pi_\lambda) \\ & \text{subject to} \quad \forall r \in \{1, \dots, R\},\ \mathcal{C}_r(\Pi_\lambda) = 0. \end{aligned} \tag{6.23}$$

The prior $\Pi^C_{\lambda^*}$ is called the constrained VA-RP. When D_f corresponds to a δ-divergence, the maximizer of the mutual information is expected to approximate the explicit solution whose density π^C satisfies

$$\pi^C(\theta) \propto J(\theta) \left(1 + \sum_{r=1}^{R} \eta_r a_r(\theta)\right)^{1/\delta}, \tag{6.24}$$

where $\eta_1, \dots, \eta_R \in \mathbb{R}$ are constants determined by the constraints.

According to the results stated in chapter 5, when the constraint is defined from only a function a that satisfies eq. (6.4), the reference prior Π^C under the constraint $\mathcal{C}(\Pi^C) < \infty$ has a density of the form $\pi^C \propto J \cdot a^{1/\delta}$ and is proper. We propose the following method to approximate the VA-RP under such constraint:

- Denote by $\mathbf{J}$ the Jeffreys prior. Compute the VA-RP $\Pi_{\hat{\lambda}} \approx \mathbf{J}$, in the same manner as for the unconstrained case.
- Estimate the constants $\mathcal{K}$ and c using Monte Carlo samples from the VA-RP, as:

$$\begin{aligned} \mathcal{K}_{\hat{\lambda}} &= \int_\Theta (\theta) a(\theta)^{1/\delta} d\Pi_{\hat{\lambda}}(\theta) \approx \int_\Theta J(\theta) a(\theta)^{1/\delta} d\theta = \mathcal{K}, \\ c_{\hat{\lambda}} &= \int_\Theta a(\theta)^{1+(1/\delta)} d\Pi_{\hat{\lambda}}(\theta) \approx \int_\Theta J(\theta) a(\theta)^{1+(1/\delta)} d\theta = c. \end{aligned} \tag{6.25}$$

- Since we have the equality for the constrained reference prior Π^C:

$$\mathbb{E}_{\theta \sim \Pi^C}[a(\theta)] = \frac{1}{\mathcal{K}} \int_\Theta J(\theta) a(\theta)^{1+(1/\delta)} d\theta = \frac{c}{\mathcal{K}}, \tag{6.26}$$

  we compute the constrained VA-RP using the constraint : $\mathbb{E}_{\theta \sim \Pi_\lambda}[a(\theta)] = c_{\hat{\lambda}} / \mathcal{K}_{\hat{\lambda}}$ to approximate $\Pi^C_{\lambda^*} \approx \Pi^C$.

The idea is to solve the constrained optimization problem as an unconstrained problem but with a Lagrangian as the objective function. We take the work of Nocedal and Wright (2006) as support.

We denote η the Lagrange multiplier. Instead of using the usual Lagrangian function, Nocedal and Wright (2006) suggest adding two terms in the optimization problem: (i) a term denoted $\tilde{\eta}$, which a vector with positive components that serves as penalization coefficients, and (ii) a vector η', which can be thought of an initial estimate of η. The objective is to find a saddle point (λ^*, η^*) which is a solution of the updated optimization problem:

$$\max_\lambda \left(\min_\eta \mathcal{O}^k_{D_f}(\Pi_\lambda) + \sum_{r=1}^{R} \eta_r \mathcal{C}_r(\Pi_\lambda) + \sum_{r=1}^{R} \frac{1}{2\tilde{\eta}_r} (\eta_r - \eta'_r)^2 \right). \tag{6.27}$$

One can see that the third term serves as a penalization for large deviations from η'. The minimization on η is feasible because it is a convex quadratic, and we get $\eta = \eta' - \tilde{\eta} \cdot \mathcal{C}(\Pi_\lambda)$. Replacing η by its expression leads to the resolution of the problem:

$$\max_\lambda \mathcal{O}^k_{D_f}(\Pi_\lambda) + \sum_{r=1}^R \eta'_r \mathcal{C}_r(\Pi_\lambda) - \sum_{r=1}^R \frac{\tilde{\eta}_r}{2} \mathcal{C}_r(\Pi_\lambda)^2. \tag{6.28}$$

This motivates the definition of the augmented Lagrangian:

$$\mathcal{L}_A(\lambda, \eta, \tilde{\eta}) = \mathcal{O}^k_{D_f}(\Pi_\lambda) + \sum_{r=1}^R \eta_r \mathcal{C}_r(\Pi_\lambda) - \sum_{r=1}^R \frac{\tilde{\eta}_r}{2} \mathcal{C}_r(\Pi_\lambda)^2. \tag{6.29}$$

Its gradient has a form that is acceptable (see definition 6.1) while $\mathcal{O}^k_{D_f}$ is acceptable as well:

$$\begin{aligned} \nabla_\lambda \mathcal{L}_A(\lambda, \eta, \tilde{\eta}) &= \nabla_\lambda \mathcal{O}^k_{D_f}(\Pi_\lambda) + \mathbb{E}_{\varepsilon \sim \mathbb{P}_\varepsilon} \left[\sum_{j=1}^d \left(\sum_{r=1}^R \partial_{\theta_j} a_r(g(\lambda, \varepsilon))(\eta_k - \tilde{\eta}_k \mathcal{C}_k(\Pi_\lambda)) \right) \nabla_\lambda g_j(\lambda, \varepsilon) \right] \\ &= \mathbb{E}_{\varepsilon \sim \mathbb{P}_\varepsilon} \left[\sum_{j=1}^d \left(\partial_{\theta_j} \tilde{\mathcal{O}}^k_{D_f}(g(\lambda, \varepsilon)) + \sum_{r=1}^R \partial_{\theta_j} a_r(g(\lambda, \varepsilon))(\eta_r - \tilde{\eta}_r \mathcal{C}_r(\Pi_\lambda)) \right) \nabla_\lambda g_j(\lambda, \varepsilon) \right]. \end{aligned} \tag{6.30}$$

In practice, the augmented Lagrangian algorithm takes the sequential form:

$$\begin{cases} \lambda^{t+1} = \arg\max_{\lambda \in \Lambda} \mathcal{L}_A(\lambda, \eta^t, \tilde{\eta}) \\ \forall r \in \{1, \dots, R\},\ \eta_r^{t+1} = \eta_r^t - \tilde{\eta}_r \cdot \mathcal{C}_r(\pi_{\lambda^{t+1}}). \end{cases} \tag{6.31}$$

In our implementation, η is updated every 100 epochs. The penalty parameter $\tilde{\eta}$ can be interpreted as the learning rate of η, we use an adaptive scheme inspired by Basir and Senocak, 2023 where we check if the largest constraint value $||\mathcal{C}(\Pi_\lambda)||_\infty$ is higher than a specified threshold TH or not. If $||\mathcal{C}(\Pi_\lambda)||_\infty >$ TH, we multiply $\tilde{\eta}$ by v, otherwise we divide by v. We also impose a maximum value $\tilde{\eta}_{\max}$.

6.2.5 Posterior sampling from the implicit definition

Although our main object of study is the prior distribution, one needs to find the posterior distribution given an observed dataset $\mathbf{y} \in \mathcal{Y}^k$ in order to do the inference on θ. If the prior Π_λ admits a density π_λ, the posterior density is of the form :

$$p_\lambda(\theta|\mathbf{y}) = \frac{\pi_\lambda(\theta)\ell_k(\mathbf{y}|\theta)}{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})}. \tag{6.32}$$

In practice, we cannot evaluate from this density, as the prior density is not known in general. This complicates the approximation of the posterior distribution and the computation of *a posteriori* quantities. To circumvent this problem, we suggest considering the posterior distribution of the latent variable ε, which admits the density:

$$q_\lambda(\varepsilon|\mathbf{y}) = \frac{q_\varepsilon(\varepsilon)\ell_k(\mathbf{y}|g(\lambda, \varepsilon))}{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})}, \tag{6.33}$$

where q_ε is the probability density function of ε. The latter is known, so that the posterior density is known up to a constant. Therefore, it is possible to sample θ *a posteriori* by sampling ε w.r.t. its posterior distribution $\mathbb{P}_{\varepsilon|\mathbf{y}}$:

$$\theta \sim \mathbb{P}_{T|\mathbf{y}} \iff \theta = g(\lambda, \varepsilon) \quad \text{with} \quad \varepsilon \sim \mathbb{P}_{\varepsilon|\mathbf{y}}. \tag{6.34}$$

This statement can be simply verified by deriving, for any bounded measurable function φ on Θ:

$$\mathbb{E}_{\varepsilon \sim \mathbb{P}_{\varepsilon|\mathbf{y}}} \varphi(g(\lambda, \varepsilon)) = \mathbb{E}_{\varepsilon \sim \mathbb{P}_\varepsilon} \left[\varphi(g(\lambda, \varepsilon)) \frac{\ell_k(\mathbf{y}|g(\lambda, \varepsilon))}{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})} \right] = \mathbb{E}_{\theta \sim \Pi_\lambda} \left[\varphi(\theta) \frac{\ell_k(\mathbf{y}|\theta)}{p_{\lambda,\mathbf{Y}_k}(\mathbf{y})} \right] = \mathbb{E}_{\theta \sim \mathbb{P}_{T|\mathbf{y}}} \varphi(\theta) \tag{6.35}$$

Samples of ε distributed w.r.t. its posterior distribution can be obtained using different methods from the expression of the density in eq. (6.33). For instance, variational inference techniques could be implemented. In this work, we choose MCMC sampling, namely an adaptive Metropolis-Hastings sampler.

6.3 Numerical experiments on different models

In this section, we apply our algorithm to different statistical models, and we assess the correct approximation of the Jeffreys prior by the obtained VA-RP.

As for the output of the neural networks, the activation function just before the output is different for each statistical model, the same can be said for the learning rate. In some cases, we apply an affine transformation on the variable θ to avoid divisions by zero during training. In every test case, we consider simple networks for an easier fine-tuning of the hyperparameters and also because the precise computation of the loss function is an important bottleneck. The neural networks weights are initialized randomly w.r.t. a Gaussian distribution, and their biases are initially set at 0.

6.3.1 Multinomial model

Description of the model

The multinomial model is a case study that is simple in the sense that the target distributions —the Jeffreys prior and posterior— have explicit expressions and are part of a usual parametric family of proper distributions. This study allows testing and validating our algorithm on a multidimensional problem.

The multinomial distribution can be interpreted as the generalization of the binomial distribution for higher dimensions. In this case, $\mathcal{Y} = \{y \in \{0, \dots, n\}^d, \sum_{i=1}^d y^i = n\}$ for a fixed $n \in \mathbb{N}^*$, and $\Theta = (0,1)^d$. The data follow, conditionally to θ, a multinomial distribution characterized by the likelihood:

$$\ell(y|\theta) = \frac{n!}{y^1! \dots y^d!} \prod_{i=1}^d \theta_i^{y^i}; \tag{6.36}$$

whose log-gradient w.r.t. θ equals

$$\partial_{\theta_j} \log \ell(y|\theta) = \frac{y^j}{\theta_j}; \tag{6.37}$$

for all $\theta \in \Theta$, $y = (y^1, \dots, y^d) \in \mathcal{Y}$. In this model, an explicit expression of the maximum likelihood estimator can be derived as

$$\ell_k(\mathbf{y}|\hat{\theta}^{\mathrm{MLE}}) = \ell_k\left(\mathbf{y} \,\middle|\, \tfrac{1}{nk}\textstyle\sum_{i=1}^k y_i\right) \quad \text{for all} \quad \mathbf{y} = (y_1, \dots, y_k) \in \mathcal{Y}^k; \tag{6.38}$$

where $\frac{1}{nk}\sum_{i=1}^k y_i$ is the d-dimensional vector whose j-th coordinate is $\frac{1}{nk}\sum_{i=1}^k y_i^j$.

The reference prior (the prior maximizing the mutual information) is the Jeffreys prior which is a Dirichlet distribution for this model: $\mathbf{J} = \mathrm{Dir}_d\left(\frac{1}{2}, ..., \frac{1}{2}\right)$, which is proper. The Jeffreys posterior is a conjugate Dirichlet distribution:

$$\mathbf{J}_{\mathrm{post}}(T|\mathbf{Y}_k) = \mathrm{Dir}_d(\zeta) \quad \text{with} \quad \zeta = (\zeta_j)_{j=1}^d,\ \zeta_j = \frac{1}{2} + \sum_{i=1}^k Y_i^j. \tag{6.39}$$

We recall that the probability density function $p_{\mathrm{Dir}_d(\zeta)}$ of a Dirichlet distribution is the following:

$$p_{\mathrm{Dir}_d(\zeta)}(t) = \frac{\Gamma(\sum_{j=1}^d \zeta_j)}{\prod_{j=1}^d \Gamma(\zeta_j)} \prod_{j=1}^d t_j^{\zeta_j - 1} \quad \text{for all} \quad t \in (0,1)^d. \tag{6.40}$$

In the chapter 3, we mentioned that although the Jeffreys prior is the prior that maximizes the mutual information, Berger and Bernardo (1992b) and Berger, Bernardo, and Sun (2015) argue that other priors for the multinomial model are more suited in terms of non-informativeness as the dimension of θ increases. According to them, an appropriate reference prior should result from a sequential construction, referred to as the r-group reference prior. This prior is derived through hierarchical maximization of the mutual information where the parameter coordinates are grouped into r groups. Their process is explained in the chapter 3 (section 3.3.2). According to the cited authors, while the Jeffreys prior corresponds to the $r = 1$-group reference prior using this construction, the d-group one should be favored for this model. Nevertheless,

our approach consists in approaching the maximizer of the mutual information, hence, the Jeffreys prior remains our target in this regard.

Regarding the neural network that defines the VA-RP in this study, we opt for a simple neural network with one linear layer and a Softmax activation function assuring that all components are positive and sum to 1. Explicitly, we have that:

$$\theta = \text{Softmax}(W^\top \varepsilon + b), \tag{6.41}$$

with $W \in \mathbb{R}^{p \times d}$ the weight matrix and $b \in \mathbb{R}^d$ the bias vector. The density function of θ does not have a closed expression. The following results are obtained with $d = 4$ and $\delta = 0.5$ for the divergence and the lower bound is used as the objective function.

Results

We notice (fig. 6.1) that the D_{f_δ}-mutual information lies between 0 and $1/\delta(1-\delta) = 4$ during the training, which is coherent with the theory, the confidence interval is rather large because it is computed with few samples, but the mean value has an increasing trend. In order to obtain more reliable values for the mutual information, one can use more samples in the Monte Carlo estimates at the cost of heavier computations.

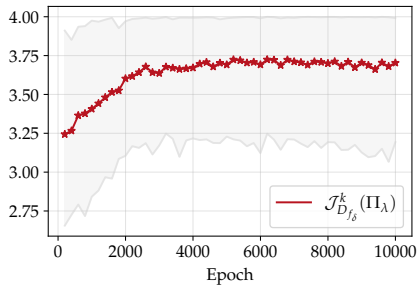

Figure 6.1: Monte Carlo estimation of the D_{f_δ}-mutual information with $\delta = 0.5$ and $k = 200$ of Π_λ through the training of the neural network. The red curve is the mean value and the gray zone is the 95% confidence interval of the Monte-Carlo estimate. The learning rate used in the optimization is 0.0025.

Figures 6.2 and 6.3 propose qualitative comparisons between the obtained VA-RP at the end of the training and the Jeffreys prior. The first comparison takes the form of histograms of samples distributed w.r.t. each of the priors. The second differs from the fact that the samples are drawn w.r.t. the posterior distributions for a given observed dataset of size 10. We notice that although the shape of the fitted prior resembles the one of the Jeffreys prior, one can notice that it tends to put more weight towards the extremities of the interval (fig. 6.2). The posterior distribution however is quite similar to the target Jeffreys posterior on every component (fig. 6.3).

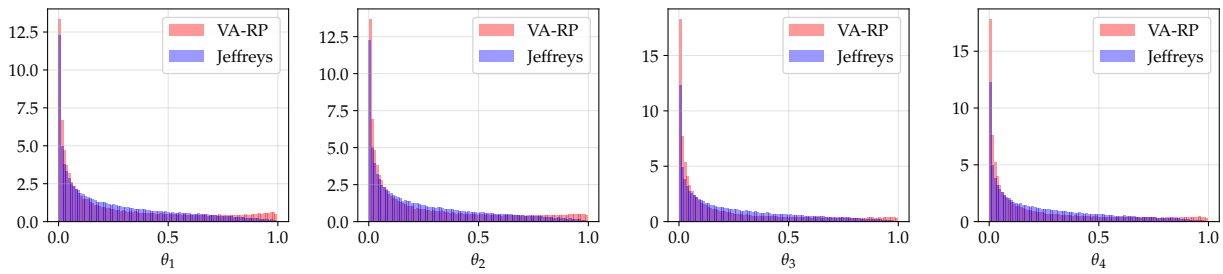

Figure 6.2: Histograms of the VA-RP and the Jeffreys prior for each dimension of θ, each one is obtained from 10^5 samples.

Since the multinomial model is simple and computationally practical, we would like to quantify the effect of the divergence with different δ values on the output of the algorithm. In order to do so, we utilize the

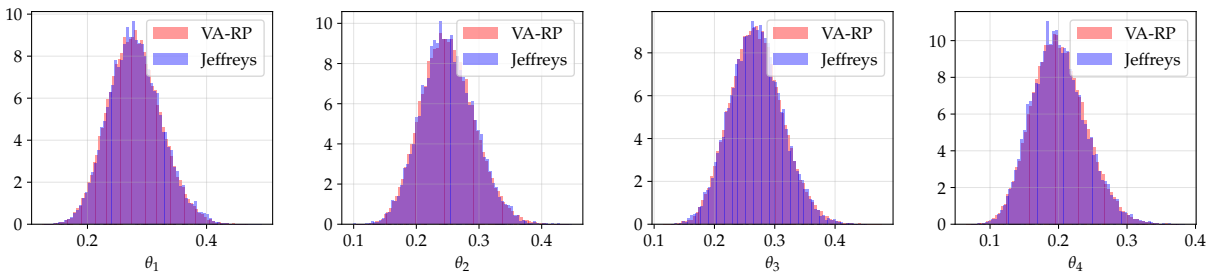

Figure 6.3: Histograms of the posteriors obtained from the VA-RP and the Jeffreys prior for each dimension of θ, each one is obtained from 10^5 samples. The posteriors are derived given a random dataset of size 10, and distributed using $\theta_{\text{true}} = (\frac{1}{4}, \dots, \frac{1}{4})$.

maximum mean discrepancy (MMD) defined as :

$$\text{MMD}^2(\mathcal{H};\, P||Q) = \mathbb{E}_{X,X'\sim P\otimes P}[K(X,X')] + \mathbb{E}_{Y,Y'\sim Q\otimes Q}[K(Y,Y')] - 2\mathbb{E}_{X,Y\sim P\otimes Q}[K(X,Y)]; \tag{6.42}$$

for any distributions P, Q, and where $\mathcal{H}$ denotes a reproducible kernel Hilbert space whose K is a reproducing kernel. The MMD is used for instance in the context of two-sample tests (Gretton et al., 2012), whose purpose is to compare distributions. We use in our computations the Gaussian or RBF kernel :

$$K(\theta, \theta') = \exp(-0.5 \cdot ||\theta - \theta'||_2^2) \tag{6.43}$$

for which the MMD is a metric, this means that the following implication:

$$\text{MMD}(\mathcal{H};\, P||Q) = 0 \implies P = Q \tag{6.44}$$

is verified with the other axioms. In practice, we consider an unbiased estimator of the MMD^2 given by:

$$\widehat{\text{MMD}^2}(P||Q) = \frac{1}{m(m-1)} \sum_{i\neq j} K(x_i, x_j) + \frac{1}{m'(m'-1)} \sum_{i\neq j} K(y_i, y_j) - \frac{2}{mm'} \sum_{i,j} K(x_i, y_j), \tag{6.45}$$

where $(x_1, ..., x_m)$ and $(y_1, ..., y_{m'})$ are sampled from P and Q respectively. In our case, P is the VA-RP and Q is the target Jeffreys prior. Since the MMD can be time-consuming or memory inefficient to compute in practice for very large samples, we consider only the last $2 \cdot 10^4$ entries of our priors and posterior samples. According to Table 6.1, the difference between δ values in terms of the MMD criterion is essentially inconsequential. One remark is that the mutual information tends to be more unstable as δ gets closer to 1. The explanation is that when δ tends to 1, we have the approximation :

$$\hat{f}_\delta(x) \approx \frac{x-1}{\delta(\delta-1)} + \frac{x\log(x)}{\delta}, \tag{6.46}$$

which diverges for all x because of the first term. Hence, we advise the user to avoid δ values that are too close to 1. In the following, we use $\delta = 0.5$ for the divergence.

The last quantitative study conducted for this model validates the result developed in chapter 4 about the asymptotic expansion of the D_{f_δ}-mutual information. In the case $d = 1$, we show in fig. 6.4 that $l^k_{D_{f_\delta}}(\mathbf{J})$ and $\overline{\mathscr{I}}^k_{D_{f_\delta}}(\Pi_\lambda) := \mathscr{I}^k_{D_{f_\delta}}(\Pi_\lambda) + \frac{1}{\delta(1-\delta)}$ are close, where $l^k_{D_{f_\delta}}(\mathbf{J})$ denotes the theoretical asymptotic rate w.r.t. k of $\mathscr{I}^k_{D_{f_\delta}}(\mathbf{J})$. According to the result of chapter 4, the latter admits the following asymptotic expansion:

$$\mathscr{I}^k_{D_{f_\delta}}(\mathbf{J}) \underset{k\to\infty}{\sim} \frac{-1}{\delta(1-\delta)} + l^k_{D_{f_\delta}}(\mathbf{J}), \quad \text{with} \quad l^k_{D_{f_\delta}}(\mathbf{J}) = \frac{-1}{\delta(1-\delta)^{3/2}} \left(\frac{\pi k}{2}\right)^{-\delta/2}. \tag{6.47}$$

This comparison supports two claims: (i) the D_{f_δ} mutual information derived for a fixed k resembles to its asymptotic form w.r.t. k, which is the form that is maximized by the reference prior by definition; (ii) as the VA-RP maximizes the former, it is close to the maximal argument of the latter.

δ	**Prior**	**Posterior**
0.1	7.07×10^{-2}	2.09×10^{-3}
0.25	7.42×10^{-2}	3.39×10^{-3}
0.5	5.26×10^{-2}	1.96×10^{-3}
0.75	7.80×10^{-2}	1.50×10^{-3}
0.9	6.15×10^{-2}	4.84×10^{-4}

Table 6.1: MMD values for different δ-divergences at prior and posterior levels. As a reference on the prior level, when computing the criterion between two independent Dirichlet $\mathrm{Dir}_d(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ distributions (ie the Jeffreys prior) on $2 \cdot 10^4$ samples, we obtain an order of magnitude of 10^{-3}.

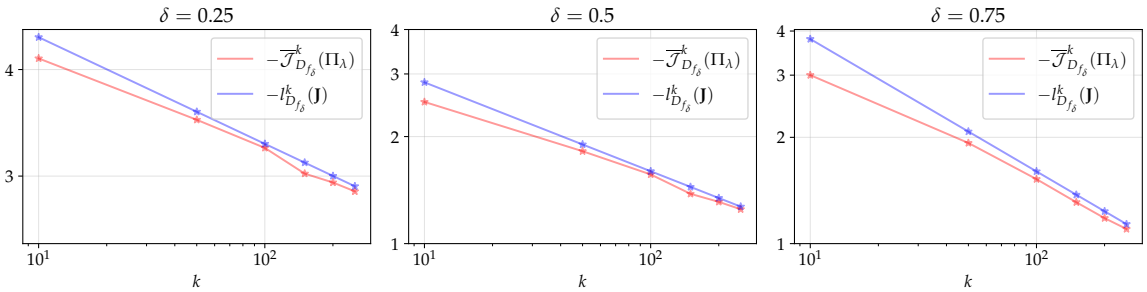

Figure 6.4: Values of the quantities $-\overline{\mathscr{I}}^k_{D_{f_\delta}}(\Pi_\lambda)$ and $-l^k_{D_{f_\delta}}(\mathbf{J})$ when $d = 1$ as a function of k, in log scale, for different values of δ: $\delta = 0.25$ (left), $\delta = 0.5$ (middle), and $\delta = 0.75$ (right).

6.3.2 Normal model

Description of the model and results without constraints

The second case study that is developed in this chapter is the Gaussian model with known mean. Explicitly, $\mathcal{Y} = \mathbb{R}$, $\Theta = (0, \infty)$, and the model is characterized by the likelihood

$$\ell(y|\theta) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{1}{2\theta}(y-\mu)^2\right), \tag{6.48}$$

with $\mu \in \mathbb{R}$ being fixed. The gradient of the log-likelihood equals

$$\partial_{\theta_j} \log \ell_k(\mathbf{y}|\theta) = -\frac{k}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{k}(y_i-\mu)^2 \quad \text{for all} \quad \mathbf{y} = (y_1, \ldots, y_k) \in \mathcal{Y}^k. \tag{6.49}$$

There exists an explicit expression of the MLE as well: $\ell_k(\mathbf{Y}_k|\hat{\theta}^{\mathrm{MLE}}) = \ell_k(\mathbf{Y}_k|\frac{1}{k}\sum_{i=1}^k Y_i)$. In this model, the Jeffreys prior admits a simple analytical density $J(\theta) \propto 1/\theta$, which is improper. It is conjugate with the likelihood: the distribution of the Jeffreys posterior is an inverse-gamma distribution:

$$\mathbf{J}_{\mathrm{post}}(T|\mathbf{y}) = \Gamma^{-1}\left(\frac{k}{2}, \frac{1}{2}\sum_{i=1}^{k}(y_i-\mu)^2\right) \quad \text{for all} \quad \mathbf{y} \in \mathcal{Y}^k, \tag{6.50}$$

which is proper.

It is possible to implement the training of the VA-RP to approximate the improper Jeffreys prior. However, it becomes complicated to sample from the obtained prior, and one cannot directly compare it with the target. Yet, we can circumvent this problem by comparing the posterior distributions. In fig. 6.5, we present the result obtained from training a neural network with one layer and a Softplus activation function. The dimension of ε is set at $p = 10$. We observe that the evolution of the D_{f_δ}-mutual information (fig. 6.5-left) remains stable, and the posterior distribution obtained from the VA-RP is close to the one obtained from Jeffreys (fig. 6.5-right).

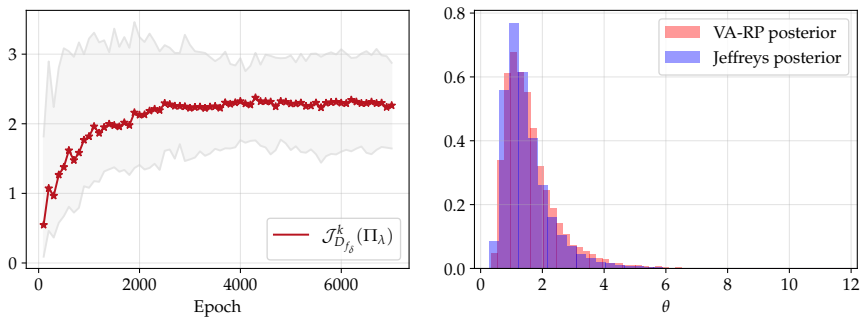

Figure 6.5: Left: Monte-Carlo estimation (mean in red, 95% confidence interval of the estimate in gray) of the D_{f_δ}-mutual information with $\delta = 0.5$ and $k = 200$ of Π_λ through the training of the neural network. Right: histograms of the posteriors obtained from the VA-RP and from the Jeffreys prior. The posteriors are derived given a random dataset of size 10 and distributed using $\theta_{\text{true}} = 1$.

Constraint incorporation

We also propose the introduction of constraints in this model, in order to ensure the VA-RP approximates a proper prior. We define $a(\theta) = \frac{1}{\theta^{e_1}+\theta^{e_2}}$ with $e_1 < 0 < e_2$. The constant $\mathcal{K}$ and c defined in section 6.2.4 are finite:

$$\forall \xi \geq 1, \int_0^\infty \frac{1}{\theta}\left(\frac{1}{\theta^{e_1}+\theta^{e_2}}\right)^\xi d\theta \leq \frac{1}{\xi}\left(\frac{1}{e_2}-\frac{1}{e_1}\right). \tag{6.51}$$

Thus, the constrained reference prior that is the target of the constrained VA-RP is the prior whose density π^C verifies $\pi^C(\theta) \propto J(\theta)a(\theta)^{1/\delta}$ and is proper.

We implement the above taking $\delta = 1/2$, $e_1 = -1$, $e_2 = 1$, so that $\mathcal{K} = 1/2$ and $c = \pi/16$, making the explicit constraint being $\mathcal{C}(\Pi) = \mathbb{E}_{\theta\sim\Pi}[a(\theta)] = \pi/8$. In fig. 6.6-left, we compare the prior distributions, which are both proper. The figure supports the robustness of the algorithm to approximate the reference prior even with added constraints. The posterior distributions are compared as well in fig. 6.6-right, which still reinforce the correctness of the approximation.

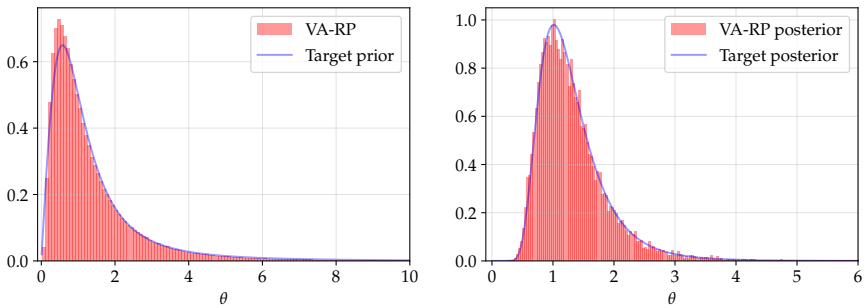

Figure 6.6: Comparisons between the VA-RP and the Jeffreys prior. Left: histogram of the VA-RP (red) and density of the target constrained reference prior π^C (blue). Right: Histogram of the posterior obtained from the VA-RP (red) and density of the posterior obtained from the target constrained reference prior (blue). The posteriors are derived given a random dataset of size 10 and distributed using $\theta_{\text{true}} = 1$.

6.4 Conclusion

In this work, we developed an algorithm to perform variational approximation of reference priors using the generalized definition of mutual information based on f-divergences that was developed in the chapter 4.

Additionally, because the reference priors of interest are often improper and often yield improper posteriors, we adapted the variational definition of the problem to incorporate constraints that ensure the posteriors are proper, following the development conducted in the chapter 5.

Numerical experiments have been carried out on two test cases of different complexities in order to validate our approach. The results demonstrate the usefulness of our approach in estimating both prior and posterior distributions across the considered problems. This study thus provides a flexible and robust solution for approximating the theoretical solutions that were proposed in this part I of the manuscript. We argue that the implementation can be adapted to a wide range of statistical models, including the ones where the target prior is not known explicitly. In the part II, an example of such statistical model will be studied.

Challenges remain...

II

Seismic fragility curve estimation

7

Seismic fragility curves in probabilistic risk assessment studies

Abstract Seismic fragility curves are key quantities of interest in the seismic probabilistic risk assessment framework because they efficiently describe the fragility of a mechanical system of interest under a seismic excitation. They are studied since the 1980s on various types of data and characterization of the seismic motion and of the structure's response. In this chapter, we propose a review of the numerous methods that exist to estimate the curves. We particularly describe the role played by the characteristics of the available data to select a modeling of the fragility curve. This chapter is also the occasion to present some mechanical equipments that will be studied in the following chapters of this manuscript.

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7.1 Introduction

The Seismic Probabilistic Risk Assessment (SPRA) defines a framework and a methodology for the study of seismic structural reliability. It has been introduced since 1968 (Cornell, 1968) to incorporate the seismic risk evaluation in the probabilistic risk assessment studies. The SPRA has been mostly developed and carried out in the 1980s on nuclear facilities (see e.g. Kennedy, Cornell, et al., 1980; Kennedy and Ravindra, 1984). It includes: the determination of the seismic hazard, the analysis of the seismic fragility of structural components, the evaluation of the risks combination and their consequences in the system, as described in the report of the Electric Power Research Institute (EPRI, 2013), which provides guidelines for its implementation. It is since widely used in the nuclear industry (e.g. Ellingwood, 1990; Park, Hofmayer, and Chokshi, 1998; Kennedy, 1999) and its consideration is established among the safety standards adopted by the International Atomic Energy Agency (IAEA, 2020).

Within the SPRA, seismic fragility curves represent a key asset that play a prominent role in the analysis of structural components' fragility step (see EPRI, 2011). They express in probabilistic terms the fragility of structures under seismic excitation. We can mention that they are also a tool of interest of performance-based earthquake engineering (PBEE; Ghobarah, 2001; Noh, Lallemant, and Kiremidjian, 2014), which aims at relating performance objectives to level of damage to the structure. In any case, they are defined as a function of the seismic hazard, which —driven by the magnitude (M), the source-site distance (R), and other earthquake parameters— is reduced to a scalar value derived from the seismic signal: the intensity measure (IM), under the so-called "sufficiency assumption" (Cornell, 2004; Luco and Cornell, 2007). In practice, a fragility curve, denoted P_f, therefore expresses the probability of failure of a mechanical structure as a function of an IM value of interest such as peak ground acceleration (PGA) or pseudo-spectral acceleration (PSA), among others:

$$P_f(a) = \mathbb{P}(\text{“failure”}|\text{IM} = a). \tag{7.1}$$

Within the SPRA framework, the fragility curve expression is expected to be combined with the seismic hazard frequency to compute the damage frequency of the component C_f:

$$C_f = \int_0^\infty P_f(a) dH(a). \tag{7.2}$$

If H is the mean annual distribution of the IM, C_f represents the annual damage frequency of the component. It should be noted that the sufficiency assumption was introduced to reduce estimation costs since it assumes that the fragility curve of a given structure is identical regardless of the seismic scenario. Actually, this assumption embeds some uncertainty in the problem since, as shown in Radu and Grigoriu, 2018; 2021, different seismic scenarios can lead to identical distributions of some IMs, despite having significantly different frequency contents. The uncertainty rooted in the fragility curve by this assertion can be classified as an *aleatoric* (irreducible) kind of uncertainty (we refer to the chapter 1 for the definition of the uncertainty quantification framework).

In this second part of the manuscript, we focus on the estimation of seismic fragility curves. Thus, we do not address the determination of the hazard distribution H, which is the responsibility of seismologists, nor the derivation of the damage frequency of the component C_f. Nevertheless, estimating the fragility curve itself remains a daunting task. It is commonly done statistically, using different kind of methods depending on the source of data available, their characteristics, and their quantity. This chapter proposes a review of these methods. In the next section, we define the different kind of data involved when estimating seismic fragility curves. In particular, we present a stochastic seismic signal generator, and we present how the failure of mechanical equipment is generally defined. In section 7.3, different models and methods are presented, they are grouped in two categories: the non-parametric ones and the parametric ones. Explicit examples of case studies are then given in section 7.4, from a theoretical one to an experimental one for which really few data are available. Conclusive thoughts are proposed in section 7.5.

7.2 Data: from seismic signals to equipments failures

In the literature, different sources of data are exploited to estimate seismic fragility curves. We can cite, for instance: (i) expert assessments supported by test data (e.g. Kennedy, Cornell, et al., 1980; Kennedy and Ravindra, 1984; Zentner, Gündel, and Bonfils, 2017), (ii) experimental data (e.g. Park, Hofmayer, and Chokshi, 1998), (iii) empirical data from past earthquakes (e.g. Shinozuka, Feng, et al., 2000; Straub and Der Kiureghian, 2008; Lallemant, Kiremidjian, and Burton, 2015; Buratti et al., 2017; Laguerre, Salehi, and Desroches, 2024), and (iv) analytical results obtained from various numerical models using artificial or natural seismic excitations (e.g. Ellingwood, 2001; S.-H. Kim and Shinozuka, 2004; Zentner, 2010; Koutsourelakis, 2010; Mai, Konakli, and Sudret, 2017; Trevlopoulos, Feau, and Zentner, 2019; F. Wang and Feau, 2020; Mandal, S. Ghosh, and Pujari, 2016; Z. Wang, Pedroni, et al., 2018; Z. Wang, Zentner, and Zio, 2018; Zhao, Yu, and Mo, 2020; Katayama et al., 2021; Gauchy, Feau, and Garnier, 2021; Khansefid et al., 2023; Lee, Kwag, and Ju, 2023). In every of these studies, each sample in the dataset regroups:

1. Information about a seismic ground motion. The considered ground motions are sometimes natural

and sometimes artificial. The information can take several forms, as stated in the introduction it is generally used to derive one or several scalars called intensity measures (IMs).

2. Information about the response of the structure or the equipment of interest to the seismic excitation. To define the fragility curve, this information, must permit to characterize the failure of the studied system.

Since available records of real seismic excitations for a given site are often scarce, it is common to construct a dataset of artificial earthquake accelerograms using a seismic signal generator. Various techniques exist for this purpose. According to Rezaeian and Der Kiureghian (2008), the methods can be categorized among the "source-based" ones, which model the occurrence of earthquake rupture at some sources and the propagation of seismic waves to the studied site, and the "site-based" ones, which model the seismic signals for the site of interest from the consideration of its characteristics and historical recorded earthquakes. A review of the first kind is proposed in Zerva, 1988, and a review of the second can be found in Shinozuka and Deodatis, 1988. As more recent examples for the latter, we can also cite Trevlopoulos, Feau, and Zentner, 2019; and Zentner and Poirion, 2012. In the following, we present a stochastic seismic signal generator that is proposed by Rezaeian and Der Kiureghian (2010). It lies among the site-based models, and has been implemented by Sainct et al. (2020), who calibrated it using 97 real accelerograms selected in the European Strong Motion Database for a magnitude M such that $5.5 \leq M \leq 6.5$, and a source-to-site distance $R < 20$ km (Ambraseys et al., 2000). An example of one of these real signals is plotted in fig. 7.1.

As said in the introduction, this thesis does not seek to question thoroughly the seismic hazard, and we aim at providing methods that can be applied to any modeling of the seismic signals. Nevertheless, in the following chapters, our methods are applied and validated on different case studies that are presented later on. These case studies take the form of mechanical equipments that have been submitted to artificial seismic signals generated using the generator implemented by Sainct et al. (2020), and presented below.

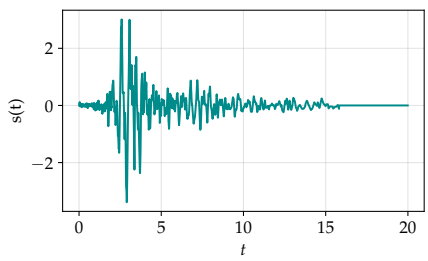

Figure 7.1: Example of an accelerogram record of a real seismic signal.

Seismic signals generator and IMs

The seismic excitation generated takes the form of a temporal signal that is the realization $s : t \in [0, T] \mapsto s(t) \in \mathbb{R}$ of a random process defined on a probability space $(\Omega, \Xi, \mathbb{P})$. The generator presented here corresponds to a modulated filtered stochastic white noise with time dependent parameters:

$$s(t) = s(t; w, \boldsymbol{\rho}, \boldsymbol{\lambda}) = q(t, \boldsymbol{\rho}) \left[\frac{1}{\sigma_f(t)} \int_{-\infty}^{t} h(t - \tau, \boldsymbol{\lambda}(\tau)) w(\tau) d\tau \right], \tag{7.3}$$

where w is a realization of a Gaussian white noise process W. In other terms $W : [0, T] \times \Omega \longrightarrow \mathbb{R}$ is such that for all $t_1 \neq t_2$, $\mathbb{E}W(t_1) = 0$, $\mathbb{E}W(t_1)^2 = \infty$, and $\mathbb{E}W(t_1)W(t_2) = \varsigma(t_1 - t_2)$.

The integral in the above equation corresponds to the filtering of w, $h(t, \boldsymbol{\lambda})$ being to the impulse response function (IRF) of the linear filter and σ_f^2 being its variance: $\sigma_f^2(t) = \int_{-\infty}^{t} h^2(t - \tau, \boldsymbol{\lambda}(\tau)) d\tau$. The IRF is defined by

$$h(t - \tau, \boldsymbol{\lambda}(\tau)) = \frac{\omega_f(\tau)}{\sqrt{1 - \zeta_f^2}} \exp[-\zeta_f \omega_f(\tau)(t - \tau)] \sin\left(\omega_f(\tau)\sqrt{1 - \zeta_f^2}(t - \tau)\right) \mathbb{1}_{t \geq \tau}, \tag{7.4}$$

where $\boldsymbol{\lambda}(\tau) = (\omega_f(\tau), \zeta_f)$ with $\omega_f(\tau) := \omega_0 + \frac{\tau}{T}(\omega_n - \omega_0)$ being the natural frequency and $\zeta_f \in [0,1]$ being a constant damping ratio. The quantities ω_0 and ω_n are parameters of the IRF.

The function $q(t, \boldsymbol{\rho})$ is the non-negative modulating function, it is defined by

$$q(t, \boldsymbol{\rho}) = \begin{cases} \rho_1 t^2 / T_1^2 & \text{if } 0 \leq t < T_1 \\ \rho_1 & \text{if } T_1 \leq t < T_2 \\ \rho_1 \exp\left[-\rho_2 (t - T_2)^{\rho_3}\right] & \text{if } T_2 \leq t \end{cases} \tag{7.5}$$

where $\boldsymbol{\rho} = (\rho_1, \rho_2, \rho_3, T_1, T_2) \in (0, \infty)^5$. Therefore, the signal s depends on w and $\boldsymbol{\phi} := (\boldsymbol{\rho}, \omega_0, \omega_n, \zeta_f) \in \boldsymbol{\Phi} := (0, \infty)^7 \times [0,1]$. The generator consists in deriving realizations of $s(\cdot; W, \boldsymbol{\phi})$ where $\boldsymbol{\phi}$ is stochastic, its distribution being identified using real acceleration records. As previously announced, the records considered are $N_r = 97$ real acceleration records form the European Strong Motion Database for a magnitude M such that $5.5 \leq M \leq 6.5$ and a source-to-site distance $R < 20$ km. Each of the records corresponds to a realization of $s(\cdot; W, \overline{\boldsymbol{\phi}}_i)$ using a tuple $\overline{\boldsymbol{\phi}}_i$ as parameters. The distribution of $\boldsymbol{\phi}$ is given by the Gaussian kernel density estimation (see Kristan, Leonardis, and Skočaj, 2011) of the one of the $(\overline{\boldsymbol{\phi}}_i)_{i=1}^{N_r}$, i.e. its density $p_{\boldsymbol{\phi}}$ is given by

$$p_{\boldsymbol{\phi}}(x) \propto \sum_{i=1}^{N_r} \exp\left(-\frac{1}{2}(x - \overline{\boldsymbol{\phi}}_i)^\top \Sigma^{-1} (x - \overline{\boldsymbol{\phi}}_i)\right) \mathbb{1}_{x \in \boldsymbol{\Phi}}, \tag{7.6}$$

where Σ is derived from the $(\overline{\boldsymbol{\phi}}_i)_{i=1}^{N_r}$ (see Kristan, Leonardis, and Skočaj, 2011).

From a seismic signal s that is a realization of the process described above, different intensity measure indicators can be derived. The choice of the appropriate IM to estimate seismic fragility curves remains a complex question. According to Giovenale, Cornell, and Esteva (2004), the appropriateness of an IM must be defined in terms of efficiency, sufficiency, and hazard compatibility. However, the most efficient or sufficient IM is not the same for two different case studies (see Mackie and Stojadinović, 2001; Hariri-Ardebili and Saouma, 2016). Moreover, while we do not thoroughly research the best IM in this thesis, we will see in the following chapters that the best choice is not necessarily simply the one that is the most correlated with the structure's response. Below we propose a non-exhaustive list of common IMs, a more complete one can be found in Luco and Cornell, 2007:

- the peak ground acceleration (PGA) is defined as $\text{PGA} = \max_{t \in [0,T]} |s(t)|$;
- the peak ground velocity (PGV) is defined as $\text{PGV} = \max_{t \in [0,T]} \left| \int_0^t s(\tau) d\tau \right|$;
- the peak ground displacement (PGD) is defined as $\text{PGD} = \max_{t \in [0,T]} \left| \int_0^t \int_0^\tau s(u) du d\tau \right|$;
- the pseudo spectral acceleration (PSA) at frequency f_L and damping ratio ξ is defined as $\text{PSA} = (2\pi f_L)^2 \max_{t \in [0,T]} |x(t)|$, where x is the solution of the linear equation

$$x''(t) + 2\xi 2\pi f_L x'(t) + (2\pi f_L)^2 x(t) = -s(t). \tag{7.7}$$

  This IM is component-dependent since, in practice, the damping ratio ξ and the frequency f_L are evaluated from the mechanical characteristic of the studied system. Generally, the frequency f_L considered for deriving the PSA corresponds to the first mode of the structure's response under excitation. Equation (7.7) corresponds to the response equation of the linear single degree of freedom system associated to the structure.

For conducting the numerical experiments of this thesis, 10^5 seismic signals have been computed using this generator. In fig. 7.2, we draw the distribution of the PGA and the PSA (at frequency 5 Hz and damping ratio 1%) given by this generator. It is compared with the kernel density approximation of the PGA and PSA values of the 97 real seismic signals that have been used for fitting the generator. These comparisons show that the synthetic signals have realistic features.

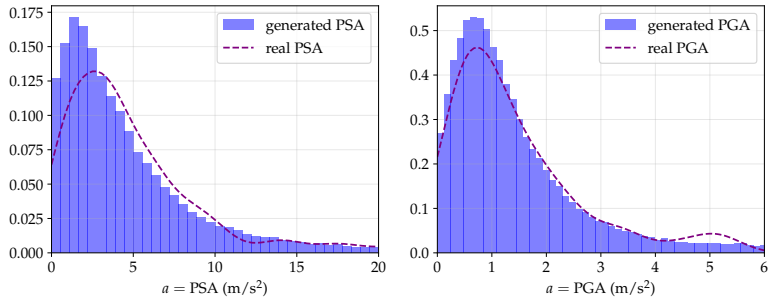

Figure 7.2: Histograms of IMs derived from 10^5 generated synthetic signals : the IM is the PSA (left) and the PGA (right). They are compared with the densities of IMs coming from real accelerograms estimated by Gaussian kernel estimation (dashed lines).

Engineering demand parameter and failure

As evoked in the preamble of this section, numerous data sources exist for identifying the fragility (and the failure) of mechanical structures and components. In most cases a particular engineering demand parameter (EDP) of the system is focused on. The EDP is a scalar quantity that is observed (numerically or practically) during the seismic excitation, it can be the maximal displacement of a specific part of mechanical equipment for instance. More explicit examples will be given when specific case studies will be presented later on in this chapter. In those cases, the failure is defined when the EDP exceeds a threshold limit.

All in all, available datasets often take the form of tuples $((\mathcal{S}_i, \mathcal{D}_i))_{i=1}^k$, where $\mathcal{S}_i$ is the i-th seismic signal submitted to the system and $\mathcal{D}_i$ is the EDP that was observed during the experiment. In the case studies that are treated in this manuscript a specific IM is chosen, and a reduced dataset of the form $((a_i, z_i))_{i=1}^k$ is studied, where a_i is the IM of the i-th seismic signal, and $z_i = \mathbb{1}_{\mathcal{D}_i > C}$, where C is the threshold that defines the failure of the system. In other words, $z_i = 1$ if the system has failed when submitted to the i-th seismic signal, and $z_i = 0$ otherwise. The dataset permits to statistically estimate the fragility curve of the system of interest. In this thesis, we restrict the study to cases where the observations can be considered independent. Naturally, this restriction excludes certain case studies, such as experimental campaigns carried out on certain structures subjected to a series of seismic solicitations of increasing level, such as reinforced concrete structures, because these systems are damaged with each seismic solicitation. The advantage of considering binary data (failure or non-failure) is that it allows to deal with cases for which (i) the failure is defined by multiple criteria (e.g. stacked structures) and (ii) an EDP is not directly observed (e.g. electrical devices).

7.3 Modeling of the fragility curve

Different models and methods have been suggested in the literature to estimate seismic fragility curves. In the genesis of the SPRA, seismic fragility curves were estimated based on expert judgments, principally due to the scarcity of available data to conduct an accurate statistical estimation (see Kennedy, Cornell, et al., 1980). Of course, the accuracy of such an estimation is not guaranteed either, and its reliability depends on that of the experts.

Nowadays, estimates of seismic fragility curves are mostly derived using statistical techniques, leveraging (i) dataset that are sometimes less scarce and more precise, and also (ii) statistical techniques that are more efficient. The expert judgment still complement those in some works in the literature.

Non-parametric and parametric methods are both used in the literature to estimate seismic fragility curves. In the former, the probabilistic relation between the failure and the IM is sought. When an EDP is available, authors seek to estimate first the probabilistic relation between the IM and the EDP, to deduce the resulting fragility curve. In the latter, the fragility curve is assumed to belong to a parameterized set, reducing the problem to the estimation of these parameters. While the first method is more general in the sense that it covers a wider range of possible estimates, it is also often less efficient than the second. Indeed, reducing the problem to a finite-dimensional one allows providing satisfying estimates with fewer data. However,

their reliability depends on the trustworthiness of the chosen parametric modeling.

In the following, we review briefly the state-of-the art on the estimation of seismic fragility curves. We start by a review of non-parametric methods in section 7.3.1. In that section we also describe a non-parametric estimation that is based on Monte-Carlo estimates, and that can serve as a reference when a large dataset is available. In section 7.3.2 we discuss the parametric modelings of the fragility curves, and we present the most common one: the probit-lognormal model.

7.3.1 Non-parametric modeling

Most of non-parametric models for seismic fragility curves estimation leverage —if possible— the knowledge of an EDP that describes more precisely the response of the structure under seismic excitation than just the binary outcome —failure or non-failure. In this case it is possible to estimate the conditional distribution of the EDP to the IM (or IMs) via a surrogate modeling of the system EDP $= \mathcal{M}(\text{IM})$.

Among common surrogates we can mention the Gaussian processes (GPs). As examples of their use for seismic fragility curves estimation, we can cite Gidaris, Taflanidis, and Mavroeidis, 2015, in which GPs are used to estimate the EDP given multiple parameters characterizing the ground motion; and Gauchy, Feau, and Garnier, 2024, in which the GPs-based estimation of the EDP is coupled with a sensitivity analysis of the fragility curve to the mechanical parameters of the studied system.

We also mention polynomial chaos expansion (PCE) as a surrogate numerously used for seismic fragility curves estimation. For instance, PCEs are combined in Mai, Spiridonakos, et al., 2016 with non-linear autoregressive with exogenous input models to estimate the temporal response of the mechanical system as a function of the seismic signal. In X. Zhu, Broccardo, and Sudret, 2023, a stochastic PCE is conducted, consisting in the addition of a latent variable and an additive noise to the deterministic PCE expression.

Outside surrogate modeling, what one would call machine-learning-based techniques are also implemented to estimate seismic fragility curves. For examples, we quote linear regression or generalized linear regression (Lallemant, Kiremidjian, and Burton, 2015), classification-based methods (see the review suggested in Kiani, Camp, and Pezeshk, 2019) such as logistic regression (as in Bernier and Padgett, 2019) or support vector machine (as in Sainct et al., 2020). In the latter work, support vector machine is used to classify EDPs whether they led to failures or non-failures as a function of combinations of IMs. We also mention artificial networks based methods, such as used in Mitropoulou and Papadrakakis, 2011; Z. Wang, Pedroni, et al., 2018.

A reference fragility curve constructed with a large dataset

All the methods cited above provide estimations whose efficiency will depend on the size of the available dataset and the correctness of the assumptions they involve on the probability relation between the system's response and the IMs. It is possible to minimize such assumptions considerably, trying to estimate directly the probabilities $\mathbb{P}(\text{“failure”}|\text{IM} = a)$, for all value of a, via Monte-Carlo estimates for instance. Such a methodology must give the best estimation of the fragility curve in terms of robustness, in the sense that it does not rely on any assumption. Of course, one does not have infinitely many data for all value of a to provide an exact estimation of the whole curve, yet it is possible to approximate the curve locally in different sub-areas of the domain $\mathcal{A}$ in which a lives. More explicitly consider a dataset $((a_i, z_i))_{i=1}^N$ of IMs and binary outcomes ($z_i = 1$ if the i-th seismic signal led to a failure, and 0 otherwise), and choose N_c clusters of the $(a_i)_{i=1}^N$ that we denote $(K_j)_{j=1}^{N_c}$: $\bigsqcup_{j=1}^{N_c} K_j = \{a_i, i = 1, \dots, N\}$. Then it is possible to approximate the fragility curve evaluated at the centroids $(c_j)_{j=1}^{N_c}$ of the clusters:

$$P_f^{\text{MC}}(c_j) = \frac{1}{n_j} \sum_{i, a_i \in K_j} z_i, \tag{7.8}$$

where n_j is the sample size of cluster K_j. This non-parametric estimation is implemented by Trevlopoulos, Feau, and Zentner (2019), who suggest defining the clusters $(K_j)_{j=1}^{N_c}$ using K-means since the IM values in available datasets are generally not uniformly distributed.

When the number of available data is small and when they are poorly diverse in terms of values of the IM, this estimation method becomes limited for estimating seismic fragility curves. However, in the other case, we consider that it provides a robust result. For this reason, while this thesis addresses the estimation of fragility curves when few data are available, we will evaluate our methods on case studies for which a large validation dataset exists. The latter is used to derive what we call a reference fragility curve that is identified to P_f^{MC}, and which will be compared with the estimates provided by our methods.

7.3.2 Parametric modeling: probit-lognormal model

The probit-lognormal model

Although based on stronger assumptions on the structure's response than non-parametric methods, parametric fragility curves were historically considered, especially in cases with small dataset limited to binary outcome (i.e. no EDP is available). In the SPRA and PBEE frameworks, the so-called probit-lognormal model was chosen, and it remains prevalent to this day (see e.g. Shinozuka, Feng, et al., 2000; Straub and Der Kiureghian, 2008; Zentner, 2010; F. Wang and Feau, 2020; Mandal, S. Ghosh, and Pujari, 2016; Zhao, Yu, and Mo, 2020; Ellingwood, 2001; S.-H. Kim and Shinozuka, 2004; Mai, Konakli, and Sudret, 2017; Trevlopoulos, Feau, and Zentner, 2019; Katayama et al., 2021; Lee, Kwag, and Ju, 2023). This model consists in introducing a parameter $\theta = (\alpha, \beta)$ and defining the fragility curve as follows

$$P_f(a) = \mathbb{P}(\text{“failure”}|\mathrm{IM} = a) = \Phi\left(\frac{\log a - \log \alpha}{\beta}\right), \tag{7.9}$$

where Φ is the cumulative distribution function of a standard normal variable. An example of such curves is given in fig. 7.3. We mention that, sometimes, the parametrization of the model is slightly different. As recalled by Zentner, Gündel, and Bonfils, 2017, the distinguishing of epistemic and aleatoric uncertainty in β is commonly suggested, leading to rewriting eq. (7.9) with $\beta = \sqrt{\beta_U^2 + \beta_R^2}$. The literature suggests that the uncertainty embedded in α is only epistemic.

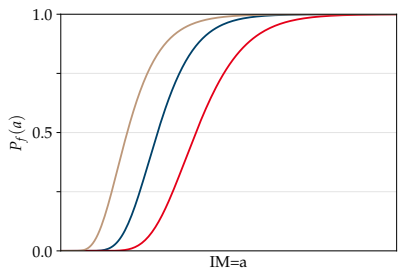

Figure 7.3: Examples of probit-lognormal fragility curves.

Different strategies exist to estimate the two parameters, namely the median α and the log standard deviation β. Historically, maximum likelihood estimation (MLE) techniques are recommended. When the data are independent, the bootstrap technique can be used to obtain confidence intervals relating to the size of the sample considered (e.g. Shinozuka, Feng, et al., 2000; Zentner, 2010; F. Wang and Feau, 2020). To describe this technique, we define first the associated statistical model: the observations are modeled as realizations of the random variable $(A, Z) \in \mathcal{A} \times \{0, 1\}$. It is supposed that observations are independent conditionally to θ and that (A, Z) are distributed conditionally to $\theta \in \Theta = (0, \infty)^2$ as:

$$A|\theta \sim A \sim H, \quad \text{and} \quad Z|A, \theta \sim \mathcal{B}\left(\Phi\left(\beta^{-1} \log \frac{A}{\alpha}\right)\right), \tag{7.10}$$

where H is the distribution of A and $\mathcal{B}(p)$ denotes a Bernoulli distribution of mean p. Thus, the likelihood

ℓ_k given observations $(\mathbf{a}^k, \mathbf{z}^k)$ with $\mathbf{a}^k = (a_i)_{i=1}^k$ and $\mathbf{z}^k = (z_i)_{i=1}^k$ is expressed as

$$\ell_k(\mathbf{z}^k, \mathbf{a}^k|\theta) = \prod_{i=1}^k \ell(z_i, a_i|\theta) = \prod_{i=1}^k \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)^{z_i} \left(1 - \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)\right)^{1-z_i} h(a_i), \tag{7.11}$$

where h is the p.d.f. of A. Since the $(h(a_i))_i$ are constants of θ and are not known in general, it is common to consider the following alternative form of the likelihood:

$$\ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta) = \prod_{i=1}^k \ell(z_i|a_i, \theta) = \prod_{i=1}^k \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)^{z_i} \left(1 - \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)\right)^{1-z_i}. \tag{7.12}$$

The maximum likelihood estimator is $\theta^{\mathrm{MLE}}(\mathbf{z}^k, \mathbf{a}^k) = \arg\max_{\theta\in\Theta} \ell_k(\mathbf{z}^k, \mathbf{a}^k|\theta)$. The bootstrap technique consists in deriving a stochastic estimator $\hat{\theta}_k^{\mathrm{BMLE}}(\mathbf{z}^k, \mathbf{a}^k)$ whose distribution is defined from expressing it as: $\hat{\theta}^{\mathrm{BMLE}}(\mathbf{z}^k, \mathbf{a}^k) = \theta^{\mathrm{MLE}}((z_{U_i}, a_{U_i})_{i=1}^k)$, where $U_1, \ldots, U_k$ are i.i.d. random variables distributed w.r.t. a uniform distribution in $\{1, \ldots, k\}$.

About the implementation of the Bayesian framework to estimate $\theta = (\alpha, \beta)$

As a strategy that allows to estimate the parameters defining the fragility curve, the Bayesian framework has recently become increasingly popular in seismic fragility analysis (see e.g. Gardoni, Der Kiureghian, and Mosalam, 2002; Z. Wang, Zentner, and Zio, 2018; Katayama et al., 2021; Koutsourelakis, 2010; Damblin et al., 2014; Tadinada and A. Gupta, 2017; Kwag and A. Gupta, 2018; Jeon et al., 2019; Tabandeh, Asem, and Gardoni, 2020). It is praised for its capacity to solve the irregularity issues encountered when estimating fragility curves with classical methods. For instance, the MLE-bootstrap method is known to lead to unrealistic estimates such as unit-step functions when few data are available. However, a challenge remains when implementing the Bayesian framework, and lies in the selection of the prior. As a matter of fact, as announced since the chapter 1, selecting a prior is a critical step in Bayesian analysis, and when exploring the literature, a wide range of different consideration can be found regarding its construction.

For example, in Tadinada and A. Gupta, 2017 and Kwag and A. Gupta, 2018, the median prior values come from equivalent linearized mechanical models. In Z. Wang, Zentner, and Zio, 2018, both aleatory and epistemic uncertainties are taken into account in the parametric model originally introduced in Kennedy, Cornell, et al., 1980: an artificial neural network is trained and used to characterize (i) the aleatory uncertainty and (ii) the prior median value of α, while the associated epistemic uncertainty is taken from the existing literature. The log-normal prior distribution of α is then updated with empirical data. In Katayama et al., 2021, the results of incremental dynamic analysis are used to obtain a prior value of α, whereas the prior value of β is determined through a parametric study. This results in satisfactory convergence, whatever its target value, before application to practical problems. In Straub and Der Kiureghian, 2008, the authors mainly focus on the implications for fragility analyses of statistical dependencies within the data. The prior is defined as the product of a normal distribution for $\ln(\alpha)$, and the improper distribution $1/\beta$ for β. The definition of the normal distribution is based on engineering assessments. This prior was preferred to $1/\alpha$ on the grounds that it led to unrealistically large posterior values of α. A sensitivity analysis is further performed to examine the impact of the choice of the prior distribution on the final results. Finally, we note that the Bayesian framework is also relevant for fitting numerical models (e.g., mathematical expressions based on engineering assessments or physics-based models) to experimental data in order to estimate fragility curves (Gardoni, Der Kiureghian, and Mosalam, 2002; Tabandeh, Asem, and Gardoni, 2020) or meta-models such as logistic regressions (Koutsourelakis, 2010; Jeon et al., 2019).

7.4 Example of case studies

7.4.1 An elasto-plastic oscillator

The first case study that we present is a single-degree-of-freedom elasto-plastic oscillator with kinematic hardening. This simple mechanical system illustrates the essential features which can be found in the non-linear responses of some real-world structures under seismic excitation and has, for this reason, already been

used in several studies (Trevlopoulos, Feau, and Zentner, 2019; Sainct et al., 2020; Gauchy, Feau, and Garnier, 2021). In addition, it provides reference results as a reasonable numerical cost. It is depicted in fig. 7.4, and its equation of motion when submitted to an excitation s is the following:

$$x''(t) + 2\xi 2\pi f_L x'(t) + f^{\mathrm{nl}}(t) = -s(t), \tag{7.13}$$

with x'' and x' respectively the relative acceleration and velocity of the mass, ξ the damping ratio, f_L the circular frequency, and f^{nl} the nonlinear resisting force.

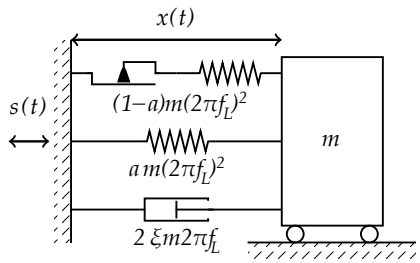

Figure 7.4: Elasto-plastic oscillator of mass m with kinematic hardening, with parameters $f_{\mathrm{L}} = 5$ Hz and $\xi = 2\%$. The yield limit is $Y = 5.10^{-3}$ m, and the post-yield stiffness is 20% of the elastic stiffness, i.e., $a = 0.2$.

The relevant EDP is the absolute maximum value of the mass' displacement, i.e., EDP $= \max_{t\in[0,T]} |x(t)|$, where T is the duration of the seismic excitation. In fig. 7.5, we plot that EDP that has been derived for the 10^5 seismic signals generated and presented in section 7.2.

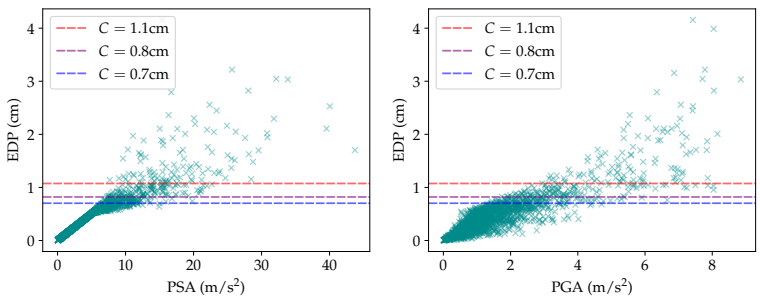

Figure 7.5: Results of the 10^5 simulations conducted on the elasto-plastic oscillator. Each cross is an element of the dataset (IM, EDP) where the IM is the PSA (left) and the PGA (right). Different critical rotation thresholds C are plotted in dashed lines. They yield different proportions of failures in the dataset: respectively 95% (red), 90% (purple) and 85% (blue).

Since a very large number of data are available for this case study, it is possible to derive a reference fragility curve using the non-parametric method that is described in section 7.3.1. Such reference fragility curves are plotted in fig. 7.6, for different thresholds C that define the failure. In general, the failure criterion that corresponds to the 90%-level quantile of the maximum displacements calculated with the 10^5 artificial signals is chosen, i.e., $C = 8.0\ 10^{-3}$ m. The reference curves are compared with the probit-lognormal fragility curves (described in section 7.3.2) where θ is estimated by MLE using the full dataset as well. This comparison is presented for both the PSA and the PGA as IM. It demonstrates that the probit-lognormal model allows a good approximation of the reference curve for this case study.

7.4.2 A piping system from a pressurized water reactor

The case study presented in this section is a piping system that was tested on the Azalee shaking table at the EMSI laboratory of CEA/Saclay, as shown in fig. 7.7-left. Figure 7.7-right depicts the finite element model (FEM), based on beam elements and implemented through the proprietary FE code CAST3M (CEA,

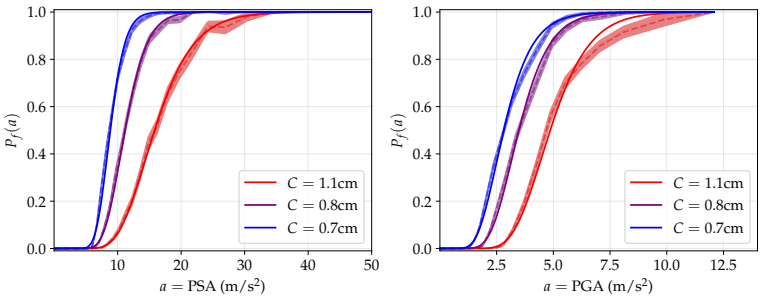

Figure 7.6: Reference non-parametric fragility curves of the elasto-plastic oscillator obtained via Monte Carlo estimates (dashed lines) surrounded by their 95% confidence intervals, for different critical displacement threshold C with (left) the PSA and (right) the PGA as IM. The thresholds yield different proportions of failures in the dataset: respectively 95% (red), 90% (purple) and 85% (blue). For each value of C are plotted (same color, solid line) the corresponding probit-lognormal MLE.

2019). The validation of the FEM was carried out thanks to an experimental campaign described in Touboul, Sollogoub, and Blay, 1999.

The mock-up comprises a carbon steel TU42C pipe with an outer diameter of 114.3 mm, a thickness of 8.56 mm, and a 0.47 elbow characteristic parameter. This pipe, filled with water without pressure, includes three elbows, with a valve-mimicking mass of 120 kg, constituting over 30% of the mock-up's total mass. One end of the mock-up is clamped, while the other is guided to restrict displacements in the X and Y directions. Additionally, a rod is positioned atop the specimen to limit mass displacements in the Z direction (refer to fig. 7.7-right). During testing, excitation was applied exclusively in the X direction.

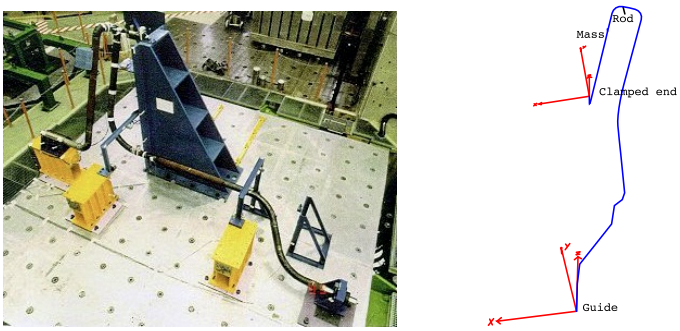

Figure 7.7: (left) Overview of the piping system —which is part of a French pressurized water reactor— on the Azalee shaking table and (right) associated finite element model.

In order to conduct comparative performance studies, numerous simulations have been performed. They were carried out from a subset of $8 \cdot 10^4$ of the 10^5 artificial seismic signals. Nevertheless, as in practice the piping system is located in a building, the artificial signals were filtered using a fictitious 2% damped linear single-mode building at 5 Hz, which corresponds to the first eigenfrequency of the 1% damped piping system. In such a situation, for some seismic signals, the behavior of the piping system is nonlinear. Regarding the nonlinear constitutive law of the material, a bilinear law exhibiting kinematic hardening was used to reproduce the overall nonlinear behavior of the mock-up with satisfactory agreement compared to the results of the seismic tests (Touboul, Sollogoub, and Blay, 1999). Following the recommendation in Touboul, Blay, et al., 2006, we consider that the engineering demand parameter (EDP) is the out-of-plane rotation of the elbow near the clamped end of the mock-up. As a result we have a dataset of $8 \cdot 10^4$ independent tuples of the form (IM, EDP) for different IMs. In fig. 7.8 are plotted the elements of the datasets (PSA, EDP) and (PGA, EDP), along with different critical thresholds.

The complete set of $8 \cdot 10^4$ simulations provides a satisfactory dataset to derive a reference of the seismic fragility curve for this case study, still using the non-parametric method depicted in section 7.3.1. In fig. 7.9,

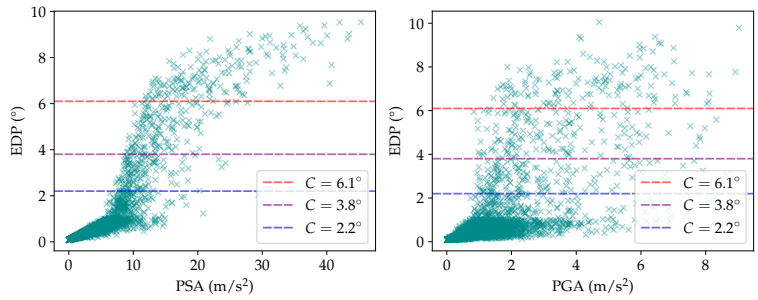

Figure 7.8: Results of the $8 \cdot 10^4$ numerical simulations conducted on the piping system. Each cross is an element of the dataset (IM, EDP) where the IM is the PSA (left) and the PGA (right). Different critical rotation thresholds C are plotted in dashed lines. They yield different proportions of failures in the dataset: respectively 95% (red), 90% (purple) and 85% (blue).

we compare this non-parametric reference with the parametric fragility curve using the probit-lognormal model (section 7.3.2) where α and β are estimated by MLE using the $8 \cdot 10^4$ data items as well. This comparison is presented considering different critical rotation thresholds C of the equipment for both the PSA and the PGA. First, it should be noted that, with the PGA as IM, it is not possible to completely describe the fragility curve. For the maximum PGA values observed, the failure probabilities stagnate between 0.5 and 0.8 depending on the failure criterion considered. Therefore, the PGA is not the most suitable IM of the two. For the type of structure considered here, this point is well documented in the literature. Then, the comparisons demonstrate, in our setting, the adequacy of a probit-lognormal modeling of the fragility curves, even with high failure thresholds. Its bias with non-parametric fragility curve exists but remains limited. When the number of observations is small, this model bias is negligible in front of the uncertainty on the estimates.

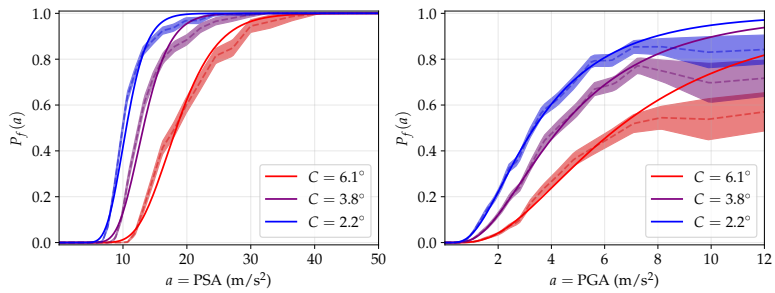

Figure 7.9: Reference non-parametric fragility curves of the piping system obtained via Monte Carlo estimates (dashed lines) surrounded by their 95% confidence intervals, for different critical rotation threshold C with (left) the PSA and (right) the PGA as IM. The thresholds yield different proportions of failures in the dataset: respectively 95% (red), 90% (purple) and 85% (blue). For each value of C are plotted (same color, solid line) the corresponding probit-lognormal MLE.

7.4.3 Stacked structure for storage of packages

The case study considered hereafter concerns a freestanding stacked structure composed of three pallets intended for the storage of packages. As shown in fig. 7.10, it is a square-based structure 3 meters high with a slenderness of 2.4.

The mass of one package is equal to 265 kg, whereas the mass of one pallet is equal to 60 kg. The total mass of the structure is equal to 3,360 kg. The pallets are made of 3 mm-thick hollow-section aluminum square tubes. These tubes are welded and form the base and uprights of the pallets. The base of a pallet supports four freestanding packages, whereas the uprights support the upper freestanding pallet(s). The stacked structure was studied in Beylat, 2020 by means, among others, of a complete experimental campaign,

which was carried out on the 1D shaking table "Vesuve" of CEA/Saclay. In particular, the stack was subjected to 21 of the 97 real signals that were used to fit the generator depicted in section 7.2. For this structure, the EDP corresponds to the maximal displacement of the top of the stack. An example of a test result is shown in fig. 7.10, which depicts the horizontal displacements over time of the top of the stack. The initial position is indicated in red, while the different positions in time are indicated in blue. Due to uplift, sliding, and rotation motions, the top of the stack exceeds, for 2 of the 21 tests, the admissibility criterion, which is materialized in black. Only a little number of experimental results are available for this case study, so that it is not possible to derive a reference fragility curve here.

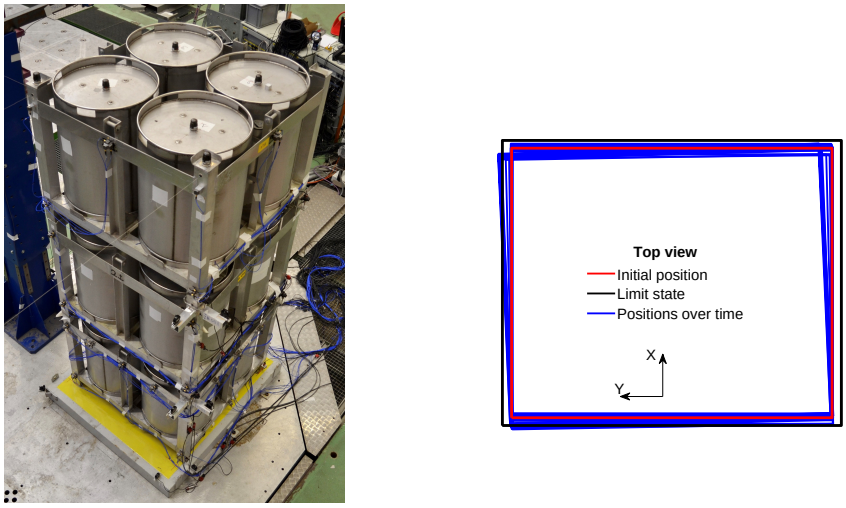

Figure 7.10: (left) Overview of the stacked structure placed on the Vesuve 1D shaking table and (right) example of test result: horizontal displacements of the top of the stack when subjected to seismic excitation in the X direction.

7.5 Conclusion: which case study and which data for a Bayesian estimation of fragility curves?

In the literature, many works address the estimation of seismic fragility curves, introducing many approaches to handle that task. Nevertheless, the appropriate method to provide robust estimates heavily depends on the characteristics of the studied equipments, on the type of data that are available, and on their number. In this chapter, we have described standard forms of datasets considered in the literature, and we have depicted how they are used to providing approximations of fragility curves using different approaches.

In the worst case, the dataset is made of tuples composed by (i) a value of the IM, and (ii) a binary outcome about the system's response: failure or non-failure. In those cases, the non-parametric models provide limited estimations, and a parametric model, such as the prominent probit-lognormal one has to be favored. If the dataset is also scarce, then most classical frequentist methods become unsatisfying as well, letting the Bayesian framework to arise as a cornerstone.

However, a major issue connects all the studies suggesting a Bayesian estimation of seismic fragility curves that we reviewed: the prior selection is not thoroughly questioned. Moreover, the choices done regarding its construction are hard to justify, and one could question the reliability of the provided estimates.

All in all, this chapter and that last thought introduce the essential interplay that exists between the part I and the part II of this manuscript. Since our work is sought to be applied on case studies taken from the nuclear industry, we aim at providing a reliable and auditable methodology when we estimate seismic fragility curves. The expected auditability also concerns the choice of the prior, which hence must be constructed in a way that ensures its objectivity.

8

Bayesian estimation of seismic fragility curves using a reference prior

This chapter draws mainly on the published works: Antoine Van Biesbroeck, Clément Gauchy, Cyril Feau, and Josselin Garnier (2024). "Reference prior for Bayesian estimation of seismic fragility curves". In: *Probabilistic Engineering Mechanics* 76, p. 103622. DOI: 10.1016/j.probengmech.2024.103622

Abstract For many structures of interest, estimating seismic fragility curves is a daunting task because of the limited amount of data available, and the limited amount of information they contain (such as binary data, i.e. only describing the structure as being in a failure or non-failure state). In such a context, a probit-lognormal modeling of the fragility curve is prominent, and among various methods that exist to estimate the parameters of the model, the Bayesian inference became popular in the literature. However, the latter's trustworthiness relies on the prior selection, jeopardizing the approach if the selection is motivated by subjective considerations. This chapter proposes a comprehensive objective Bayesian estimation of seismic fragility curves, implementing the Jeffreys prior for the probit-lognormal model. This prior is completely studied along with the decay rates of the likelihood to ensure the validity of our methodology. It is also implemented on three different case studies, including industrial examples. Our theoretical and practical results show that the Jeffreys prior leads to more robust and efficient fragility curves estimates in comparison with other methods of the literature.

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8.1 Introduction

Seismic fragility curves are key quantities of interest of the seismic probabilistic risk assessment (SPRA) framework. They are defined as the probability of failure of a mechanical system conditionally to a scalar value that is derived from a seismic signal, coined intensity measure (IM). A detailed review of those curves and of the methods that exist to estimate them is proposed in the chapter 7.

Although various data sources can be exploited to evaluate the curves, most of them suffer from their scarcity, complicating the uncertainty quantification in the estimates. Moreover, we focus in this work on cases where the dataset contains information of the mechanical system's response under seismic excitation that is limited to binary outcome, i.e. failure or non-failure. In this sense, this work will mainly address equipment problems for which only binary results of seismic qualification tests (e.g., tests of electrical relays, etc.) or empirical data such as presented in Straub and Der Kiureghian, 2008 are available. However, the methodology developed here could perfectly be applied to simulation-based approaches as well.

In those cases, the Bayesian framework became heavily popular in the literature, being praised for its capability to regularize the estimates. As a matter of fact, the framework is known to avoid the generation of unrealistic fragility curves such as unit-step functions, which are common with classical frequentist methods. In these settings where only the binary outcome about the system's response are available, Bayesian analysis is used to infer a parameterized fragility curve to the data. The parametric model generally takes the form of a probit-lognormal fragility curve, which is prominent in the field of earthquake engineering. However, the estimates given by the posterior distribution in the Bayesian paradigm are significantly influenced by the prior, compounding their reliability since there is no consensus on some rule for the choice of the latter. Actually, there are plethora of different considerations (which are sometimes questionable because of the subjectivity they introduce) to define the prior in the literature. As an example, Straub and Der Kiureghian (2008) consider independent prior distributions for the parameters defining the fragility curve, namely the median α and the log standard deviation β. The prior is defined as the product of a normal distribution for $\ln(\alpha)$, and the improper distribution $1/\beta$ for β. The definition of the normal distribution is based on engineering assessments, assuming that, for the relevant component and considering the peak ground acceleration (PGA) as IM for example, the median lies between 0.02 g and 3 g with a probability of 90%. This prior was preferred to $1/\alpha$ on the grounds that it led to unrealistically large posterior values of α.

In this study, the goal is to choose the prior while eliminating, insofar as it is possible, any subjectivity which would unavoidably lead to open questions regarding the impact of the prior on the final results. To achieve that goal, the reference prior theory permits constructing priors that can be qualified as objective, because they are sought to maximize the influence of the real observations over that of the prior on the posterior distribution. We remind that a complete review of the theory is proposed in chapter 3. This allows us to focus on the well-known Jeffreys prior, the asymptotic optimum of the mutual information w.r.t. the size of the data set (see chapter 3 or chapter 4), and which will be explicitly derived and studied for the first time in this context.

Of course, from a subjectivity perspective, the choice of a parametric model for the fragility curve is debatable. However, numerical experiments based on the seismic responses of mechanical systems suggest that the choice of an appropriate IM makes it possible to reduce the potential biases between reference fragility curves (that can be obtained by massive Monte-Carlo methods) and their log-normal estimations (Gauchy, Feau, and Garnier, 2021). This observation is reinforced by recent studies on the impact of IMs on fragility curves (Sainct et al., 2020; Ciano, Gioffrè, and Grigoriu, 2020; Ciano, Gioffrè, and Grigoriu, 2022). In this paper, we will ensure the relevance of the estimations by comparing them to the results of massive

Monte-Carlo methods on academic examples. The results are illustrated with the PGA, yet we recall that our method can be implemented with any other IM of interest, without additional complexity. We also indicate that the influence of the IM —PGA vs. PSA— on the convergence of the estimates is questioned in the chapter A.

This chapter is organized as follows. In the next section, we recall the definition of the probit-lognormal model, and we express the implementation of reference prior theory framework in this context. Then, the Jeffreys prior is derived and completely studied in section 8.3. The conduction of Bayesian inference for estimating fragility curves using the Jeffreys prior is the proposed approach of this study, it is compared with other approaches taken from the literature that are depicted in section 8.4. In section 8.5, we discuss the limitation of the different methods implemented, that results from the form of the likelihood of the probit-lognormal model, whose decay rates are derived. The comparison of the approaches is done by computing the performance evaluation metrics proposed in section 8.6, and by evaluating them on three case studies in section 8.7. Eventually, an appendix precedes the conclusion in section 8.8. It proposes a deeper study of the competing prior considered in this chapter.

8.2 Probit-lognormal model and Bayesian framework

We remind that the probit-lognormal model of the fragility curve has been described in the chapter 7. It defines the fragility curve of the mechanical system of interest as

$$P_f(a) = \mathbb{P}(\text{“failure”}|\text{IM} = a) = \Phi\left(\frac{\log a - \log \alpha}{\beta}\right), \tag{8.1}$$

where Φ is the c.d.f. of a standard Gaussian, and α, β are parameters that we seek to estimate. To be precise, $\alpha \in (0, \infty)$ is the median and $\beta \in (0, \infty)$ is the log standard deviation of the curve. We denote $\theta = (\alpha, \beta) \in \Theta = (0, \infty)^2$.

In statistical terms, we consider that the failure of the equipment is the realization of a random variable Z, which takes values in $\mathcal{Z} = \{0, 1\}$ (1 for failure, 0 for non-failure). We also denote by A the random variable of the IM. It takes value in a set $\mathcal{A} \subset (0, \infty)$ and is supposed to follow a distribution H. Conditionally to θ, the tuple $Y = (Z, A)$ follows a distribution defined by $A \sim H$ and $Z|A, \theta \sim \mathcal{B}(P_f(A))$, where $\mathcal{B}(p)$ denotes the Bernoulli distribution of parameter p, and P_f is defined in eq. (8.1).

We recall that given realizations $(\mathbf{z}^k, \mathbf{a}^k)$, where $\mathbf{z}^k = (z_i)_{i=1}^k$, $\mathbf{a}^k = (a_i)_{i=1}^k$, of the r.v. Y, this model admits the following likelihood:

$$\ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta) = \prod_{i=1}^k \ell(z_i|a_i, \theta) = \prod_{i=1}^k \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)^{z_i} \left(1 - \Phi\left(\frac{\log a_i - \log \alpha}{\beta}\right)\right)^{1-z_i}. \tag{8.2}$$

To conduct a Bayesian estimation of the fragility curve, we have to consider a prior distribution on θ. From a prior whose density is π, and given observations $(\mathbf{z}^k, \mathbf{a}^k)$ the posterior density $p(\theta|\mathbf{z}^k, \mathbf{a}^k)$ is derived using the Bayes' theorem:

$$p(\theta|\mathbf{z}^k, \mathbf{a}^k) = \frac{\ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta)\pi(\theta)}{\int_\Theta \ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta')\pi(\theta')d\theta'}. \tag{8.3}$$

The construction of that prior is done using the reference prior theory. We recall that this theory is comprehensively introduced in the chapter 3. We remind that it consists in choosing the prior that maximizes the mutual information $\mathscr{I}^k$ as $k \longrightarrow \infty$. The mutual information for a prior whose density is π is defined here as

$$\mathscr{I}^k(\pi) = \int_\Theta \int_{\mathcal{A}^k} \sum_{\mathbf{z}^k \in \{0,1\}^k} f\left(\frac{\int_\Theta \ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta')\pi(\theta')d\theta'}{\ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta)}\right) \ell_k(\mathbf{z}^k|\mathbf{a}^k, \theta)\mathbf{h}(\mathbf{a}^k)\pi(\theta)d\mathbf{a}^k d\theta, \tag{8.4}$$

where $f = -\log$ classically, or follows the suggestion we made in the chapter 4; and $\mathbf{h}(\mathbf{a}^k) := \prod_{i=1}^k h(a_i)$, h being the density of A.

The solution of this asymptotic optimization problem is the Jeffreys prior (see the chapters 3 and 4). The Jeffreys prior is already praised in Bayesian inference of its property to be invariant by re-parametrization of the model. Its density J is defined up to a constant as follows

$$J(\theta) \propto \sqrt{|\det \mathcal{I}(\theta)|} \quad \text{with} \quad \mathcal{I}(\theta) = -\sum_{z\in\{0,1\}} \int_{\mathcal{A}} \nabla^2_\theta \log \ell(z|a,\theta) h(a) da. \tag{8.5}$$

8.3 Jeffreys prior construction in the probit-lognormal model

Based on the elements discussed in the previous section, the Jeffreys prior seems to be the best objective candidate for this problem. In this section, we will therefore study the Jeffreys prior in order to estimate probit-lognormal seismic fragility curves with binary data. To our knowledge, the application of the reference prior theory to this field of study is completely new. The explicit calculation of this prior is carried out in section 8.3.1. It is followed in section 8.3.2 by an explanation about the practical implementation suggested. In section 8.3.3, we propose a thorough study of the prior's decay rates and its proper characteristic.

8.3.1 Derivation of the Jeffreys prior

The first step consists in computing the Fisher information matrix $\mathcal{I}(\theta)$ in our model, defined in eq. (8.5). Here, $\theta = (\theta_1, \theta_2) = (\alpha, \beta) \in \Theta$ and

$$\mathcal{I}(\theta)_{i,j} = -\sum_{z\in\{0,1\}} \int_{\mathcal{A}} \ell(z|a,\theta) \partial^2_{\theta_i\theta_j} \log \ell(z|a,\theta) h(a) da \tag{8.6}$$

for $i, j \in \{1, 2\}$, with

$$\log \ell(z|a,\theta) = z \log \Phi\left(\frac{\log a - \log \alpha}{\beta}\right) + (1-z) \log\left(1 - \Phi\left(\frac{\log a - \log \alpha}{\beta}\right)\right). \tag{8.7}$$

Denoting $\gamma = \gamma(a) = \beta^{-1}\log(a/\alpha)$, the first-order partial derivatives of $\log \ell(z|a,\theta)$ with respect to θ are:

$$\partial_\alpha \log \ell(z|a,\theta) = -\frac{1}{\alpha\beta} z \frac{\Phi'(\gamma)}{\Phi(\gamma)} + \frac{1}{\alpha\beta}(1-z)\frac{\Phi'(\gamma)}{1-\Phi(\gamma)}, \tag{8.8}$$

$$\partial_\beta \log \ell(z|a,\theta) = -\frac{\log\frac{a}{\alpha}}{\beta^2} z \frac{\Phi'(\gamma)}{\Phi(\gamma)} + \frac{\log\frac{a}{\alpha}}{\beta^2}(1-z)\frac{\Phi'(\gamma)}{1-\Phi(\gamma)}, \tag{8.9}$$

and the second-order partial derivatives are:

$$\begin{aligned}\partial^2_{\alpha\beta} \log \ell(z|a,\theta) = &-\frac{1}{\beta}\partial_\alpha \ell(z|a,\theta) + \frac{\log\frac{a}{\alpha}}{\alpha\beta^3} z \frac{\Phi''(\gamma)\Phi(\gamma) - \Phi'(\gamma)^2}{\Phi(\gamma)^2} \\ &- \frac{\log\frac{a}{\alpha}}{\alpha\beta^3}(1-z)\frac{\Phi''(\gamma)(1-\Phi(\gamma)) + \Phi'(\gamma)^2}{(1-\Phi(\gamma))^2},\end{aligned} \tag{8.10}$$

$$\begin{aligned}\partial^2_{\alpha\alpha} \log \ell(z|a,\theta) = &-\frac{1}{\alpha}\partial_\beta \log \ell(z|a,\theta) + \frac{1}{\alpha^2\beta^2} z \frac{\Phi''(\gamma)\Phi(\gamma) - \Phi'(\gamma)^2}{\Phi(\gamma)^2} \\ &- \frac{1}{\alpha^2\beta^2}(1-z)\frac{\Phi''(\gamma)(1-\Phi(\gamma)) + \Phi'(\gamma)^2}{(1-\Phi(\gamma))^2},\end{aligned} \tag{8.11}$$

and

$$\begin{aligned}\partial^2_{\beta\beta} \log \ell(z|a,\theta) = &-\frac{2}{\beta}\partial_\beta \log \ell(z|a,\theta) + \frac{\log^2\frac{a}{\alpha}}{\beta^4} z \frac{\Phi''(\gamma)\Phi(\gamma) - \Phi'(\gamma)^2}{\Phi(\gamma)^2} \\ &- \frac{\log^2\frac{a}{\alpha}}{\beta^4}(1-z)\frac{\Phi''(\gamma)(1-\Phi(\gamma)) + \Phi'(\gamma)^2}{(1-\Phi(\gamma))^2}.\end{aligned} \tag{8.12}$$

The expressions in eqs. (8.10) to (8.12) of the second-order partial derivatives of $\ell(z|a,\theta)$ need to be integrated over $\mathcal{Z}$ and $\mathcal{A}$. Summing over the discrete variable z first replaces z by $\Phi(\gamma)$ in the equations. Finally, if we denote A_{01}, A_{02}, A_{11}, A_{12}, A_{21}, A_{22} by

$$\begin{aligned} A_{0u} &= \int_{\mathcal{A}} \frac{\Phi'(\gamma(a))^2}{\Phi((-1)^{u+1}\gamma(a))} h(a)da, \\ A_{1u} &= \int_{\mathcal{A}} \log\frac{a}{\alpha} \frac{\Phi'(\gamma(a))^2}{\Phi((-1)^{u+1}\gamma(a))} h(a)da, \\ A_{2u} &= \int_{\mathcal{A}} \log^2\frac{a}{\alpha} \frac{\Phi'(\gamma(a))^2}{\Phi((-1)^{u+1}\gamma(a))} h(a)da, \end{aligned} \tag{8.13}$$

for $u \in \{1,2\}$, then the Fisher information matrix $\mathcal{I}(\theta)$ takes on the following form:

$$\mathcal{I}(\theta) = \begin{pmatrix} \frac{1}{\alpha^2\beta^2}(A_{01}+A_{02}) & \frac{1}{\alpha\beta^3}(A_{11}+A_{12}) \\ \frac{1}{\alpha\beta^3}(A_{11}+A_{12}) & \frac{1}{\beta^4}(A_{21}+A_{22}) \end{pmatrix}. \tag{8.14}$$

The integrals in eq. (8.13) are computed using Simpson's rule on a regular grid. The distribution $h(a)$ is approximated by Gaussian kernel density estimation based on seismic signals that we present in the next subsection. Finally, the Jeffreys prior density is obtained by taking the square root of the determinant of the matrix defined in eq. (8.14).

8.3.2 Practical implementation

The derivation conducted in the above showed that knowing the probability distribution of the IM is required in order to calculate the Fisher information matrix. In this thesis we have derived the Jeffreys prior using both the PGA and the PSA as IM. Still, it is important to emphasize that this choice is purely illustrative and bears no consequence for the proposed methodology. The distributions of both IMs are approximated given their values for 10^5 artificial signals generated using the generator presented in the chapter 7 (section 7.2). The empirical distribution of the PGA and of the PSA of these signals is recalled in fig. 8.1, and is compared with a log-normal distribution with same mean and same log deviation. The density h of their distribution can be identified as that log-normal density. In fig. 8.1, we also show that the Jeffreys prior resembles to that log-normal approximation w.r.t. α.

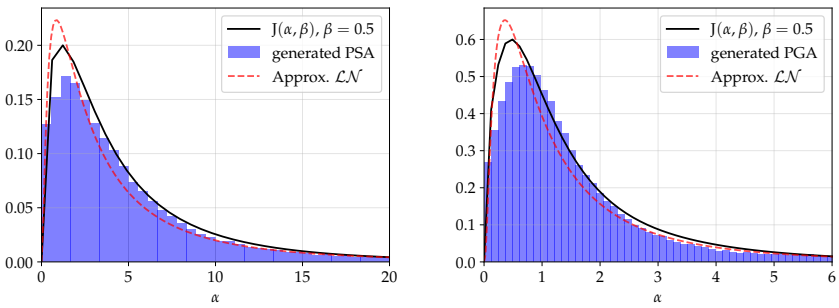

Figure 8.1: Comparison of a sectional view of the Jeffreys' prior w.r.t. α (black) with the histograms of the generated signals (blue) and the log-normal approximations (red) for both IMs (PSA in the left figure, PGA in the right figure).

In practice, the use of Markov Chain Monte-Carlo (MCMC) methods (see section 8.4) to sample the *a posteriori* distribution means that the prior must be evaluated (up to a multiplicative constant) many times during the calculations. Considering the computational complexity stemming from the integrals that need to be calculated, it was decided to perform the evaluations of the prior on an experimental design based on a fine-mesh grid of Θ (here $(0,+\infty)^2$) and to build an interpolated approximation of the Jeffreys prior matching this design. This strategy is more suitable for our numerical applications and very tractable because Θ is a

two-dimensional domain. Figure 8.2 shows plots of the Jeffreys prior for the two IMs considered. To be precise, 2000×2000 prior values were computed for $(\alpha, \beta) \in [10^{-5}, 50] \times [10^{-5}, 2]$ in the case of the PSA, and $(\alpha, \beta) \in [10^{-5}, 10] \times [10^{-3}, 2]$ in the case of the PGA. These values are then processed in order to obtain a linear interpolation.

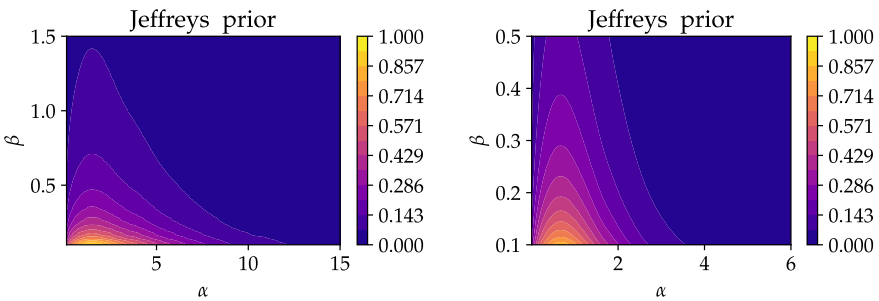

Figure 8.2: The Jeffreys priors calculated from PSA (left) and PGA (right) on subdomains of $\Theta = (0, +\infty)^2$.

The computational complexity of the Jeffreys prior is not in itself a major drawback. Since it depends exclusively on the distribution of the IM, the initial cost of the Jeffreys prior's complex calculation would quickly be recovered in installations on the scale of a nuclear power plant, where it is necessary to determine the fragility curves of many Structures and Components (SCs). Compared to methodologies that aim to define a prior based on mechanical calculations which are, by definition, specific to SCs, the generic character of the Jeffreys prior is a clear advantage. This will be explored in the applications section of this paper (section 8.7). Moreover, the Jeffreys prior is completely defined and does not depend on any additional subjective choices.

8.3.3 Thorough study of the prior's decay rates

Since the Jeffreys prior is known to be often improper, we propose in this section to determine its asymptotic decay rates. In the section 8.5, the decay rates of the likelihood will be computed as well, leading us to elucidate whether the posteriors yielded by Jeffreys are propers or not.

We focus on the asymptotic behavior of Jeffreys in four bounds of the space Θ, the study is compiled in the following propositions. The statements rely on the assumption below about the distribution of the IM. The comparison in fig. 8.1 illustrates the correctness of this assumption.

Assumption 8.1. The IM is distributed according to a log-normal distribution, i.e., there exists $\mu \in \mathbb{R}$ and $\sigma \in (0, +\infty)$ such that

$$h(a) = \frac{1}{\sqrt{2\pi\sigma^2 a}} \exp\left(-\frac{(\log a - \mu)^2}{2\sigma^2}\right). \tag{8.15}$$

Proposition 8.1. *Fixing $\alpha > 0$, there exists a $D'(\alpha) > 0$ such that*

$$J(\theta) \underset{\beta \to 0}{\sim} \frac{D'(\alpha)}{\beta}. \tag{8.16}$$

Proposition 8.2. *There exists a constant $E' > 0$ such that for any $\alpha > 0$*

$$J(\theta) \underset{\beta \to \infty}{\sim} \frac{E'}{\alpha\beta^3}. \tag{8.17}$$

Proposition 8.3. *Fixing $\beta > 0$, there exists a $G''(\beta) > 0$ such that*

$$J(\theta) \underset{|\log\alpha|\to\infty}{\sim} G''(\beta)\frac{|\log\alpha|}{\alpha}\exp\left(-\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}\right). \tag{8.18}$$

To prove these statements, we rely on the two following lemmas, which provide upper bounds for the function $\gamma \mapsto [\Phi(\gamma)(1-\Phi(\gamma))]^{-1}$.

Lemma 8.1. *For any $\gamma \in \mathbb{R}$, $[\Phi(\gamma)(1-\Phi(\gamma))]^{-1} \leq 4\exp\left(2\gamma^2/\pi\right)$.*

Lemma 8.2. *For any $\gamma \in \mathbb{R}$,*

$$\Phi(\gamma)(1-\Phi(\gamma)) \geq \frac{\sqrt{2/\pi}\exp(-\gamma^2/2)}{(|\gamma|+\sqrt{\gamma^2+4})}. \tag{8.19}$$

Proof of lemma 8.1. From the following inequality about the erf function (Chu, 1955):

$$\forall\gamma > 0,\ \sqrt{1-e^{-\frac{\gamma^2}{2}}} \leq \mathrm{erf}(\gamma/\sqrt{2}) \leq \sqrt{1-e^{-2\frac{\gamma^2}{\pi}}}, \tag{8.20}$$

we can deduce that, for any $\gamma > 0$,

$$\begin{aligned} e^{-\frac{2\gamma^2}{\pi}} &\leq 1-\mathrm{erf}(\gamma/\sqrt{2})^2 \leq e^{-\frac{\gamma^2}{2}}, \\ \frac{1}{4}e^{-\frac{2\gamma^2}{\pi}} &\leq \frac{1}{4}\left(1-\mathrm{erf}\left(\gamma/\sqrt{2}\right)\right)\left(1+\mathrm{erf}\left(\gamma/\sqrt{2}\right)\right) \leq \frac{1}{4}e^{-\frac{\gamma^2}{2}}, \end{aligned} \tag{8.21}$$

the middle term being equal to $\Phi(\gamma)(1-\Phi(\gamma))$. This implies that:

$$[\Phi(\gamma)(1-\Phi(\gamma))]^{-1} \leq 4e^{\frac{2\gamma^2}{\pi}}, \tag{8.22}$$

hence the result for $\gamma > 0$.

While it is clear that the inequality still stands for $\gamma = 0$, notice that from $\Phi(-\gamma) = 1-\Phi(\gamma)\ \forall\gamma \in \mathbb{R}$ it follows that $\gamma \mapsto \Phi(\gamma)(1-\Phi(\gamma))$ is an even function. Thus, the inequality still stands for any $\gamma < 0$; this concludes the proof of the lemma. □

Proof of lemma 8.2. Komatsu's inequality (Ito and McKean, 1974, p. 17):

$$\forall\gamma > 0,\ \frac{2}{\sqrt{\gamma^2+4}+\gamma} \leq e^{\frac{\gamma^2}{2}}\int_\gamma^\infty e^{-\frac{t^2}{2}}dt \leq \frac{2}{\sqrt{\gamma^2+2}+\gamma} \tag{8.23}$$

implies

$$\forall\gamma > 0,\ \frac{2e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+4}+\gamma} \leq \sqrt{2\pi}(1-\Phi(\gamma)) \leq \frac{2e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+2}+\gamma}. \tag{8.24}$$

Since $0 < \Phi < 1$ it follows for $\gamma > 0$ that:

$$\Phi(\gamma)(1-\Phi(\gamma)) \geq \frac{2e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+4}+\gamma}\left(1-\frac{2e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+2}+\gamma}\right) \geq \frac{\sqrt{2/\pi}e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+4}+\gamma}. \tag{8.25}$$

Finally, since $\Phi(-\gamma) = 1 - \Phi(\gamma)$, $\gamma \mapsto \Phi(\gamma)(1-\Phi(\gamma))$ is an even function and we obtain for any $\gamma \in \mathbb{R}$

$$\Phi(\gamma)(1-\Phi(\gamma)) \geq \frac{\sqrt{2/\pi}e^{-\frac{\gamma^2}{2}}}{\sqrt{\gamma^2+4}+|\gamma|}. \tag{8.26}$$

□

Proofs of the propositions

Proof of proposition 8.1. Let $\alpha > 0$. For $0 \leq k \leq 2$, let us consider $A_{k1,k2} = A_{k1} + A_{k2}$ with A_{kj} defined eq. (8.13):

$$A_{k1,k2} = \int_0^\infty \log^k \frac{a}{\alpha} \frac{\Phi'(\gamma(a))^2}{\Phi(\gamma(a))(1-\Phi(\gamma(a)))} h(a)da. \tag{8.27}$$

A substitution gives

$$A_{k1,k2} = \beta^{k+1} \int_{-\infty}^{\infty} F_{A_{k1,k2}}(\gamma)d\gamma, \quad \text{with} \quad F_{A_{k1,k2}}(\gamma) = \frac{\gamma^k}{2\sqrt{\pi^3\sigma^2}} \frac{e^{-\gamma^2} e^{-\frac{(\beta\gamma-\mu+\log\alpha)^2}{2\sigma^2}}}{\Phi(\gamma)(1-\Phi(\gamma))}. \tag{8.28}$$

Using lemma 8.1 an upper bound can be derived for $F_{A_{k1,k2}}$: for any $\gamma \in \mathbb{R}$, $\beta > 0$,

$$|F_{A_{k1,k2}}(\gamma)| \leq \tilde{C}|\gamma|^k e^{-\frac{1}{3}\gamma^2}, \tag{8.29}$$

which defines an integrable function on $\mathbb{R}$, $\tilde{C}$ being a constant independent of γ and β. Hence the limit

$$\lim_{\beta\to 0} \int_{-\infty}^{\infty} F_{A_{k1,k2}}(\gamma)d\gamma = \int_{-\infty}^{\infty} \frac{\gamma^k}{2\sqrt{\pi^3\sigma^2}} \frac{e^{-\gamma^2} e^{-\frac{(\mu-\log\alpha)^2}{2\sigma^2}}}{\Phi(\gamma)(1-\Phi(\gamma))} d\gamma. \tag{8.30}$$

The last integral is null when $k = 1$ since the integrand is odd in this case. When k is even, the integrand is positive valued almost everywhere, which implies that the integral is positive. From this, we can establish that $A_{k1,k2} \underset{\beta\to 0}{\sim} D_k(\alpha)\beta^{k+1}$ for some $D_k(\alpha) > 0$ if $k = 0, 2$, and $A_{11,12} \underset{\beta\to 0}{=} o(\beta^2)$.

Looking back at the Fisher information matrix, we can state that

$$\det \mathcal{I}(\theta) \underset{\beta\to 0}{=} \frac{D_0(\alpha)D_2(\alpha)}{\alpha^4\beta^2} + o\left(\frac{1}{\beta^2}\right). \tag{8.31}$$

Finally, we obtain:

$$J(\theta) \underset{\beta\to 0}{\sim} \frac{D'(\alpha)}{\beta}, \tag{8.32}$$

where $D'(\alpha) > 0$ is a constant independent of β. □

Proof of proposition 8.2.

The asymptotic expansion of the erf function in 0 is:

$$\text{erf}(\gamma) \underset{\gamma\to 0}{=} \frac{2}{\sqrt{\pi}}\gamma + O(\gamma^2), \tag{8.33}$$

which allows us to state the behavior of $\Phi(\gamma)$ when $\gamma \longrightarrow 0$:

$$\Phi(\gamma) \underset{\gamma\to 0}{=} \frac{1}{2} + \frac{1}{\sqrt{2\pi}}\gamma + O(\gamma^2). \tag{8.34}$$

Let us now fix $\alpha > 0$ and consider $A_{k1,k2} = A_{k1} + A_{k2}$ written as:

$$A_{k1,k2} = \int_{-\infty}^{\infty} \tilde{F}_{A_{k1,k2}}(x)dx, \quad \text{with} \quad \tilde{F}_{A_{k1,k2}}(x) = \frac{x^k}{2\sqrt{\pi^3\sigma^2}} \frac{e^{-\frac{x^2}{\beta^2}} e^{-\frac{(x-\mu+\log\alpha)^2}{2\sigma^2}}}{\Phi(\beta^{-1}x)(1-\Phi(\beta^{-1}x))}. \tag{8.35}$$

Let us note the convergence of $\tilde{F}_{A_{k1,k2}}(x)$ towards an integrable function when $\beta \longrightarrow \infty$. Moreover, lemma 8.1 allows us to bound $\tilde{F}_{A_{k1,k2}}$:

$$|\tilde{F}_{A_{k1,k2}}(x)| \leq \frac{2|x|^k}{\sqrt{\pi^3\sigma^2}} e^{-\frac{(x-\mu+\log\alpha)^2}{2\sigma^2}} e^{2\frac{x^2}{\pi\beta^2}} \leq \frac{2|x|^k}{\sqrt{\pi^3\sigma^2}} e^{-\frac{(x^2-2(\mu-\log\alpha))^2}{4\sigma^2}} e^{\frac{(\mu+\log\alpha)^2}{2\sigma^2}}, \tag{8.36}$$

for any $x \in \mathbb{R}$ and $\beta > 2\sigma/\sqrt{\pi}$. This dominating function is integrable on $\mathbb{R}$. Thus, when $\beta \longrightarrow \infty$, $A_{k1,k2}$ admits a limit expressed by:

$$\lim_{\beta\to\infty} A_{k1,k2} = E_k(\alpha) = \int_{-\infty}^{\infty} \frac{2x^k}{\sqrt{\pi^3\sigma^2}} e^{-\frac{(x-\mu+\log\alpha)^2}{2\sigma^2}} dx = \frac{2\sqrt{2}}{\pi}\mathbb{E}[X^k], \tag{8.37}$$

with $X \sim \mathcal{N}(\mu - \log\alpha, \sigma^2)$. Recalling the expression of the Jeffreys prior:

$$J(\theta)^2 = \left| \frac{1}{\alpha^2\beta^6} A_{01,02}A_{21,22} - \frac{1}{\alpha^2\beta^6} A_{11,12}^2 \right|, \tag{8.38}$$

we can deduce that it is equivalent to $(E_0(\alpha)E_2(\alpha) - E_1^2(\alpha))/\alpha^2\beta^6$ when $\beta \longrightarrow \infty$. Finally,

$$J(\theta) \underset{\beta\to\infty}{\sim} \frac{E'}{\alpha\beta^3}, \tag{8.39}$$

with $E' = \sqrt{E_0(\alpha)E_2(\alpha) - E_1^2(\alpha)} = 2\sigma/\pi$.

□

Proof of proposition 8.3.

As a preliminary result, let us recall that $\mathrm{erf}(x) \underset{x\to\infty}{=} 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} + o\left(\frac{e^{-x^2}}{x}\right)$, to establish

$$\Phi(\gamma)(1-\Phi(\gamma)) \underset{|\gamma|\longrightarrow\infty}{\sim} \frac{e^{\frac{-\gamma^2}{2}}}{|\gamma|\sqrt{2\pi}}. \tag{8.40}$$

Let us consider $A_{k1,k2} = A_{k1} + A_{k2}$:

$$A_{k1,k2} = C' \int_0^{\infty} \left(\log\frac{a}{\alpha}\right)^k \frac{e^{-\frac{1}{\beta^2}\log^2\frac{a}{\alpha}} e^{-\frac{(\log a-\mu)^2}{2\sigma^2}}}{\Phi(\beta^{-1}\log\frac{a}{\alpha})\left(1-\Phi(\beta^{-1}\log\frac{a}{\alpha})\right)} \frac{da}{a}, \tag{8.41}$$

denoting $C' = \sqrt{4\pi^3\sigma^2}^{-1}$. By substituting

$$\nu = \log a - \frac{\sigma^2}{\sigma^2+\beta^2}\log\alpha - \frac{\beta^2}{\sigma^2+\beta^2}\mu = \log a - r\log\alpha - s\mu, \tag{8.42}$$

we obtain

$$A_{k1,k2} = C' \int_{-\infty}^{\infty} \hat{F}_{A_{k1,k1}}(\nu)d\nu, \tag{8.43}$$

with

$$\hat{F}_{A_{k1,k1}}(\nu) = (\nu + (r-1)\log\alpha + s\mu)^k \frac{e^{-\frac{(\nu+(r+1)\log\alpha+s\mu)^2}{\beta^2}} e^{-\frac{(\nu+r\log\alpha+(s-1)\mu)^2}{2\sigma^2}}}{[\Phi(1-\Phi)]\,(h_\beta(\nu))}, \tag{8.44}$$

and where $h_\beta(\nu) = \beta^{-1}(\nu + (r-1)\log\alpha + s\mu)$. Using eq. (8.40), we obtain

$$[\Phi(1-\Phi)]\,(h_\beta(\nu)) \underset{|\log\alpha|\to\infty}{\sim} \frac{\beta e^{-\frac{(\nu+(r-1)\log\alpha+s\mu)^2}{2\beta^2}}}{|\nu + (r-1)\log\alpha + s\mu|\sqrt{2\pi}}. \tag{8.45}$$

Then for a clear understanding of the asymptotic behavior of $\hat{F}_{A_{k1,k2}}$, let us compute

$$\begin{aligned} &-\frac{(\nu + (r-1)\log\alpha + s\mu)^2}{2\beta^2} - \frac{(\nu + r\log\alpha + (s-1)\mu)^2}{2\sigma^2} \\ &= -\left(\frac{1}{2\beta^2} + \frac{1}{2\sigma^2}\right)\nu^2 - \left(\frac{((r-1)\log\alpha + s\mu)^2}{2\beta^2} + \frac{(r\log\alpha + (s-1)\mu)^2}{2\sigma^2}\right) \\ &= -\left(\frac{1}{2\beta^2} + \frac{1}{2\sigma^2}\right)\nu^2 - \frac{(\log\alpha - \mu)^2}{2(\beta^2+\sigma^2)}. \end{aligned} \tag{8.46}$$

We expand $\hat{F}_{k1,k2}(\nu) = \sum_{j=1}^{k} C_k^j (r-1)^j \log^j\alpha(\nu + s\mu)^{k-j} g(\nu) = \sum_{j=0}^{k} \hat{F}_{k1,k2}^{(j)}(\nu)$, with $g(\nu)$ defined as

$$g(\nu) = \frac{e^{-\frac{(\nu+(r+1)\log\alpha+s\mu)^2}{\beta^2}} e^{-\frac{(\nu+r\log\alpha+(s-1)\mu)^2}{2\sigma^2}}}{[\Phi(1-\Phi)]\,(\beta^{-1}(\nu + (r-1)\log\alpha + s\mu))}. \tag{8.47}$$

Therefore, by combining eqs. (8.45) and (8.46), we obtain that $\hat{F}_{k1,k2}^{(j)}$ satisfies

$$\hat{F}_{k1,k2}^{(j)}(\nu) e^{\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} (\log\alpha)^{-j} |\log\alpha|^{-1} \underset{|\log\alpha|\to\infty}{\longrightarrow} (\sqrt{2\pi}\beta)^{-1} C_k^j (-1)^j (1-r)^{j+1} (\nu + s\mu)^{k-j} e^{-\left(\frac{1}{2\beta^2}+\frac{1}{2\sigma^2}\right)\nu^2}. \tag{8.48}$$

Using lemma 8.2, we can also define an integrable upper bound for the above function as:

$$|\hat{F}_{k1,k2}^{(j)}(\nu)| e^{\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} |\log\alpha|^{-j-1} \leq \frac{\sqrt{2/\pi} e^{-\left(\frac{1}{2\beta^2}+\frac{1}{2\sigma^2}\right)\nu^2}}{|h_\beta(\nu)| + \sqrt{h_\beta(\nu)^2 + 4}} \leq \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{1}{2\beta^2}+\frac{1}{2\sigma^2}\right)\nu^2}. \tag{8.49}$$

Thus, we can switch the limits and the integration, and the following results are obtained:

$$C''\beta A_{01,02} e^{\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} \underset{\log\alpha\to\infty}{=} (1-r)G\log\alpha - s\mu G + o(1) \tag{8.50}$$

$$C''\beta A_{11,12} e^{\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} \underset{\log\alpha\to\infty}{=} -(1-r)^2 G\log^2\alpha - s^2\mu^2 G + 2(1-r)s\mu G\log\alpha - G' + o(1), \tag{8.51}$$

$$\begin{aligned} C''\beta A_{22,22} e^{\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} \underset{\log\alpha\to\infty}{=} &(1-r)^3 G\log^3\alpha - s^3\mu^3 G - 3(1-r)^2 s\mu G\log^2\alpha \\ &- 3(1-r)s^2\mu^2 G\log\alpha + 3(1-r)G'\log\alpha + o(1), \end{aligned} \tag{8.52}$$

with $C'' = (C'\sqrt{2\pi})^{-1}$, $G = \sigma\beta\sqrt{2\pi(\beta^2+\sigma^2)^{-1}}$ and $G' = G^2/2\pi$. This way

$$\begin{aligned}(C'')^2\alpha^2\beta^8 J(\theta)^2 e^{2\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}} \underset{\log\alpha\to\infty}{=} & 3(r-1)^2 s^2\mu^2 G^2\log^2\alpha + 3(r-1)^2 s^2\mu^2 G^2\log^2\alpha \\ & + 3(r-1)^2 GG'\log^2\alpha - 4(r-1)^2 s^2\mu^2 G^2\log^2\alpha \\ & - 2(r-1)^2 s^2\mu^2 G^2\log^2\alpha - 2(r-1)^2 GG'\log^2\alpha + o(\log^2\alpha).\end{aligned} \tag{8.53}$$

Note that the above equality is still valid when $\log\alpha \longrightarrow -\infty$. Finally,

$$J(\theta) \underset{|\log\alpha|\to\infty}{\sim} G''(\beta)\frac{|\log\alpha|}{\alpha}\exp\left(-\frac{(\log\alpha-\mu)^2}{2\beta^2+2\sigma^2}\right), \tag{8.54}$$

with

$$G''(\beta) = C''^{-1}(r-1)^2 GG'\beta^{-4} = \frac{2\sigma^3\beta^3}{\sqrt{\pi}(\sigma^2+\beta^2)^{7/2}}. \tag{8.55}$$

□

This study states that the Jeffreys prior is improper, since it is not integrable w.r.t. β.

8.4 Competing approaches and estimation tools

8.4.1 Bayesian estimates of the seismic fragility curve

The most relevant method in order to benefit from the Bayesian theory and the reference prior construction presented in section 8.2 consists in deriving the posterior defined in eq. (8.3). When the latter is proper, it then becomes possible to generate, according to that distribution, samples of θ conditioned on the observed data. These *a posteriori* generations of θ can be obtained using MCMC methods. In this study, we have implemented an adaptive Metropolis-Hastings (M-H) algorithm with a Gaussian transition kernel and a covariance adaptation process (following the suggestions of Haario, Saksman, and Tamminen, 2001). Such an algorithm allows sampling from a probability density known up to a multiplicative constant. In this context, the *a posteriori* samples of θ can be used to define credibility intervals for the probit-lognormal estimations of the fragility curves.

8.4.2 Competing prior taken from the literature

For comparison purposes, we selected the prior suggested by Straub and Der Kiureghian (2008) —called the SK prior— whose density is defined as the product of a normal distribution for $\ln(\alpha)$ and the improper distribution $1/\beta$ for β, namely:

$$\pi_{SK}(\theta) \propto \frac{1}{\alpha\beta}\exp\left(-\frac{(\log\alpha-\mu)^2}{2\sigma^2}\right). \tag{8.56}$$

In Straub and Der Kiureghian, 2008, the parameters μ and σ of the log-normal distribution are chosen to generate a non-informative prior. For a fair comparison with the approach proposed in this paper, we decided to pick μ and σ as equal to the mean and the standard deviation of the logarithm of the IM. This choice is consistent with the fact that the Jeffreys prior is similar to a log-normal distribution with these parameters (see fig. 8.1). The prior $\pi_{SK}(\theta)$ is plotted in fig. 8.3 for both the PSA and the PGA as IM.

An analysis of the posterior obtained from the SK prior is given in section 8.5. It shows that the posterior is always improper, which jeopardizes the validity of any *a posteriori* estimation using MCMC methods. This could however be mitigated by truncating w.r.t. β, which we will do in the numerical application conducted in this work. This issue persists in the authors' original framework, which is slightly different from ours. This is confirmed in the appendix 8.8.

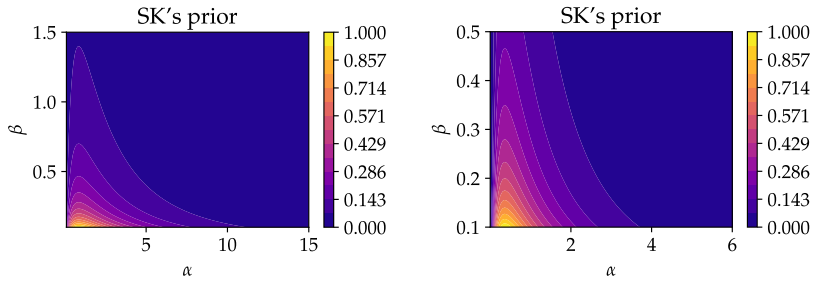

Figure 8.3: Prior suggested by Straub and Der Kiureghian (2008), expressed in eq. (8.56) and scaled to a log-normal estimation of the IM distribution, for both the PSA (left) and the PSA (right) as IM.

Asymptotic comparison of the Jeffrey and SK priors By comparing the Jeffreys and SK prior asymptotics, it can be observed that:

- Regarding the asymptotics w.r.t. β, while the divergence rates are the same when $\beta \longrightarrow 0$, the Jeffreys prior performs better when $\beta \longrightarrow \infty$:

$$J(\theta) \underset{\beta \to \infty}{\propto} \beta^{-2} \pi_{SK}(\theta). \tag{8.57}$$

  Consequently, the SK posterior will result in higher probabilities for high values of β compared to the Jeffreys prior.

- Regarding the asymptotics w.r.t. α, both are asymptotically close to a log-normal distribution, with a slight "disadvantage" for the Jeffreys prior, for which the asymptotic variance is derived by adding β^2 to the variance of the SK prior. This means that while for small values of β (smaller than σ), both priors remain comparable w.r.t. α, the Jeffreys prior gives higher probabilities to α outliers when β also has a high value. However, as seen above, the probability for large values of β is quite low for the Jeffreys prior compared to the SK prior.

8.4.3 Maximum likelihood estimation with bootstrapping

The most common non-Bayesian estimator of the seismic fragility curves under the probit-lognormal modeling is the maximum likelihood estimator (MLE). This estimator has been introduced in the chapter 7 (section 7.3). It is expressed as follows for the given of observations $(\mathbf{z}^k, \mathbf{a}^k)$:

$$\theta^{\text{MLE}}(\mathbf{z}^k, \mathbf{a}^k) = \underset{\theta \in \Theta}{\arg\max}\, \ell_k(\mathbf{z}^k | \mathbf{a}^k, \theta). \tag{8.58}$$

It is common to compute multiple MLE in order to obtain confidence intervals. Doing so, the bootstrap technique defines the stochastic estimator of θ:

$$\hat{\theta}_k^{\text{BMLE}}(\mathbf{z}^k, \mathbf{a}^k) = \theta^{\text{MLE}}((z_{U_i}, a_{U_i})_{i=1}^k), \tag{8.59}$$

where $U_1, \ldots, U_k$ are i.i.d. random variables distributed w.r.t. a uniform distribution in $\{1, \ldots, k\}$. This estimator is used to compute confidence intervals by estimating empirically its quantiles from many samples. This is a very common approach for fragility curves (e.g., Shinozuka, Feng, et al., 2000; Gehl, Douglas, and Seyedi, 2015; Baker, 2015). The convergence of the MLE and the relevance of this method are detailed in Vaart, 1992. However, the relevance of the bootstrap method is often limited by the irregularity of its results for small values of k (see e.g., Zentner, Gündel, and Bonfils, 2017).

8.5 Limits of the estimates given by the three approaches: the curse of degeneracy

In this section, we study the likelihood decay rates in order to (i) explain irregular phenomena commonly encountered with classical frequentist methods, and (ii) verify the proper characteristics of the posteriors in

the Bayesian framework. Our study of the likelihood's asymptotic behavior is summarized in the following proposition.

Proposition 8.4. *Let us consider $k > 1$ and a data sample $(\mathbf{a}^k, \mathbf{z}^k)$. Let us introduce the vectors $\mathbf{N} = (z_i \mathbb{1}_{a_i<\alpha} + (1-z_i)\mathbb{1}_{a_i>\alpha})_{i=1}^k$, $\log^2 \frac{\mathbf{a}^k}{\alpha} = (\log^2 \frac{a_i}{\alpha})_{i=1}^k$.*

- *Fixing $\alpha > 0$, then*

$$\ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) \underset{\beta\to\infty}{\longrightarrow} 2^{-k} \quad \textit{and} \quad \ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) \underset{\beta\to 0}{=} O\left(\beta^{|\mathbf{N}|} e^{-\frac{\mathbf{N}^T \log^2 \frac{\mathbf{a}^k}{\alpha}}{2\beta^2}}\right), \tag{8.60}$$

  where $|\mathbf{N}| = \sum_{i=1}^k N_i$.

- *Fixing $\beta > 0$, then*

$$\ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) \underset{\alpha\to 0}{=} O\left(|\log\alpha|^{|\mathbf{z}^k|-k} e^{-\frac{1}{2\beta^2}\sum_{i=1}^k (1-z_i)(\log a_i - \log\alpha)^2}\right) \tag{8.61}$$

  and

$$\ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) \underset{\alpha\to\infty}{=} O\left(\log(\alpha)^{-|\mathbf{z}^k|} e^{-\frac{1}{2\beta^2}\sum_{i=1}^k z_i(\log a_i - \log\alpha)^2}\right), \tag{8.62}$$

  where $|\mathbf{z}^k| = \sum_{i=1}^k z_i$ is the number of failures in the observed sample.

Proof of proposition 8.4. As a reminder, the likelihood is expressed as:

$$\begin{aligned}\ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) &= \prod_{i=1}^k \Phi\left(\frac{\log a_i - \log\alpha}{\beta}\right)^{z_i}\left(1-\Phi\left(\frac{\log a_i - \log\alpha}{\beta}\right)\right)^{1-z_i} \\ &= \exp\left[\sum_{i=1}^k (z_i \log\Phi(\gamma_i) + (1-z_i)\log(1-\Phi(\gamma_i)))\right],\end{aligned} \tag{8.63}$$

denoting $\gamma_i = \beta^{-1}\log\frac{a_i}{\alpha}$.

To treat the case where $\beta \longrightarrow \infty$ we can observe that while α is fixed, the quantities $\Phi(\gamma_i)$ all converge to $1/2$. The product of those limits gives the limit $\ell_k(\mathbf{z}^k|\mathbf{a}^k,\theta) \underset{\beta\to\infty}{\longrightarrow} 2^{-k}$.

For the other cases, it should be reminded that $\Phi(x) = \frac{1}{2}(1+\mathrm{erf}(x/\sqrt{2}))$ and $\mathrm{erf}(x) \underset{x\to\infty}{=} 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} + o\left(\frac{e^{-x^2}}{x}\right)$, which leads to

$$\Phi(x) \underset{x\to\infty}{=} 1 - \frac{e^{-\frac{x^2}{2}}}{x\sqrt{2\pi}} + o\left(\frac{e^{-\frac{x^2}{2}}}{x}\right). \tag{8.64}$$

Let us consider an $i \in \{1, \dots, k\}$ and compute

$$\begin{aligned} & z_i \log \Phi(\gamma_i) + (1 - z_i) \log(1 - \Phi(\gamma_i)) \\ & \underset{\gamma_i \to \infty}{=} z_i \log \left(1 - \frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i \sqrt{2\pi}} + o\left(\frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i} \right) \right) + (1 - z_i) \log \left(\frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i \sqrt{2\pi}} + o\left(\frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i} \right) \right) \\ & \underset{\gamma_i \to \infty}{=} -z_i \frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i \sqrt{2\pi}} + (1 - z_i) \log \left(\frac{e^{-\frac{\gamma_i^2}{2}}}{\gamma_i \sqrt{2\pi}} \right) + o(1) \\ & \underset{\gamma_i \to \infty}{=} -(1 - z_i) \frac{\gamma_i^2}{2} - (1 - z_i) \log(\gamma_i \sqrt{2\pi}) + o(1). \end{aligned} \tag{8.65}$$

Using the relation $\Phi(-x) = 1 - \Phi(x)$, it follows that, in the case $a_i < \alpha$:

$$z_i \log \Phi(\gamma_i) + (1 - z_i) \log(1 - \Phi(\gamma_i)) \underset{\gamma_i \to -\infty}{=} -z_i \frac{\gamma_i^2}{2} - z_i \log(-\gamma_i \sqrt{2\pi}) + o(1). \tag{8.66}$$

Going back to the likelihood asymptotics, let us first suppose $\alpha > 0$ and $\beta \longrightarrow 0$. Thus, denoting the vectors $\mathbf{N} = (z_i \mathbb{1}_{a_i < \alpha} + (1 - z_i) \mathbb{1}_{a_i > \alpha})_{i=1}^k$ and $\log^2 \frac{\mathbf{a}^k}{\alpha} = (\log^2 \frac{a_i}{\alpha})_{i=1}^k$, we obtain

$$\ell_k(\mathbf{z}^k | \mathbf{a}' \theta) \underset{\beta \to 0}{=} \frac{C(\alpha)}{\sqrt{2\pi}^{|\mathbf{N}|}} \beta^{|\mathbf{N}|} e^{-\frac{\mathbf{N}^T \log^2 \frac{\mathbf{a}^k}{\alpha}}{2\beta^2} + o(1)} \underset{\beta \to 0}{=} O\left(\beta^{|\mathbf{N}|} e^{-\frac{\mathbf{N}^T \log^2 \frac{\mathbf{a}^k}{\alpha}}{2\beta^2}} \right), \tag{8.67}$$

where $|\mathbf{N}| = \sum_{i=1}^k N_i$ and $C(\alpha) = \prod_{i=1}^k \left| \log \frac{a_i}{\alpha} \right|^{N_i}$.

Let us then fix $\beta > 0$ to get

$$\begin{aligned} \ell_k(\mathbf{z}^k | \mathbf{a}^k, \theta) & \underset{\alpha \to \infty}{=} \frac{\beta^{|\mathbf{N}|}}{\sqrt{2\pi}^{|\mathbf{N}|}} \left(\prod_{i=1}^k \log \left(\frac{\alpha}{a_i} \right)^{-z_i} \right) \exp \left(-\frac{1}{2\beta^2} \sum_{i=1}^k z_i (\log a_i - \log \alpha)^2 + o(1) \right) \\ & \underset{\alpha \to \infty}{=} O\left(\log(\alpha)^{-|\mathbf{z}^k|} e^{-\frac{1}{2\beta^2} \sum_{i=1}^k z_i (\log a_i - \log \alpha)^2} \right), \end{aligned} \tag{8.68}$$

where $|\mathbf{z}^k| = \sum_{i=1}^k z_i$ is the number of failures in the observed sample. Similarly, we obtain

$$\begin{aligned} \ell_k(\mathbf{z}^k | \mathbf{a}^k, \theta) & \underset{\alpha \to 0}{=} \frac{\beta^{|\mathbf{N}|}}{\sqrt{2\pi}^{|\mathbf{N}|}} \left(\prod_{i=1}^k \log \left(\frac{a_i}{\alpha} \right)^{-(1 - z_i)} \right) \exp \left(-\frac{1}{2\beta^2} \sum_{i=1}^k (1 - z_i)(\log a_i - \log \alpha)^2 + o(1) \right) \\ & \underset{\alpha \to 0}{=} O\left(|\log \alpha|^{|\mathbf{z}^k| - k} e^{-\frac{1}{2\beta^2} \sum_{i=1}^k (1 - z_i)(\log a_i - \log \alpha)^2} \right). \end{aligned} \tag{8.69}$$

□

This statement expresses that, under general circumstances, the likelihood converges rapidly to 0 in every direction but the one where $\beta \longrightarrow \infty$. The latter is a concern, because it requires that the decay rate of the prior in that direction is proper to yield proper posteriors. Moreover, under some circumstances, the vector $\mathbf{N}$ is null. It is the case when we fall in one the three scenarios described in the following definition. In that case, we say that the likelihood is degenerate, and it does not converge towards 0 when $\beta \longrightarrow 0$.

Definition 8.1 (Likelihood degeneracy)**.** If the observed samples $(\mathbf{z}^k, \mathbf{a}^k)$ belong to one of the following three types:

type 1 : no failure is observed: $z_i = 0$ for any i;

- type 2 : only failures are observed: $z_i = 1$ for any i;

- type 3 : the failures and non-failures are partitioned into two disjoint subsets when classified according to their IM values: there exists $a \in \mathcal{A}$ such that for any i, j, $a_i < a < a_j \iff z_i \neq z_j$ (see the illustration in fig. 8.4);

then the likelihood is degenerate.

An example of degenerate likelihood is presented in fig. 8.4 with an illustration of this phenomenon. In the following, we deduce from this study the decay rates of the posterior densities yielded by the two prior considered in this work (the SK prior and the Jeffreys prior). In particular, we show that the Jeffreys prior may produce improper posteriors in degenerate scenarios. For this reason, this chapter addresses the estimation of seismic fragility curves from dataset issuing non-degenerate likelihoods.

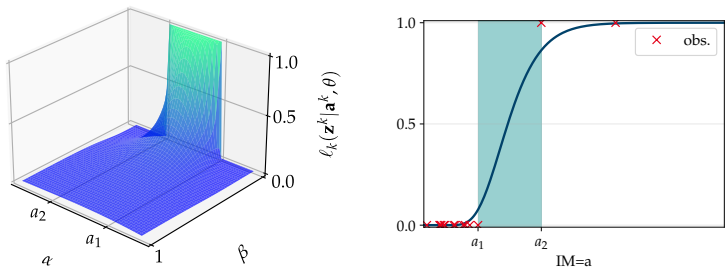

Figure 8.4: Example of a type 3 data sample $(\mathbf{z}^k, \mathbf{a}^k)$ which gives a degenerate likelihood. Graphs of (left) the likelihood given the degenerate data sample as a function of the tuple (α, β) and (right) the fragility curve (blue curve) according to which the points (red crosses) are sampled : each red cross is a tuple (z, a), where its X-axis equals a and its Y-axis equals z. On both figures, a_1 is the maximal observed IM value among "non-failures", and a_2 is minimal one among "failures".

Decay rates of the posterior densities and properness

The Jeffreys and SK priors are not proper with respect to β (see section 8.3 for the Jeffreys prior and eq?? for the SK prior). For the Jeffreys prior, the divergence and convergence rates with respect to β only make the resulting posterior proper when the prior is coupled with non-degenerate likelihoods. However, one can see that this is not the case for the SK posterior, which is not integrable w.r.t. β because of a convergence rate that is too low at $+\infty$. This prevents the validation of the MCMC estimates for this posterior, unless a truncation of the distribution is considered. Note that Straub and Der Kiureghian (2008) considered this prior within a Bayesian framework that slightly differs from ours. In the appendix 8.8, we confirm that the posterior is not proper even when derived in the exact framework of their paper.

8.6 Performance evaluation metrics

In order to assess and compare the performances of the proposed approaches, we propose different quantitative metrics. Beforehand, we suppose we have access to a reference fragility curve P_f^{ref}. In two of the case studies that we treat in the following, a large validation dataset is available and can be used to derive such reference using non-parametric Monte-Carlo estimates following the suggestion of Trevlopoulos, Feau, and Zentner (2019). This method was described in the chapter 7 (section 7.3) and implemented in the chapter 7 onto two case studies for which a large validation dataset was available.

Given observations $(\mathbf{z}^k, \mathbf{a}^k)$, we denote by $a \mapsto P_f^{|\mathbf{z}^k, \mathbf{a}^k}(a)$ the random process defined as the fragility curve conditional to the $(\mathbf{z}^k, \mathbf{a}^k)$. Its distribution inherits from the posterior distribution of θ in the case of the Bayesian method, or it inherits from the distribution of the bootstrap estimator $\hat{\theta}^{\text{BMLE}}(\mathbf{z}^k, \mathbf{a}^k)$ in the case of the MLE. For each value a, the r-quantile of the random variable $P_f^{|\mathbf{z}^k, \mathbf{a}^k}(a)$ is denoted by $q_r^{|\mathbf{z}^k, \mathbf{a}^k}(a)$. We then define:

- The quadratic error: $\mathcal{E}^{|\mathbf{z}^k,\mathbf{a}^k} = \mathbb{E}_{\theta|\mathbf{z}^k,\mathbf{a}^k}\left[\|P_f^{|\mathbf{z}^k,\mathbf{a}^k} - P_f^{\text{ref}}\|_{L^2}^2\right]$.
- The $1-r$ square credibility width: $\mathcal{W}^{|\mathbf{z}^k,\mathbf{a}^k} = \|q_{1-r/2}^{|\mathbf{z}^k,\mathbf{a}^k} - q_{r/2}^{|\mathbf{z}^k,\mathbf{a}^k}\|_{L^2}^2$.

The L^2 norm is taken on the domain $[a_{\min}, a_{\max}]$ that covers the available seismic signals:

$$\|P\|_{L^2}^2 = \int_{a_{\min}}^{a_{\max}} P(a)^2 da. \tag{8.70}$$

The L^2 norms are approximated using Simpson's interpolation, and the expectation are estimated by Monte-Carlo sums, using samples following either the posterior distributions or the distribution of the bootstrap estimator.

8.7 Numerical applications

In this section, we will examine three case studies. Two of them leverage the many simulation datasets available that have been previously computed for validation purposes. They will be used in the derivation of a reference fragility curve (as explained in section 8.6), and allow us to validate the corresponding probit-lognormal models. The three case studies were presented in the chapter 7. The first one is a nonlinear oscillator for which 10^5 nonlinear simulations have been implemented for validation purposes. The second case study deals with a piping system which is part of the secondary cooling system of a French Pressurized Water Reactor. Due to the high computational cost, only 10^4 simulations have been performed for this case. In both cases, estimations are performed using different testing data sets of a size k chosen as negligible compared to the size of the validation dataset. These testing datasets are taken from the set of available nonlinear dynamical simulation results. A third case study is eventually presented in order to showcase how our method could be applied to practical experiments.

8.7.1 Case study 1: the elasto-plastic oscillator

The first case study is a single-degree-of-freedom elasto-plastic oscillator with kinematic hardening. This mechanical system was presented in the chapter 7 (section 7.4.1). For this case study, 10^5 nonlinear dynamical simulations have been conducted to constitute a validation dataset. In fig. 8.5, we recall (i) a scheme of the oscillator, (ii) the validation dataset in the plan (PGA,EDP), (iii) reference fragility curves derived using non-parametric estimation from the validation dataset (see section 8.6). In this study, the failure is defined when the EDP exceeds a threshold $C = 0.8$ cm.

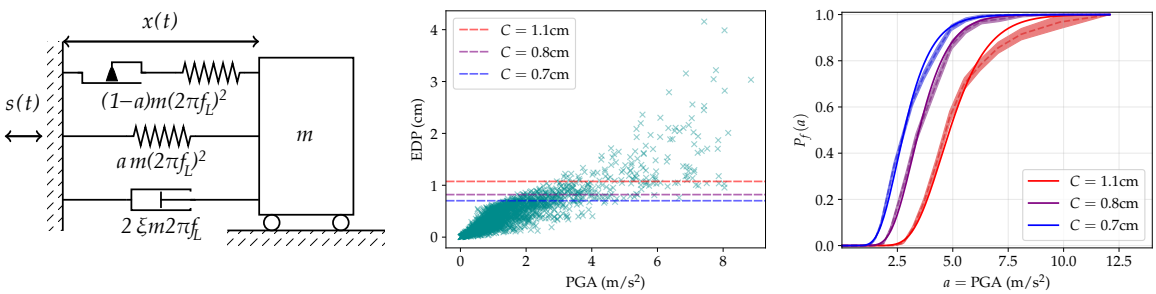

Figure 8.5: Illustration of the first case study. Left: scheme of an elasto-plastic oscillator of mass m. Middle: results of 10^5 non-linear dynamical simulations. Right: reference non-parametric fragility curves (dashed lines) surrounded by their 95%-confidence intervals, compared with the probit-lognormal fragility curves derived by MLE from the validation dataset of size 10^5. Different threshold C are considered, each yields different proportions of failures in the validation dataset: resp. 95% (red), 90% (purple) and 85% (blue).

Examples of fragility curve estimations are shown in fig. 8.6. They are obtained from 5000 samples of θ generated with the implemented statistical methods (see section 8.4), which are based on two observation

sets of sizes $k = 20$ and $k = 30$. Although the nature of the two intervals compared is different —credibility interval for the Bayesian framework and confidence interval for the MLE—, these results clearly illustrate the advantage of the Bayesian framework over the MLE for small samples. With the MLE, irregularities characterized by null estimates of β appear, resulting in "vertical" confidence intervals. In section 8.5, we established that the likelihood is easily maximized for $\beta = 0$ when samples are partitioned into two disjunct subsets when classified according to IM values: one subset for which there is no failure and one for which there is failure. Moreover, when few failures are observed in the initial sample, the bootstrap technique can lead to the consideration of many sub-samples that maximize the likelihood at $\beta = 0$. This is better evidenced by an examination of the raw values of θ generated in fig. 8.7. The degenerate β values resulting from the MLE appear clearly but, although it should theoretically also be affected, the Bayesian framework shows no evidence of a similar phenomenon for this type of samples.

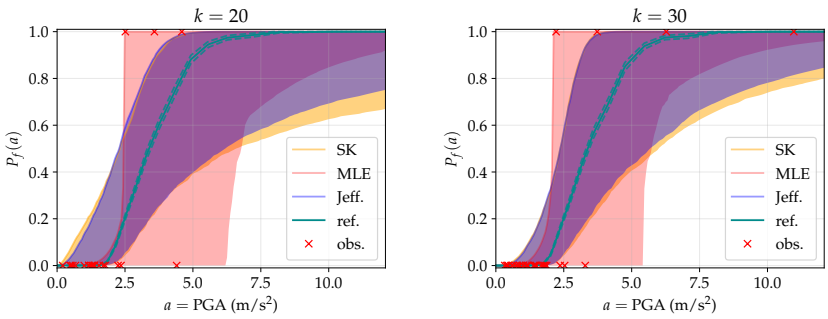

Figure 8.6: 95% credibility (for Bayesian estimation) or confidence (for the MLE) intervals of fragility curve estimations for the elasto-plastic oscillator, using the methods described in section 8.4. The cyan curve is the reference P_f^{ref} suggested in section 8.6, surrounded by its 95%-confidence interval (dashed lines). The red crosses represent the observed data.

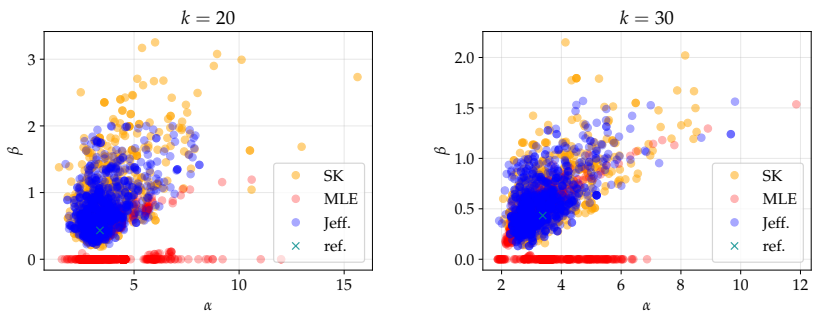

Figure 8.7: Scatter plots of the generated θ for the estimation of the fragility curves presented in fig. 8.6 for the elasto-plastic oscillator. For all three statistical methods, we plotted 5000 $\theta = (\alpha, \beta)$ estimated with two observed datasets of size $k = 20$ (left figure) and $k = 30$ (right figure). The cyan cross represents θ^{MLE} derived from the 10^5 samples composing the validation dataset.

Since the SK prior is calibrated to look like the Jeffreys prior, fig. 8.6 shows a strong similarity between the Bayesian estimations of the fragility curves obtained with these two priors. However, fig. 8.7 shows that many outliers are obtained with the SK prior. These values explain why the credibility intervals of the fragility curves are larger with the SK prior when $k = 20$. This observation is supported theoretically by the calculation provided in section 8.3. There is actually a better convergence of the Jeffreys prior toward 0 when $\beta \longrightarrow \infty$. This superior asymptotic behavior obviously results in posteriors that happen to assign a lower probability to outlier points (a phenomenon particularly noticeable when the data sample is small) as well as to the weight of the likelihood within the posterior.

For a better understanding of this phenomenon, we calculated the quantitative metrics defined in section 8.6. For any k ranging from 15 to 100, we considered $m = 200$ different draws of observations

$(\mathbf{z}^k_{(j)}, \mathbf{a}^k_{(j)})_{j=1}^m$ in order to derive the average values and 95%-confidence intervals of the metrics $\mathcal{E}^{|\mathbf{z}^k_{(j)},\mathbf{a}^k_{(j)}}$, $\mathcal{W}^{|\mathbf{z}^k_{(j)},\mathbf{a}^k_{(j)}}$, $j \in \{1, \ldots, m\}$, for each of the three methods. They are plotted in fig. 8.8. Firstly, these diagrams demonstrate the benefits of the Bayesian framework compared to the MLE approach for small observation sets. Secondly, the compared performance of the Jeffreys and SK posteriors is highlighted by the confidence interval endpoints of the quadratic error and the credibility interval. Specifically, the latter highlights the effect of the superior asymptotic behavior of the Jeffreys prior along the width of the credibility interval. It shows variations similar to but smaller than the SK prior, thus highlighting its capacity to generate fewer outliers for the pair (α, β), as expected.

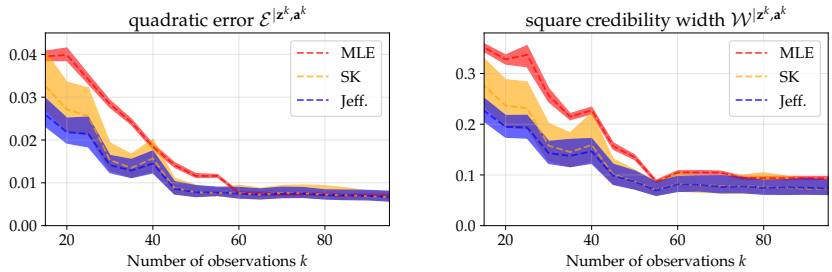

Figure 8.8: Average values of the performance evaluation metrics $\mathcal{E}^{|\mathbf{z}^k,\mathbf{a}^k}$ (left) and $\mathcal{W}^{|\mathbf{z}^k,\mathbf{a}^k}$ (right) for the elasto-plastic oscillator computed by replications of the experiment with different observation sets and for different sample sizes, ranging from $k = 15$ to 100. In each figure, the shaded areas show the 95%-confidence intervals of the metrics.

8.7.2 Case study 2: the piping system from a pressurized reactor

This second case study deals with a piping system which is part of the secondary cooling system of a French Pressurized Water Reactor. This piping system was presented in details in the chapter 7 (section 7.4.2). Nonlinear dynamical simulations of this system's response have been carried out for $8 \cdot 10^4$ of the seismic signals we dispose. The resulting validation dataset in the plan (PGA,EDP) is depicted in fig. 8.9, along with a picture of the system's mock-up and reference fragility curves derived using non-parametric estimation from the validation set (as explicated in section 8.6). In this study, the failure is defined when the EDP exceeds a threshold $C = 3.8°$.

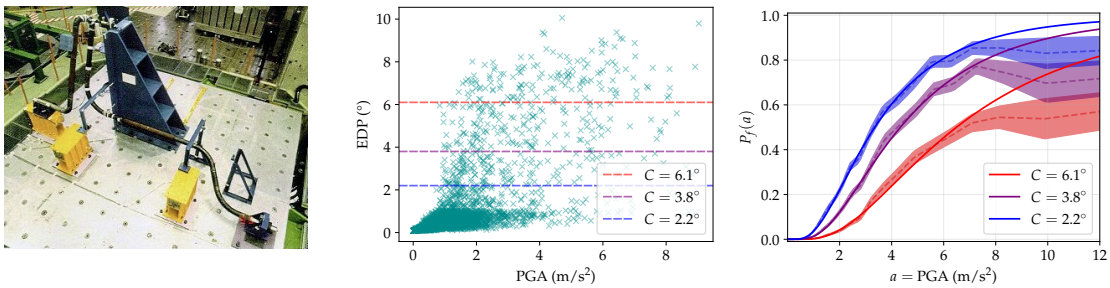

Figure 8.9: Illustration of the second case study. Left: picture of the mock-up on the Azalee shaking table of CEA. Middle: results of $8 \cdot 10^4$ non-linear dynamical simulations. Right: reference non-parametric fragility curves (dashed lines) surrounded by their 95%-confidence intervals, compared with the probit-lognormal fragility curves derived by MLE from the validation dataset of size $8 \cdot 10^4$. Different thresholds C are considered, each yields different proportions of failures in the validation dataset: resp. 95% (red), 90% (purple) and 85% (blue).

Estimations similar to the first case study's were performed here, and were found to highlight the same trends. As expected, for sets of 5000 values of θ —generated with each statistical method considered in this work— and for two sample sizes $k = 20$ and $k = 30$ of nonlinear dynamical simulations, fig. 8.10 shows

the superiority of the Bayesian framework over the coupled MLE and bootstrap approach. Just like in the case study with the oscillator, irregularities appear with the MLE-based approach: the confidence intervals are similarly "quasi-vertical", reflecting the fact that many estimations of β are equal to 0. Moreover, the credibility intervals are wider with the SK prior than with the Jeffreys prior, which here too can be interpreted as an increased number of outliers of θ being generated with the SK prior. These observations are clearly supported by the results presented in fig. 8.11.

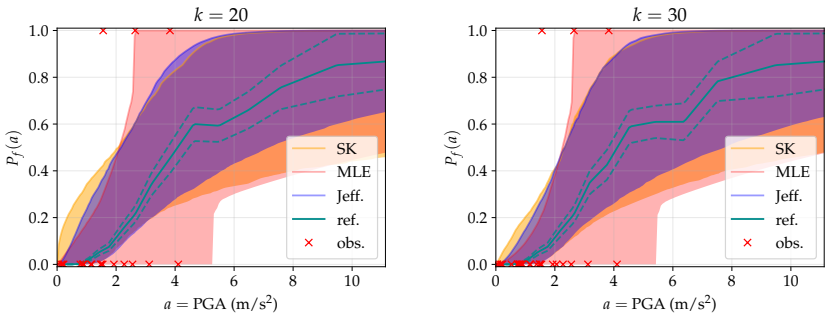

Figure 8.10: 95% credibility (for Bayesian estimation) or confidence (for the MLE) intervals of fragility curve estimations for the piping system, using the methods described in section 8.4. The cyan curve is the reference P_f^{ref} suggested in section 8.6, surrounded by its 95%-confidence interval (dashed lines). The red crosses represent the observed data.

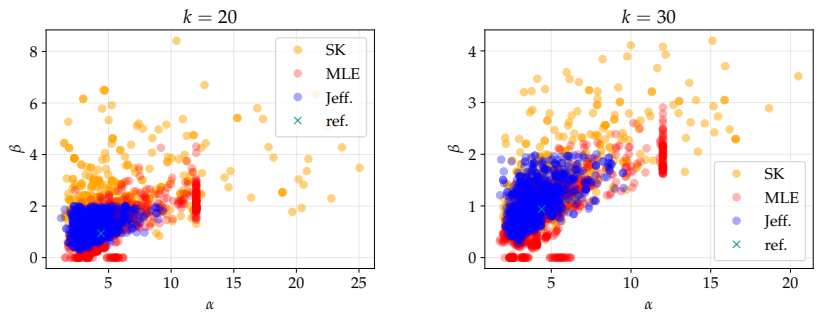

Figure 8.11: Scatter plots of the generated θ for the estimation of the fragility curves presented in fig. 8.6 for the piping system. For all three statistical methods, we plotted 5000 $\theta = (\alpha, \beta)$ estimated with two observed datasets of size $k = 20$ (left figure) and $k = 30$ (right figure). The cyan cross represents θ^{MLE} derived from the 10^5 samples composing the validation dataset.

For a more complete overview of their relative performances, the evaluation metrics described in section 8.6 have been computed in the same way as for the first case study: $m = 200$ draws of data samples $(\mathbf{z}^k_{(j)}, \mathbf{a}^k_{(j)})_{j=1}^m$ have been randomly chosen to compute, for any value of k ranging from 15 to 100, the average values and 95%-confidence intervals of the metrics $\mathcal{E}^{|\mathbf{z}^k_{(j)}, \mathbf{a}^k_{(j)}}$, $\mathcal{W}^{|\mathbf{z}^k_{(j)}, \mathbf{a}^k_{(j)}}$, for each of the three methods. They are presented in fig. 8.12. These results confirm the superior performance of the Jeffreys prior compared to the other two methods.

8.7.3 Case study 3: the stacked structure for storage of packages

The case study considered hereafter concerns a freestanding stacked structure composed of three pallets intended for the storage of packages. A complete description of the system was proposed in the chapter 7. In fig. 8.13-left. For this case study, only $k = 21$ experimental results have been performed on a shaking table. An example of a result is shown in fig. 8.13-right, which depicts the horizontal displacements over time of the top of the stack. The initial position is indicated in red, while the different positions in time are indicated in blue. Due to uplift, sliding, and rotation motions, the top of the stack exceeds, for 2 of the 21 tests, the

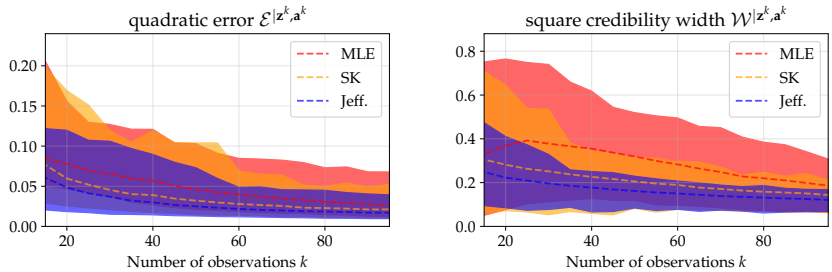

Figure 8.12: Average values of the performance evaluation metrics $\mathcal{E}^{|\mathbf{z}^k,\mathbf{a}^k}$ (left) and $\mathcal{W}^{|\mathbf{z}^k,\mathbf{a}^k}$ (right) for the piping system computed by replications of the experiment with different observation sets and for different sample sizes, ranging from $k = 15$ to 100. In each figure, the shaded areas show the 95%-confidence intervals of the metrics.

admissibility criterion, which is materialized in black. Due to the signals used for these tests, the Jeffreys prior is suitable for estimating a fragility curve of the stack, as is the SK prior calibrated to approximate the Jeffreys prior.

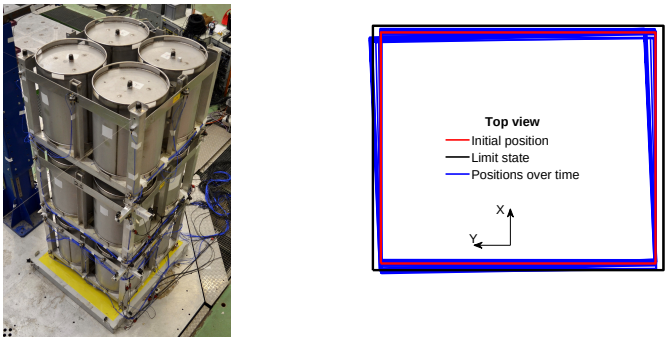

Figure 8.13: (left) Overview of the stacked structure placed on the Vesuve 1D shaking table and (right) example of test result: horizontal displacements of the top of the stack when subjected to seismic excitation in the X direction.

The small number of data points does not allow comparison with a reference. Therefore, only a qualitative study can be carried out. Figure 8.14 shows the results from 5000 estimations of θ performed with the three statistical methods considered in this study. These results do not call into question the previous ones because they, once again, highlight the same phenomena, namely: (i) some irregularities of the MLE-based approach characterized by "quasi-vertical" confidence intervals; (ii) wider credibility intervals with the SK prior than with the Jeffreys prior; and (iii) the generation of many outliers of θ with the SK posterior compared to the Jeffreys posterior, which explains the slightly larger credibility intervals on the log-normal estimates of the fragility curves.

8.8 Appendix: a review of the properties of the SK posterior

In this paper, we have compared our approach with the one that results from an adaptation of the prior suggested by Straub and Der Kiureghian (2008). We proved in section 8.5 that within our framework, this prior results in an improper posterior. This puts the validity of the MCMC estimations into question, and could explain the lower performance of the SK prior compared to the Jeffreys prior. In Straub and Der Kiureghian, 2008, the authors use the Bayesian methodology the same way we do, yet the consideration of uncertainties over the observed earthquake intensity measures and the equipment capacities leads to a slightly different likelihood. In order to verify that the drawbacks of their prior highlighted in this paper are not due to our statistical choices, we dedicated this appendix to the study of the asymptotic expansions of

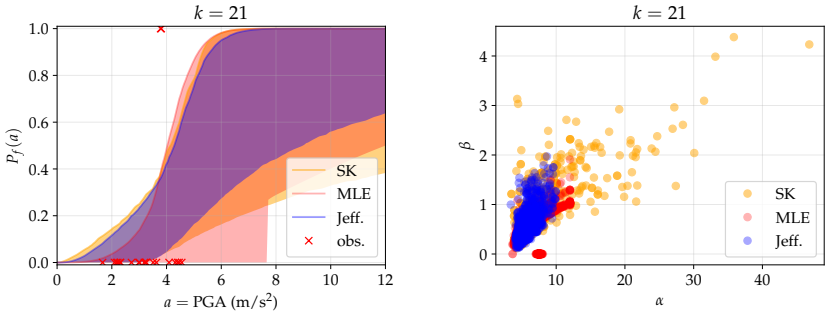

Figure 8.14: Left: 95% credibility (for Bayesian estimation) or confidence (for the MLE) intervals of fragility curve estimation for the stacked structure. The red crosses represent the observed data. Right: scatter plot of the 5000 generated θ used for the computation of the credibility or confidence zones depicted in the left figure.

the posterior in the exact framework presented in Straub and Der Kiureghian, 2008. We shall first introduce the exact model of their paper for the estimation of seismic fragility curves in section 8.8.1, using notations consistent with our study. We will then derive the likelihood and its asymptotics in section 8.8.2. Finally, we will express the convergence rates of the posterior in section 8.8.3, which will allow us to conclude that the SK posterior is indeed improper.

8.8.1 Statistical model and likelihood

Let us consider the observations of earthquakes labeled $l = 1, \dots, L$ at equipment labeled $i = 1, \dots, I_j$ located in substations labeled $j = 1, \dots, J$. The observed items are $(\mathbf{z}_{jl}, \hat{a}_{jl})_{j,l}$, $\mathbf{z}_{jl} = (z_{ijl})_i$ being the failure occurrences of the I_j pieces of equipment at substation j during earthquake l ($z_{ijl} \in \{0, 1\}$), and $\hat{a}_{jl}$ being the observed IM at substation j during earthquake l ($\hat{a}_{jl} \in (0, +\infty)$). They are assumed to follow the latent model presented below.

At substation j the l-th earthquake results in an IM value a_{jl} that is observed with an uncertainty multiplicative noise: $\log \hat{a}_{jl} = \log a_{jl} + \varepsilon_{jl}$ where $\varepsilon_{jl} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. The noise variance σ_ε^2 is supposed to be known. The uncertain intrinsic capacity of equipment i at substation j is $r_{ij} \sim \mathcal{N}(\mu_r, \sigma_r^2)$ and $y_{jl} \sim \mathcal{N}(0, \sigma_y^2)$ is the uncertain factor common to all equipment capacities at substation j during earthquake l. The random variables r_{ij}, y_{jl} and ε_{jl} are supposed to be independent.

A failure of equipment i at substation j during earthquake l is considered when the performance of the structural component g_{ijl} satisfies $g_{ijl} > 0$.This performance can be expressed as

$$g_{ijl} = \log \hat{a}_{jl} + \varepsilon_{jl} - y_{jl} - r_{ij} = x_{jl} - r_{ij}$$

with $x_{jl} = \log \hat{a}_{jl} + \varepsilon_{jl} - y_{jl}$.

This establishes the following conditional relation between the observed data:

$$p(z_{ijl}|\hat{a}_{jl}, \Sigma) = \int_{\mathbb{R}} p(z_{ijl}|x_{jl}, \hat{a}_{jl}, \Sigma) \frac{\exp\left(-\frac{(x_{jl} - \log \hat{a}_{jl})^2}{2(\sigma_\varepsilon^2 + \sigma_y^2)}\right)}{\sqrt{2\pi(\sigma_\varepsilon^2 + \sigma_y^2)}} dx_{jl}, \tag{8.71}$$

denoting $\Sigma = (\sigma_r, \sigma_y, \mu_r)$, and with

$$p(z_{ijl}|x_{jl}, \hat{a}_{jl}, \Sigma) = \Phi\left(\frac{x_{jl} - \mu_r}{\sigma_r}\right)^{z_{ijl}} \left(1 - \Phi\left(\frac{x_{jl} - \mu_r}{\sigma_r}\right)\right)^{1 - z_{ijl}}, \tag{8.72}$$

when substation j is only affected by one earthquake. The method proposed in Straub and Der Kiureghian, 2008 actually considers cases in which a substation may be impacted by two successive earthquakes and takes into account the fact that its response to the second would be correlated to its response to the first one. This

would lead to a different likelihood. However, it is mentioned that this possibility only concerns a small number of data points. We can therefore limit our calculations to the simplest case and assume $l = L = 1$. The subscript l will therefore be dropped in what follows.

Finally, the likelihood for this model can be expressed as:

$$\ell_J(\mathbf{z}|\hat{\mathbf{a}},\Sigma) = \prod_{j=1}^{J} \int_{\mathbb{R}} \prod_{i=1}^{I_j} p(z_{ij}|x_j, \log \hat{a}_j, \Sigma) \frac{\exp\left(-\frac{(x_j - \log \hat{a}_j)^2}{2(\sigma_\varepsilon^2 + \sigma_y^2)}\right)}{\sqrt{2\pi(\sigma_\varepsilon^2 + \sigma_y^2)}} dx_j, \tag{8.73}$$

denoting $\mathbf{z} = (\mathbf{z}_j)_{j=1}^J$, $\hat{\mathbf{a}} = (\hat{a}_j)_{j=1}^J$, and with the integrated conditional distribution defined in eq. (8.72).

In the Bayesian framework introduced in Straub and Der Kiureghian, 2008, the model parameter is Σ. Let us denote $\alpha = \exp \mu_r$, $\beta = \sqrt{\sigma_r^2 + \sigma_y^2}$ and $\rho = \sigma_y^2/\beta^2$. Denoting $\theta = (\alpha, \beta, \rho)$, the knowledge of Σ then becomes equivalent to the one of θ and the likelihood of eq. (8.73) can be expressed conditionally to θ instead of Σ:

$$\ell_J(\mathbf{z}|\hat{\mathbf{a}},\theta) = \prod_{j=1}^{J} \int_{\mathbb{R}} \prod_{i=1}^{I_j} \Psi^{z_{ij}}\left(\frac{x - \log \alpha}{\beta\sqrt{1-\rho}}\right) \frac{\exp\left(-\frac{(x - \log \hat{a}_j)^2}{2(\sigma_\varepsilon^2 + \rho\beta^2)}\right)}{\sqrt{2\pi(\sigma_\varepsilon^2 + \rho\beta^2)}} dx, \tag{8.74}$$

where the notation $\Psi^{z_{ij}}(\gamma)$ is used to denote $\Phi(\gamma)^{z_{ij}}(1 - \Phi(\gamma))^{1-z_{ij}}$.

Straub and Der Kiureghian (2008) propose the following improper prior distribution for the parameter θ:

$$\pi_{SK}(\theta) \propto \frac{1}{\beta\alpha} \exp\left(-\frac{(\log \alpha - \mu)^2}{2\sigma^2}\right) \mathbb{1}_{0 \le \rho \le 1}. \tag{8.75}$$

A posteriori estimations of θ are consequently generated from MCMC methods

$$p(\theta|\mathbf{z},\hat{\mathbf{a}}) \propto \ell_J(\mathbf{z}|\hat{\mathbf{a}},\theta)\pi_{SK}(\theta). \tag{8.76}$$

8.8.2 Likelihood asymptotics

In this appendix, we will study the asymptotics of the likelihood defined in eq. (8.74) when $\beta \longrightarrow \infty$. Let us first consider the substitution $u = (x - \log \hat{a}_j)/\sqrt{\sigma_\varepsilon^2 + \rho\beta^2}$ to express the likelihood as

$$\ell_J(\mathbf{z}|\hat{\mathbf{a}},\theta) = \prod_{j=1}^{J} \int_{\mathbb{R}} f_j^\beta(u) du, \quad \text{with} \quad f_j^\beta(u) = \prod_{i=1}^{I_j} \Phi(h_j^\beta(u))^{z_{ij}} (1 - \Phi(h_j^\beta(u)))^{1-z_{ij}} \frac{e^{-\frac{u^2}{2}}}{\sqrt{2\pi}}, \tag{8.77}$$

with

$$h_j^\beta(u) = \frac{(u + \log \hat{a}_j)\sqrt{\sigma_\varepsilon^2 + \rho\beta^2} - \log \alpha}{\beta\sqrt{1-\rho}}. \tag{8.78}$$

This way, remembering that $0 \le \Phi(1 - \Phi) \le 1$, an upper bound $u \mapsto e^{-u^2/2}/\sqrt{2\pi}$ can be found for f_j^β for any β, u. It converges when $\beta \longrightarrow +\infty$ as follows:

$$\lim_{\beta\to\infty} f_j^\beta(u) = \prod_{i=1}^{I_J} \Psi^{z_{ij}}\left(\frac{(u + \log \hat{a}_j)\sqrt{\rho}}{\sqrt{1-\rho}}\right) \frac{e^{-\frac{u^2}{2}}}{\sqrt{2\pi}}. \tag{8.79}$$

This gives the following limit for the likelihood:

$$\lim_{\beta\to\infty} \ell_J(\mathbf{z}|\hat{\mathbf{a}},\theta) = \prod_{j=1}^{J} \int_{\mathbb{R}} \prod_{i=1}^{I_j} \Psi^{z_{ij}}\left(\frac{(u + \log \hat{a}_j)\sqrt{\rho}}{\sqrt{1-\rho}}\right) \frac{e^{-\frac{u^2}{2}}}{\sqrt{2\pi}} du, \tag{8.80}$$

which is a positive quantity.

8.8.3 Posterior asymptotics

By combining eqs. (8.75), (8.76) and (8.80) we obtain the posterior asymptotics

$$p(\theta|\mathbf{z}, \hat{\mathbf{a}}) \underset{\beta \to \infty}{\sim} \frac{C}{\beta}, \tag{8.81}$$

with C being a positive constant. This makes the posterior improper w.r.t. β, with the same convergence rate as the one derived in our framework.

8.9 Conclusion

Assessing the seismic fragility of Structures and Components (SCs) is a daunting task when data is limited. The performance of the Bayesian framework in this kind of situation is well-known. Nevertheless, choosing a prior remains difficult because its impact on the *a posteriori* distribution cannot be neglected, and therefore neither can its impact on the estimation of any relevant element linked to the fragility curves.

Elaborating on the reference prior theory in order to define an objective prior, we derived, for the first time in this field of study, the Jeffreys prior for the log-normal model, with binary data that indicates the state of the structure (i.e. failure or non-failure). This prior is completely defined and thoroughly studied, and does not depend on any additional subjective choice.

This work is also an opportunity to develop a better theoretical understanding of the conditions that result in irregular fragility curves such as unit-step functions in practice. This issue is quite inevitable when data is limited, since such curves are a result of the very composition of the sample. Although this issue affects every approach, our results prove the robustness of the proposed approach over the traditional ones in terms of regularization (i.e. the absence of irregular estimation of the fragility curve) and stability (i.e. the absence of outliers when sampling the *a posteriori* distribution of the parameters). Nevertheless, we recall that the present study is limited to the generation of estimates when the observed data do not yield what we call a degenerate likelihood. In the latter case, we have proven theoretically that all the methods carried out in this chapter —including ours— are jeopardized. That statement paves the way for further work that could (i) reinforce the prior construction to suppress its posteriors' weaknesses under degenerate likelihoods; and (ii) study the probability and the methods to reduce that of the existence of degenerate likelihoods.

Although its numerical implementation is complex —more so than a prior defined as the product of two classical distributions such as log-normal distributions, for instance— it is not a major issue. As a matter of fact, since it depends solely on the distribution of the IM, the "cost" of the initial calculation would quickly be recovered on the scale of an industrial installation containing several SCs whose fragility curves must be estimated. For example, compared to methodologies that aim to define a prior based on mechanical calculations for a given SC, the advantage of the Jeffreys prior lies in its generic nature. The fact that it can be applied to all SCs subjected to the same seismic scenario largely compensates for the implementation of mechanical studies dedicated to each relevant SC. Additionally, the methodology can be implemented with any relevant IM without creating additional complexity.

9

Integration of a properly constrained reference prior

This chapter draws mainly on the published work: Nils Baillie, Antoine Van Biesbroeck, Cyril Feau, and Clément Gauchy (2025). "Bayesian estimation of seismic fragility curves based on variational reference priors using neural networks". In: *Proceedings of the 6th UNCECOMP Conference*

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intro

The rsults in the appendix... the curse of degeneracy prevents from using a more correlated IM. Which is paradoxal since, ..

10

Design of experiments in a Bayesian framework

This chapter draws mainly on the submitted work: Antoine Van Biesbroeck, Clément Gauchy, Cyril Feau, and Josselin Garnier (Mar. 10, 2025b). *Robust a posteriori estimation of probit-lognormal seismic fragility curves via sequential design of experiments and constrained reference prior*. DOI: 10.48550/arXiv.2503.07343

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III

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Appendix

A

Influence of the choice of the IM on fragility curves estimation in a Bayesian framework based on reference prior

??. *Influence of the choice of the IM on fragility curves estimation in a Bayesian framework based on reference prior*

B

Design of experiments on a low fidelity model of seismic fragility curves

This appendix is a postprint of the published work: Antoine Van Biesbroeck, Clément Gauchy, Cyril Feau, and Josselin Garnier (2025a). "Design of experiment based on a low fidelity model for seismic fragility estimation". In: *ESAIM: ProcS* ((to appear)). URL: https://hal.science/hal-04719458v1

??. *Design of experiments on a low fidelity model of seismic fragility curves*

C

Non-asymptotic reference priors, a brief study

??. *Non-asymptotic reference priors, a brief study*

D

Reminder of useful mathematical notions

??. *Reminder of useful mathematical notions*

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